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List of Symbols, Abbrevations and Station Numbers

Table A - Symbols

Symbol	Explanation	Unit
A	Cross-sectional area	[m^2]
a	Acceleration	[m/s^2]
a	Speed of Sound	[m/s]
c_p	Specific heat at constant pressure	[J/(kg K)], [J/(kmol K)]
c_v	Specific heat at constant volume	[J/(kg K)]
C	Heat capacity	[J/kg]
D	Drag	[N]
E	Emission index	[g/kg]
F	(Resultant) force	[N]
F_G	Gross thrust	[N]
F_N	Net thrust	[N]
H	Absolute enthalpy	[J]
h	Specific enthalpy	[J/kg]
h_0	Total enthalpy	[J/kg]
h_f^0	Enthalpy of formation at standard state	[J/kg], [J/kmol]
h_i	Mass or molar specific enthalpy of species i	[J/kg], [J/mol]
Δh_c	Specific enthalpy of combustion	[J/kg], [J/kmol]
Δh_s	Sensible enthalpy change	[J/kg], [J/kmol]
k	Ratio of specific heats	[-]
L	Lift	[N]
LHV	Lower heating value	[J/kg]
M	Mach number	[-]
M_i	Molecular weight of species i	[kg/kmol]
m	Mass	[kg]
$\dot{m}$	Mass flow	[kg/s]
n_i	Molar ratio of species i per mole of fuel	[kmol/kmol]
n	Number of compressor stages	[-]
n_{air}	Polytropic exponent for air	[-]

p	Pressure	[Pa], [atm]
p_0	Total pressure	[Pa]
Δp	Pressure loss	[Pa], [atm]
$\dot{Q}$	Heat	[W]
Q_s	Specific heat	[W/(kg/s)]
R	Universal gas constant	[J/(kg K)]
s	Specific entropy	[J/(kg K)]
sfc	Specific fuel consumption	[kg/(W s)]
T	Temperature	[K]
T_0	Total temperature	[K]
TSFC	Thrust specific fuel consumption	[kg/(N s)]
v	Velocity	[m/s]
$\dot{W}$	Power	[W]
$\dot{W}_{gg}$	Gas power	[W]
$\dot{W}_{s,gg}$	Specific gas power	[W/(kg/s)]
X_i	Mole fraction of species i	[-]
x/y	Number of carbon/ hydrogen atoms in a fuel molecule	[-]
Λ	Degree of reaction	[-]
Π	Pressure ratio	[-]
Π_{opt}	Optimum pressure ratio	[-]
Π_{cc}	Combustion chamber pressure loss	[-]
Θ	Temperature ratio	[-]
η_{th}	Thermodynamic efficiency	[-]
η_{thm}	Thermal efficiency	[-]
$\eta_{complete}$	Factor of complete combustion	[-]
λ	Percent excess air	[%]
ν	Specific volume	[kg/m^3]
φ	Fuel-to-air equivalence ratio	[-]
ρ	Density	[kg/m^3]
σ	Stress	[N/m^2]

Table B - Subscripts and Acronyms

Subscript/Acronym	**Explanation**
0	Entry
1EO	First engine order
∞	Polytropic
ad	Adiabatic
AC	Alternating Current
AFR	Air-to-Fuel Ratio
ARP	Aerospace Recommended Practice
amb	Ambient
cc	Combustion chamber
comb	Combustion (process)
comp	Compressor
CTE	Coefficient of thermal expansion
CVD	Chemical vapour deposition
d	Diffuser
DS	Directionally solidified
DT	Damage tolerance
EGT	Exhaust gas temperature
EIFS	Effective initial flaw sizes
f	Fuel
FAR	Fuel-to-air ratio
FOD	Foreign object damage
g	Gas
g	Post-combustion gases
gg	Gas generator
HCF	High Cycle Fatigue
HIP	Hot Isostatic Pressing
HPC	High pressure compressor
HPC	High pressure compressor
HVOF	High velocity oxygen fuel
ICAO	International Civil Aviation Organization
is	Isentropic
l	Liner
LCF	Low Cycle Fatigue
LCV	Low Calorific Value
LHV	Lower Heating Value

LNG	Liquefied natural gas
LPC	Low pressure compressor
LPC	Low pressure compressor
LPPS	Low pressure plasma spraying
mix	Mixture
NDI	Non-destructive inspection
NOx	Oxides of nitrogen
ODS	Oxide dispersion strengthened alloy
opt	Optimum
PH / SH / DH	Primary / Secondary / Dilution hole
pinch	Pinch point
Ppm	Parts per million
prod	Products
prop	Propulsion/ propulsive
PT	Power turbine
PT	Power turbine
PVD	Physical vapour deposition
PZ / SZ / DZ	Primary / Secondary / Dilution zone
PZ, SZ, DZ	Primary, Secondary and Dilution zones
react	Reactants
ref	Reference state
RFC	Retirement for Cause
s	Specific
SFC	Specific fuel consumption
stoich	Stoichiometric
sw	Swirler
TBC	Thermal barrier coatings
th	Thermodynamic
thm	Thermal
TIT	Turbine inlet temperature
tot	Total quantity (opposed to static properties)
tot	Total
TSFC	Thrust specific fuel consumption
turb	Turbine
UHC	Unburned hydrocarbons

The subsequent stations in a gas turbine are numbered according to the ARP755 published by the Society of Automotive Engineers (SAE). A list of the station numbers is given in Table C while a more throughout explanation on station numbering can be found in Appendix A. Please note that all italic items are optional to a core gas turbine.

Table C - Station Numbers (according to SAE manual ARP 755)

Station Number	**Location in Gas Turbine**
0	Ambient/ undisturbed
1	Aircraft-engine interface/ inlet face
2	First compressor inlet
21	*Inner stream fan exit*
13	*Outer stream fan exit*
16	*Bypass exit*
161	*Cold side mixer inlet*
163	*Cold side mixing plane*
18	*Bypass nozzle throat*
24	*Intermediate compressor exit*
25	*High-pressure compressor inlet*
3	Last compressor exit/ cold side heat exchanger inlet
31	*Burner inlet*
35	*Cold side heat exchanger exit*
4	Burner exit
41	*First turbine stator exit = rotor inlet*
*43**	*High-pressure turbine exit before addition of cooling air*
*44**	*High-pressure turbine exit after addition of cooling air*
*45**	*Low pressure turbine inlet*
*49**	*Low-pressure turbine exit before addition of cooling air*
*42***	*High-pressure turbine exit before addition of cooling air*
*43***	*High-pressure turbine exit after addition of cooling air*
*44***	*Intermediate turbine inlet*
*45***	*Intermediate turbine stator exit*
*46***	*Intermediate turbine exit before addition of cooling air*

*47***	*Intermediate turbine exit after addition of cooling air*
*48***	*Low-pressure turbine inlet*
*49***	*Low-pressure turbine exit before addition of cooling air*
5	Low-pressure turbine exit after addition of cooling air
6	Jet pipe inlet, reheat entry for turbojet, hot side heat exchanger inlet
61	*Hot side mixer inlet*
63	*Hot side mixing plane*
64	*Mixed flow, reheat entry*
7	Reheat exit, hot side heat exchanger exit
8	Nozzle throat
9	Nozzle exit (convergent-divergent nozzle only)

(* two spool engines)

(** three spool engines)

1. Introduction

(Prof. Ir. Jos P. van Buijtenen, Dr. Wilfried P.J. Visser and Dr. Arvind Gangoli Rao)

1.1 The Gas Turbine Engine Concept

The gas turbine engine is a machine delivering mechanical power (or thrust in case of a jet engine) using a gaseous working fluid. It is an *internal combustion engine* like the *reciprocating* Otto- and Diesel piston engines with the major difference being that the working fluid flows through the gas turbine continuously and not intermittently. The continuous flow of the working fluid requires the compression, heat input, and expansion to take place in separate components. For that reason a gas turbine consists of at least a compressor, a combustion chamber and a turbine.

Even though a gas turbine engine consists of more components than just a turbine, it is named after that single component. This is for historical reasons because the gas turbine was developed as an alternative for the steam turbine. The compression component of a steam cycle, the water pump, usually receives far less attention than the gas expansion component (i.e. the turbine). More obvious designations for the gas turbine and its components would be *turbo compressor*, and *turbo expander* respectively for the compression and the expansion part, and *turbo engine* for the whole engine.

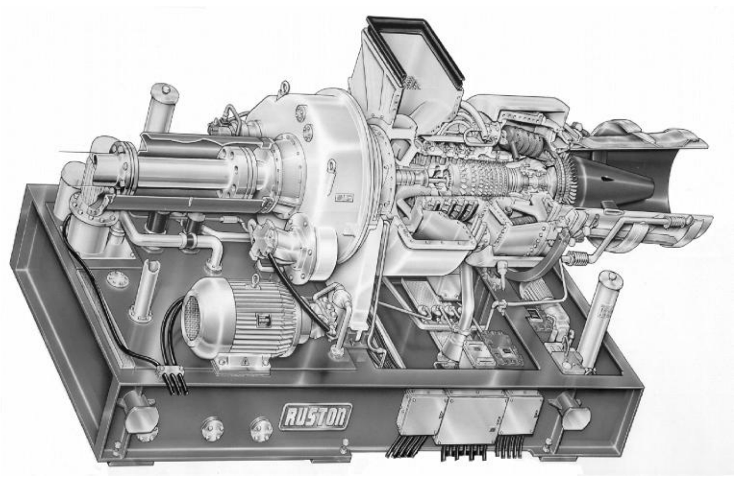


Figure 1-1 - Alstom Typhoon (previously Ruston) 4900 kW single shaft gas turbine for generator drive

Figure 1-1 shows a gas turbine delivering shaft power, consisting of a single compressor, combustion chamber and turbine. Figure 1-2 shows a “turbofan” jet engine used for aircraft propulsion.

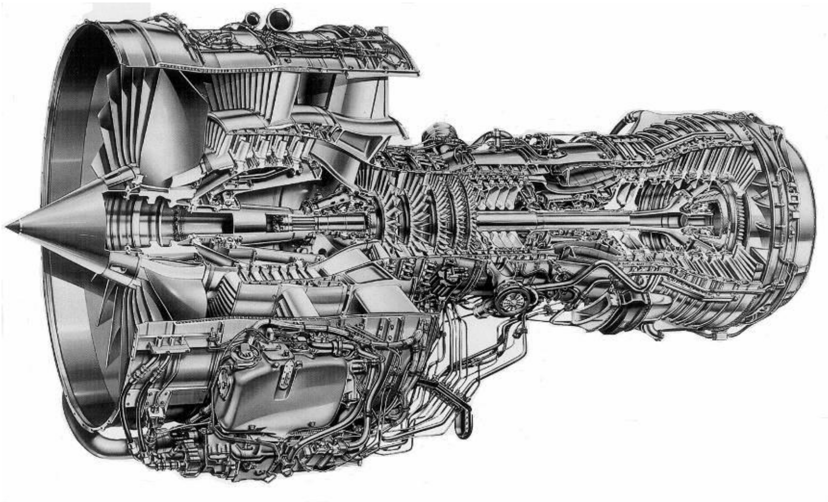


Figure 1-2 - IAE V2500 turbofan engine (application: Airbus A320 and other aircraft)

Gas turbine configurations may differ due to the use of different types of components. There are both axial and radial compressors and turbines, referring to the main direction of flow inside the component. In axial components, the airflow is axial (parallel to the rotor drive shaft) through the component. While in radial components, the flow is diverted from an axial to a radial direction (in case of a radial compressor), and vice versa for radial turbines / turbine components. Also, combustion chambers come in various types: multiple small combustion chambers or annular type combustion chambers (for example Figure 1-6). The different types of compressors, turbines, and combustion chambers will be discussed in more detail in the following chapters.

Since the compressor is driven by the turbine, a large part of the energy available from the high pressure and high temperature gas (at the end of the combustion chamber) is extracted by the turbine to drive the compressor. The remaining energy of the gas is primarily extracted by two means, either by futher expanding the hot gases in a turbine and using the mechanical work for usefule purposes or by expanding the hot gases in a nozzle (thereby converting the energy contained in the gas it into kinetic energy and using it for producing thrust).

The free power turbine, shown in Figure 1-3, converts the potential energy of the gas generator exhaust gas into mechanical work. The shaft of the free power turbine can be used to drive a propeller (aircraft or ship), a pump, or a helicopter rotor (Figure 1-4), etc.

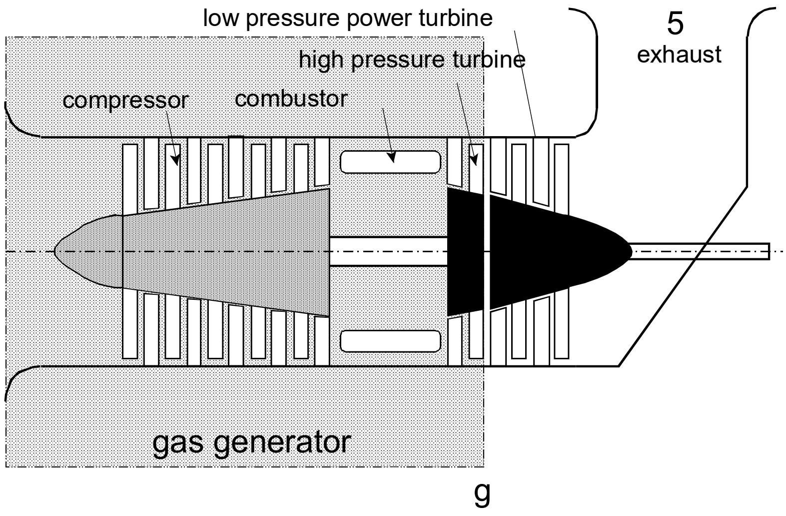


Figure 1-3 - Free power turbine configuration

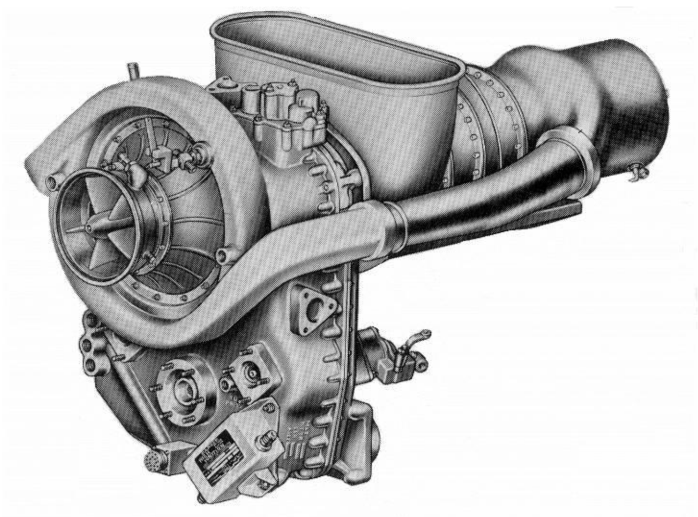


Figure 1-4 - Allison C250 485 kW free power turbine configuration for helicopter propulsion (Bo107/115 helicopter)

The high-pressure gas can also be converted into kinetic energy by expansion in a nozzle or jet pipe for aircraft propulsion (Figure 1-6). The various power conversion processes will be further addressed in the following chapters.

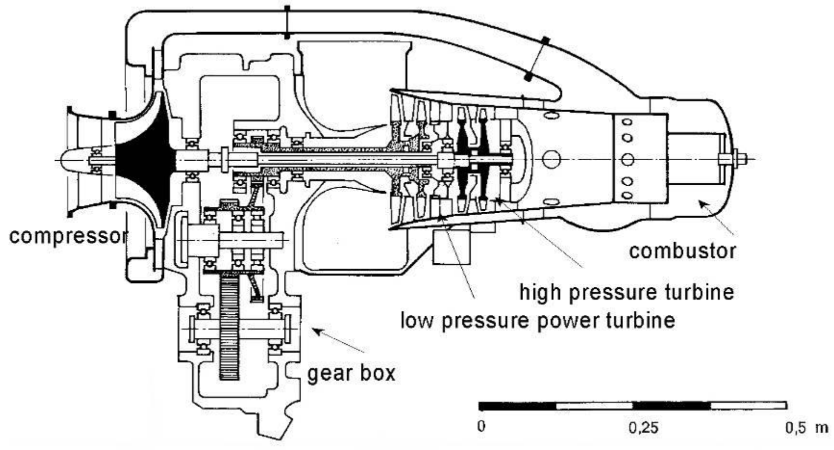


Figure 1-5 - Longitudinal cross-section of Allison C250 gas turbine

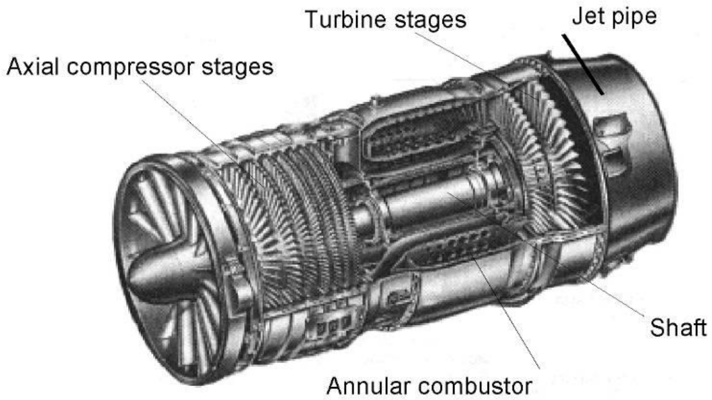


Figure 1-6 - General Electric J-85 turbojet engine

1.2 History

The history of the gas turbine is, when compared to the steam turbine and the Otto- and Diesel piston engines, relatively young. The first (usable) steam turbines were already built during the second half of the 19th century by De Laval, Parsons, Curtis and others. The first practically useful gas turbine engines emerged at the beginning of the 20th century but large-scale application only started after WWII. The reason is the specific nature of the gas turbine thermodynamic process. All gas or steam cycle processes, produce useful power only if the power required for compression is less than the power delivered by expansion. In a steam cycle the compression power of the feed water is relatively low (as water is an incompressible fluid) and losses do not play a significant role. The highest process (steam) temperature is limited, but when using a condenser the pressure ratio for expansion of the steam is high. The compression power of the gas turbine cycle however, is very high. For the expansion of the gas, a pressure ratio equal to the compression pressure ratio minus some pressure losses is available. This means any surplus turbine power (the difference between compression and expansion power) can only be the result of the higher temperature level (compared to compressor entry temperature) at the start of the expansion in the turbine. Gas turbine compression power approximately is 2/3rd of the expansion power used for driving the compressor. This means useful power is the difference between two large values and this makes losses in the compression- and expansion processes very significant for overall efficiency.

1.2.1 The First Industrial Gas Turbines

The first experimental gas turbine engines were not able to run self-sustained, but required an external power source. Only in 1905, the Frenchman Rateau built a gas turbine that actually delivered shaft power with 25 centrifugal compressor stages delivering a pressure ratio of 3. This pressure ratio would normally not suffice for a gas turbine to deliver power, but with an extremely high combustion temperature combined with water-cooled turbine blades, Rateau managed to generate some useful power. However, the thermal efficiency of this gas turbine was only 3.5%. Further development of the gas turbine continued, especially in Switzerland by Prof. Stodola of the University of Zurich and manufacturer Asea Brown Boveri (currently named ABB). Brown Boveri pioneered in the development of gas turbines for electrical power generation and other industrial applications. The first gas turbine for power generation became operational in 1939 in Neufchateau, Switzerland (Figure 1-7).

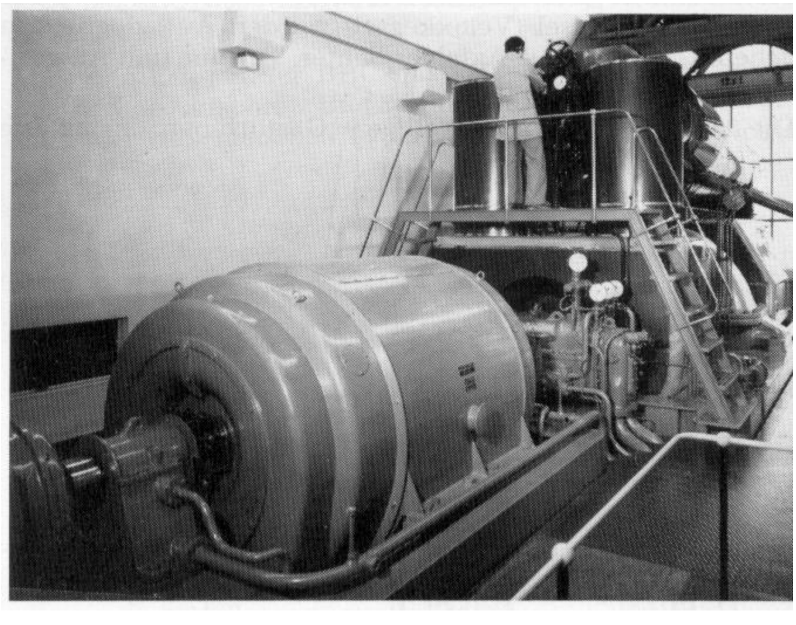


Figure 1-7 - Brown-Boveri industrial 4 MW gas turbine in 1939

The gas turbines of the early years were mainly used to provide power at peak loads. This is because the gas turbine can start up relatively quickly, requires relatively low investment costs and short production times. The low thermal efficiency as compared to steam turbines was of less concern due to the relatively small number of peak load operating hours.

Only during the 1980's, the gas turbine had its breakthrough in the power generation application. This happened due to the availability of natural gas as a fuel, which made the gas turbine particularly attractive for integration in existing natural gas fired power stations into a combined cycle unit. Also in cogeneration installations for industries consuming large amounts of heat, the gas turbine became very popular. The modern combined cycle power plants are one of the most efficient heat engines with an overall thermal to electric efficiency at design conditions exceeding 62%. Most of the industrial gas turbine manufacturers are now aiming at achieving 65% efficiency in the next few years.

1.2.2 The First Jet Engines

In the same period that the gas turbine developed for power generation and industrial applications, Sir Frank Whittle (England), Hans von Ohain, Herbert Wagner, and Helmut Schelp (Germany) independently started the development of a jet engine gas turbine for aircraft propulsion.
Frank Whittle, working as a flying officer in the Royal Air Force at that time, first considered the concept of the gas turbine as a jet engine in 1929 and was the first to claim a patent on the concept in 1930. Whittle set a target to design an aircraft engine capable of operating at higher altitudes and speeds (up to 900 km/h), which were far beyond the operating limits of piston engines and propellers. The British government as well as the British aircraft engine manufacturers did not share Whittle's enthusiasm and did not support Whittle financially nor technically. In 1936, Whittle along with some friends and investors established a company called "Power Jets Limited". In spite of many technological problems and a lack of funds, Frank Whittle eventually managed to build his first gas turbine. During the late 30's, Whittle drew attention with an engine running on a test bed and suddenly got financial support from the British government. After getting the financial support, Sir Frank Whittle was able to rapidly solve technological difficulties and finally built his first jet engine for the Gloster E28 aircraft in the year 1941. This successful achievement resulted in further development of Whittle's jet engine design by others (Rover, Rolls Royce and General Electric). The first operational British jet fighter, the Gloster Meteor, flew in August 1944 and was initially used for intercepting German V-1 missiles.

Although Sir Frank Whittle was the first to register a patent for the jet engine concept, it was Hans von Ohain who first built a gas turbine in a jet engine configuration. After completion of his study in physics and aerodynamics from the University of Göttingen in 1935, Von Ohain started to work for aircraft constructor Ernst Heinkel. Due to Heinkel's desire to build the world's fastest aircraft, Von Ohain received substantial support needed to develop a jet engine. In 1937, Von Ohain designed a simple gas turbine with a radial compressor, a combustor running on hydrogen and a radial turbine. After a number of successful tests, Von Ohain received more support from Heinkel, enabling him to demonstrate the historic first flight of the jet engine powered Heinkel He-178 aircraft in 1939. Von Ohain not only proved the concept of jet propulsion, but also proved that with a jet engine, very favorable thrust-to-weight ratios can be achieved when compared to piston engines with propellers.

During the same time period in Germany, Herbert Wagner and Helmut Schelp worked on the development of gas turbine jet engines. Helmut Schelp contributed to the development of the successful and first operational Messerschmidt Me-262 jet fighter and Helmut Wagner worked for Junkers on a gas turbine driving a propeller.

1.2.3 Gas Turbine Research and Development

After the WWII, the gas turbine rapidly developed as a powerful new alternative for industrial and aircraft applications. The development of high-temperature materials and later supported by sophisticated cooling techniques, enabled the gas turbine to operate at higher turbine inlet temperatures. Extensive research in the aerodynamics improved the efficiencies of compressors and turbines. With the development of new gas turbine configurations (e.g. turbofan aircraft engines and combined-cycle concepts for stationary applications), which further improved performance and efficiency, the gas turbine engiens have become the primary choice for many applications.

Currently, gas turbine research and development is focused on many different disciplines. The most important ones are:

- **Aerodynamics:** compressor and turbine stage efficiency and loading, cooling techniques, clearance control, noise, etc.
- **Materials:** high-temperature alloys, strength, life, coatings, ceramics matrix composite for high temperatures, composite fan blades, etc.
- **Combustion:** high-efficient, stable operation, low-emission, combustion in short and small combustors.
- **System performance:** cycle optimization, combined cycle concepts, new engine concepts & architectures.

1.3 Application Areas

In section 1.1 the concept of the gas turbine has been explained as a gas generator providing hot, high-pressure gas. The way the energy in the hot gas (i.e. the ‘gas power’) is used depends on the application. This means that in general, the gas generator may be considered a subsystem that all gas turbine engines have in common while the systems converting the gas power can be very different. Although all gas generators have the same function and most will have the same/similar configuration, significant differences exist also for the gas generator depending on the applications. These usually result from requirements with respect to

- Power output (ranging from several tens of megawatts for the larger aircraft gas turbines to small gas turbines for RC aircraft. For industrial gas turbines ranging from several hundreds of megawatts for large power generation heavy-duty gas turbines to small kilowatts range gas turbines for domestic combined heat and power applications)
- Volume and weight (e.g. for aerospace applications).

- Operating profile (e.g. electricity base load generation with almost constant operating conditions and power setting or the usually large variations in power setting in a helicopter or a fighter aircraft).
- Fuel type.
- Emissions of pollutant exhaust gasses and noise.
- Operating conditions (corrosion, erosion), etc.

The diversity in requirements and consequences for the design has led to a division into separate groups of gas turbine manufacturers for aircraft gas turbines and industrial gas turbines.

1.4 Gas Turbine Engine Manufacturers

The largest manufacturer for industrial gas turbines at the moment is General Electric – USA (GE). GE's share of the market is almost 70 percent. The other manufacturers share the remaining part of the market; among them are Alstom (several European countries, includes former Asea Brown Boveri (ABB), Alsthom, European Gas Turbines), Siemens from Germany (includes KWU and Westinghouse from USA), Mitsubishi Heavy Industries in Japan and several other small manufacturers. Recently, a major part of Alstom was purchased by GE and the remaining part was purchased by Ansaldo Enrgia. Due to the rise in the renewables, the market for industrial gas turbines has shrunk in the recent years.

GE is also the largest manufacturer of aircraft gas turbines, followed by Rolls Royce (UK, includes Allison), Pratt & Whitney (USA/Canada), Honeywell (USA, includes Allied Signal and Garret), Snecma (France, includes Turbomeca), MTU (Germany), Avio (Italy), Japanese Aero Engine Corporation (JAEC), and some other small manufacturers.

Big:	General Electric	USA
	Pratt & Whitney	USA
	Rolls-Royce	UK
'Medium':	Garret Airesearch (currently Honeywell)	USA
	Allison (owned by Rolls-Royce)	USA
	Honeywell (was Allied Signal)	USA
	MTU	Germany
	Snecma (Safran Aircraft Engines)	France
	Turbomeca (Safran helicopter engines)	France
	AVIO (Now GE)	Italy
	Japanese Aeroengine Corp.	Japan
	Volvo (Now GKN)	Sweden

Overview of the western aircraft engine manufacturers

The costs and also the risks of R&D for new advanced gas turbines are very high and have forced many manufacturers to collaborate with other manufacturers. Sometimes a manufacturer develops a new engine, and other companies develop one or more modules. Sometime joint ventures have been established with several partners and engines are designed and produced under the new joint venture name. Examples of collaborations are:

- CFM with GE and Safran Aircraft Engines: Products include CFM-56 engine and Leap engines.
- GE with Safran Aircraft Engines, IHI and FiatAvio (GE90 turbofan engine for the B777),
- IAE (International Aero Engines, Rolls-Royce, Pratt & Whitney (USA), JAEC, FiatAvio and MTU united in 1983 to develop the IAE-V2500 engine, see Figure 1-2),
- Turbo-Union (Rolls-Royce, FiatAvio and MTU (RB199 for the Panavia Tornado),
- BWM-RR (Rolls Royce and BMW (regional and business jet BR700 series engines).

The Russian industrial and aircraft gas turbine industry is significant in size, but, since the end of the Soviet Union is still struggling to become competitive with the other manufacturers.

The CFM international has been one of the most successful engine manufacturer in the single aisle aircraft market. Some of the highlights of CFM have been mentioned below

- CFM International is a 50:50 joint venture between General Electric and Snecma (Safran Aircraft Engines)
- It was founded in 1974
- The company is most famous for building CFM56 turbofans
- CFM engine is derived from the GE F101 engine.
- 12,000 commercial and military aircraft including the Airbus A320 and Boeing 737 Families.
- More than 28,000 CFM56s have been built since its introduction to the market in 1982.
- This is the most successful engine in the history.
- GE makes the engine core (HPC, combustor and HPT)
- Safran Aircraft Engines make the Low pressure systems (Fan, LPC, LPT).
- The CFM56 line has six variants CFM56-2, CFM56-3, CFM56-5A, CFM56-5C,CFM56-5B and CFM56-7B
- The Leap-X Engine is the successor of CFM 56, used for A320 neo, 737 max and C919.

1.5 Gas Turbine Performance

Aircraft gas turbines are manufactured in a wide thrust range. From small gas turbines for remotely piloted aircraft with 40 to 100 Newtons of thrust up to about 400 kN (Rolls-Royce Trent, GE90). Industrial gas turbines range from 200 kW (Kawasaki) up to 240 MW (ABB).

Several aircraft gas turbine designs have derivatives for stationary applications on the ground. These usually are referred to as ‘aeroderived’ industrial gas turbines. Examples are the aeroderived versions of the Rolls-Royce Avon, Spey, Olympus, RB211 and Trent engines. The

GE LM2500 and LM6000 industrial gas turbines are 'aeroderivatives' of the CF6-50 and CF6-80 engines respectively.

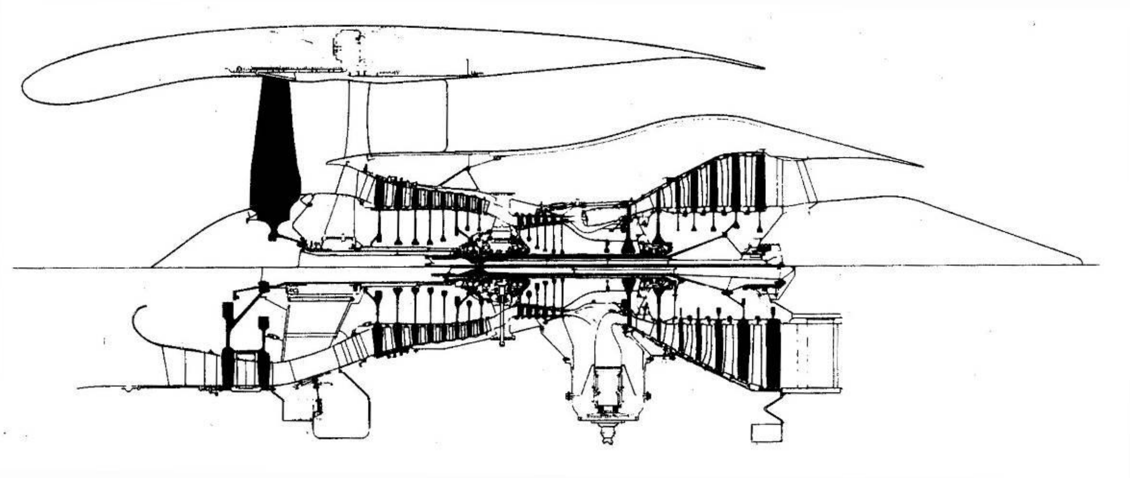


Figure 1-8 - Rolls-Royce Trent turbofan (top) and 'aeroderived' turboshaft (bottom)

If the large fan at the front and the exhaust nozzle at the end of the turbofan in Figure 1-8 would be removed, a gas generator or 'core engine' remains capable of providing gas power applications other than providing thrust to an aircraft. The lower half of Figure 1-8 is an image of the 'aeroderived' industrial version of the RB211 engine: with a suitable inlet and the low-pressure turbine is coupled to a drive shaft, a *turboshaft* engine is created for delivering shaft power. The low-pressure turbine, which originally drove the fan that consumed most of the available power for generating thrust, now is used for proving shaft power. The removal of the fan, which also contributes to the compression of the gas generator, results in a small decrease in overall compression ratio. The low-pressure speed often is in the range suitable for generator drive (3000/3600 rpm for 50/60 Hz electrical AC power).

For jet engines, power output generally is specified in terms of thrust (kN or lbs). Gas turbines have emerged as the choice of propulsion system for most of the aircraft used today. Only small general aviation aircraft use reciprocating engines. This choice has been driven primarily by following factors which are in favor of gas turbines as compared to other types of prime mover

- high power to weight ratio of gas turbines
- high specific thrust as compared to other heat engines
- wide operating envelope (operation at higher altitudes and speeds)
- high reliability: as there are no reciprocating parts
- better balancing of the machine as there are no reciprocating parts
- high efficiency (modern commercial turbofans can achieve more than 50% thermal efficiency)
- architectural flexibility (single spool, twin spool, three spool, turbofans, turboshaft, turboprop, open rotor, geared turbofan, etc)

- fuel flexibility (kerosene, blends of biofuels, LNG, etc)

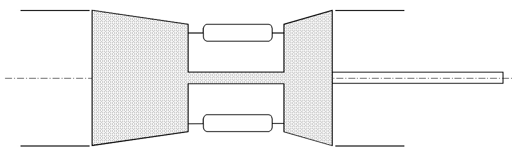
The table below shows a brief comparison between various heat engines. It can be seen that gas turbines are better than other types of prime movers on several criteria

Table 1.1 : Comparison of various prime movers

Type	Power Range	Speed Range (Mach No)	Power to weight (KW/kg)	Approx. Efficiency (%)
Military Engines	**< 15 MW**	**Upto Mach 2**	**20**	**30**
Large Turbofans	**< 50 MW**	**< Mach 1**	**10**	**40**
Micro gas turbines	**< 50 KW**	**< Mach 1**	**6**	**10**
Turbo Props	**< 10 MW**	**< Mach 0.7**	**6**	**40**
Electric Motors	**< 150 KW**	**< Mach 0.7**	**5**	**90**
Wankel Engine	**< 200 KW**	**< Mach 0.7**	**2.5**	**25**
Radial engines	**< 3000 KW**	**< Mach 0.7**	**1.8**	**30**

1.6 Gas Turbine Configurations

In the previous sections it was explained that the configuration of the gas turbine is highly dependent on the type of application. Figure 1-9 and Figure 1-10 show some common turboshaft configurations for providing shaft power. **Error! Not a valid bookmark self-reference.** and Figure 1-12 show some jet engine configurations.



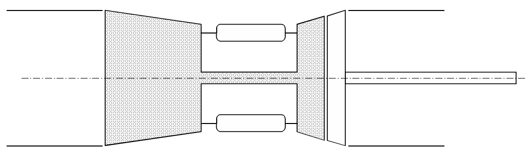


Figure 1-9 -.Single-spool turboshaft Single-spool gas generator with free power turbine

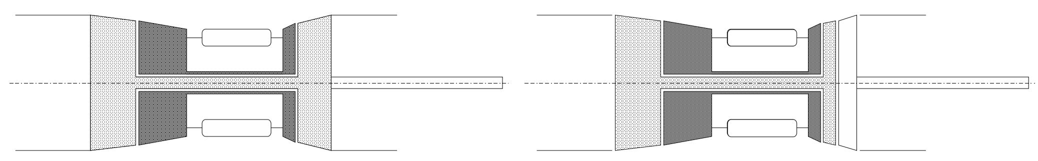


Figure 1-10 - Twin-spool turboshaft Twin-spool turboshaft with free power turbine

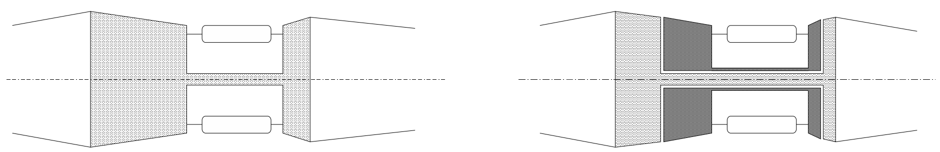


Figure 1-11 - Single-spool turbojet Twin-spool turbojet

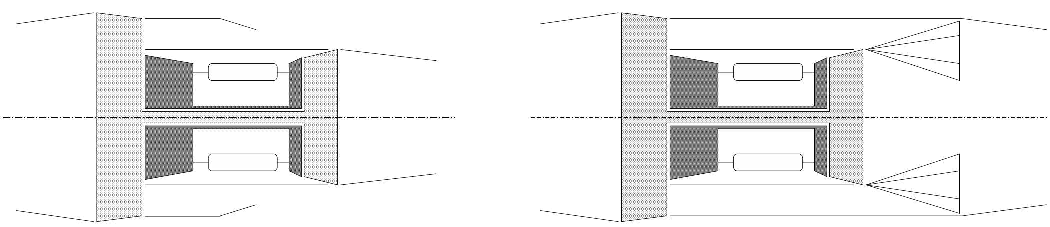


Figure 1-12 - Twin-spool turbofan Twin-spool mixed turbofan

It should be noted that each of the configuration has a specific application. More examples and explanation regarding various engine configuration and their applications will be provided in later chapters.

There are several ways of classifying air breathing engines. One such classification is shown in the Figure below. It can be seen from the figure that gas turbine engines are a subset of air breathing engines.

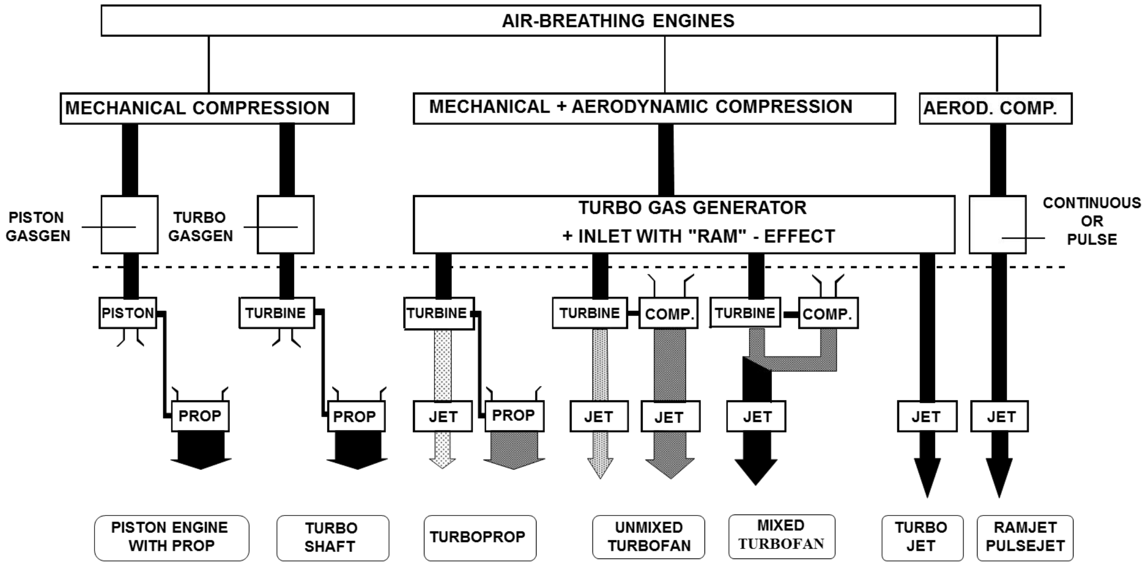


Figure 1-13 Classification of air breathing engines

1.7 Aero Engine Market

The aero engine market is distributed over several segments. The most important being the civil aircraft market. The civil aviation aero engine market can be further divided into regional aircraft market (< 130 seater aircraft, range < 3000km), short to medium range aircraft (narrow body aircraft, 130 to 250 passengers, range 3000 to 5500 km), wide body market (typically long range aircraft, > 250 passengers). The figure below shows the variation of aircraft range and passenger capacity for several aircraft

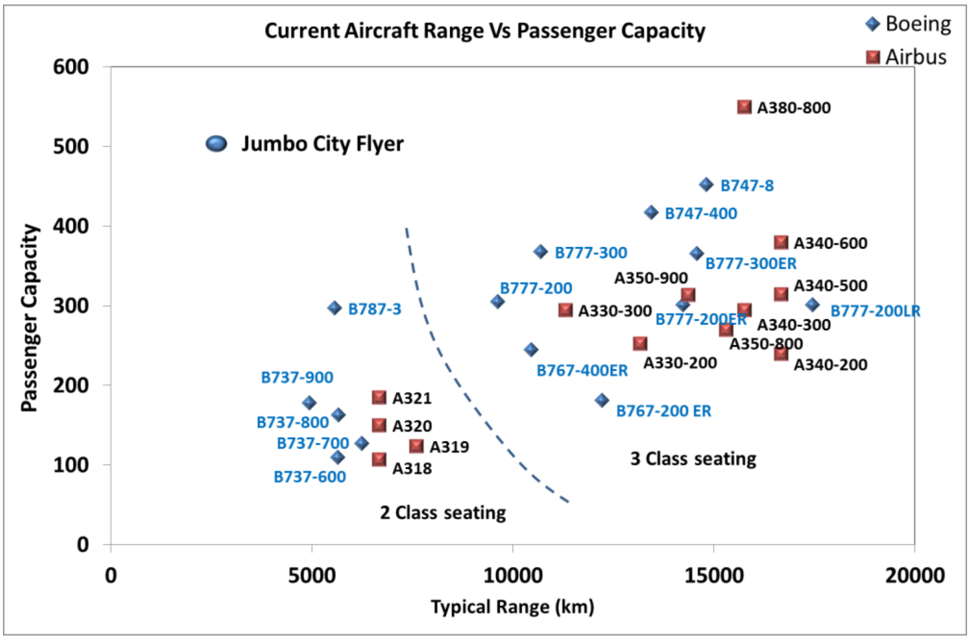


Figure 1-14 Aircraft range Vs aircraft range

The breakdown of the regional aircraft market is shown in the figure below. It can be seen that GE is the dominant player in this market with almost 65% of the total market.

COMMERCIAL REGIONAL JET

MANUFACTURER	AIRCRAFT	ENGINES
General Electric	2,330	4,660
Rolls-Royce	809	1,618
Honeywell	102	408
Lycoming	58	232
Powerjet	46	92
Pratt & Whitney	13	26
TOTAL	3,358	7,036

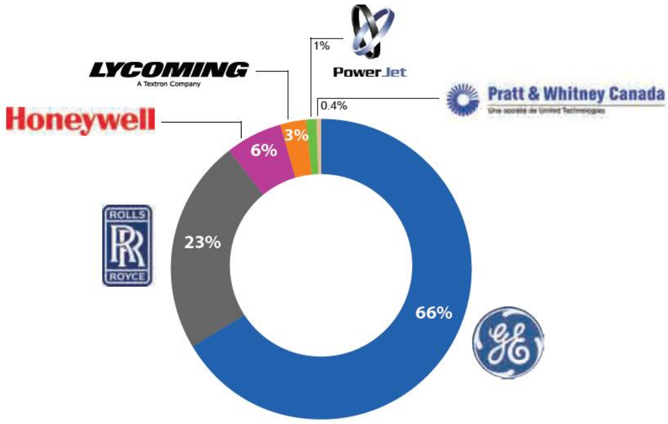


Figure 1-15 Overview of the regional aircraft market (source: *Flight Global Insight Analysis*)

The short to medium range single aisle aircraft market is the most important market of civil aviation as this is the workhorse of civil aviation. This market is a competitive market with new aircraft and engine manufacturer trying to grab a share of this market. It can be seen from the figure below that CFM international (a joint venture between GE and Safran Aircraft Engines) dominates this market with a share of around 70%, followed by International Aero Engines (a consortium consisting of Pratt & Whitney, MTU Aero Engines and the Japanese Aero Engine Corporation).

COMMERCIAL NARROWBODY AIRCRAFT

MANUFACTURER	AIRCRAFT	ENGINES
CFM International	9,248	18,500
International Aero Engines	2,658	5,316
Pratt & Whitney	921	1,912
Rolls-Royce	600	1,200
TOTAL	13,427	26,928

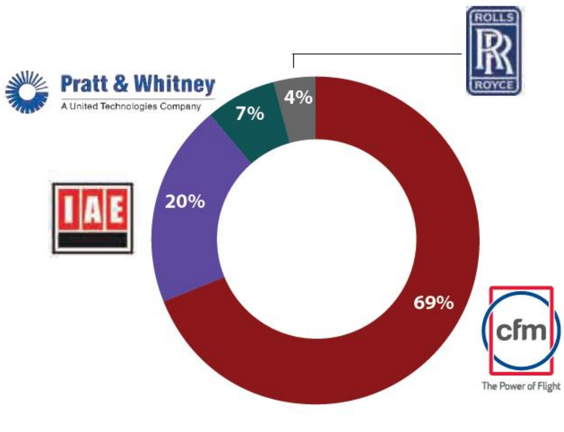


Figure 1-16 Overview of the narrow body aircraft market (source: *Flight Global Insight Analysis*)

The third significant market in civil aviation is the wide body market. This market is dominated by wide body twin engine aircraft like B-777, B787, A350, etc. The four engine aircraft like A340, A380 and B747 are on a decline. The wide body market is also dominated by GE with a market share of almost 50%, followed by Rolls Royce and Pratt & Whitney.

COMMERCIAL WIDEBODY AIRCRAFT		
MANUFACTURER	AIRCRAFT	ENGINES
General Electric	2,437	5,654
Rolls-Royce	1,182	2,812
Pratt & Whitney	910	2,212
CFM International	147	588
Engine Alliance	86	344
TOTAL	4,762	11,610

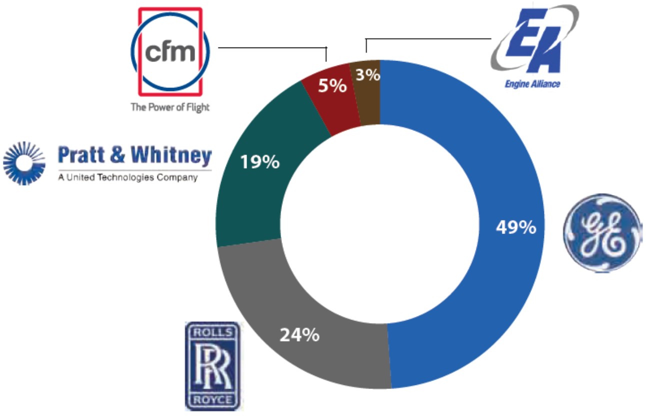


Figure 1-17 Overview of the wide body aircraft market (source: *Flight Global Insight Analysis*)

2. Ideal Cycles

(Prof. Ir. Jos P. van Buijtenen, Dr. Wilfried P.J. Visser)

2.1 The Joule-Brayton Cycle

The Joule-Brayton cycle represents the thermodynamic process in the gas turbine. Apart from the continuous flow of the medium through the gas turbine (see the previous chapter), another distinctive property of the Joule-Brayton cycle is that heat input (usually combustion) is taking place at constant pressure rather than at constant volume, as is the case with a piston engine. Also, the cycle can either be *open* or *closed*. In an *open* cycle, atmospheric air is drawn into the gas turbine compressor continuously and heat is added, usually by the combustion of fuel. The hot combustion gas is expanded in a turbine and ejected into the atmosphere, as shown in Figure 2-1(a). In a *closed* cycle, the same working fluid, be it air or some other gas, is circulated through the gas turbine and heat is usually added by a heat exchanger, as shown in Figure 2-1(b). An open or closed cycle gas turbine process, as depicted in Figure 2-1(a) and (b), would ideally be represented by the cycle depicted in Figure 2-2. Ignoring irreversibility, meaning ignoring pressure drops due to friction and heat losses to the surroundings, the ideal cycle is composed of two isentropic (lines *2-3* and *4-5*) and two isobaric (lines *2-3* and *4-1*) processes. The cycle resulting from these idealizations is called the Joule (or Brayton) cycle, often also referred to as *ideal simple cycle*.

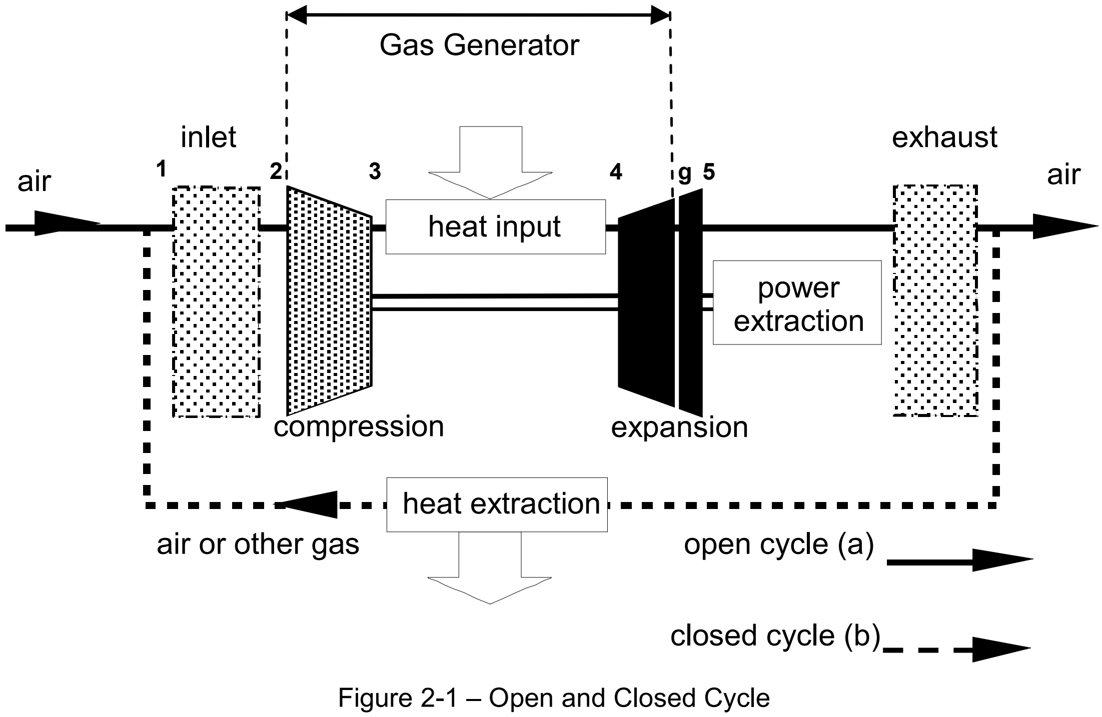


Figure 2-1 – Open and Closed Cycle

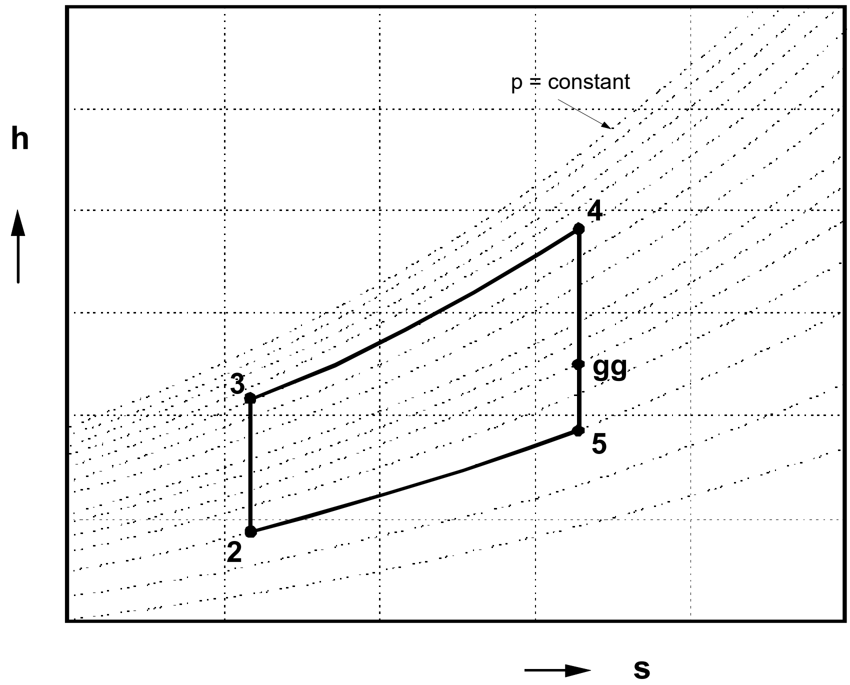


Figure 2-2 - The ideal gas turbine cycle h-s (enthalpy – entropy) diagram

With respect to the real gas turbine process, the ideal cycle assumes the following simplifications:

1. The ideal cycle's working fluid is considered an ideal gas having constant specific heats C_p & C_v and constant composition. For numerical calculations, values for specific heat C_p and specific heat ratio k are obtained from air at atmospheric conditions. Because of the "ideal" air working fluid, the cycle is called the "ideal air cycle".
2. Changes in kinetic and potential energy between inlet and exit of the various components can be ignored.
3. The compression and expansion processes are isentropic (i.e. reversible and adiabatic).
4. In a *closed* cycle, there is heat transfer during transition 5-2 (see Figure 2-2) to arrive at condition 2. In an *open* cycle, the atmosphere can be considered as a heat exchanger that cools down the exhaust gases at the inlet pressure (see Figure 2-1(a)). Both processes can be modeled using the same cycle in Figure 2-2
5. Pressure losses in the heat exchanger 3-4 (the combustion chamber), in the heat exchanger 5-2, in the connections between the components, in the inlet and exit are ignored.
6. Constant mass flow rate of the circulating medium
7. Mechanical losses with transmission of expansion power to the compression process are ignored.

Between stations 4 and 5 (i.e. the expansion process), station 'gg' can be identified in the h-s diagram (see Figure 2-2). The position of this point is such that the distance 4-gg equals distance 2-3, representing the required specific compression power. The process 2-3-4-gg represents the process that takes place in the *gas generator*. The residual power, represented by gg-5, is the specific *gas power*. *Gas power* is defined as the power that can be extracted from the hot

pressurized gas with 100 % isentropic efficiency (i.e. the maximum mechanical shaft or thrust power that would be obtained under ideal conditions with an ideal 100 % efficiency turbine). *Specific* gas power is gas power per unit of mass flow.
With the above-defined simplifications, the cycle variable parameters are ambient conditions p_2 and T_2, end-compression pressure p_3, maximum cycle temperature T_4 and mass flow.

2.2 Performance Analysis of an Ideal Simple Cycle

In this section the physical relations of the cycle parameters with specific gas power and efficiency are explained. These relations indicate how an ideal cycle can be optimized in terms of power output and efficiency. For a real cycle, the cycle relations show significant deviations from the ideal cycle, but they still roughly point in the same direction. Therefore, for a preliminary assessment of gas turbine cycle configurations, analysis of the ideal cycle equations provides valuable information.

The exchange of mechanical power and heat among the various components of the ideal cycle gas turbine can be calculated using the following equations:

Compressor power:

$$\dot{W}_{2-3} = \dot{m} c_p \left(T_3 - T_2\right) \tag{2.1}$$

Heat input rate:

$$\dot{Q}_{3-4} = \dot{m} c_p \left(T_4 - T_3\right) \tag{2.2}$$

Turbine power:

$$\dot{W}_{4-g} = \dot{m} c_p \left(T_4 - T_g\right) \tag{2.3}$$

Gas power:

$$\dot{W}_{gg} = \dot{W}_{g-5} = \dot{m} c_p \left(T_{gg} - T_5\right) \tag{2.4}$$

Waste heat:

$$\dot{Q}_{5-2} = \dot{m} c_p \left(T_5 - T_2\right) \tag{2.5}$$

For any isentropic process between two states for ideal gas holds:

$$\frac{p_3}{p_2} = \left(\frac{T_3}{T_2}\right)^{\frac{k}{k-1}} \tag{2.6}$$

Since the compression and the expansion are isentropic and k is constant, the pressure ratio of the compression process (*2-3*) equals the pressure ratio of the expansion process (*4-5*):

$$\Pi = \frac{p_3}{p_2} = \frac{p_4}{p_5} = \left(\frac{T_3}{T_2}\right)^{\frac{k}{k-1}} = \left(\frac{T_4}{T_5}\right)^{\frac{k}{k-1}} \tag{2.7}$$

Also applicable for gg-4

$$\frac{p_g}{p_4} = \left(\frac{T_{gg}}{T_4}\right)^{\frac{k}{k-1}} \tag{2.8}$$

The obtained work of 4-gg equals the work of 2-3, $W_{4\text{-}gg} = W_{2\text{-}3}$, meaning $T_{gg} = T_4 - T_3 + T_2$. Using equation (2.7):

$$T_{gg} = T_4 - T_2\left(\Pi^{\frac{k-1}{k}} - 1\right) \tag{2.9}$$

Using equation (2.8) it follows:

$$p_{gg} = p_3\left(\frac{T_{gg}}{T_4}\right)^{\frac{k}{k-1}} = p_2\Pi\left[1 - \frac{T_2}{T_4}\left(\Pi^{\frac{k-1}{k}} - 1\right)\right]^{\frac{k}{k-1}} \tag{2.10}$$

Substituting equation (2.7) and (2.9)into equation (2.4), and dividing the gas power W_{gg} by the mass flow, the **specific gas power** is obtained:

$$W_{s,gg} = c_p\left(T_{gg} - T_5\right) = c_p T_4\left[1 - \frac{1}{\Pi^{\frac{k-1}{k}}}\right] - c_p T_2\left[\Pi^{\frac{k-1}{k}} - 1\right] \tag{2.11}$$

In dimensionless form:

$$\frac{W_{s,gg}}{c_p T_2} = \frac{T_4}{T_2}\left[1 - \left(\frac{1}{\Pi^{\frac{k-1}{k}}}\right)\right] - \left[\Pi^{\frac{k-1}{k}} - 1\right] \tag{2.12}$$

Specific gas power can be used as a measure for the *compactness* of the gas generator (i.e. diameter). Gas generator dimensions together with maximum power output are important properties for the gas turbine application. A large specific gas power means a relatively *small* mass flow and for a certain flow velocity (because of $\frac{\dot{m}}{v} = \frac{1}{4}\pi\rho D^2$) a relatively small flow passage. The relation between specific gas power and volume or weight of the gas generator is more complex. The length of the gas generator is determined by pressure ratio ε and compressor technology level (pressure ratio achieved per compressor stage). For a certain stage pressure ratio, the number of compressor stages increases with cycle pressure ratio. For the turbine, this relation is less severe since turbine stage pressure ratios do not suffer from aerodynamic limitations as the compressor does (see chapter 7 on turbomachinery).

Thermodynamic efficiency is defined as the ratio of gas power over heat added to the process:

$$\eta_{\text{th}} = \frac{W_{s,gg}}{Q_{s,3-4}} = \frac{T_{gg} - T_5}{T_4 - T_3} \tag{2.13}$$

Substituting T_g from equation (2.9)and T_2 and T_4 from (2.7) the following equation is obtained:

$$\eta_{th} = \left[1 - \frac{T_2}{T_3}\right] = \left[1 - \frac{1}{\Pi^{\frac{k-1}{k}}}\right] \tag{2.14}$$

Ideal cycle thermodynamic efficiency only depends on pressure ratio ε and specific heat ratio k. k depends on the type and temperature of the fluid used in the cycle; in a gas turbine usually air. In simplified calculations and also in this text book k is considered a constant in the equations derived above.

Figure 2-3 shows the relation between the specific gas power and the thermodynamic efficiency as function of the temperature ratio T_4/T_2 and the pressure ratio Π (equation (2.12) and (2.14)). The figure shows there is a tradeoff between lower pressure ratio (with benefits in terms of low weight and small volume) and higher-pressure ratio (high thermal efficiency, i.e. low specific fuel consumption).

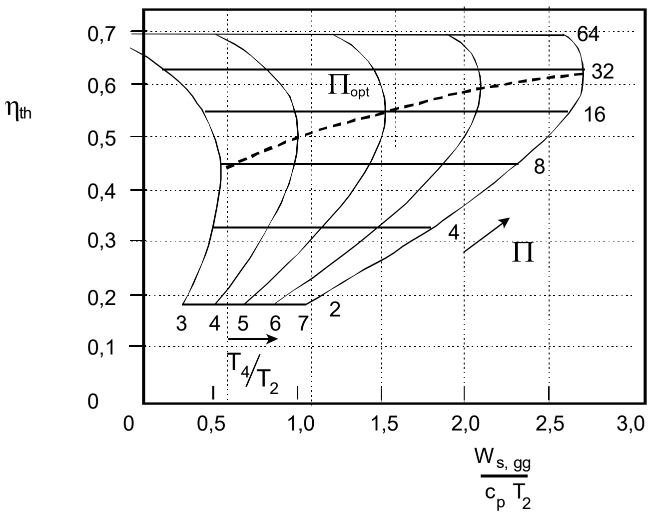


Figure 2-3 - Ideal cycle performance

The peak value of specific power for a given temperature ratio T_4/T_2 is called the optimum pressure ratio, Π_{opt} (see the dashed curve in Figure 2-3). One way to obtain the optimum pressure ratio is to differentiate the equation (2.12) using the Π as variable. Another method is to differentiate equation (2.4) using T_3 (which has a direct relation with Π via equation (2.6)) as a variable as follows:

$$W_{s,gg} = c_p\left(T_{gg} - T_5\right) = c_p\left[\left(T_4 - T_5\right) - \left(T_3 - T_2\right)\right] \qquad (2.15)$$

Since the following equation holds from the isentropic gas equation

$$\Pi^{\frac{k-1}{k}} = \frac{T_3}{T_2} = \frac{T_4}{T_5} \quad \text{then} \quad T_5 = \frac{T_4}{T_3}T_2 \qquad (2.16)$$

Equation (2.15) can be written to

$$W_{s,gg} = c_p\left(T_4 - \frac{T_4T_2}{T_3} - T_3 + T_2\right) \qquad (2.17)$$

Differentiate equation (2.17) using T_3 as variable for a given T_2 and T_4, the equation becomes:

$$\tfrac{d}{dT_3}W_{s,gg} = 0 \;\Rightarrow\; c_p\left(\frac{T_4T_2}{T_3^2} - 1\right) = 0 \;\Rightarrow\; T_3^2 = T_2T_4 \qquad (2.18)$$

Thus, T_3 for maximum gas power is:

$$T_3 = \sqrt{T_2T_4} \qquad (2.19)$$

Then Π_{opt} can be written as:

$$\Pi_{opt} = \left(\frac{T_3}{T_2}\right)^{\frac{k}{k-1}} = \left(\frac{\sqrt{T_2T_4}}{T_2}\right)^{\frac{k}{k-1}} = \left(\frac{T_4}{T_2}\right)^{\frac{k}{2(k-1)}} \qquad (2.20)$$

Using equation (2.16) and (2.19), at the optimum pressure ratio the following result is obtained:

$$T_3 = T_5 \qquad (2.21)$$

The specific power and the thermodynamic efficiency for the optimum pressure ratio are respectively:

$$\left(\frac{W_{s,gg}}{c_pT_2}\right)_{\Pi_{opt}} = \left(\sqrt{\frac{T_4}{T_2}} - 1\right)^2 \qquad (2.22)$$

$$\eta_{th} = 1 - \sqrt{\frac{T_2}{T_4}} \qquad (2.23)$$

Figure 2-4 shows why there is an optimum pressure ratio in the T-s diagram: both at very large ($\Pi >> \Pi_{opt}$) and very small ($\Pi << \Pi_{opt}$) pressure ratios the area of the cycle representing mechanical power becomes very small. Obviously, somewhere at a value in between the area is at its maximum.

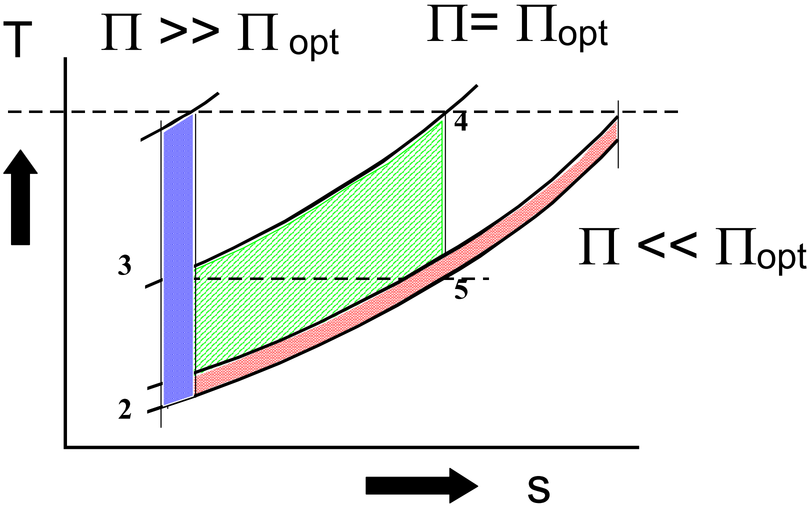


Figure 2-4 - Ideal cycle at different pressure ratios

2.2.1 Example

Consider an 'ideal cycle gas turbine' with a generator attached to the shaft as shown in Figure 2-5*. The entry temperature of the air entering the compressor is 288 K. The temperature of the air entering the turbine inlet is 1400 K. The thermodynamic properties of air are:* c_p *=1000 J/(kg K) and k = 1.4.*

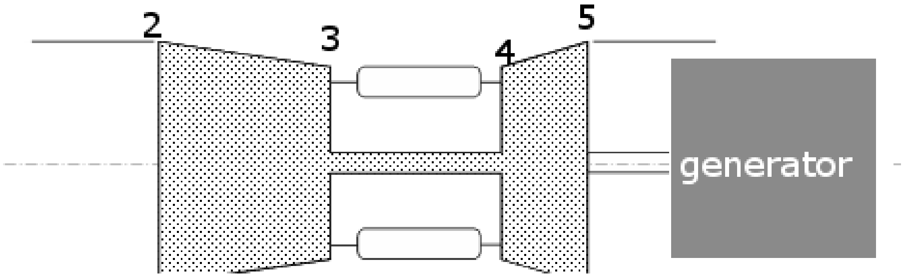


Figure 2-5 - Ideal cycle gas turbine with attached generator

Problems:

1. *Calculate the pressure ratio for maximum net power.*
2. *Calculate for this pressure ratio required specific compressor power, specific turbine power, heat input, specific delivered power and thermodynamic efficiency.*
3. *Calculate for* $\Pi = 10$ *and* $\Pi = 20$*, required specific compressor power, specific turbine power, heat input, specific delivered power and thermodynamic efficiency.*
4. *Compare in the results obtained for different* Π *in a table.*
5. *Sketch a qualitative T-s diagram in which is shown a comparison between the thermodynamic cycle evaluated at point 2 and 3.*

Solutions:

1.

Net power is at maximum if pressure ratio equals optimum pressure ratio.

$$\Pi_{opt} = \frac{p_2}{p_1} = \left(\frac{T_4}{T_2}\right)^{\frac{k}{2(k-1)}} = \left(\frac{1400}{288}\right)^{\frac{1.4}{2(1.4-1)}} = 15.9 \quad with \quad T_3 = T_5 = \sqrt{T_2\,T_4} = 635\ [K]$$

2.

Specific compressor power

$$W_{s,2-3} = c_p\left(T_3 - T_2\right) = 1000\left(635 - 288\right) = 347 \cdot 10^3\ [W/(kg/s)]$$

Total specific power delivered by the turbine

$$W_{s,4-5} = c_p\left(T_4 - T_5\right) = 1000\left(1400 - 635\right) = 765 \cdot 10^3\ [W/(kg/s)]$$

Specific heat input

$$Q_{s,3-4} = c_p\left(T_4 - T_3\right) = 1000\left(1400 - 635\right) = 765 \cdot 10^3\ [W/(kg/s)]$$

Specific delivered power

$$W_{s,delivered} = W_{s,4-5} - W_{s,2-3} = \left(765 - 347\right) \cdot 10^3 = 418 \cdot 10^3\ [W/(kg/s)]$$

Thermodynamic Efficiency

$$\eta_{th} = 1 - \frac{T_2}{T_3} = \frac{W_{s,4-5} - W_{s,2-3}}{Q_{s,3-4}} = \frac{418 \cdot 10^3}{765 \cdot 10^3} = 55\ \%$$

Note that the maximum theoretical cycle efficiency, the Carnot-efficiency, is

$$\eta_{carnot} = 1 - \frac{T_2}{T_4} = 1 - \frac{288}{1400} = 79\ \%$$

The Carnot efficiency is considerably higher. The difference is caused by not adding the heat at the highest process temperature in the Joule cycle.

3. Repeated calculations for decreased pressure ratio (Π = 10)

$$\Pi = \frac{p_3}{p_2} = \left(\frac{T_3}{T_2}\right)^{\frac{k}{(k-1)}} \Rightarrow T_3 = T_2\,\Pi^{\frac{k-1}{k}} = 288\left(10\right)^{\frac{0.4}{1.4}} = 556\ [K]$$

Specific compressor power

$$W_{s,2-3} = c_p\left(T_3 - T_2\right) = 1000\left(556 - 288\right) = 268 \cdot 10^3\,[W/(kg/s)]$$

$$\Pi = \frac{p_4}{p_5} = \left(\frac{T_4}{T_5}\right)^{\frac{k}{(k-1)}} \Rightarrow T_5 = T_4\left(\frac{1}{\Pi}\right)^{\frac{k-1}{k}} = 1400\left(\frac{1}{10}\right)^{\frac{0.4}{1.4}} = 725\ [K]$$

Total specific power delivered by the turbine

$$W_{s,4-5} = c_p\left(T_4 - T_5\right) = 1000\left(1400-725\right) = 675\cdot 10^3 \;\; [W/(kg/s)]$$

Heat input

$$Q_{s,3-4} = c_p\left(T_4 - T_3\right) = 1000\left(1400-556\right) = 844\cdot 10^3 \;\; [W/(kg/s)]$$

Specific delivered power

$$W_{s,delivered} = W_{s,4-5} - W_{s,2-3} = \left(675-268\right)\cdot 10^3 = 407\cdot 10^3 \;\; [W/(kg/s)]$$

Thermodynamic efficiency

$$\eta_{th} = \frac{W_{s,4-5} - W_{s,2-3}}{Q_{s,3-4}} = \frac{407\cdot 10^3}{844\cdot 10^3} = 48\ \%$$

4. Repeated calculations for increased pressure ratio ($\Pi = 20$)

$$\Pi = \frac{p_3}{p_2} = \left(\frac{T_3}{T_2}\right)^{\frac{k}{(k-1)}} \Rightarrow T_3 = T_2\,\Pi^{\frac{k-1}{k}} = 288\left(20\right)^{\frac{0.4}{1.4}} = 678\,\left[K\right]$$

Specific compressor power

$$W_{s,2-3} = c_p\left(T_3 - T_2\right) = 1000\left(678-288\right) = 390\cdot 10^3 \;\; [W/(kg/s)]$$

$$\Pi = \frac{p_4}{p_5} = \left(\frac{T_4}{T_5}\right)^{\frac{k}{(k-1)}} \Rightarrow T_5 = T_4\left(\frac{1}{\Pi}\right)^{\frac{k-1}{k}} = 1400\left(\frac{1}{20}\right)^{\frac{0.4}{1.4}} = 595\,\left[K\right]$$

Total specific power delivered by the turbine

$$W_{s,4-5} = c_p\left(T_4 - T_5\right) = 1000\left(1400-595\right) = 805\cdot 10^3 \;\; [W/(kg/s)]$$

Specific heat input

$$Q_{s,3-4} = c_p\left(T_4 - T_3\right) = 1000\left(1400-678\right) = 722\cdot 10^3 \;\; [W/(kg/s)]$$

Specific delivered power

$$W_{s,delivered} = W_{s,4-5} - W_{s,2-3} = \left(805-390\right)\cdot 10^3 = 415\cdot 10^3 \;\; [W/(kg/s)]$$

Thermodynamic Efficiency

$$\eta_{th} = \frac{W_{s,4-5} - W_{s,2-3}}{Q_{s,3-4}} = \frac{415\cdot 10^3}{722\cdot 10^3} = 57\,\%$$

5. Comparison of results for different pressure ratios

	η_{th}	$W_{s,delivered}$ *[W/(kg/s)]*	Q_s *[W/(kg/s)]*	T_3 *[K]*	T_5 *[K]*
$\Pi = 10$	*48%*	$407\cdot 10^3$	$844\cdot 10^3$	*556*	*725*
$\Pi = 15.9 = \Pi_{opt}$	*55%*	$418\cdot 10^3$	$765\cdot 10^3$	*635*	*635*

$\Pi = 20$	*57%*	$415 \cdot 10^3$	$722 \cdot 10^3$	*678*	*595*

6. Qualitative sketch of a T-s diagram for the calculated cycles

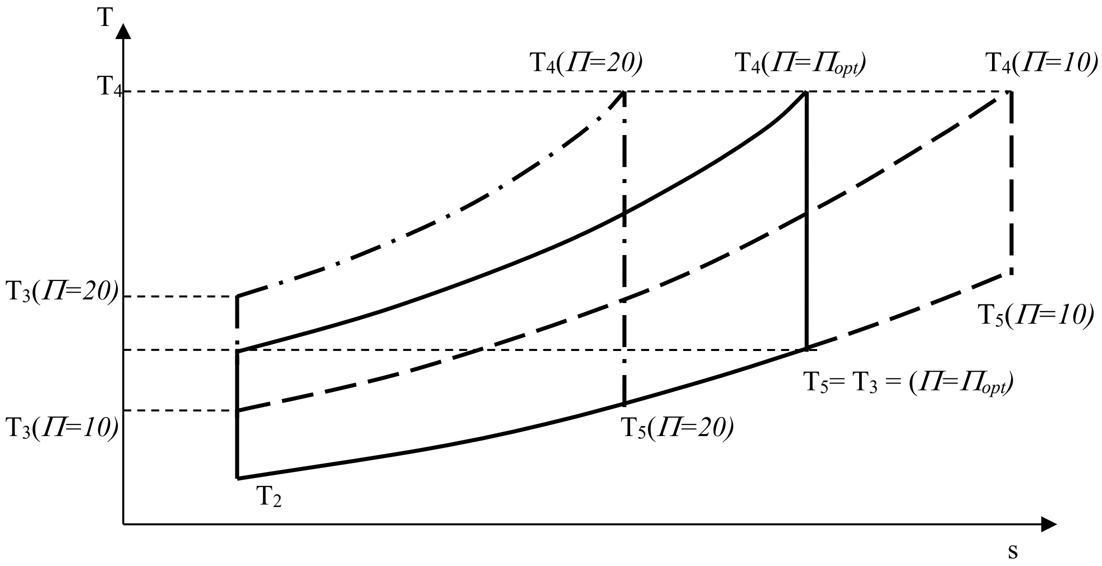


Figure 2-6 - Comparison of T-s diagrams for cycles with different pressure ratios

Remarks:

Increasing Π beyond Π_{opt}, further increases η_{th} but decrease $W_{s,delivered}$

2.3 Enhanced Cycles

The thermodynamic efficiency of the simple cycle can be improved and net power can be increased by adapting a cycle using

- heat exchangers for recovery ('recuperation') of exhaust waste heat,
- compressor intercooling,
- reheat ("afterburning").

2.3.1 Heat Exchange

An effective method to enhance the thermodynamic efficiency of the Joule cycle is to recuperate the waste heat from the exhaust, using a heat exchanger or *recuperator*. Figure 2-7 shows a gas turbine configuration with a heat exchanger to recover exhaust waste heat and Figure 2-8 shows the accompanying h-s diagram.

As long as $\Pi < \Pi_{opt}$ ($T_3 < T_5$, see Figure Figure 2-6) part of the heat added to the cycle can be taken from the flue gas of the heat rejection phase (5-2). The thermal efficiency of the *recuperated* cycle increases, because less heat (thus fuel) needs to be added to the cycle, while specific power is maintained.

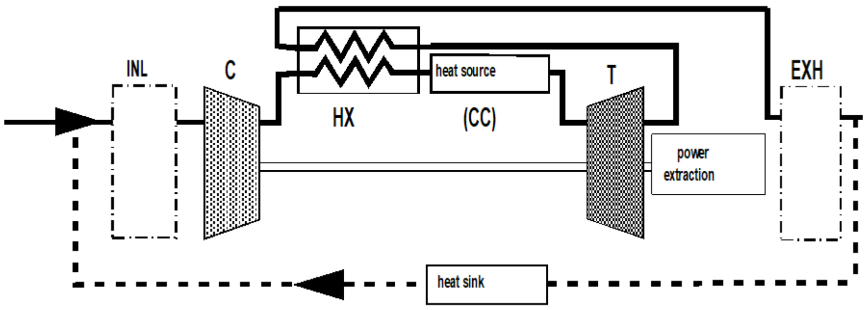


Figure 2-7 – Gas turbine cycle with heat exchanger ('recuperator')

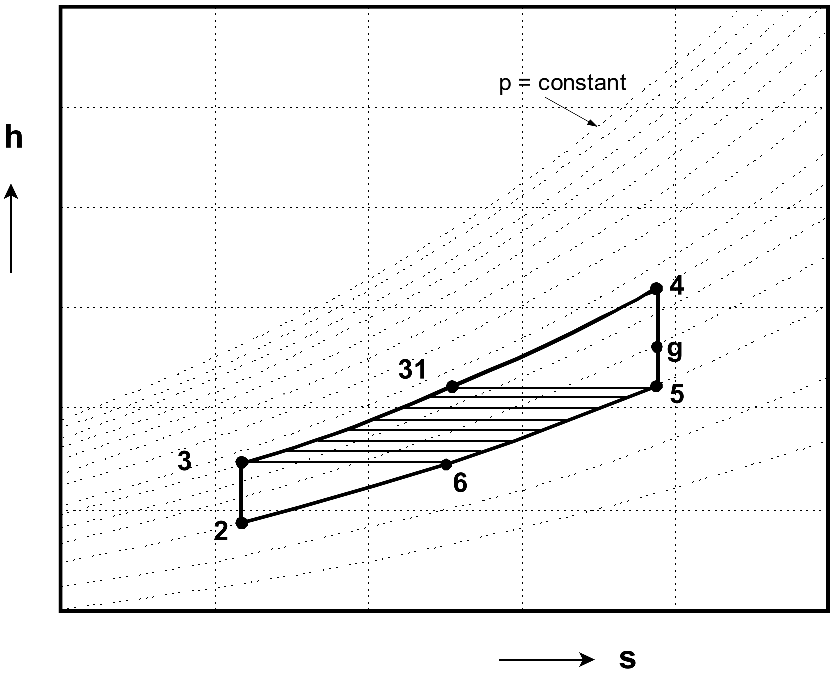


Figure 2-8 - h-s diagram of the recuperated cycle

The hot exhaust gas can not be cooled below T_6. The maximum amount of heat that can be transferred is $Q_{s,5\text{-}6} = Q_{s,3\text{-}31}$, implying $T_{31} = T_5$. Equation (2.12) can be used to determine specific power of a process using maximum heat exchange. The net supplied heat will become $Q_{s,4\text{-}5} = c_p (T_4 - T_5)$. Thermodynamic efficiency then becomes:

$$\eta_{th} = \frac{W_{s,gg}}{Q_{s,4-5}} = \frac{c_p T_4 \left[1 - \frac{1}{\Pi^{\frac{k-1}{k}}}\right] - c_p T_2 \left[\Pi^{\frac{k-1}{k}} - 1\right]}{c_p T_4 \left[1 - \frac{1}{\Pi^{\frac{k-1}{k}}}\right]} = 1 - \frac{T_2}{T_4} \Pi^{\frac{k-1}{k}} \tag{2.24}$$

This equation shows that the efficiency **increases** for **decreasing** pressure ratio. Figure 2-9 shows the thermodynamic efficiency as function of the T_4/T_2 and the pressure ratio *Π*.

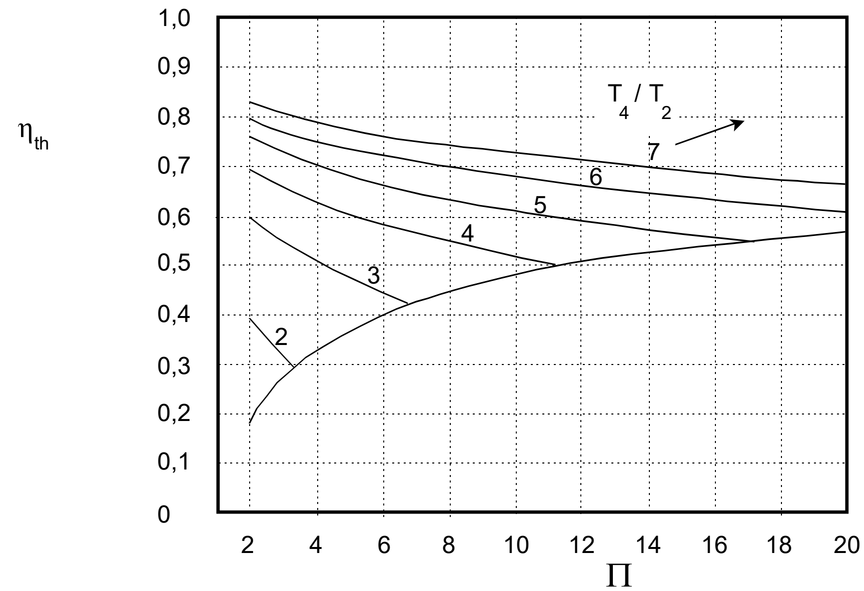


Figure 2-9 - Recuperated cycle thermodynamic efficiency

Figure 2-9 shows a number of curves representing the recuperated cycle efficiency η_{th} for different values of T_4/T_2 and varying Π. For each curve, Π can be increased up to a point (intersection with the lower right curve) where T_5=$T_3$and heat exchange from the hot exhaust to the compressor exit air becomes impossible.

2.3.2 Example

Consider the 'ideal cycle gas turbine' of the previous example and suppose it to work at Π= *10, which is less than the optimum pressure ratio* Π_{opt} *= 15.9.*

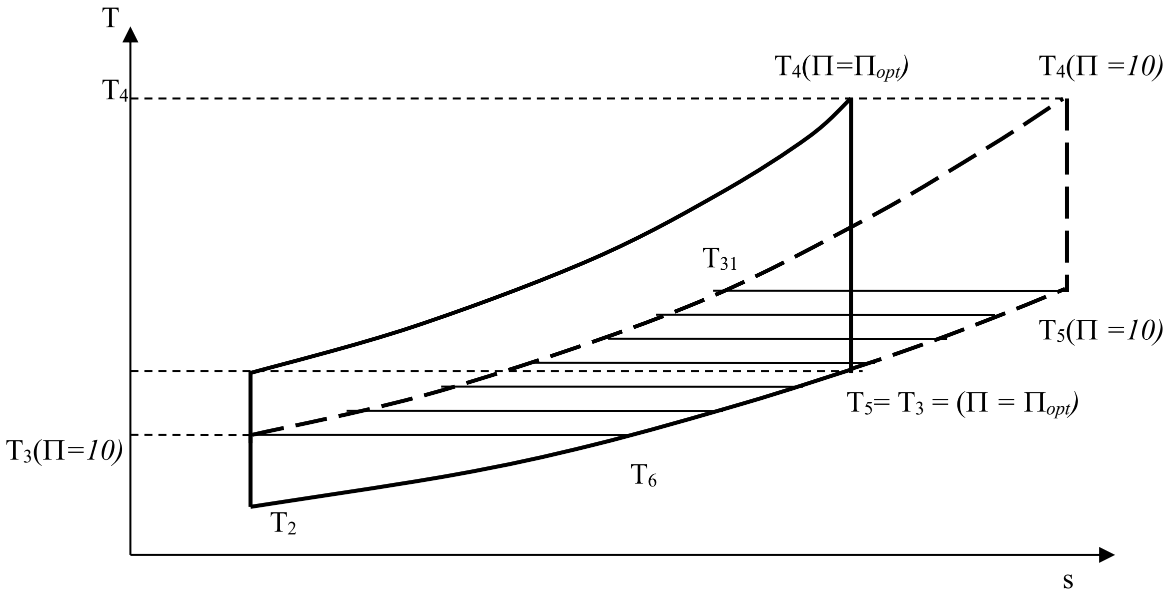
Problems:

Add a heat exchanger and evaluate specific gas power and thermodynamic efficiency.
Compare the results with the previous calculation for Π *= 10 and* Π_{opt} *= 15.9 and present the results in a table.*

Solutions:

1. From previous calculations

	Π *=10*	Π = Π_{opt} *=15.9*
T_2 [K]	*288*	*288*
T_3 [K]	*556*	*635*
T_4 [K]	*1400*	*1400*
T_5 [K]	*725*	*635*



Adding the heat exchanger

$T_6 = T_3 = 556\ K$ $T_{31} = T_5 = 725\ K$ $T_2 = 288\ K$ $T_4 = 1400\ K$

$$\eta_{th} = 1 - \frac{T_2}{T_4}\Pi^{\frac{k-1}{k}} = 1 - \frac{288}{1400}10^{\frac{0.4}{1.4}} = 60\%$$

Heat input

$$Q_{s,31-4} = c_p\left(T_4 - T_{31}\right) = 1000\left(1400 - 725\right) = 675\cdot 10^3\ \left[W/(kg/s)\right]$$

Specific delivered power

$$W_{s,delivered} = W_{s,4-5} - W_{s,2-3} = \left(675 - 288\right)\cdot 10^3 = 407\cdot 10^3\ \left[W/(kg/s)\right]$$

2. Comparison of results

	η_{th}	$W_{s,delivered}$ *[W/(kg/s)]*	Q_s *[W/(kg/s)]*
$\Pi = 10$	*48%*	$407\cdot 10^3$	$844\cdot 10^3$
$\Pi = \Pi_{opt} = 15.9$	*55%*	$418\cdot 10^3$	$765\cdot 10^3$
$\Pi = 10$ + *heat exchanger*	*60%*	$407\cdot 10^3$	$675\cdot 10^3$

It is possible to see that using a heat exchanger in the cycle with lower pressure ratio, η_{th} is higher than the one in the cycle with optimum pressure ratio, the heat supplied is lower, but the specific work is not changed. It is worth to remember that the ratio T_4/T_2 is kept constant in this numerical example.

2.3.3 Intercooling

Consider the equation for specific compressor work for an adiabatic reversible process with constant mass flow:

$$W_{2-3} = \int V\, dp \tag{2.25}$$

This equation shows that in order to limit the compressor work the increase of the specific volume V must be kept as low as possible. The ideal case would be *isothermal* compression, but would be very complex to implement in a gas turbine. A more practical approach is to split up the compression process into multiple parts and cool the airflow between two consecutive compression phases using an 'intercooler'. Figure 2-10 and Figure 2-11 show the configuration and the h-s diagram for a cycle with an intercooler between two compressors. The air cooled by the intercooler is cooled down to T_{25}.

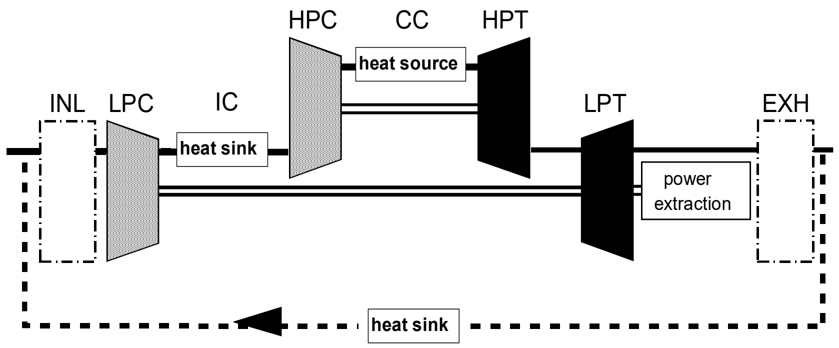


Figure 2-10 - Intercooled cycle

Note that the divergence of the isobars indicates that compression work required from p_2 to p_3 in Figure 2-11 decreases with decreasing initial compression temperature (isentropic compression work is corresponding to the vertical distance between p_2 and p_3).

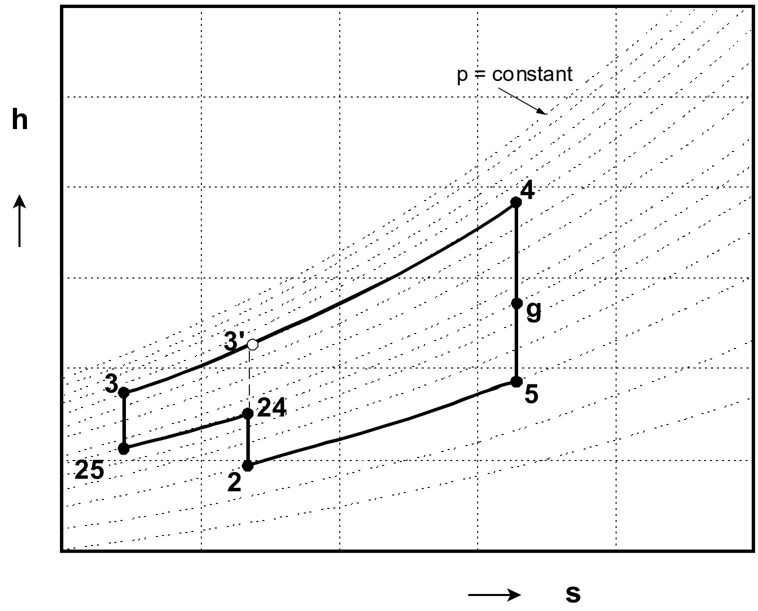


Figure 2-11 - Enthalpy - Entropy diagram for an intercooled cycle

The increase of net power output resulting from the decrease of compressor work is diminished by the additional heat (3 - 3' in Figure 2-11) required to realize maximum cycle temperature T_4. As a result, intercooling will provide more power with a somewhat decreased efficiency. This can be explained as follows: to obtain the (single) intercooled gas turbine cycle, the ideal cycle (2-3'-4-5) will be extended with a small cycle 25-3-3'-24. The pressure ratio of this additional small cycle will always be lower than the pressure ratio of the ideal cycle. The efficiency of the additional cycle will therefore be smaller.

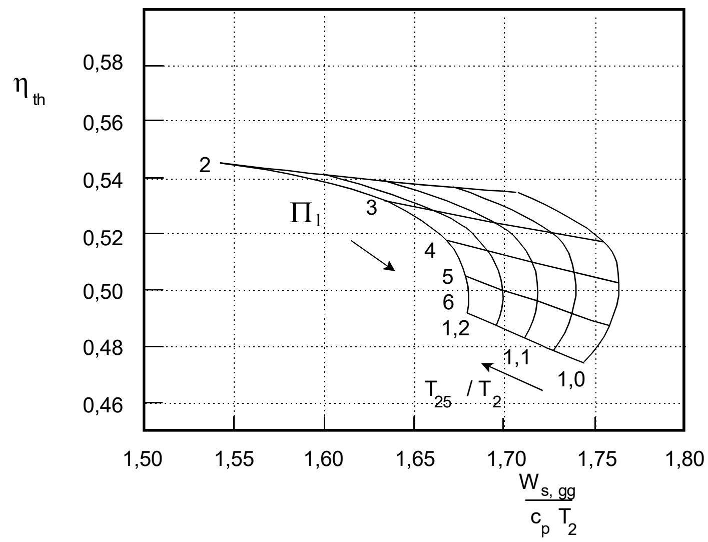


Figure 2-12 - Effects of intercooler pressure ratio distribution and degree of cooling

($\Pi_1 = p_{25}/p_2$, $\Pi_{tot} = p_3/p_2 = 16$ and $T_4/T_2=5$)

Figure 2-12 shows the effect of pressure ratio distribution before and after intercooling in terms of Π_1 ($\Pi_1 = p_{25}/p_2$) and temperature T_{25} on the performance of the cycle. Π_1 represents the point where the medium is cooled in the compression phase. A low value for T_{25} is apparently favorable for a high specific power. For the pressure ratio Π_1 an optimum exists with regard to specific power. Thermodynamic analysis learns that with $T_{25} = T_2$ this optimum is:

$$\Pi_1 = \frac{p_{25}}{p_2} = \sqrt{\Pi_{tot}} \tag{2.26}$$

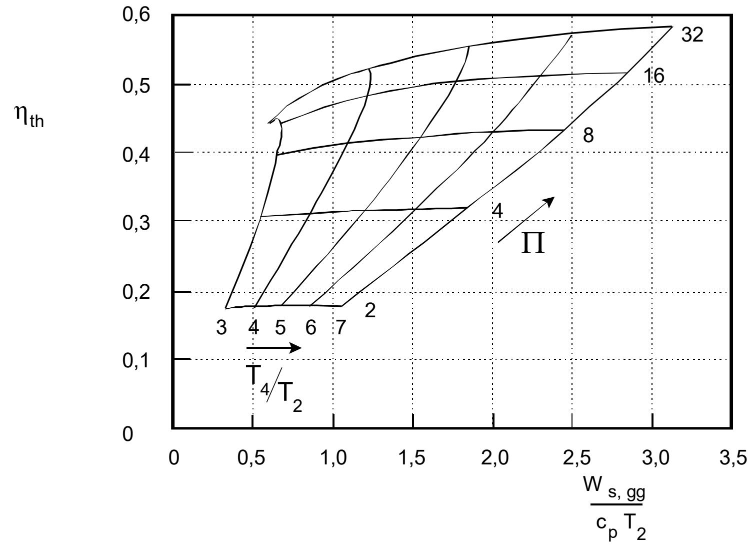


Figure 2-13 - Ideal intercooled cycle performance with T_{25}=T_2 and $\Pi_1 = \sqrt{\Pi_{tot}}$

Figure 2-13 shows cycle performance at the optimal intercooling configuration with $T_{25} = T_2$ and optimal Π_1. When compared with the simple cycle (Figure 2-3), the intercooled cycle has a higher specific power at lower efficiency. Note that the lines for constant pressure ratio ε are not horizontal anymore. With intercooling, at constant cycle pressure ratio, T_4/T_2 has an effect on efficiency.

Intercoolers are not widely applied for gas turbines because they are bulky, increase system complexity and require large quantities of cooling water. Moreover, its advantages in terms of being compact and self-contained would then be compromised.

2.3.4 Example

Consider an 'ideal cycle gas turbine' of the previous example and for the case in which $\Pi = 20$ *add an intercooling system that takes down the intermediate compressor temperature to the value of the entering condition* $T_2 = 288$ *K. Referring to* Figure 2-10*,* $T_{25} = T_2 = 288$ *K and assume that the intercooling stage starts when the pressure ratio in the compressor is* $p_{24}/p_2 = \sqrt{\Pi_{tot}} = \sqrt{20}$

Problems:

Evaluate specific gas power and thermodynamic efficiency of the intercooled system.

Compare in a table the results with the previous calculation for Π = 20.

Solutions:

1.

$$T_{24} = T_2 \left(\frac{p_{24}}{p_2} \right)^{\frac{k-1}{k}} = 288 \left(\sqrt{20} \right)^{\frac{0.4}{1.4}} = 442 [K]$$

$$T_3 = T_{25} \left(\frac{p_3}{p_{25}} \right)^{\frac{k-1}{k}} = 288 \left(\sqrt{20} \right)^{\frac{0.4}{1.4}} = 442 [K]$$

Specific compressor power

$$W_{s,2-24} = c_p \left(T_{24} - T_2 \right) = 1000 \left(442 - 288 \right) = 154 \cdot 10^3 \ \left[W/(kg/s) \right]$$

$$W_{s,25-3} = c_p \left(T_3 - T_{25} \right) = 1000 \left(442 - 288 \right) = 154 \cdot 10^3 \ \left[W/(kg/s) \right]$$

Specific heat input

$$Q_{s,3-4} = c_p \left(T_4 - T_3 \right) = 1000 \left(1400 - 442 \right) = 958 \cdot 10^3 \ \left[W/(kg/s) \right]$$

Total specific power delivered by the turbine

$$W_{s,4-5} = c_p \left(T_4 - T_5 \right) = 1000 \left(1400 - 595 \right) = 805 \cdot 10^3 \ \left[W/(kg/s) \right]$$

Specific delivered power

$$W_{s,delivered} = W_{s,4-5} - W_{s,2-24} - W_{s,25-3} = \left(805 - 154 - 154 \right) \cdot 10^3 = 497 \cdot 10^3 \ \left[W/(kg/s) \right]$$

Thermodynamic Efficiency

$$\eta_{th} = \frac{Specific\ Delivered\ Power}{Heat\ Input} = \frac{W_{s,4-5} - W_{s,2-24} - W_{s,25-3}}{Q_{s,3-4}} = \frac{497 \cdot 10^3}{958 \cdot 10^3} = 52\ \%$$

2. Comparison of results

	η_{th}	$W_{s,delivered}$ *[W/(kg/s)]*	Q_s *[W/(kg/s)]*
Π = *20*	*57%*	$415 \cdot 10^3$	$722 \cdot 10^3$
Π = *20 + intercooler*	*52%*	$497 \cdot 10^3$	$958 \cdot 10^3$

It is possible to see that using an intercooling system, higher specific gas power can be obtained in a cycle, but the thermodynamic efficiency is lower. It is worth noting that in the cycle with Π*=20 a heat exchanger cannot be used, because T_5 is lower than T_3 (look at Example 2.2.1) and the intercooling system is the only available device to improve the specific gas power.*

2.3.5 Reheat

A similar effect on the specific power and the thermodynamic efficiency as the intercooled cycle can be obtained with *reheat*. Reheat of the working medium can be applied between the turbine stages, resulting in an increase of the net specific power. The effect of reheat on the thermodynamic efficiency is dependent on the process parameters. Again an optimum pressure ratio for the expansion process before and after the reheater exists.

Figure 2-14 shows the configuration schematic overview of an ideal cycle (2-3-4-5') supplemented with a small reheat cycle 44-45-5-5'. Figure 2-15 shows the h-s diagram.

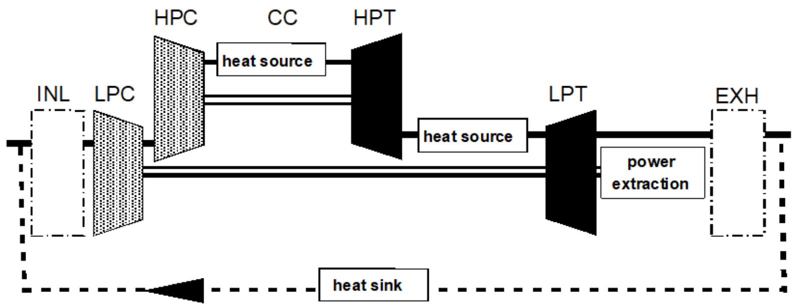


Figure 2-14 - Ideal cycle with reheat

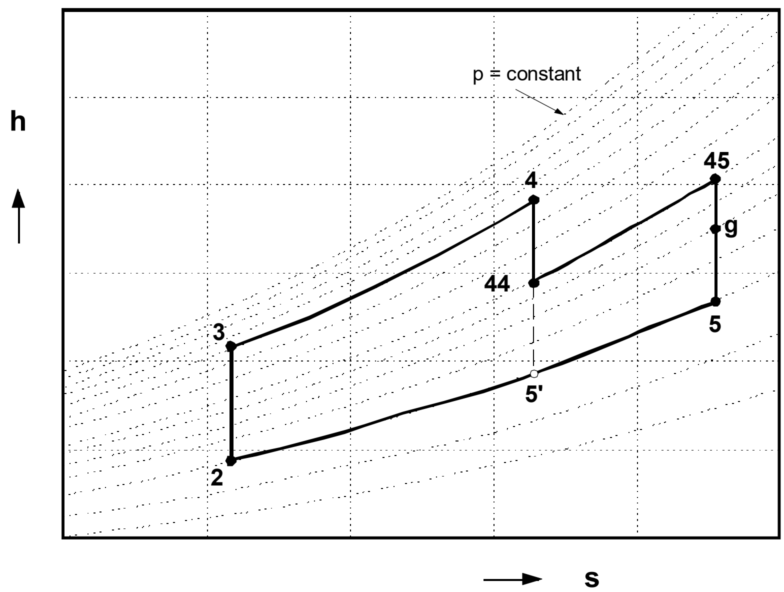


Figure 2-15 - Ideal cycle with reheat h-s diagram

Figure 2-16 shows the effect of the distribution of expansion pressure ratio before and after reheat in terms of Π_1 ($\Pi_1 = p_{45}/p_4$) and reheat end-temperature temperature T_{45} on cycle performance.

Π_1 represents the point where reheat takes place in the expansion phase. As with intercooling, for the pressure ratio Π_1 an optimum exists with regard to specific power. Analysis of the thermodynamic relations learns that with $T_{45} = T_4$ this optimum is:

$$\Pi_1 = \frac{p_{45}}{p_4} = \sqrt{\Pi_{tot}} \tag{2.27}$$

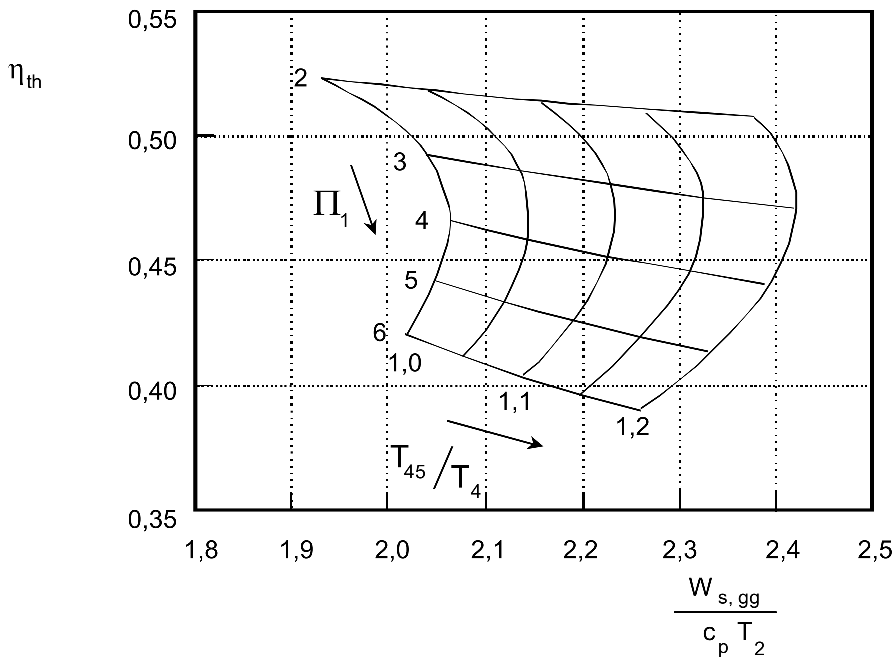


Figure 2-16 - Ideal cycle with reheat – effect of expansion pressure ratio distribution

It is important to point out that efficiency decreases as the ratio T_{45}/T_4 increases. This aspect can be explained by looking at diagram in Figure 2-17 with higher T_{45}, the 'second cycle' at the lower pressure ratio becomes more significant and as a result the whole cycle will have lower efficiency.

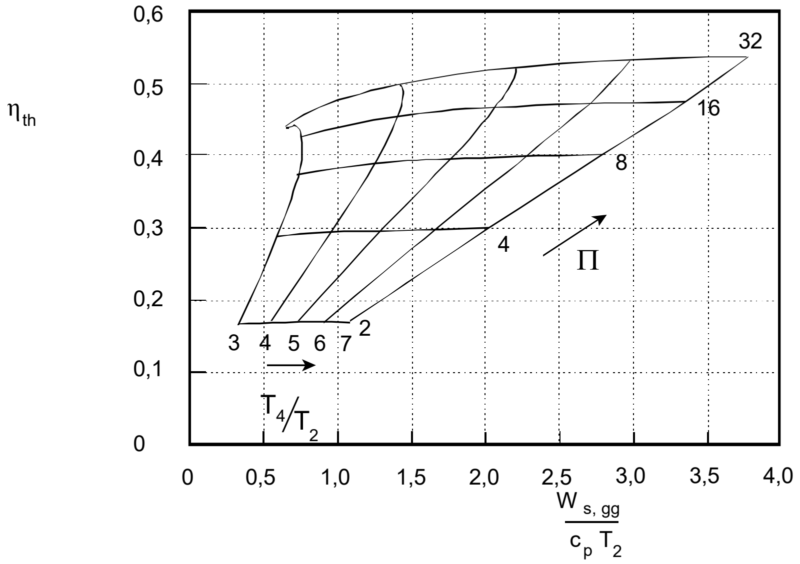


Figure 2-17 - Ideal reheated cycle performance with T_{45}=T_4 and $\Pi_1 = \sqrt{\Pi_{tot}}$

Figure 2-17 shows cycle performance at the optimal reheat configuration with $T_{45} = T_4$ and optimal Π_1. When compared with the simple cycle (Figure 2-3), the reheated cycle has a higher specific power at lower efficiency and the lines with constant pressure ratio Π are not horizontal anymore. It should be noted that heat addition between every two turbine stages is very complex to implement in hardware. The ideal objective to achieve isothermal expansion is not feasible. For turboshaft gas turbines, reheat is sometimes applied between the gas generator and the free power turbine, which usually is located in a separate assembly.

For high-speed (military) jet engines, reheat is applied between the turbines and the exhaust nozzle in an “afterburner”. The final expansion phase then takes place in the exhaust nozzle, see chapter 5. Since no rotating parts come after the afterburner, T_{45} (in the jet engine usually referred to as T_7) can be set much higher (around 2500 K) than T_4 to obtain maximum thrust for a short period.

2.3.6 Combined Intercooling, Reheat and Recuperation

The combination of cycle enhancements mentioned in the previous sections offers a means to improve both specific power and efficiency. With the application of both intercooling and reheat, the temperature at the end of compression decreases and the temperature at the end of the expansion process increases. Then the total cycle pressure ratio at which recuperation still is useful will become higher (see section 2.3.1), which will improve the thermodynamic efficiency. The combination of all three enhancements is depicted in Figure 2-18

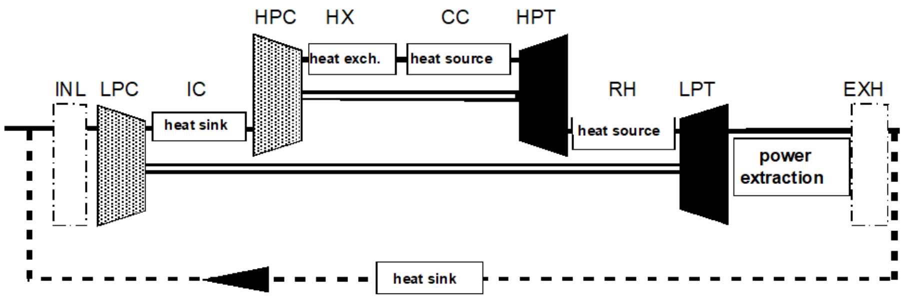


Figure 2-18 - Combined intercooling, reheat and recuperation in the ideal cycle

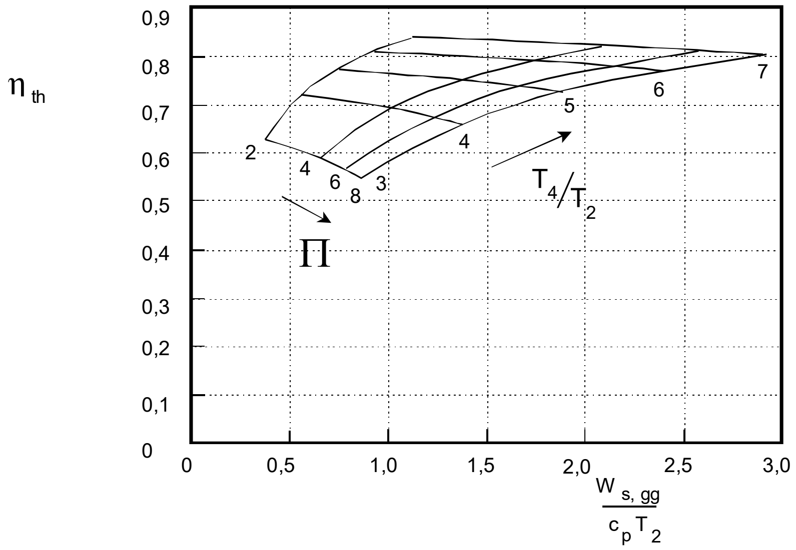


Figure 2-19 - Combined intercooling, reheat and recuperation cycle performance

Figure 2-19 shows that the combination of intercooling, reheat and recuperation indeed results in a significant improvement of both specific power and efficiency when compared with the simple ideal cycle depicted in Figure 2-3. The next table provides a summary of the different cycle variants and effects on thermodynamic efficiency η_{th}, specific gas power $W_{s,gg}$ and optimal Π value.

Modification	***Effect***	***optimal*** Π
Heat exchange / recuperation	$\uparrow \eta_{th} \downarrow W_{s,gg}$	low Π
Intercooling	$\downarrow \eta_{th} \uparrow W_{s,gg}$	high Π
Afterburning	$\downarrow \eta_{th} \uparrow W_{s,gg}$	high Π
Heat exchange & intercooling	$\uparrow \eta_{th} \sim W_{s,gg}$	high Π

3. **Real Cycles**

(Prof. Ir. Jos P. van Buijtenen, Dr. Wilfried P.J. Visser, Prof. Dr. A. Gangoli Rao)

3.1 Deviations with Respect to the Ideal Process

The processes taking place in a real gas turbine deviate from the ideal cycle as presented in section 0. The simplifications listed in section 2.1 are reviewed again to verify whether, or under which conditions, these can be accepted.

1. **The ideal cycle's working fluid is considered as an ideal gas having constant specific heats c_p and c_v and constant composition.**
   This simplification can partially be preserved, because the working fluid can be considered to behave like an ideal gas. The effect of the pressure on the specific heat values for different pressure values can be ignored as the pressure encountered in current gas turbines is not extremely high. However, the effects of both temperature and changing composition on specific heat are significant and cannot be ignored. This will be explained in section 3.2.
2. **Changes in kinetic and potential energy between inlet and exit of the various components can be ignored.**
   This simplification cannot be preserved for kinetic energy because there are considerable differences between the inlet and exit kinetic energy levels, as shown in section 3.3. In general, differences in potential energy can be safely ignored.
3. **The compression and expansion processes are isentropic (i.e. reversible and adiabatic).**
   This simplification cannot be preserved in a real gas turbine because friction and other losses occur as a result of which these processes can no longer be considered reversible and the entropy of working fluid increases (discussed in section 3.4). However, the compression and expansion processes can be considered adiabatic, because steady-state heat exchange between working fluid and the surroundings is very small and hardly affects performance.
4. **There is heat transfer during transition 5-2 (see Figure** 2-2**) to arrive at condition 2. The "open" process can be modeled as a "closed").**
   This simplification can be preserved since the cycle entry condition 2 is the same for both the closed and open cycles. With the possibility to expand exhaust gas to ambient pressure there also are no pressure losses between 5 and 2.
5. **Pressure losses in the combustion chamber and other components with heat addition or extraction are ignored.**
   This simplification cannot be preserved since friction between working fluid and walls of the gas path induces significant pressure losses. In the combustor for example, there is a decrease in pressure for two reasons. Firstly, for high combustor efficiencies in compact combustion chambers, air and injected fuel must be mixed intensively. The energy required for mixing air and fuel is obtained from the pressure. Secondly, *fundamental* pressure loss is caused by the heat addition itself. See section 3.5 for more on pressure losses.

6. **Constant mass flow rate throughout the whole engine**
   This simplification often can be applied because the fuel mass flow (added in the combustion chamber) is small relative to the air mass flow (1-2%). Usually part of the compressor air is used to cool various hot parts. The cooling flow does not (or only to a small extent) contribute to generating turbine power, and this in many cases more or less 'compensates' for the effect of omitted fuel mass flow on turbine power. Note that in case of high fuel mass flow rates (for example if low calorific value (LCV) fuel is used), or significant amounts of air extracted from the compressor for use outside the engine, the effects do not compensate. Then fuel flow and/or bleed airflow values need to be included in the calculations. In this text book mass flow can be kept constant (and fuel flow ignored) unless stated otherwise.
7. **Mechanical losses with transmission of expansion power to the compressor are ignored.**
   These losses are usually represented by shaft mechanical efficiency, which usually is more than 99%. Although mechanical losses can be easily included in the calculations they can safely be ignored if errors of 1% are accepted.

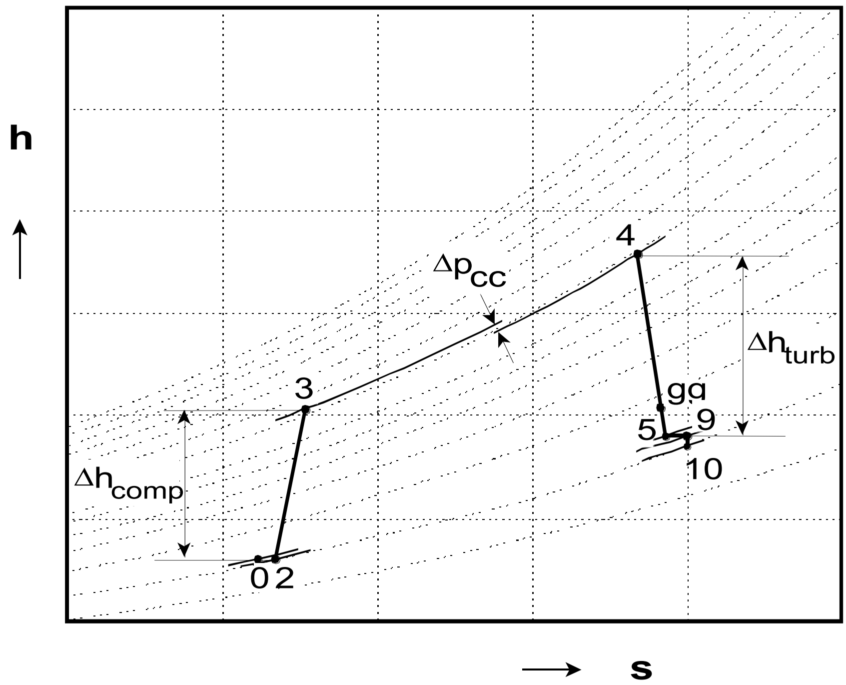


Figure 3-1 - Enthalpy-entropy diagram of an real industrial gas turbine cycle

As a result of above-mentioned effects, the real cycle significantly deviates from the ideal cycle. Figure 3-1 shows the enthalpy-entropy diagram for an industrial gas turbine. Because of losses in the inlet system (diffuser and filters), station *2* is somewhat on the right of station *0*. Total temperature and enthalpy remain constant in the (adiabatic) inlet. Station *0* denotes the plant's ambient conditions; station *9* denotes the exhaust exit. Losses in the compression phase cause station *3* to move somewhat to the right of station *2* due to the associated increase in entropy. Station *4*, the turbine entry, is located on a constant pressure curve (isobar) at a level lower than station *3*. This is the pressure loss in the combustor described above. Station *gg* is located somewhat to the right of station *4* again due to the non-isentropic expansion in the turbine.

3.2 Specific Heat c_p and Specific Heat Ratio k

The thermodynamic properties of the fluid, represented by specific heat c_p and specific heat ratio k, are dependent on the temperature, pressure and gas composition. The effect of pressure on c_p and k can usually be considered negligible. The effect of the temperature on on c_p and k is much larger and cannot be neglected. For air from 300 K to 2000 K, c_p increases from 1000 to 1300 J/kg/K while k decreases from 1.4 down to 1.28. The effect of gas composition is mainly caused by the differences in CO_2 and H_2O concentrations, which are the products of combustion from hydrocarbon fuels. The c_p values of flue gas (or gas downstream of the combustor) are higher than those for air due to the different c_p values for CO_2 and H_2O (840 and 1870 J/kg/K respectively).

A minimal requirement for accurate gas turbine cycle calculations is to take the temperature dependence effect into account. In case alternate fuels (other than the "standard" fuels, natural gas or kerosene) are used, such as low colorific fuels (synthesis gas), or steam injection is used in the combustion chamber, the effect of the composition needs to be taken into account. As a result, the cycle calculation becomes far more complex and requires extra iterations. With manual calculations of gas turbine cycles it is sufficient to use mean values for c_p and k. In this textbook there are two separate sets of 'mean' values for c_p and k, one set for air and one for flue gas. The universal gas constant R is kept constant:

Specific heat and specific heat ratio of air:

$$c_{p\,air} = \mathbf{1000} \text{ [J/(kg K)]}, \qquad k_{air} = \mathbf{1.4}$$

Specific heat and specific heat ratio of flue gas:

$$c_{p\,gas} = \mathbf{1150} \text{ [J/(kg K)]}, \qquad k_{gas} = \mathbf{1.33}$$

Universal gas constant:

$$R = \mathbf{287} \text{ [J/(kg K)]}$$

For engines with mixed exhaust, an additional set of values for c_p and k need to be defined for the mixture of the bypass mass flow (cold flow) and the core mass flow (hot flow):

$$c_{p\,mix} = \mathbf{1150} \text{ [J/(kg K)]}, \qquad k_{mix} = \mathbf{1.33}.$$

Specific heat ratio k is related to c_p and R using equation (3.1):

$$k = \frac{c_p}{c_v} = \frac{c_p}{c_p - R} \tag{3.1}$$

The variation of c_p and k as a function of temperature and fuel to air ratio is shown in the figures below. It can be seen that while c_p increases with temperature and fuel to air ratio, k decreases with increasing temperature and fuel to air ratio.

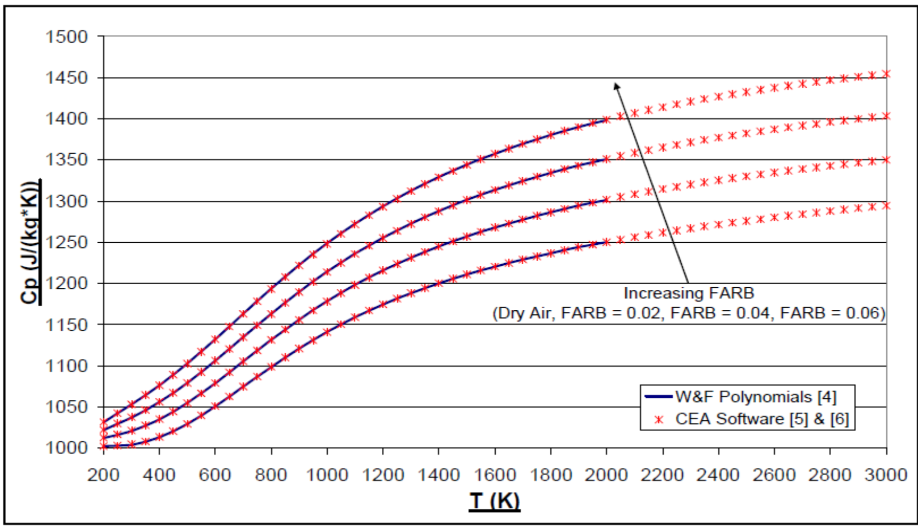


Figure 3-2 – Variation of C_p as a function of temperature and composition
(Source: Vishal Sethi "Advanced Performance Simulation of Gas Turbine Components and Fluid Thermodynamic Properties" 2008, Cranfield University)

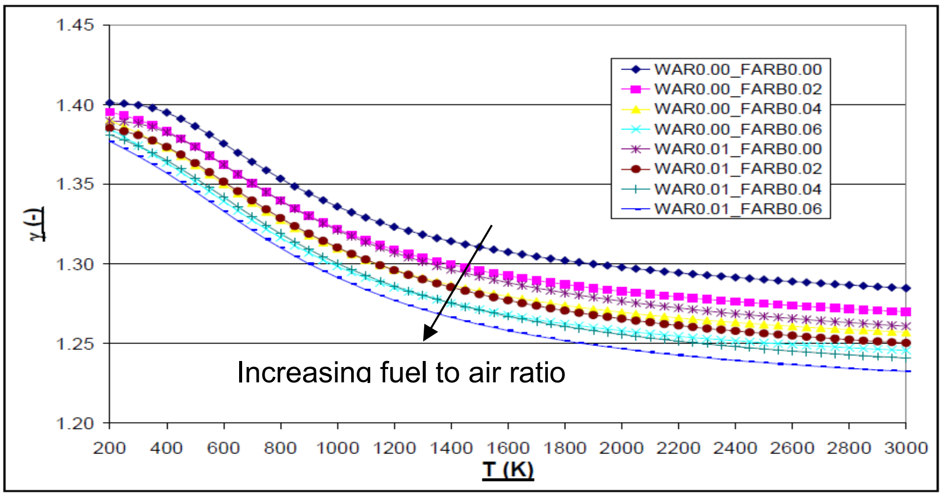


Figure 3-3 – Variation of C_p as a function of temperature and composition
(Source: Vishal Sethi "Advanced Performance Simulation of Gas Turbine Components and Fluid Thermodynamic Properties" 2008, Cranfield University)

The mean values for c_p and k are averaged for the usual temperature ranges (compression, combustion and expansion trajectories) for air and for flue gas in the gas turbine cycle. With this set of constants, reasonably accurate manual calculations can be performed. The errors remain limited due to mutual compensation of the effects of deviations from the real values for c_p and k. Still, with manual cycle calculations at high turbine entry temperatures, accuracy will deteriorate, and significant deviations in pressure and temperature from reality will occur (typically more than 5%) downstream the combustor. This will also cause larger error in calculated power output or thrust.

3.3 Total Enthalpy, Temperature and Pressure

With the first law of thermodynamics (conservation of energy) and when ignoring potential energy, the energy balance becomes:

$$\dot{Q} = \dot{m}\left(h_2 - h_1\right) + \tfrac{1}{2}\dot{m}\left(v_2^2 - v_1^2\right) + \dot{W} \tag{3.2}$$

For gas turbine cycle calculations, fluid velocity inside the components (i.e. the difference between component inlet and exit velocity) is of little concern. Therefore the concept of **total enthalpy** h_0, **total temperature** T_0 and **total pressure** p_0, is introduced. The **total enthalpy** (also known as the stagnation enthalpy) is defined as:

$$h_0 = h + \tfrac{1}{2}v^2 \tag{3.3}$$

The qualification ***total*** is opposed to ***static***, which refers to the state of the fluid without taking velocity into account (p, T and h are the static properties).

Physically, total enthalpy is defined as the enthalpy level the fluid would obtain if a moving fluid were adiabatically brought to a standstill without the addition or extraction of work. One of the assumptions in this textbook is that the fluid in the cycle is considered as an ideal gas with constant c_p and k (i.e. independent of temperature). This simplifies the relation between total enthalpy and temperature and then **total temperature** T_0 can be defined as:

$$h_0 = c_p \cdot T_0 = c_p \cdot T + \tfrac{1}{2}v^2 \tag{3.4}$$

$$T_0 = T + \frac{v^2}{2 \cdot c_p} \tag{3.5}$$

Deceleration of the fluid results in an increase in both temperature and pressure. When assuming isentropic change of state (i.e. thermodynamically reversible deceleration), **total pressure** p_0 is defined as:

$$p_0 = p\left(\frac{T_0}{T}\right)^{\frac{k}{k-1}} \tag{3.6}$$

Using total enthalpy, total pressure and total temperature, the energy level of the fluid can be determined at any station in the gas turbine cycle. Cycle calculations can be performed without explicit specification of kinetic energy in the equations.

3.4 Compressor and Turbine Efficiency

The compression and expansion in the real process are not irreversible and adiabatic, which means the relation between temperature and the pressure ratio is not fixed. Figure 3-4 shows the compression and expansion processes in the temperature-entropy diagram. The relation between temperature and pressure can be expressed in terms of the ratio of work for the ideal versus the real process in the form of the **isentropic efficiency.**

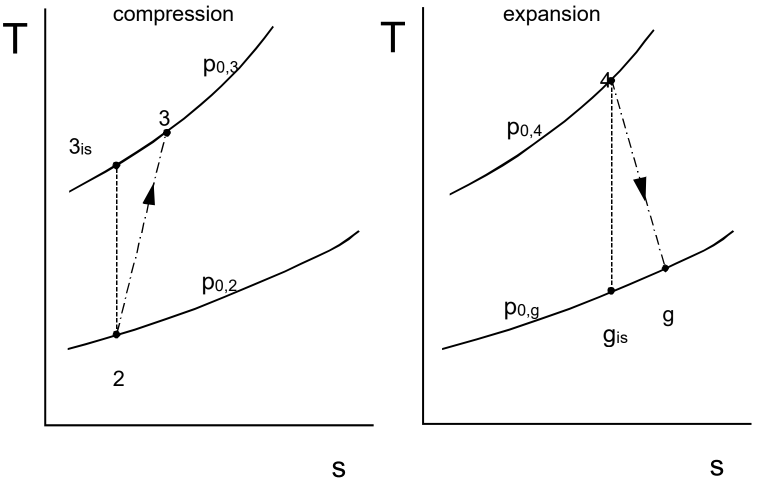


Figure 3-4 - Non-isentropic compression and expansion respectively

Using the concept of total enthalpy or temperature, which accounts for kinetic energy changes of the fluid between inlet and exit of the component, isentropic efficiency for a compressor can be defined as:

$$\eta_{comp} = \frac{h_{0,3,is} - h_{0,2}}{h_{0,3} - h_{0,2}} = \frac{T_{0,3,is} - T_{0,2}}{T_{0,3} - T_{0,2}} \tag{3.7}$$

Isentropic efficiency for a turbine is defined as:

$$\eta_{turb} = \frac{h_{0,4} - h_{0,g}}{h_{0,4} - h_{0,g,is}} = \frac{T_{0,4} - T_{0,g}}{T_{0,4} - T_{0,g,is}} \tag{3.8}$$

Substituting:

$$\frac{T_{0,3,is}}{T_{0,2}} = \left(\frac{p_{0,3}}{p_{0,2}}\right)^{\frac{k_{air}-1}{k_{air}}} \quad \text{resp.} \quad \frac{T_{0,4}}{T_{0,g,is}} = \left(\frac{p_{0,4}}{p_{0,g}}\right)^{\frac{k_{gas}-1}{k_{gas}}} \tag{3.9}$$

into equation (3.7) and (3.8) gives:

$$\eta_{is,comp} = \frac{\left(\dfrac{p_{0,3}}{p_{0,2}}\right)^{\frac{k_{air}-1}{k_{air}}} - 1}{\dfrac{T_{0,3}}{T_{0,2}} - 1} \quad \text{resp.} \quad \eta_{is,turb} = \frac{\dfrac{T_{0,g}}{T_{0,4}} - 1}{\left(\dfrac{p_{0,g}}{p_{0,4}}\right)^{\frac{k_{gas}-1}{k_{gas}}} - 1} \tag{3.10}$$

In a gas generator, specific power required by the compressor and specific power delivered by the turbine are defined by respectively:

$$\dot{W}_{s,comp} = c_{p_{air}}\left(T_{0,3} - T_{0,2}\right) = \frac{c_{p_{air}} T_{0,2}}{\eta_{is,comp}}\left[\left(\frac{p_{0,3}}{p_{0,2}}\right)^{\frac{k_{air}-1}{k_{air}}} - 1\right] \tag{3.11}$$

$$\dot{W}_{s,turb} = c_{p_{gas}}\left(T_{0,4} - T_{0,g}\right) = c_{p_{gas}} T_{0,4}\eta_{is,turb}\left[1 - \left(\frac{p_{0,g}}{p_{0,4}}\right)^{\frac{k_{gas}-1}{k_{gas}}}\right] \tag{3.12}$$

A problem arises when gas turbine cycles are analyzed with varying compression ratios. In practice, varying pressure ratio means varying the number of compressor or turbine stages. Typical compression ratios that can be achieved with a single compressor stage range between 1.2 to 1.4. Assuming a compression ratio of 20 is necessary for a specific gas turbine cycle, the amount of stages (n) would vary between $n_1 = ln(20) / ln(1.4) \approx 9$ stages and $n_2 = ln(20) / ln(1.2) \approx 17$ stages depending on single stage pressure ratio. A pressure ratio variation between 5 and 30 means a variation in number of stages between 5 – 11 or 9 – 19 respectively.

A compressor generally has a number of successive stages in series with similar characteristics, i.e. similar isentropic efficiencies. Geometry is changing and blade length is decreasing, because of increasing density with increasing pressure of the medium downstream. Design rules for the variation of geometry usually make flow losses and thereby also stage isentropic efficiency remain rather constant.

Figure 3-5 shows a T-s diagram for a compressor with three stages. This figure will be used to show that the overall isentropic efficiency of a series of compressor stages, all having the same isentropic efficiency per stage, is smaller than the isentropic efficiency of an individual compressor stage.

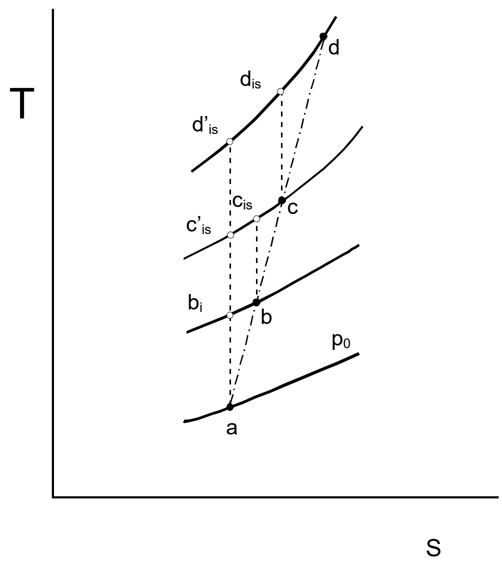


Figure 3-5 - Three stage compression

For the individual stages with the same isentropic efficiencies the following formula is applicable:

$$\eta_{stage} = \frac{T_{0,b,is} - T_{0,a}}{T_{0,b} - T_{0,a}} = \frac{T_{0,c,is} - T_{0,b}}{T_{0,c} - T_{0,b}} = \frac{T_{0,d,is} - T_{0,c}}{T_{0,d} - T_{0,c}} \quad [-] \tag{3.13}$$

From which follows that:

$$\eta_{stage} = \frac{\left(T_{0,b,is} - T_{0,a}\right) + \left(T_{0,c,is} - T_{0,b}\right) + \left(T_{0,d,is} - T_{0,c}\right)}{T_{0,d} - T_{0,a}} \tag{3.14}$$

The overall isentropic efficiency for the three-stage compressor becomes:

$$\eta_{is,total} = \frac{T_{0d',is} - T_{0,a}}{T_{0,d} - T_{0,a}} = \frac{\left(T_{0,b,is} - T_{0,a}\right) + \left(T_{0,c',is} - T_{0,b,is}\right) + \left(T_{0d',is} - T_{0,c',is}\right)}{T_{0,d} - T_{0,a}} \tag{3.15}$$

Due to the divergent nature of the lines of constant pressure, interval $T_{0ds} - T_{0c}$ is larger than interval $T_{0d',is} - T_{0c',is}$, and interval $T_{0c,is} - T_{0b}$ is larger than interval $T_{0c',is} - T_{0b',is}$. Comparing equation (3.13) to (3.14) the isentropic efficiency of the stage is larger than the overall isentropic efficiency.

If we divide the compression phase *a – d* into an infinite number of infinitely small compression stages, with equal isentropic efficiencies, the result is a polytropic compression process with a constant value for the polytropic exponent n_{air}. The relation between the pressure and temperature then is:

$$\frac{T_0}{T} = \left(\frac{p_0}{p}\right)^{\frac{n_{air}}{n_{air}-1}} \tag{3.16}$$

For an infinitely small step the relation between the temperature and the pressure can be written as:

$$\frac{dT_0}{T_{0,a}}=\frac{(n_{air}-1)}{n_{air}}\left(\frac{p_0}{p_{0,a}}\right)^{\frac{n_{air}-1}{n_{air}}}\frac{dp_0}{p_0}=\frac{(n_{air}-1)}{n_{air}}\left(\frac{T_0}{T_{0,a}}\right)\frac{dp_0}{p_0}=\frac{(n_{air}-1)}{n_{air}}\frac{dp_0}{p_0} \tag{3.17}$$

A similar derivation for an isotropic change of state leads to:

$$\frac{dT_{0,s}}{T_0}=\frac{(k_{air}-1)}{k_{air}}\frac{dp_0}{p_0} \tag{3.18}$$

As an alternative for the **isentropic efficiency** we now define the **polytropic efficiency** as the isentropic efficiency of an infinitely small compression step with the assumption that it is constant for throughout the compression phase. The polytropic efficiency can be calculated by the quotient of equation (3.18) and (3.17):

$$\eta_\infty=\frac{dT_{0,s}}{dT_0}=\frac{\frac{(k_{air}-1)}{k_{air}}}{\frac{(n_{air}-1)}{n_{air}}} \tag{3.19}$$

For a compressor polytropic efficiency can be expressed as:

$$\eta_{\infty,comp}=\frac{\ln\left(\frac{p_{0,3}}{p_{0,2}}\right)^{\frac{k_{air}-1}{k_{air}}}}{\ln\left(\frac{T_{0,3}}{T_{0,2}}\right)} \tag{3.20}$$

For the expansion process in a turbine a similar relation can be formulated:

$$\eta_{\infty,turb}=\frac{\ln\left(\frac{T_{0,gg}}{T_{0,4}}\right)}{\ln\left(\frac{p_{0,gg}}{p_{0,4}}\right)^{\frac{k_{gas}-1}{k_{gas}}}} \tag{3.21}$$

Note that for a turbine, isentropic stage efficiency is smaller than overall isentropic efficiency.

A compression or expansion process can be characterized by either isentropic or polytropic efficiencies. The relation between the two can be derived combining (3.10) with equation (3.20) and (3.21):

$$\eta_{comp} = \frac{\left(\frac{p_{0,3}}{p_{0,2}}\right)^{\frac{k_{air}-1}{k_{air}}} - 1}{\frac{T_{0,3}}{T_{0,2}} - 1} = \frac{\left(\frac{p_{0,3}}{p_{0,2}}\right)^{\frac{k_{air}-1}{k_{air}}} - 1}{\left(\frac{p_{0,3}}{p_{0,2}}\right)^{\frac{k_{air}-1}{k_{air}\,\eta_{\infty,comp}}} - 1} \tag{3.22}$$

$$\eta_{turb} = \frac{\frac{T_{0,g}}{T_{0,4}} - 1}{\left(\frac{p_{0,g}}{p_{0,4}}\right)^{\frac{k_{gas}-1}{k_{gas}}} - 1} = \frac{\left(\frac{p_{0,g}}{p_{0,4}}\right)^{\eta_{\infty,turb}\frac{k_{gas}-1}{k_{gas}}} - 1}{\left(\frac{p_{0,g}}{p_{0,4}}\right)^{\frac{k_{gas}-1}{k_{gas}}} - 1} \tag{3.23}$$

In Figure 3-6 the relation between isentropic- and polytropic efficiency is plotted against pressure ratio:

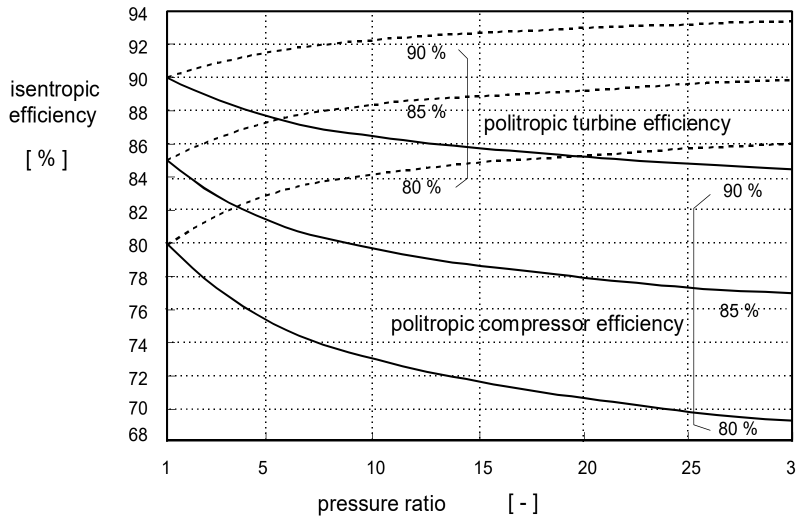


Figure 3-6 – Relation between isentropic- and polutropic efficiency

Figure 3-6 confirms and quantifies the effects of the previously mentioned observations:

- the difference between polytropic efficiency and isentropic efficiency increases with increasing pressure ratio,
- polytropic efficiency is always higher than isentropic efficiency for compression,
- polytropic efficiency is always smaller than isentropic efficiency for expansion.

In case of calculating gas turbine cycle performance for a range of compression ratio values as is typical for cycle analysis and optimization (initial gas turbine design phase), using polytropic efficiency is most practical.

3.5 Pressure Losses

3.5.1 Combustion Chamber Pressure Loss

Total pressure loss in the combustion chamber is caused by addition of heat and flow losses. Pressure loss usually is taken as a percentage of the combustor inlet pressure. The combustor pressure loss factor is defined as:

$$\Pi_{cc} = \frac{p_{0,4}}{p_{0,3}} = \frac{p_{0,3} - \Delta p_{cc}}{p_{0,3}} \tag{3.24}$$

3.5.2 Inlet Pressure Losses in Industrial Gas Turbines

The inlet (or intake) of an industrial gas turbine has multiple functions. Not only does the inlet guide the air to the compressor inlet, it also filters or conditions (humidity) the air and muffles the noise of the combustion and the rotary equipment.

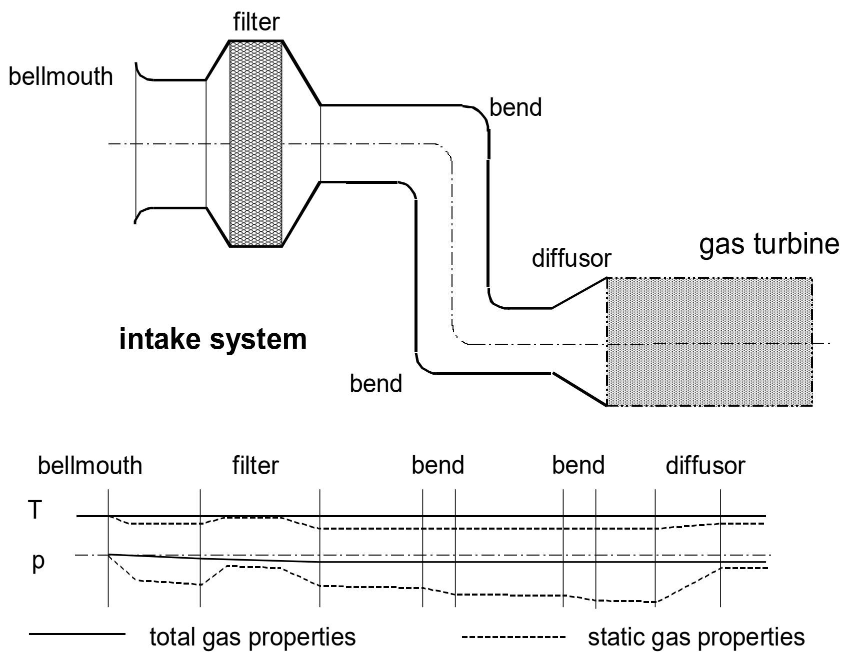


Figure 3-7 - Gas properties in the intake system

Total pressure of the air in the intake system drops due to the inflow losses, pressure loss over the filter and duct friction losses. Total temperature usually can be assumed constant throughout the inlet system, except when case air-conditioning systems or (wall) heater systems are present in the intake system. Figure 3-7 schematically shows the layout of a typical industrial gas turbine intake system. The bottom chart shows the variation of total and static temperature and pressure, indicating static temperature and pressure depend on the momentary velocity of the fluid. The figure further shows that when the total pressure for instance decreases it is possible that static pressure increases. As a measure for pressure loss in the inlet system, the difference between the ambient pressure and compressor face total pressure is chosen.

$$\Delta p_{0,inlet} = p_{amb} - p_{0,1} \tag{3.25}$$

3.5.3 Inlet Pressure Losses in Aircraft Gas Turbines

The pressure losses and ram recovery effects in aircraft gas tubines are discussed separately in the chapter on aircraft propulsion.

3.5.4 Exhaust System Pressure Losses in Industrial Gas Turbines

The purpose of the exhaust of an industrial gas turbine is similar but opposite to the inlet: to guide the gasses to the environment, cleaning the exhaust flue gases (if necessary) and muffling the noise of the combustion process and the rotary equipment. The pressure loss of the exhaust system includes duct, filter and silencer friction losses. For industrial gas turbines using a boiler in the exhaust gas system, the additional boiler pressure loss needs to be added to the overall exhaust pressure loss. Another additional loss is the kinetic energy of the flue gas leaving the exhaust system.

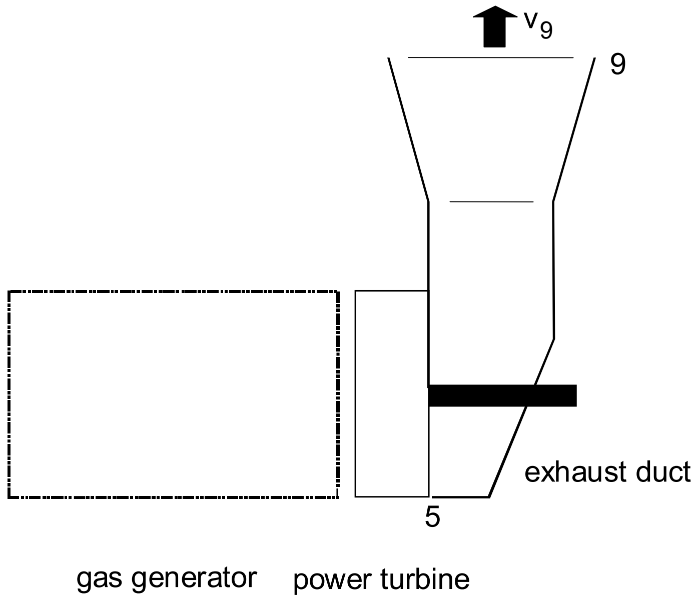


Figure 3-8 - Station numbering of an industrial gas turbine

Using Figure 3-8 as a reference for station numbering (see also section Appendix A on page 151), the pressure loss of the exhaust system can be written as:

$$\Delta p_{0,exhaust} = p_{0,5} - p_{0,9} = p_{0,5} - p_{amb} \tag{3.26}$$

and:

$$T_{0,5} = T_{0,9} = T_9 + \frac{v_9^2}{2c_{p_{gas}}} \tag{3.27}$$

3.5.5 Exhaust System Pressure Losses in Aircraft Gas Turbines

Exhaust pressure losses in aircraft gas turbines are discussed separately in the chapter on aircraft propulsion.

3.6 Mechanical Losses

Losses due to friction of bearings, seals, windage, and power needed for fuel-, oil- and control systems (accessories mounted on the "accessory gearbox") are usually accounted for in one single transmission efficiency, called mechanical efficiency η_{mech}. This efficiency is related to the turbine power:

$$\eta_{mech} = \frac{turbine\ power - mechanical\ losses}{turbine\ power} \tag{3.28}$$

If a gas turbine has multiple turbines, the mechanical losses need to be accounted to the turbines connected to the particular shaft or driving the specific accessories.

3.7 Combustor Efficiency

The maximum heat that can be extracted from a fuel is characterized by the **lower heating value** LHV. The LHV can be determined by **full** (ideal) combustion of fuel in air and then cooling the flue gas to the temperature of the fuel-air mixture it had before combustion. The heat of condensation of the water vapor created by the combustion is **not** included in the LHV. The real combustion process is not ideal but **incomplete** and then the following additional combustion products can be formed: carbon monoxide (CO), soot and hydrocarbon compounds (unburned fuel).

Not all the heat released by combustion can be used for the cycle since some of it will "escape" to the immediate surroundings of the combustion chamber, either by conduction through the metal or by radiation. Heat losses usually are very small and can be quantified by **combustor efficiency** η_{cc}, which is defined as:

$$\eta_{cc} = \frac{\dot{m}_{air} c_{p_{gas}} \left(T_{0,4} - T_{0,3}\right)}{\dot{m}_f LHV_f} \tag{3.29}$$

For a gas turbine running at full power, combustor efficiency usually is higher than 99% At partial power the efficiency may well drop to 97 – 98%.

3.8 Calculation Scheme to Determine Gas Generator Power and Efficiency

The following will show a calculation scheme to calculate power and efficiency of the gas generator. If, for a given industrial gas turbine cycle, parameters such as mass flow, pressure ratio, component efficiencies and ambient conditions are known, power and efficiency can be determined as follows:

a) Inlet/intake

For an adiabatic inlet process, the total temperature and total pressure at the inlet of the compressor are:

[a.1] $T_{0,2} = T_{0,amb}$

[a.2] $p_{0,2} = p_{amb} - \Delta p_{0,amb}$

b) Compressor

For the compressor exit temperature can be written as:

[b.1] $$T_{0,3} = T_{0,2} + \frac{T_{0,2}}{\eta_{is,comp}}\left(\Pi^{\frac{k_{air}-1}{k_{air}}} - 1\right)$$

using the isentropic compressor efficiency or alternatively using the polytropic efficiency as:

[b.2] $$T_{0,3} = T_{0,2}\Pi^{\frac{k_{air}-1}{k_{air}\cdot\eta_{\infty,comp}}}$$

the compressor exit pressure becomes:

[b.3] $$p_{0,3} = p_{0,2}\,\Pi_{comp}$$

and finally, the power required to drive the compressor becomes:

[b.4] $$P_{comp} = \dot{m}\,c_{p_{air}}\left(T_{0,3} - T_{0,2}\right)$$

c) Combustor

Under the assumption that the mass flow through the combustor is considered to be equal to the mass flow through the compressor and turbine, the heat balance of the combustor becomes:

[c.1] $$\dot{m}_f LHV_f \eta_{cc} = \dot{m}\,c_{p_{gas}}\left(T_{0,4} - T_{0,3}\right)$$

combustor exit pressure:

[c.2] $$p_{0,4} = p_{0,3}\,\Pi_{cc}$$

d) Turbine

The power delivered by the turbine is:

[d.1] $$P_{turb} = \dot{m}\,c_{p_{gas}}\left(T_{0,4} - T_{0,g}\right)$$

the power balance:

[d.2] $$P_{turb}\,\eta_{mech} = P_{comp}$$

gas generator exit pressure, using the isentropic turbine efficiency:

[d.3] $$p_{0,g} = p_{0,4}\left[1-\frac{1}{\eta_{is,turb}}\left(1-\frac{T_{0,g}}{T_{0,4}}\right)\right]^{\frac{k_{gas}}{k_{gas}-1}}$$

or as an alternative using the polytropic efficiency:

[d.4] $$p_{0,g} = p_{0,4}\left[\frac{T_{0,g}}{T_{0,4}}\right]^{\frac{k_{gas}}{\eta_{\infty,turb}\left(k_{gas}-1\right)}}$$

e) Gas power and thermodynamic efficiency

[e.1] $$P_{gg} = \dot{m}c_{p_{gas}}T_{0,g}\left[1-\left(\frac{p_0}{p_{0,g}}\right)^{\frac{k_{gas}-1}{k_{gas}}}\right]-\tfrac{1}{2}\dot{m}v_0^2$$

where v_0 denotes the airflow speed at the inlet.

[e.2] $$\eta_{th} = \frac{P_{gg}}{\dot{m}c_{p_{gas}}\left(T_{0,4}-T_{0,3}\right)}$$

From these 12 equations (14 equations have been stated, but includes 2 functions written alternatively) 23 parameters are counted, excluding the constant values k_{air}, k_{gas}, $c_{p\ air}$ *and* $c_{p\ gas}$. Knowing 13 of these parameters (T_0, p_0, v_0, $\dot{m}$, ε_{comp}, ε_{cc}, η_{comp} or $\eta_{\infty,comp}$, η_{turb} or $\eta_{\infty,turb}$, η_{inlet}, η_{cc}, η_{mech}, *LHV and* $T_{0,4}$) solves the system since 23 – 13 = 12 equations. Checking the amount of parameters and equations is useful when dealing with a limited amount of measured parameters that are needed to determine other parameters such as the component efficiencies, pressure ratios, pressure losses, etc.

In the next chapters, the number of equations and parameters is expanded with respect to the type of application (propulsion or shaft power).

3.9 Performance Characteristics of the Gas Generator

Using the calculation scheme from section 3.8 for a range of pressure ratio's and firing temperatures (Π_{comp} respectively $T_{0,4}/T_{amb}$), a graphical presentation shown in Figure 3-9 can be obtained. The outlines of the ideal cycle have been added to compare the ideal cycle to the real cycle. The figure shows that (in comparison to the ideal cycle; see also Figure 2-3) the real cycle:

a) has lower values of specific gas power and thermodynamic efficiency,
b) has a thermodynamic efficiency that no longer depends on the firing temperature ratio $T_{0,4}/T_{amb}$ (note that the constant pressure lines are no longer horizontal),
c) has an optimum pressure ratio (i.e. the Π_{comb} for which $P_{s,gg}/c_p\,T_0$ is maximal for given $T_{0,4}/T_{amb}$) that is smaller than the ideal cycle optimal pressure.

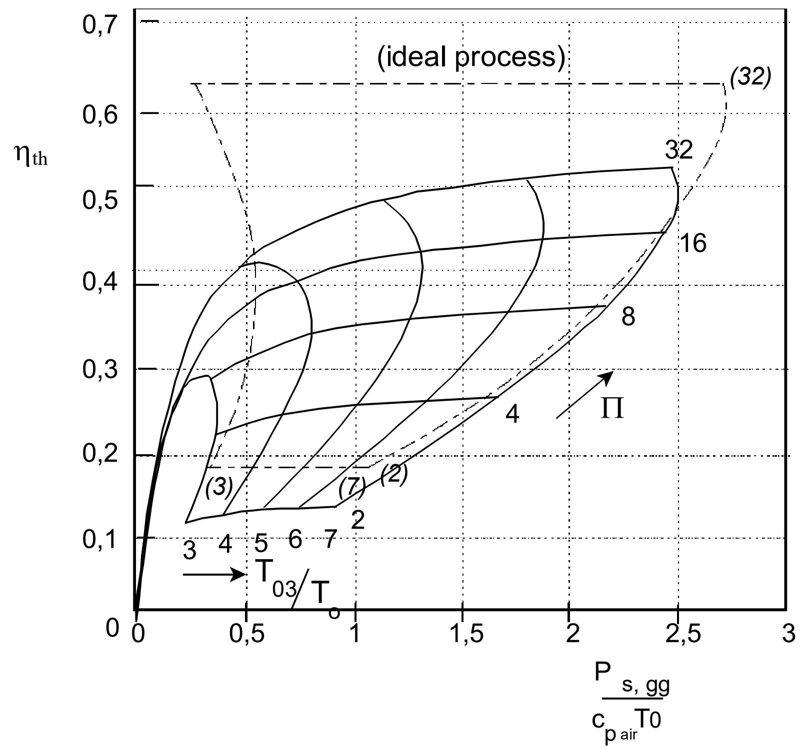


Figure 3-9 - Specific power and thermodynamic efficiency of an real gas generator for several combinations of Π and T_{04}/T_{02} ($\eta_{\infty,comp} = \eta_{\infty,turb} = 0.9$, $\Pi_{cc} = 0.98$, $\eta_{mech} = 0.98$, $\Pi_{inlet} = 0.98$, $v_0 = 0$, $c_{p\ air} = 1000$ J/kg/K, $c_{p\ gas} = 1150$ J/kg/K)

The main reason for the differences in specific power and efficiency between the ideal cycle and the real cycle is caused by the fact that the compression and expansion process are not isentropic. The effect of the compressor and turbine efficiency on the specific power and thermodynamic efficiency is shown in Figure 3-10 and Figure 3-11. The figures show that the compressor efficiency has a large effect on the specific power and thermodynamic efficiency, especially for low values.

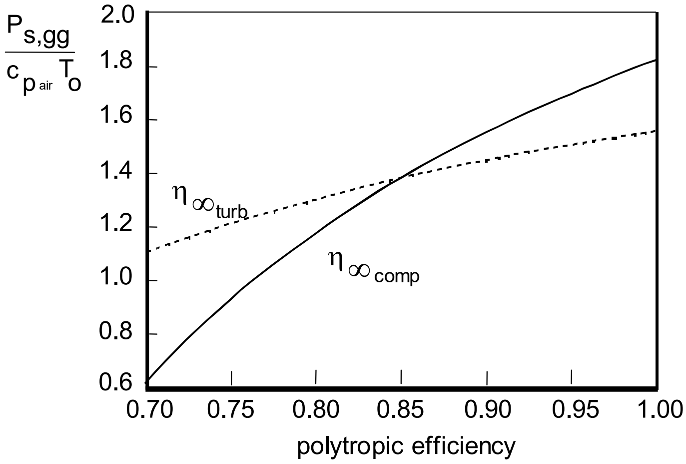


Figure 3-10 – Effect of component efficiency on specific gas generator power

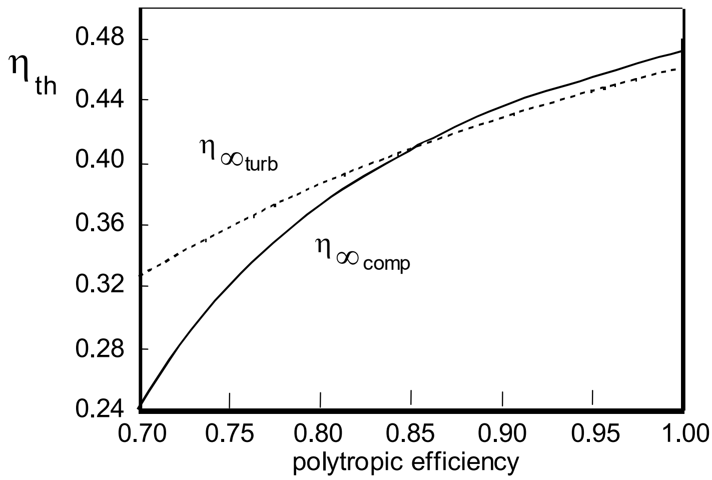


Figure 3-11 – Efficiency effects on thermodynamic efficiency

3.10 Example: Real Gas Generator

Consider a gas generator of an industrial gas turbine (see Figure 3-12). The ambient conditions are: $T_{amb} = 288\ K$, $p_{amb} = 1.013\ bar$. Assume isentropic component efficiencies: $\eta_{comp} = 87\%$ and $\eta_{turb} = 89\%$. The pressure ratio equals $\Pi_{comp} = 16$, and the turbine inlet temperature equals $T_{0,4} = 1400\ K$. The pressure loss ratio over the combustor equals $\Pi_{cc} = 99\%$ and the combustor efficiency equals $\eta_{cc} = 98\%$. Mechanical losses are $\eta_{mech} = 98\%$. Bleed air (10%) is extracted from the engine at the end of the compressor, and will not be inserted back into the gas turbine. Inlet and exhaust losses are negligible. The contribution of the fuel flow cannot be ignored due to the low calorific value of 12 MJ/kg.

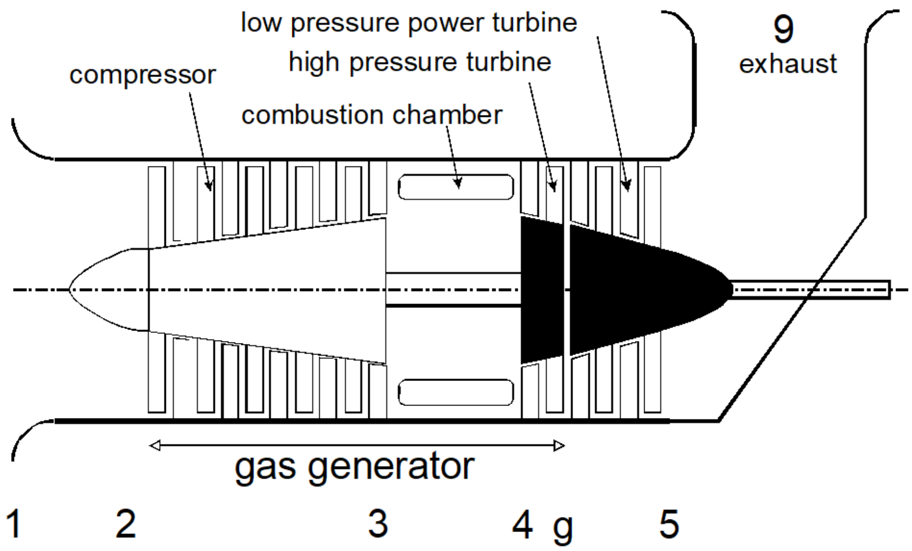


Figure 3-12 - Gas generator with free power turbine

Calculate the following:

1. *The specific compressor power*
2. *The specific turbine power delivered by the gas generator*
3. *The specific amount of heat added to the gasturbine*
4. *The thermodynamic efficiency of the gas generator*

All solutions need to be based on the compressor mass flow.

Solution:

1. *Isentropic compression temperature*

$$T_{is,0,3} = T_{0,2}\,\Pi_{comp}^{\frac{k_{air}-1}{k_{air}}} = 288 \cdot 16^{\frac{0.4}{1.4}} = 636\ [K]$$

2. *Real compressor temperature*

$$T_{0,3} = T_{0,2} + \frac{T_{is,0,3} - T_{0,2}}{\eta_{is,comp}} = 288 + \frac{636 - 288}{0.87} = 688\ [K]$$

3. *Compressor power*

$$P_{s,comp} = c_{p_{air}} \left(T_{0,3} - T_{0,2}\right) = 1.0 \cdot \left(688 - 288\right) = 400 \cdot 10^3\ \left[W/(kg/s)\right]$$

4. *Heat addition*

$$Q_{cc} = \dot{m}_f LHV_f \cdot \eta_{cc} = 0.9\, \dot{m}_{comp} c_{p_{gas}} \left(T_{0,4} - T_{0,3}\right) \qquad \Rightarrow$$

$$\frac{\dot{m}_f}{\dot{m}_{comp}} = 0.0614$$

5. *The specific heat addition*

$$Q_{s,cc} = \frac{\dot{m}_f}{\dot{m}_{comp}} LHV_f = 0.0614 \cdot 12 \cdot 10^6 = 737 \cdot 10^3\ \left[W/(kg/s)\right]$$

6. *Expansion exit temperature is determined by the power balance between the compressor and turbine*

$$\left(0.9\dot{m}_{comp} + \dot{m}_f\right) c_{p_{gas}} \left(T_{0,4} - T_{0,g}\right) \eta_{mech} = \dot{m}_{comp} c_{p_{gas}} \left(T_{0,3} - T_{0,2}\right) \qquad \Rightarrow$$

$$T_{0,g} = 1031\ [K]$$

7. *The isentropic expansion exit temperature becomes*

$$T_{is,0,g} = T_{0,4} - \frac{T_{0,4} - T_{0,g}}{\eta_{is,turb}} = 1400 - \frac{1400 - 1031}{0.89} = 985\ [K]$$

8. *Real expansion exit pressure*

$$p_{0,g} = p_{0,4} \left(\frac{T_{0,g}}{T_{0,4}}\right)^{\frac{k_{air}-1}{k_{air}}} = 0.98 \cdot 16 \cdot 1.013 \cdot \left(\frac{985}{1400}\right)^{\frac{0.33}{1.33}} = 3.85\ [\mathrm{bar}]$$

9. *At the exit of the gas generator, the following specific power will be available*

$$P_{s,gg} = \frac{0.9\dot{m}_{comp} + \dot{m}_f}{w_c} c_{p_{gas}} T_{0,g} \left[1 - \left(\frac{p_{0,2}}{p_{0,g}}\right)^{\frac{k_{gas}-1}{k_{gas}}}\right] = \left(0.9 + 0.0614\right) \cdot 1.15 \cdot 1031 \cdot \left[1 - \left(\frac{1.013}{3.85}\right)^{\frac{0.33}{1.33}}\right] =$$

$$P_{s,gg} = 322 \cdot 10^3\ \left[W/(kg/s)\right]$$

10. *The thermodynamic efficiency of the process*

$$\eta_{th} = \frac{P_{s,gg}}{Q_{s,cc}\eta_{cc}} = \frac{322}{737 \cdot 0.99} \cdot 100 = 44.1\ [\%]$$

3.11 Real Enhanced Cycles

3.11.1 Recuperated Cycles and Heat Exchanger Effectiveness

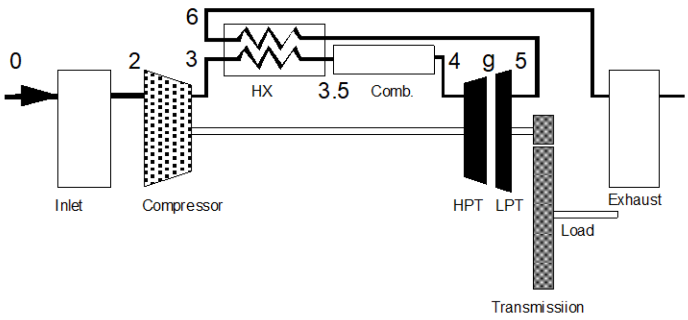


Figure 3-13 – Recuperated gas turbine

In section 2.3.1 the application of heat exchangers to increase the thermal efficiency of the gas turbine was described. The calculations in that chapter were based the ideal case of maximum possible heat exchange. Compressor exit temperature is then raised to $T_{0,5}$ (i.e. $T_{0,35}$) using the heat from the gas at expansion exit corresponding to a temperature drop from $T_{0,5}$ down to $T_{0,6}$ (which is then equal to $T_{0,3}$). This ideal case cannot be realised due to two reasons.

1. The specific heat of hot exhaust gas $c_{p\ gas}$ is higher than $c_{p\ air}$. When the exhaust gas would be cooled to $T_{0,6}$ (equal to $T_{0,3}$), from the enthalpy balance ($h_{0,5} - h_{0,3} = h_{0,35} - h_{0,3}$) it would follow that $T_{0,35} = T_{0,5} + (c_{p\ gas}/c_{p\ air} - 1)\cdot(T_{0,5} - T_{0,3})$. This would mean that $T_{0,35}$ exceeds $T_{0,5}$, which is impossible.
2. The heat exchanger dimensions and weight are limited for economic reasons. A heat exchanger with maximum efficiency ($T_{0,35} = T_{0,4}$) would require an infinitely large heat exchanging area!

The parameter used to indicate the quality of a heat exchanger is 'effectiveness' E.

$$\begin{gathered} E = \frac{C_{hot}}{C_{\min}} \cdot \frac{T_{in\_hot} - T_{out\_hot}}{T_{in\_hot} - T_{in\_cold}} = \frac{C_{cold}}{C_{\min}} \cdot \frac{T_{out\_cold} - T_{in\_cold}}{T_{in\_hot} - T_{in\_cold}} \\ \text{with:} \\ C_{hot} = c_{p_{in\_hot}} \cdot \dot{m}_{in\_hot} \quad ; \quad C_{cold} = c_{p_{in\_cold}} \cdot \dot{m}_{in\_cold} \\ C_{\min} = MIN(C_{hot}, C_{cold}) \; ; \quad C_{\max} = MAX(C_{hot}, C_{cold}) \end{gathered} \tag{3.30}$$

When used as a recuperator in a gas turbine and $\dot{m}_{cold}$ is assumed equal to $\dot{m}_{hot}$ and $c_{p,gas}$ equal to $c_{p,air}$ then E is defined as:

$$E = \frac{T_{0,35} - T_{0,3}}{T_{0,5} - T_{0,3}} \tag{3.31}$$

Note that maximum effectiveness in (3.30) is 1, while maximum E in (3.31) is $c_{p,gas}/c_{p,air}$.

Figure 3-14 shows the effect of heat exchanger effectiveness on thermal efficiency of a gas turbine with recuperator (heat exchanger). The curve E = 0 % represents a simple cycle gas turbine.

Figure 3-14 shows that an optimal cycle pressure ratio Π can be derived for different values of E. The optimal cycle pressure ratio decreases with increasing heat exchanger effectiveness. See the example of a real cycle with heat exchange in section 4.4.1.

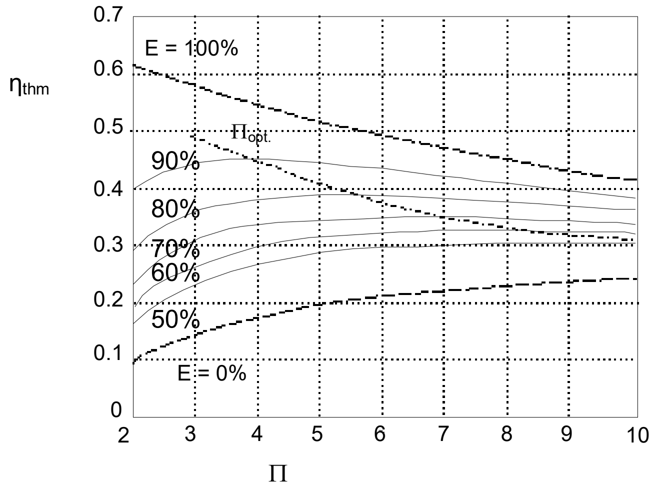


Figure 3-14 Heat exchanger effectiveness effect on η_{thm}

($\eta_{\infty,comp} = \eta_{\infty,turb} = 0.9$, $T_{0,4}/T_{0,2} = 5$)

3.11.2 Combined Intercooling and Heat Exchange

In section 2.3, ideal cycles with intercooling and heat exchange were addressed. Figure 3-15 represents the performance of a cycle with both heat exchange and intercooling, calculated with real gasses and component losses. The effect of Π_{comp} on η_{thm} is made clearer in Figure 3-16.

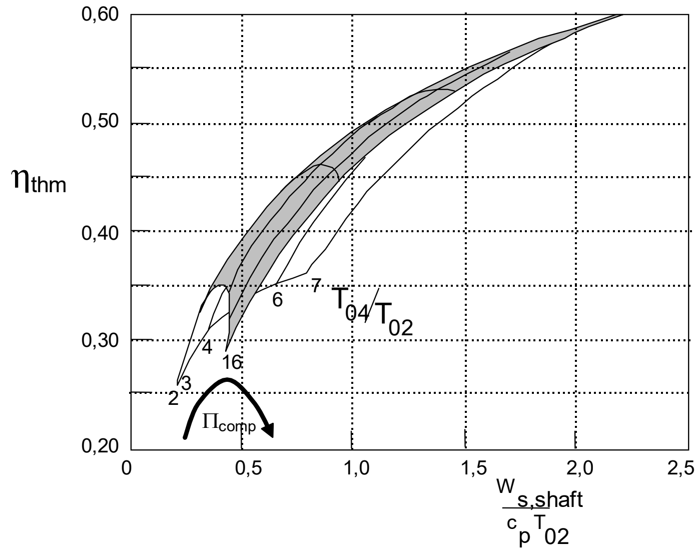


Figure 3-15 η_{thm} and specific power of a real cycle with heat

($\eta_{\infty,comp} = \eta_{\infty,turb} = 0.9$, $\eta_{mech} = 0.98$, $\Pi_{inlet} = \Pi_{exhaust} = 0.98$, $\eta_{comb} = 0.98$, $\eta_{LPC} = \eta_{HPC} = \sqrt{\Pi_{comp}}$)

Figure 3-16 shows that for lower values of $T_{0,4}/T_{0,2}$ an optimum exists for cycle pressure ratio. This effect is caused by the heat exchanger. For $T_{0,4}/T_{0,2}$ values higher than 5 the thermal efficiency becomes less sensitive to cycle pressure ratio due to the compensating effect on efficiency of the heat exchanger.

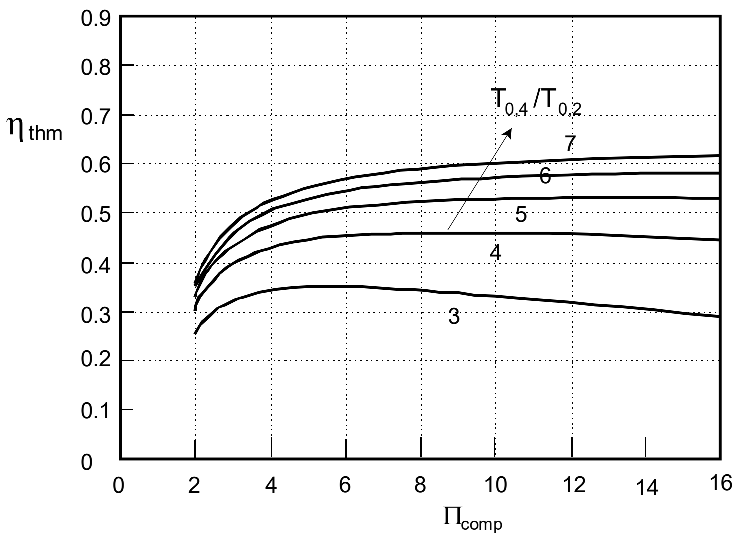


Figure 3-16 Recuperated-intercooled real cycle; pressure ratio and turbine entry temperature effect on thermal efficiency

($\eta_{\infty,comp} = \eta_{\infty,turb} = 0.9$, $\eta_{mech} = 0.98$, $\Pi_{inlet} = \Pi_{exhaust} = 0.98$, $\eta_{comb} = 0.98$, $\eta_{LPC} = \eta_{HPC} = \sqrt{\Pi_{comp}}$)

3.11.3 Reheated Cycles

For jet engines reheat (or "afterburning") is an effective means to increase thrust at the cost of high fuel consumption (chapter 1). The effect is described fairly accurately by the ideal cycle calculations in section 2.3.5 since the losses due to the real cycle are relatively small. Jet engine reheat however is limited to the point where all oxygen is used for combustion.

As will be explained in section 4.5, high efficiency gas turbines (high cycle pressure ratios) have relatively low exhaust gas temperatures. For industrial gas turbines, this makes combination with steam cycles unattractive. Reheat of the exhaust gas is a relatively simple solution for this problem. Figure 3-17 corresponds to Figure 2-17, with the only difference being the real component efficiencies. With mathematical analysis it can be shown that also for the real reheated cycle maximum power is obtained with pressure ratios equal for both parts of the expansion (before and after reheat) if the inlet temperatures of both turbines are equal. This is the case in Figure 3-17.

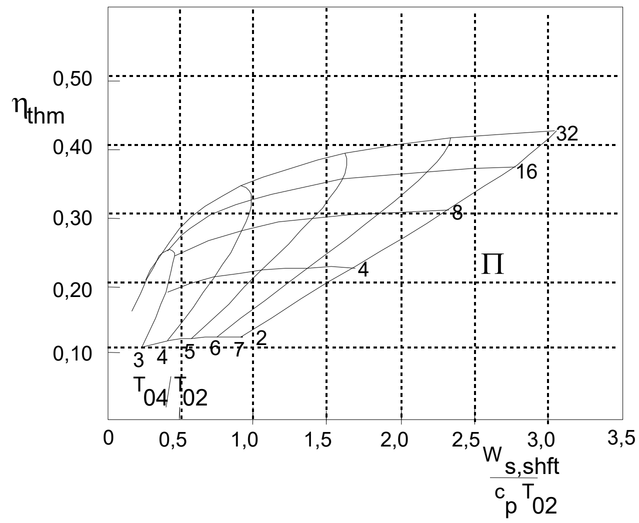


Figure 3-17 Reheated cycle thermal efficiency and shaft power

($\eta_{\infty,comp} = \eta_{\infty,turb} = 0.9$, $\eta_{mech} = 0.98$, $\Pi_{inlet} = \Pi_{exhaust} = 0.98$, $\eta_{comb} = 0.98$, $\eta_{LPC} = \eta_{HPC} = \sqrt{\Pi_{comp}}$)

Comparing Figure 3-17 to Figure 3-9 (simple cycle with equal component efficiencies) shows an increase of specific power and decrease of thermal efficiency occurs (e.g. for $\Pi_{comp} = 16$ and $T_{0,4}/T_{0,2} = 5$, the increase of specific power is about 20% and the decrease of the thermal efficiency about 10%).

Figure 3-18 shows dimensionless exhaust gas temperature $T_{0,5}/T_{0,2}$ for a cycle with reheat compared to without reheat. The significant increase in exhaust gas temperature obtained with reheat improves suitability for combined cycle configurations (i.e. a more efficient steam cycle).

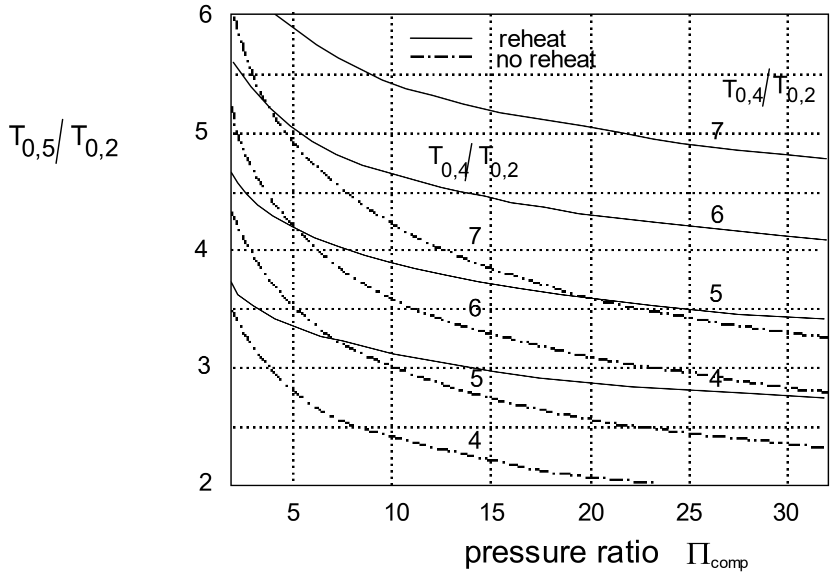


Figure 3-18 Exhaust gas temperature $T_{0,5}$ with and without reheat

($\eta_{\infty,comp} = \eta_{\infty,turb} = 0.9$, $\eta_{mech} = 0.98$, $\Pi_{inlet} = \Pi_{exhaust} = 0.98$, $\eta_{comb} = 0.98$, $\eta_{LPC} = \eta_{HPC} = \sqrt{\Pi_{comp}}$)

4. Shaft Power Gas Turbines

(Prof. Ir. Jos P. van Buijtenen, Dr. Wilfried P.J. Visser)

4.1 Introduction

One of the options to obtain mechanical power from the cycles described in chapters 2 and 3 is to expand the flue gas exiting the gas generator in a turbine driving a shaft that is connected to an external load. The loads may be generators, pumps, vehicle drive systems and (for aircraft, see the next chapter) helicopter rotors and propellers. Gas turbines delivering shaft power are generally referred to as 'turboshaft' engines.

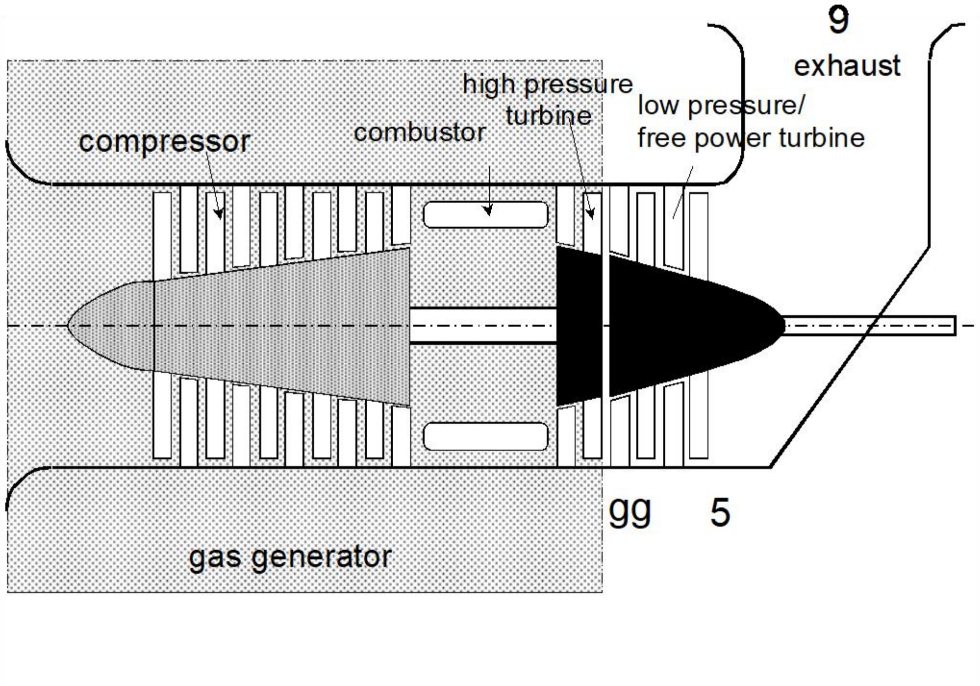


Figure 4-1 - Turbo shaft engine; gas generator and free power turbine

4.2 Single or Multi Spool Configurations

For the cycle, it does not make a difference if the expansion takes place in a single or multiple turbines. However, depending on the application, either single or multiple spool configurations are often preferred. For power generation, the single spool option is often used in view of the constant speed of the generator and relatively moderate load variations. For applications where the output shaft speed varies such as traction, pumps, fixed pitch ship propellers etc. a multi spool engine with a separate 'power turbine' often is preferred because then gas generator speed (and thus power) is independent of output shaft (i.e. power turbine) speed. Also (rapid) load variations and stringent part load requirements usually require a separate power turbine. Some gas turbine engines can be delivered in either a single or multi-spool arrangement (e.g. General Electric Frame 3 and 5).

4.3 Specific Power and Thermal Efficiency as Function of the Process Parameters

The temperature drop in the power turbine can be calculated using:

$$T_{0,g} - T_{0,5} = T_{0,g} \cdot \eta_{is,PT} \cdot \left[1 - \left(\frac{p_{0,5}}{p_{0,g}} \right)^{\frac{k_{gas}-1}{k_{gas}}} \right] \tag{4.1}$$

(the index “PT” indicates ‘power turbine’)

Using polytropic efficiency the equation becomes:

$$T_{0,g} - T_{0,5} = T_{0,g} \cdot \left[1 - \left(\frac{p_{0,5}}{p_{0,g}} \right)^{\frac{k_{gas}-1}{k_{gas}} \cdot \eta_{\infty,PT}} \right] \tag{4.2}$$

Where:

$$\Delta p_{exhaust} = p_{0,5} - p_{0,9} \tag{3.26}$$

$$T_{0,5} = T_{0,9} = T_9 + \frac{v_9^2}{2 \cdot c_{p_{gas}}} \tag{3.27}$$

The power extracted from the power turbine shaft (accounting for mechanical losses) can be written as:

$$P_{shaft} = \dot{m} \cdot c_{p_{gas}} \cdot \left(T_{0,g} - T_{0,5} \right) \cdot \eta_{mech,PT} \tag{4.3}$$

Overall turboshaft engine efficiency or ‘thermal efficiency’ is defined as:

$$\eta_{thm} = \frac{P_{shaft}}{\dot{m}_f \cdot LHV_f} \tag{4.4}$$

Note that thermal efficiency is lower than thermodynamic efficiency because the expansion after the gas generator (behind station g) is now included with associated losses and also combustor chamber heat losses are included.

Alternative indicators for fuel efficiency are specific fuel consumption (sfc) and heat rate. These are defined as:

$$sfc = \frac{\dot{m}_f}{P_{shaft}} \tag{4.5}$$

$$\text{heat rate} = \frac{\dot{\mathrm{m}}_\mathrm{f} \cdot L\mathrm{HV_f}}{\mathrm{P_{shaft}}} \tag{4.6}$$

From which follows:

$$sfc = \frac{1}{LHV_f \cdot \eta_{thm}} \tag{4.7}$$

$$heat\ rate = \frac{1}{\eta_{thm}} \tag{4.8}$$

Turboshaft engine performance characteristics (shown in Figure 4-2) are similar to individual gas generator characteristics (Figure 3-10) if the same component efficiencies are used. However, turboshaft specific shaft power and thermal efficiency levels are lower due to additional losses in the power turbine,

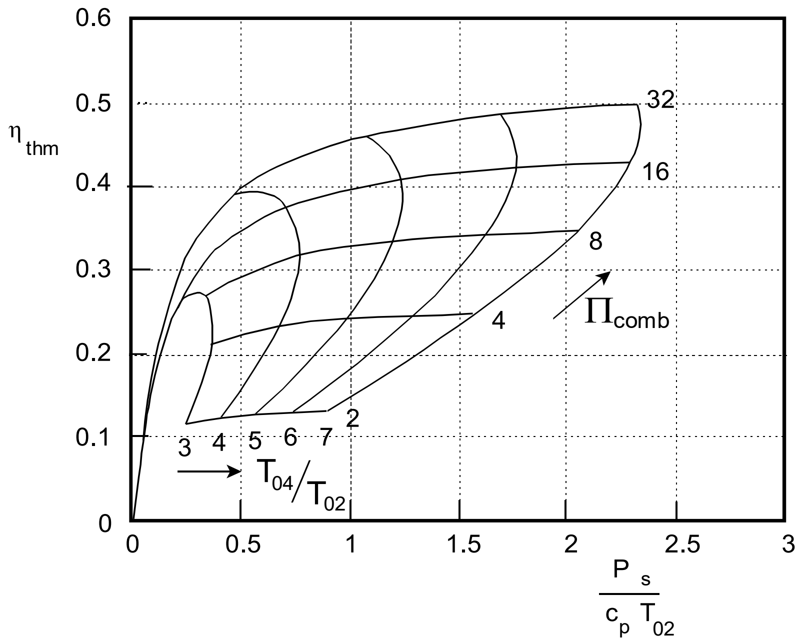


Figure 4-2 - Specific power and thermal efficiency of a turboshaft engine

($\eta_{\infty,\ comp} = \eta_{\infty,\ turb} = \eta_{\infty,\ PT} = 0.9$, $\eta_{mech} = \Pi_{inlet} = \Pi_{exhaust} = \Pi_{cc} = 0.98$)

Figure 4-3 and Figure 4-4 show the effect of compressor and turbine efficiencies. When compared to Figure 3-11 the effects of compressor and turbine efficiencies now nearly become equal. This is due to the fact that the turboshaft expansion is now completed over a pressure drop equal to compressor pressure rise.

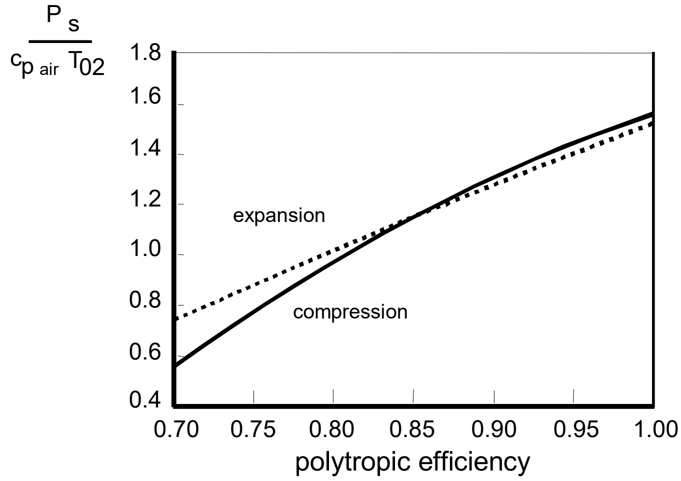


Figure 4-3 - Specific power of a turboshaft engine as a function of the polytropic efficiency of the compression and the expansion ($T_{0,4}$ = 1400 K, $\Pi_{\infty,comp}$ = 15, $\eta_{\infty,\ turb} = \eta_{\infty,\ PT}$ = 0.9, $\eta_{mech} = \Pi_{inlet} = \Pi_{exhaust} = \Pi_{cc}$ = 0.98)

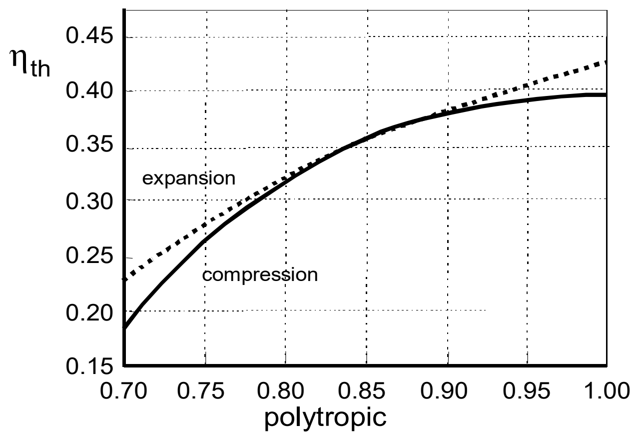


Figure 4-4 - Thermodynamic efficiency of a turboshaft engine as a function of efficiency for the compression and the expansion. ($T_{0,4}$ = 1400 K, $\Pi_{\infty,comp}$ = 15, $\eta_{\infty,turb} = \eta_{\infty,PT}$ = 0.9, $\eta_{mech} = \Pi_{inlet} = \Pi_{exhaust} = \Pi_{cc}$ = 0.98)

Single shaft turboshaft engines do not have a separate power turbine but use a single turbine for both driving the compressor and the external load. In section 3.4 the difference between polytropic and isentropic efficiency has been explained. Polytropic efficiency is more convenient to use if efficiencies of turbines need to be combined or split. The polytropic efficiencies of a gas generator and a power turbine are equal to the efficiency of both turbines combined as a single turbine. For this reason in Figure 4-2, Figure 4-3 and Figure 4-4 polytropic efficiency has been used instead of isentropic efficiency (similarly to Figure 3-10, Figure 3-11). For example, when analyzing gas turbine performance data from some source, a prime requisite is to find out which efficiency definition has been used.

4.4 Enhanced Cycles

In section 2.3 enhancements to the simple cycle including recuperation, intercooling and reheat were addressed. It was shown that intercooling and reheat primarily are used to increase specific power, while recuperation primarily increases thermal efficiency. A combination of all increases both specific power and thermal efficiency.

4.4.1 Recuperators and Regenerators

Two types of heat exchangers exist; recuperators and regenerators.

A recuperator is a heat exchanger in which the cold and hot airflow are strictly separated. Heat is transferred through a (metal) separation wall.

A regenerator is usually made of a porous/channelled ceramic disc, through which alternately hot flue gas and (cold) air flows. The heat will be accumulated in the disc when hot gas flows through it and will be passed to the cold airflow. These types of heat exchangers are very effective and the dimensions are very small, which makes regenerators very suitable for small gas turbines. However, the construction of the regenerator is quite complicated. The disc needs to be driven and requires complicated seals (slip rings) to prevent leakage of high-pressure compressor air to the low-pressure flue gas.

4.4.2 Intercooling

Several modern turboshaft engine designs also use intercoolers, such as the Rolls-Royce WR-21 for ship propulsion. Usually, intercoolers in gas turbines are combined with recuperators.

4.4.3 Reheat

Reheat between the turbines of a turboshaft engine is applied in some recent designs, such as the Alstom GT24 and GT26 engines. These engines have been designed to obtain high combined cycle efficiency due to the relatively high exhaust gas temperature.

4.5 Using exhaust Gas Waste Heat

4.5.1 Configurations

Gas turbines for power generation are often combined with installations that use exhaust gas heat to produce hot water/steam (**cogeneration)** or additional power by expanding the steam in a steam turbine (**combined cycles**).

4.5.1.1 Cogeneration

Many industrial processes require both (electrical) power and heat. The heat is required to obtain hot water or steam. Steam is required in many chemical processes for example and hot water can be used for heating systems. Figure 4-5 shows a gas turbine in a cogeneration configuration. The exhaust gas from the gas turbine is used to heat up and evaporate water and overheat steam.

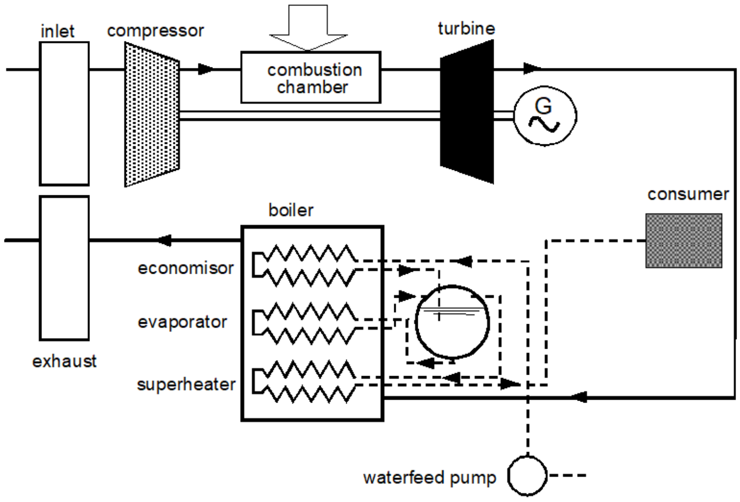


Figure 4-5 - Cogenerated gas turbine cycle

4.5.1.2 Combined Cycles

The steam produced in the previous example can also be expanded in a separate steam turbine. When combining the gas- and steam turbine to drive a single load such as an electric power generator, the cycle is called a "combined cycle" (see Figure 4-6).

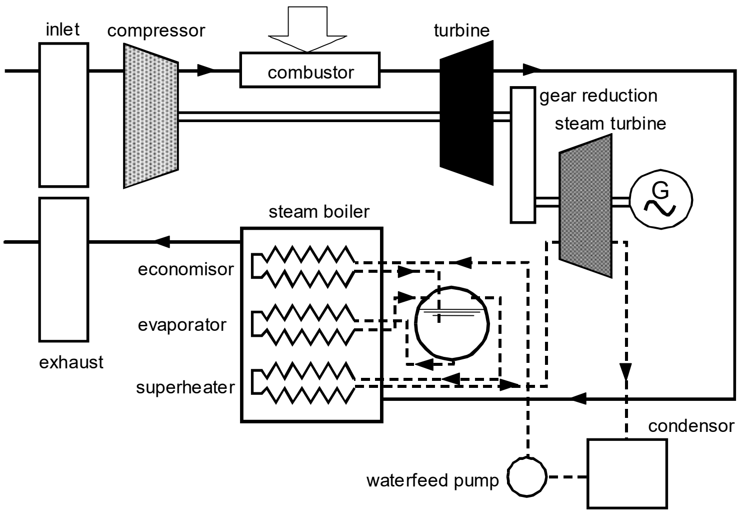


Figure 4-6 - Combined cycle

Gas turbines are very suitable for cogeneration and combined cycles because most of the waste heat is left the exhaust gas, which can then be the single source of recoverable heat. Piston engines for example also lose a lot of heat through the oil system making recovery of waste heat more complex. Another advantage of the gas turbine is that the exhaust gas still contains a lot of oxygen (circa 15%). The presence of oxygen makes it possible to reheat the exhaust gas by combusting additional fuel downstream of the exhaust.

The modern combined cycle power plants are one of the most efficient heat engines with an overall thermal to electric efficiency at design conditions exceeding 62%. Most of the industrial gas turbine manufacturers are now aiming at achieving 65% efficiency in the next few years.

4.5.2 Effects of System Parameters on Cycle Performance

One of the key aspects in the design of a cogenerated and combined cycle is tuning the gas turbine process and the water/steam process. The heat and the power demand from a cogenerated cycle for an industrial process generally do not correspond to the characteristics of the gas turbine. The demand for the heat, for instance, depends on the season (e.g. heating of buildings, fruit farms or market gardens in winter). The demand for electrical power usually is also dependent on the season, but a much stronger variation in electric power demand may be expected between day and night.

The specific gas turbine type selected for cogeneration and combined cycles are driven by either electricity or heat demand. In the case of a design for electric power, any excess heat required over the maximum that can be recovered from the exhaust gas must be obtained from an additional heat source (e.g. direct fired boiler). If the heat demand is less than the heat supply, an additional 'heat customer' (consumer) should be found. In case the heat demand is the primary design objective, a deficit of electric power must be obtained (purchased) and excess power should be delivered (sold) to the power grid. Additional to gas turbine performance characteristics, aspects related to fuel- and electricity prices, benefits from delivering to the grid and investment costs are important.

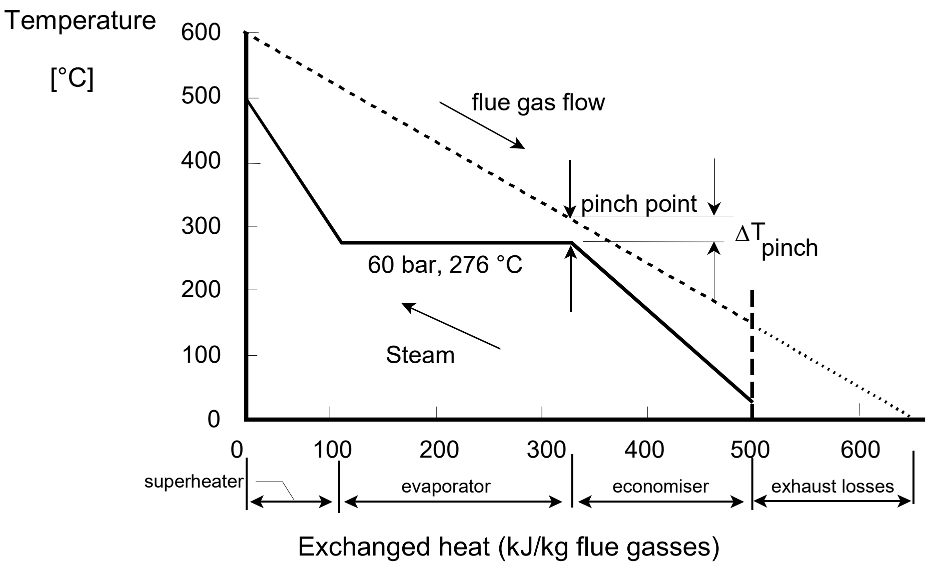


Figure 4-7 - Q,T diagram for a gas turbine with an additional boiler

Figure 4-7 shows the enthalpy - temperature diagram (Q - T diagram) of a flue gas boiler that generates steam using gas turbine exhaust gas heat. The vertical axis indicates exhaust gas and steam/water temperature; the horizontal axis indicates heat rejected by the exhaust gas and absorbed by the steam/water. For a specific steam pressure, the Q, T line for the exhaust gasses

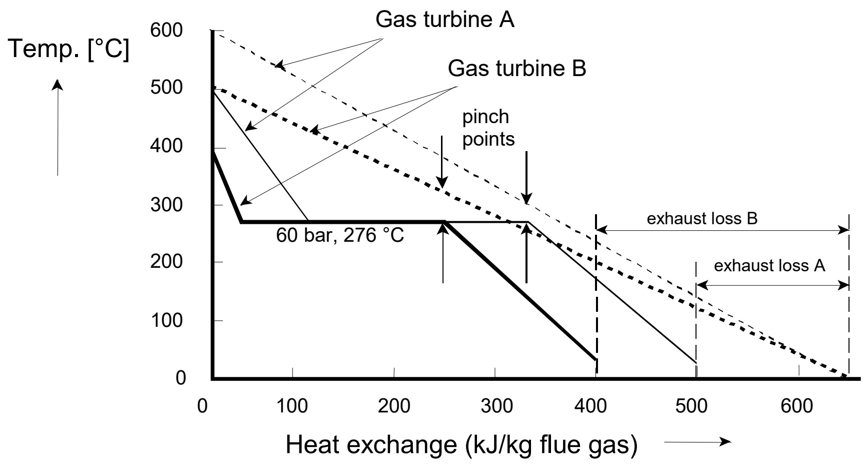


Figure 4-8 – Q,T diagram for different exhaust gas temperatures

is fixed, just as the temperature level of the saturated steam.

The steam mass flow is determined by the temperature difference ΔT_{pinch} (the “pinch point”). ΔT_{pinch} must at least be 15-20 K in order to obtain a reasonable heat flux and to avoid excessively large (i.e. economically unfavourable) heat exchangers. Figure 4-8 shows that with a lower exhaust gas temperature (gas turbine B) and the same available heat flow and steam conditions, steam production will decrease. The exhaust losses will therefore be reasonably higher for gas turbine B, and thus the overall process efficiency will be lower.

A method to increase the efficiency of the steam cycle for given exhaust gas conditions is to use multiple pressure levels for the steam. The effect is a steam line more closely following the exhaust gas cooling line (
Figure 4-9). This results in smaller exhaust losses and higher steam pressure. Disadvantages are the larger and more complex installation.

Another method to increase the efficiency of the steam cycle is “supplementary firing”. Usually, the exhaust gasses consist of more than 75% air, which allows additional combustion to increase flue gas inlet temperature. When this temperature is limited to approximately 650 ºC, a relatively simple convection boiler can still be used.

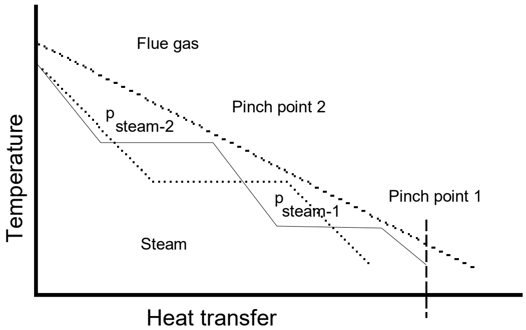


Figure 4-9 – Comparison of steam pressure levels

In chapters 2, 3 and section 4.3 it was concluded that for the simple cycle gas turbine, the high pressure ratio is an important requisite for accomplishing a high thermal efficiency.

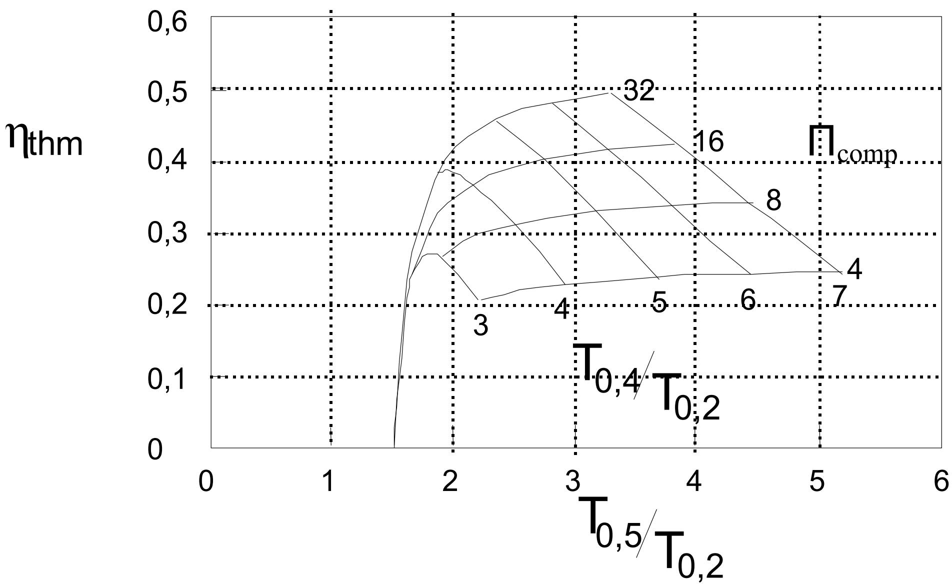


Figure 4-10 – Turboshaft engine turbine exit temperature and thermal efficiency

($\eta_{\infty,comp} = \eta_{\infty,turb} = \eta_{\infty,PT} = 0.9$, $\eta_{mech} = 0.98$, $\Pi_{inlet} = 0.98$, $\Pi_{exit} = 0.98$, $\Pi_{cc} = 0.98$)

Figure 4-10 and Figure 4-11 respectively show thermal efficiency and equivalence ratio λ as a function of temperature ratio $T_{0,5}/T_{0,2}$ ($T_{0,5}$ is power turbine exit temperature), compressor (cycle) pressure ratio Π_{comp} and temperature ratio $T_{0,4}/T_{0,2}$. Equivalence ratio φ is defined as:

$$\varphi = \frac{\dot{m}_{air} - \dot{m}_{st}}{\dot{m}_{st}} \cdot 100 \qquad (4.9)$$

where $\dot{m}_{st}$ is defined as the minimum air mass flow required for complete combustion of the fuel, with φ = 1, the combustion is stoichiometric.

The figures indicate that high $T_{0,5}$ and φ values can only be realized at relatively low values of Π_{comp} and $T_{0,4}/T_{0,2}$, which means low thermal efficiency and specific power. This means gas turbines designed for industrial applications ('heavy duty' gas turbines) usually have moderate pressure ratios (around 15 bar) and $T_{0,4}$ firing temperatures. This makes them suitable for adding a flue gas boiler to the exhaust. Gas turbines derived from large aero engines ("aero derivatives") show much higher pressure ratios (ranging from 25 to 40) meaning high thermal efficiency and relative low exhaust gas temperatures, which makes them less suitable for combined processes.

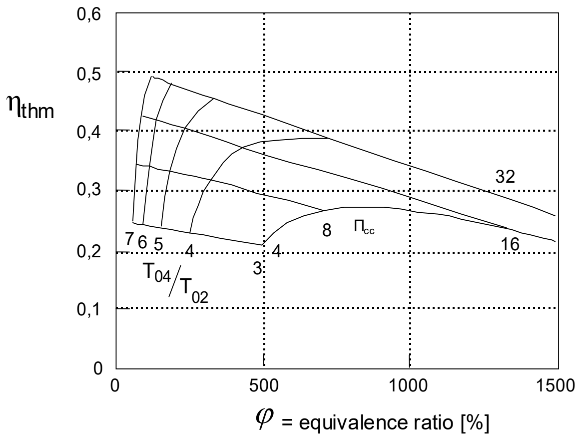


Figure 4-11 – Turboshaft engine equivalence ratio and thermal

($\eta_{\infty,comp}$ = $\eta_{\infty,turb}$ = $\eta_{\infty,PT}$ = 0.9, η_{mech} = 0.98, Π_{inlet} = 0.98, Π_{exit} = 0.98, Π_{cc} = 0.98)

5. Aero Engines

(Dr. Wilfried P.J. Visser & Dr. Arvind Gangoli Rao)

5.1 Aircraft Propulsion

In Figure 5-1 the lift and drag forces acting on an aircraft in steady-state horizontal flight are represented by vectors L and D. L and D are the vertical and horizontal components of the resultant aerodynamic force F that acts on the aircraft centre of gravity. Lift L is to compensate for the force of gravity. Total horizontal drag D needs to be compensated by the thrust of the propulsion system. Most large aircraft propulsion systems use gas turbine engines to generate a propelling jet either directly from the exhaust nozzle (jet engines) and/or by using a propeller (turboprop engines).

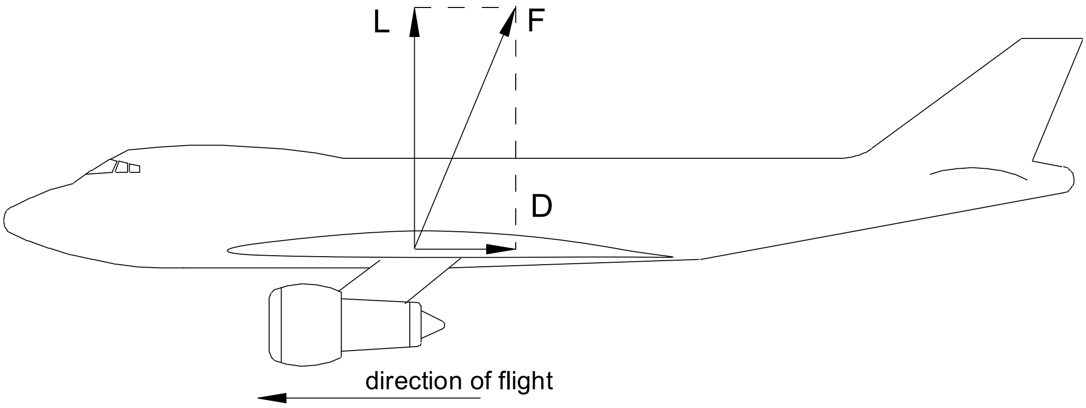


Figure 5-1 – Aircraft lift and drag forces

5.2 Thrust Equation

The basic principle of aircraft propulsion is an application of Newton's 2nd and 3rd laws of motion saying that the acceleration of an object is proportional to the net force acting upon it (F = m·a) and that for every action there always is an equal opposite reaction. All conventional aircraft propulsion systems somehow accelerate air in a direction opposite to the direction of flight. The force required for the acceleration generates an equal reactive force in the direction of flight. The equation F = m·a can easily be converted into

$$FN = \dot{m} \cdot (v_j - v_0) \tag{5.1}$$

FN is *net thrust*, v_j is the velocity of the air or gas exiting the propulsion system, v_0 is the entry velocity. Where (v_j-v_0) represents the acceleration of the flow with mass flow rate $\dot{m}$ through the propulsion system. Note this equation applies to any sort of propulsion system including jet engines, turboprop engines and piston engine driven propeller systems.

Equation (5.1) can also be considered as an expression representing the change in momentum of a mass flow, which also requires a force –FN and therefore generates a reaction force FN. Usually, with thrust, *net thrust* FN is meant. Apart from FN also *gross thrust* FG is used:

$$F_G = \dot{m} \cdot v_j \tag{5.2}$$

FG represents the force generated by the propulsion system exit or exhaust nozzle, ignoring the inlet momentum drag $\dot{m} \cdot v_0$. This inlet momentum drag is also known as "ram drag".

5.3 Determining Thrust

Jet engine thrust can be determined by either direct measurement on a testbed (using a 'load cell' measuring engine thrust) or by calculation using equations similar to those given in 5.2 based on thermodynamic data. Testbed thrust measurements play an important role in engine development programs. However, in most cases, during the engine design process thrust is calculated from other data.

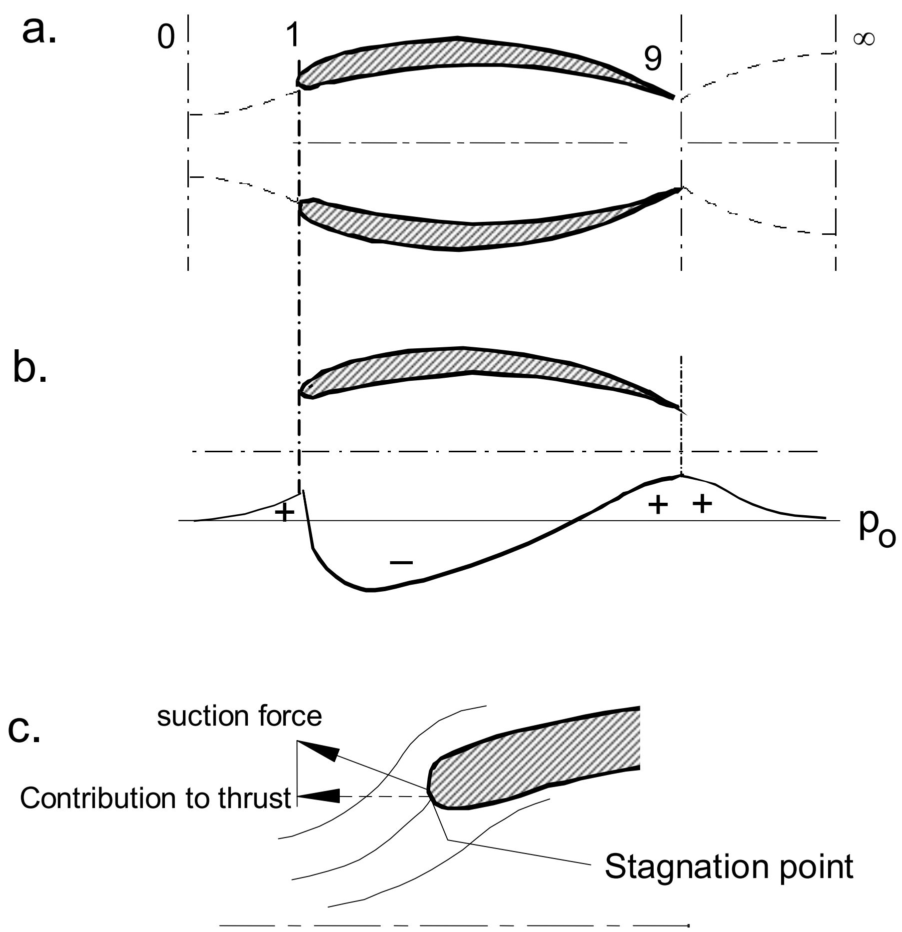


Figure 5-2 – Propulsion system boundaries (a), nacelle outside static pressure pattern (b) and forward thrust ('suction') on inlet leading edge (c)

Jet engine thrust is composed of a large number of individual forces on the engine parts. Although this approach is useful for example in calculating structural loads, it is not a practical method to determine thrust for performance calculations.

Equation (5.1) itself seems relatively simple, but determining actual engine thrust from it is not just straightforward. For a jet engine, one could assume the mass flow $\dot{m}$ to be equal to the mass flow passing through the engine only, ignoring small effects on airflow around the nacelle outside the engine. Also, v_0 can simply be considered equal to aircraft air speed (this might not be the case for an engine utilizing boundary layer ingestion or wake ingestion). However, v_j is difficult to determine, both with calculation and measurements. This is because in many cases the exhaust gas continues to accelerate beyond the exhaust nozzle exit and the point where the acceleration is complete is hard to define. Also, due to the entrainment of the ambient air, the exhaust jet is slowed down but its mass flow $\dot{m}$ is increased.

For the validity of the equation, the system boundary of the propulsion system must be considered. In Figure 5-2(a) four planes (or 'stations' in the 'gas path') are defined that can be used as system boundaries:

- 0 fully undisturbed air flow upstream of the engine
- 1 inlet entry plane
- 9 exhaust exit plane (corresponding to the j of 'jet', $v_8 = v_j$)
- ∞ fully expanded exhaust gas downstream of the engine

Note that at stations 1 and 9, static pressure deviates from the ambient pressure. At stations 0 and ∞ static pressure is equal to ambient pressure. It is clear that for equation (5.1) the system boundary must be at station 0 and station ∞.

Station 0 is always best to determine v_0 since only aircraft airspeed must be specified or measured. Determining v_j at station ∞ is difficult as explained above. Instead, it is easier to calculate v_j at station 9 (engine exhaust nozzle plane) using the appropriate equations to calculate subsonic or supersonic gas velocity. This requires exhaust pressure and temperature, usually already available from engine performance calculations. When obtaining v_9 from station 9, there remains a residual thrust effect from the acceleration between 9 and ∞. However, applying the law of conservation of momentum provides us with a convenient solution as shown in equation (5.3) (note that $p_\infty = p_0$).

$$\dot{m} \cdot (v_9 - v_\infty) = A_9 \cdot (p_9 - p_0) \tag{5.3}$$

The post-exit acceleration effect is equal to exhaust exit cross-area times the exhaust exit plane pressure delta with ambient pressure. This eventually allows us to transform equation (5.1) into equation (5.4), which enables us to calculate thrust, using data that are relatively easy to obtain at the system boundaries at stations 0 and 9.

$$F_N = \dot{m} \cdot (v_9 - v_\infty) + A_9 \cdot (p_9 - p_0) \tag{5.4}$$

Note that equation (5.3) is not fully (only approximately) valid since flow areas at stations 9 and ∞ are not equal. However, equation (5.4) provides an efficient and consistent means to define jet engine thrust. It is commonly used in combination with empirical correction factors to accurately calculate thrust.

5.4 Installed and Uninstalled Thrust

Jet engines usually are mounted in an engine nacelle, which transmits the thrust of the engines to the aircraft via its pylons or other mounting devices that connect the nacelle to the aircraft. The resulting thrust generated by a nacelle that houses an engine is usually different from bare engine thrust, mainly due to drag forces on the nacelle. Clearly, the boundary between the propulsion system and aircraft affects what is considered as thrust and as aircraft drag. Nacelles and pylons may be considered as a part of the propulsion system (and nacelle/pylon drag accounted for as a negative thrust term), depending on the bookkeeping methodology used. Also, the interaction between engine performance and nacelle aerodynamic effects makes it hard to develop a consistent definition of thrust and aircraft drag. To address this problem the term *installed* and *uninstalled* thrust are defined. Uninstalled thrust usually refers to engine thrust with ideal inlets and exhausts and no other losses due to bleed off-takes from the compressor. Uninstalled thrust data usually provide a consistent means to compare different engines. Installed thrust means the thrust produced by an engine when installed on an aircraft. Thus, the pressure losses and flow distortion caused by actual inlet, exhaust and many other losses and secondary effects are included that are specific to the particular aircraft installation. The value of the installed thrust of an engine is meaningful only for a particular aircraft and cannot be used to compare different engine types (unless installed on the same aircraft).

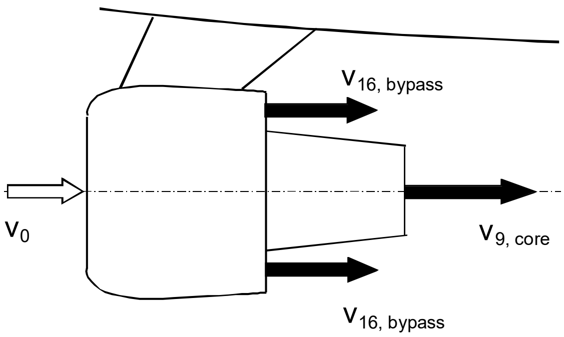


Figure 5-3 – Components of thrust in a turbofan engine

The static pressure outside the nacelle generally follows a pattern as shown in Figure 5-2(b) due to accelerations and decelerations of the airflow around the nacelle. At cruise speed, the airflow normally decelerates from the plane I until beyond plane II inside the engine inlet, resulting in a rise of static pressure (compression) inside the engine inlet. This causes a contribution to engine thrust generated by the engine inlet (see Figure 5-2(c)). The inlet is 'sucked' forward due to the lower pressure outside the front of the nacelle and high pressure inside the inlet.

Most modern commercial jet engines are *turbofan* engines. In a turbofan engine, part of the inlet air is compressed by a low pressure compressor or *fan* and flows around the engine through a *bypass*, often exiting the engine through a separate propelling exhaust nozzle (see *Figure 5-3*). For this case or other cases of multiple exhaust nozzles equation (5.5) is used.

$$F_N = \sum\left[\dot{m}\cdot(v_9 - v_\infty) + A_9\cdot(p_9 - p_0)\right] \tag{5.5}$$

Another way to express F_N is using *effective jet velocity* v_{eff}. v_{eff} represents the velocity that must theoretically be obtained after expansion downstream of the exhaust nozzle (station ∞) to obtain a thrust F_N. Due to losses during expansion, v_{eff} normally is lower than the jet velocity that could optimally be obtained.

$$F_N = \dot{m}\cdot(v_9 - v_\infty) + A_9\cdot(p_9 - p_0) = \dot{m}\cdot(v_{9,eff} - v_\infty) \tag{5.6}$$

Often, v_{eff} is calculated from F_N by solving equation (5.6) and used for comparative analysis. For conceptual design studies, often **specific thrust** is used to indicate performance relative to engine inlet air mass flow.

$$F_s = \frac{F_N}{\sum[\dot{m}]} \tag{5.7}$$

Specific thrust is useful to indicate engine performance relative to engine size, weight, frontal area and volume and also to indicate engine technology level. It is clear that the average specific engine thrust has increased drastically since the introduction of the jet engine in the 1940'ies.

5.5 Propulsion System Power and Efficiencies

The power that is used to propel the aircraft is called **thrust power** and is defined:

$$P_{thrust} = \sum F_N \cdot v_0 = \sum \left(\dot{m} \cdot (v_{9,eff} - v_0) \right) \cdot v_0 \tag{5.8}$$

Again, the summation sign is used for bypass engines having multiple jet streams. The power required to accelerate the air and gas flowing through the engine v_0 to $v_{8,\mathrm{eff}}$ is called **propulsion** power. The propulsion power equals the increase in kinetic energy the air/gas mass flow:

$$P_{prop} = \sum \left(\frac{\dot{m}}{2} \cdot (v_{8,eff}^2 - v_0^2) \right) \tag{5.9}$$

Note that the thrust power does not equal the propulsive power. The difference is the kinetic energy of the jet stream, leaving the engine, in terms of the **absolute velocity** relative to the static environment:

$$P_{loss} = \sum \left(\frac{\dot{m}}{2} \cdot (v_{9,eff}^2 - v_0^2) - \dot{m} \cdot (v_{9,eff} - v_0) \cdot v_0 \right) = \sum \left(\frac{\dot{m}}{2} \cdot (v_{9,eff} - v_0)^2 \right) \tag{5.10}$$

After leaving the engine, P_{loss} is converted to heat by vortices and turbulence. The magnitude of the energy loss is expressed in **propulsive efficiency**, also known as the Froude efficiency and can be derived as follows:

$$\eta_{prop} = \frac{P_{thrust}}{P_{prop}} = \frac{\sum \left(\left(\dot{m} \cdot \left(v_{9,eff} - v_0 \right) \right) \cdot v_0 \right)}{\sum \left(\frac{\dot{m}}{2} \cdot \left(v_{9,eff}^2 - v_0^2 \right) \right)} = \frac{\frac{2}{F_N}}{\Sigma \left(\dot{m} \cdot v_0 \right)} + 2 = \frac{2}{1 + \frac{v_{9,eff}}{v_0}} \tag{5.11}$$

The last expression immediately indicates that η_{prop} increases as $v_{8,\mathrm{eff}}$ and v_0 get closer to each other.

The propulsive efficiency needs to be distinguished from **thermal efficiency**, which indicates the efficiency of energy conversion inside the engine:

$$\eta_{thm} = \frac{P_{prop}}{\dot{m}_f \cdot LHV_f} = \frac{\sum \left(\frac{\dot{m}}{2} \cdot (v_{9,eff}^2 - v_0^2) \right)}{\dot{m}_f \cdot LHV_f} \tag{5.12}$$

Gas generator power and the propulsion power ideally are equal when no losses exist between gas generator exit and exhaust nozzle. In reality, there are some pressure and heat losses between the gas generator and the exhaust nozzle of a turbojet engine. With turbofan engines, the losses are significantly higher since the gas generator power must be converted to shaft power by a turbine driving a fan generating an additional 'cold' propulsion jet in the bypass exhaust. The turbine and the fan have isentropic efficiencies representing the associated losses. The **jet generation efficiency** is defined as:

$$\eta_{jet} = \frac{P_{prop}}{P_{gg}} \tag{5.13}$$

The total efficiency of the conversion of fuel chemical energy to thrust power then is:

$$\eta_{tot} = \frac{P_{thrust}}{\dot{m}_f \cdot LHV_f} \tag{5.14}$$

A more common parameter used to express total efficiency is **thrust specific fuel consumption**, which is fuel mass flow per unit of thrust:

$$TSFC = \frac{\dot{m}_f}{F_N} = \frac{v_0}{\eta_{tot} \cdot LHV_f} \tag{5.15}$$

6. Combustion

(**Dr. Ir.** Savad Shakariyants & Prof. Dr. A. Gangoli Rao)

6.1 Introduction

The combustion chamber (combustor) is located between the compressor and turbine in a gas turbine (Figure 6-1) and is required to convert the chemical energy of the fuel into thermal energy with the smallest possible pressure loss and with the least emission of undesirable chemicals. In other words, the combustor provides the heat (energy) input into the gas turbine cycle. It receives air from the compressor, introduces a stream of fuel into it, creates the conditions for the fuel and air to mix and react and, eventually, delivers a mixture of hot post-combustion gases to the turbine. Such a process is commonly referred to as internal combustion.

For some specific applications, the combustion process may be staged outside the gas turbine. It is therefore named external combustion or external firing. Cases like that would require equipping the gas turbine with a heat exchanger to transfer heat to the working fluid. Such design solutions do not lie within the province of this chapter.

For the sake of analysis, combustion systems are generally classified as either those of constant pressure or constant volume. Virtually, no process can be staged in an engineering system without a loss in pressure. However, this loss does not exceed few percentage points in relative terms in a gas turbine combustor. This makes us treat gas turbine combustion as constant-pressure combustion.

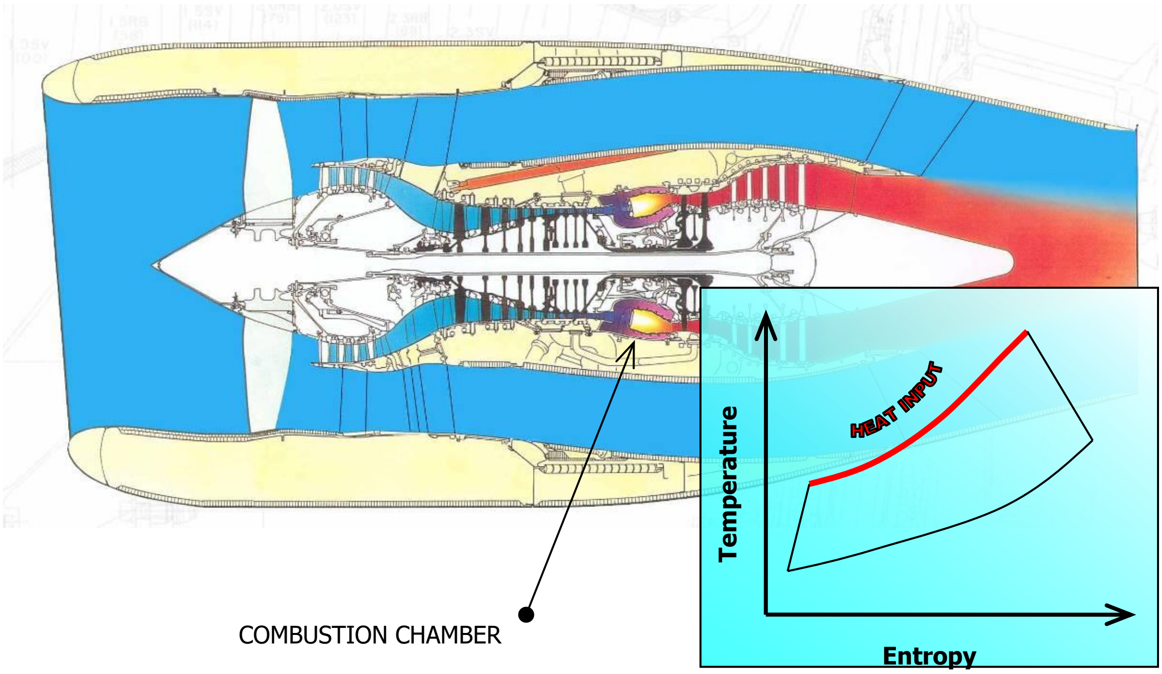


Figure 6-1 - Gasturbine combustion chamber [1]

- Fossil hydrocarbon fuels: gaseous (natural gas, propane, butane, etc.) and liquid fuels (residual oils, kerosene) for land-based power plants; and liquid kerosene-type fuels for aircraft propulsion;
- Producer gas for industrial gas turbines, which can be a product of coal, waste or biomass gasification;
- Experimental fuels: hydrogen for utility as well as aircraft gas turbines and liquefied natural gas (LNG) for aircraft propulsion.

All gas turbine fuels must be environmentally clean, have sufficient thermo-chemical properties and must comply with specific application-dependant requirements.

Fuel requirements are most stringent for aircraft propulsion. In order to comply with them, oil companies have developed special jet fuels for aviation. These fuels are very clean and have approximately 16% of hydrogen and 84% carbon. They are produced in different types with designations specific to different applications. For example, Jet A and Jet A-1 are fuels for commercial jet aircraft. JP-4 is for jet fighters. Jet B is a fuel mixed with extra light components to further lower its melting point in order to allow application at very low temperatures.

For industrial gas turbines, requirements are generally less severe, which allows the application of low-cost fuels. These fuels usually have higher density and viscosity compare to jet fuels. Contrary to aircraft propulsion, almost no size constraints are imposed on fuel tanks. This partially makes natural gas, propane, ethane, butane etc. easier to use in land-based gas turbines.

With the price of hydrocarbon fuels skyrocketing, alternative fuels such as producer gas and hydrogen are becoming ever more attractive. Besides, hydrogen combustion results only in water and small amounts of nitrogen oxides. It can be produced from water by electrolysis using electrical energy from renewable resources. However, many challenges have to be tackled before hydrogen can be introduced into revenue service. A profound R&D work is required to find the right materials, part and component designs for the combustion chamber, fuel and storage systems. The issues of safety, environmental compatibility and economic viability of using hydrogen, as an alternative fuel should be also investigated.

Figure 6-2 - Conceptual H2-powered aircraft: CRYOPLANE, [2]

In aviation, the European Union funded a project dubbed Cryoplane (Figure 6-2) to assess the applicability of liquid hydrogen in aircraft propulsion. The project was a joint effort between 35 partners from 11 European countries led by Airbus Deutschland and with the participation of TU Delft. A range of aircraft categories were considered from business jets to large long-range aircraft such as the Airbus A380. Very promising results were obtained. However, the maiden implementation of this technology is not expected earlier than in 15 to 20 years, provided that research work will continue on an adequate level [3].

Liquefied natural gas is considered as another alternative to kerosene fuels in aviation. It is estimated that LNG promises remarkable reductions in concentrations of undesirable chemicals in the exhaust [4]: up to 10 times for carbon monoxide, 2.5 to 3 times for hydrocarbons and 1.5 to 2 times for nitrogen oxides.

Remarkable progress was achieved in the former Soviet Union in the field of testing alternative fuels in aviation: The Tupolev design house built a flying Cryoplane test bed (Figure 6-3) in the 1980s based on the Tu-154 passenger aircraft. The test aeroplane, Tu-155, was provided with a second cryogenic-fuel system to feed the starboard engine. On April 15 1988, the Tu-155 made its maiden flight using liquid hydrogen. In January 1989, the aircraft already flew on LNG. R&D works on the cryogenic aeroplane continue in Russia, today.

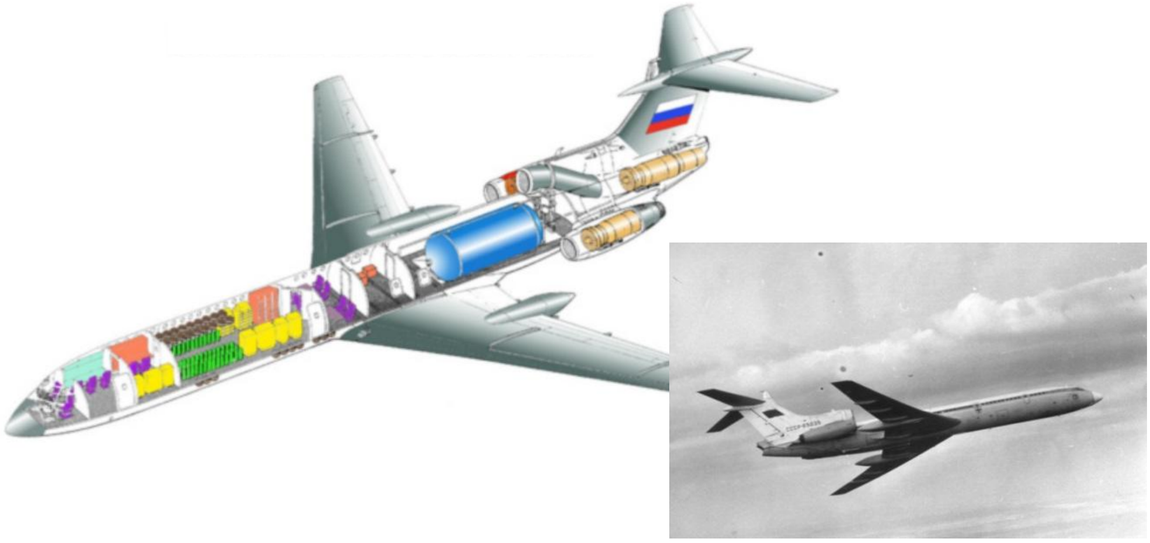


Figure 6-3 - Tupolev Tu-155 cryoplane. [4]

6.2 Heat Release

While designing a gas turbine cycle, the temperature at combustor exit $T_{0,4}$ is usually a critical design parameter. $T_{0,4}$ is very important for the power output and affects the thermal efficiency of a gas turbine (as discussed in Chapters 2 and 3). The temperature at the combustor exit is limited by turbine material properties. Apart from that, maximum $T_{0,4}$ is also dependent on the degree of cooling applied to the turbine. Exceeding this maximum must be prevented to avoid inadvertent changes in material structure or excessive corrosion and creep (see Chapter 9). The limitation of the combustor exit temperature implies that only a limited quantity of fuel should be combusted for a given quantity of compressor-delivered air.

Fuel-to-Air Ratio

Complete combustion of a hydrocarbon[1] fuel requires sufficient oxygen to convert the fuel to carbon dioxide and water vapour. That required amount of air is called stoichiometric. Such a mixture of fuel and air is therefore called stoichiometric as well, and their ratio (by mass) is referred to as stoichiometric fuel-to-air ratio (FAR_{stoich}). If more than a stoichiometric quantity of air is supplied, the mixture will be burning at a numerically smaller fuel-to-air ratio (FAR) than the FAR_{stoich}. Such a mixture is called fuel-lean, or just lean (also weak). On the contrary, if more than a stoichiometric quantity of fuel is supplied, the mixture will be burning at a numerically larger FAR than the FAR_{stoich}. Such a mixture is called fuel-rich, or simply rich. In the other words, the fuel-to-air ratio reflects the “strength” of a combustible mixture.

The stoichiometric air-to-fuel ratio can be calculated from the equation of complete ideal combustion reaction. For a hydrocarbon fuel with x atoms of carbon and y atoms of hydrogen, it writes as follows:

$$C_xH_y + \varepsilon\left(X_{O_2}O_2 + X_{N_2}N_2 + X_{CO_2}CO_2 + X_{Ar}Ar\right) \rightarrow n_{CO_2}CO_2 + n_{H_2O}H_2O + n_{N_2}N_2 + n_{Ar}Ar, \tag{6.1}$$

where

X_i - mole fraction of species i;

n_i - number of moles of species i per mole of fuel;

$$\varepsilon = \frac{x + y/4}{X_{O_2}}.$$

As for the air composition, a fairly accurate estimate can be [5, 6]:

$$\begin{aligned} X_{O_2} &= 0.209476; \quad X_{N_2} = 0.780840; \\ X_{CO_2} &= 0.000319; \quad X_{Ar} = 0.009365. \end{aligned} \tag{6.2}$$

Hence, one mole of fuel requires ε moles of air. Remembering that the product of molecular weight and number of moles results in mass, the relation for a stoichiometric fuel-to-air ratio by mass would be

$$FAR_{stoich} = \frac{1}{\varepsilon}\frac{M_{C_xH_y}}{M_a} = \frac{X_{O_2}}{x + y/4}\frac{M_{C_xH_y}}{M_a}, \tag{6.3}$$

where

M_i - molecular weight of species i

By way of example, we can find the FAR_{stoich} for methane CH_4 equal to 0.0580. Kerosene-type fuels can be considered to consist of $C_{12}H_{23}$ molecules. The FAR_{stoich} for kerosene would then be 0.0682.

1) Theoretical combustion is discussed in the chapter on the example of hydrocarbon-fuel combustion in air.

A reverse quantity to FAR, the air-to-fuel ratio (AFR), might be more convenient to use due to the typical order of its magnitude. Thus, stoichiometric air-to-fuel ratios for methane and kerosene would be 17.24 and 14.66, respectively.

For gas turbines operating on hydrocarbon fuels, the fuel flows typically account for a few percentage points compared to the airflows. The ratios of these two flows are commonly referred to as overall ratios.

In combustion analysis, it is very convenient to express the mixture strength in terms of a fuel-to-air equivalence ratio, φ. The equivalence ratio unambiguously indicates whether the mixture is rich, lean or stoichiometric. It is defined as

$$\varphi = \frac{\dot{m}_f}{\dot{m}_{air}} \frac{1}{FAR_{stoich}} = \frac{FAR}{FAR_{stoich}} = \frac{AFR_{stoich}}{AFR} \tag{6.4}$$

As the very definition suggests, for fuel-rich mixtures, $\varphi > 1$, and for fuel-lean mixtures $\varphi < 1$. $\varphi = 1$ at stoichiometric conditions. Table 6-1 summarizes overall ratios for aircraft turbofans at

Table 6-1 - Overall mixture strengths for commercial turbofans

Engine Family	zAircraft Application	Take-Off Thrust, [kN]	**Overall** *AFR* **at TO**	**Overall** φ **at TO**
CFM56-7	B737 NG	91.6	**54.0**	**0.27**
RB211-535	B757	163.3	**52.3**	**0.28**
CF6-80E1	A330	297.4	**49.3**	**0.30**
PW4000-112”	B777	396.6	**43.1**	**0.34**

take-off thrust settings (only core air flow is accounted for, no bypass flow is accounted).

The equivalence ratio is a handy parameter in comparing the combustion characteristics of different fuels and for characterizing combustor technology. Other parameters frequently used to define relative stoichiometry [7] are per cent stoichiometric air, related to φ as

$$\%\, stoichiometric\ air = \frac{100\%}{\varphi}, \tag{6.5}$$

and percent excess air,

$$\lambda = \frac{(1-\varphi)}{\varphi} 100\%. \tag{6.6}$$

The percent excess air equals zero at stoichiometry. It assumes positive values at fuel-lean conditions, indicating the surplus of air flow for a given fuel flow, and negative values at fuel-rich conditions, reflecting the shortage of air,

$$\lambda = \frac{\left(\dot{m}_{air} - \dot{m}_{air,stoich}\right)}{\dot{m}_{air,stoich}} 100\% . \tag{6.7}$$

Enthalpy of Combustion and Heating Values

The concept of absolute enthalpies is of great importance in combustion studies. The reader is therefore advised to refresh his/her understanding of this issue by consulting fundamental texts on thermodynamics. It should be noted that absolute enthalpy can be defined for any species. It is the sum of an enthalpy that takes into account the energy associated with chemical bonds - the enthalpy of formation - and an enthalpy that is solely associated with the temperature - the sensible enthalpy change. The absolute enthalpy can then be written in specific terms as

$$h_i\left(T\right) = h_{f,i}^0\left(T_{ref}\right) + \Delta h_{s,i}\left(T\right),$$

where

$$\Delta h_{s,i}\left(T\right) = h_i\left(T\right) - h_{f,i}^0\left(T_{ref}\right) = \int_{T_{ref}}^{T} c_{p,i} dT; \tag{6.8}$$

h_i - mass or molar specific enthalpy of species i.

The standard reference state designated by subscript $_{ref}$ is defined by standard-state temperature and pressure. In general, they can be chosen arbitrarily and should not vary between cross-linked computations. It is though practical to employ $T_{ref} = 298.15\left[K\right]$ and $p_{ref} = 1\left[\text{atm}\right] = 101325\left[\text{Pa}\right]$ consistent with recognized thermodynamic databases, NASA [5, 6] or Chemkin [8].

Consider now a combustion reaction described by Eq. (6.1) (complete, ideal, stoichiometric) staged in a well-stirred reactor (Figure 6-4). Assume that both the reactants and the products are at standard-state conditions. To hold this assumption, heat must be removed from the reactor. This heat can be related to the reactant and product absolute enthalpies by applying the 1st law of thermodynamics:

$$Q = h_{prod} - h_{react} . \tag{6.9}$$

The enthalpy difference quantified by Eq. (6.9) is defined as the enthalpy of combustion. Therefore, it may be written per mole of mixture[2)] as follows:

$$\Delta h_c\left[\frac{J}{kmol_{mix}}\right] \equiv Q = h_{prod} - h_{react} . \tag{6.10}$$

The specific molar enthalpy of combustion can be easily turned into a per-unit-mass basis,

$$\Delta h_c\left[\frac{J}{kg_{mix}}\right] = \Delta h_c\left[\frac{J}{kmol_{mix}}\right]\frac{1}{M_{mix}} . \tag{6.11}$$

2) By mixture, a mixture of the products, reactants, as well as both products and reactants can be considered.

The enthalpy of combustion can be graphically illustrated as shown in Figure 6-5. Note that the heat is being removed from the system (Figure 6-4), which defines a negative heat transfer.

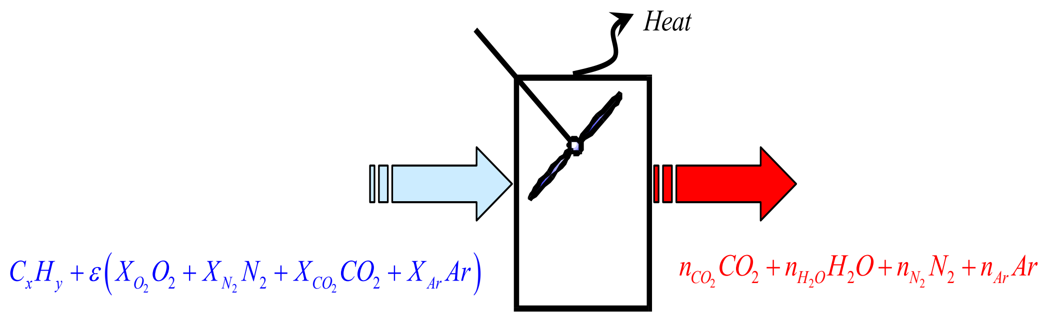


Figure 6-4 - Well-stirred reactor to determine the enthalpy of combustion

Therefore, the absolute enthalpy of the products lies below that of the reactants.

For engineering studies, it is however convenient to express the enthalpy of combustion on a per-mass-of-fuel basis,

$$\Delta h_c\left[\frac{J}{kg_f}\right] = \Delta h_c\left[\frac{J}{kg_{mix}}\right]\frac{\dot{m}_{mix}}{\dot{m}_f}. \tag{6.12}$$

The enthalpy of combustion numerically equals the heat of combustion taken with an opposite sign. The upper or higher heat of combustion - also known as the higher heating value - is calculated assuming that all the water in the products has condensed to liquid. The lower heating value, LHV, corresponds to the case where none of the water is assumed to condense. The latent heat contained in the water vapour remains un-liberated in the last scenario. This gives the designation "lower".

The heating values are also known as calorific values. They are important parameters in comparing different fuels. Practically, the fuels discussed in Section 6.2 can be sub-divided into high, medium and low calorific value fuels. There are no strict threshold values though. By way of example, Table 6-2 provides a list of LHVs for selected gas turbine fuels.

Table 6-2 - Lower heating values of fuels

Fuel	**Type**	**LHV, [MJ/kg]**
High Calorific value Fuels, [9]:		
Natural gas	gaseous	50.0300
Jet B / JP-4	liquid	43.3567
Jet A / Jet A-1	liquid	43.0310
Diesel fuel	liquid	42.0000
Medium Calorific Value Fuels, [10]	gaseous	9.3 – 10.4 [MJ / m^3]
Low Calorific Value Fuels, [10]	gaseous	4.1 – 6.4 [MJ / m^3]

Adiabatic Flame Temperature

If a fuel-air mixture burns adiabatically at constant pressure, the absolute enthalpy of the reactants at the initial state equals the absolute enthalpy of the products at the final state. The final-state temperature is defined as the constant-pressure adiabatic flame temperature. This definition is illustrated graphically in Figure 6-5 and mathematically by Eq. (6.13):

$$h_{prod}\left(T_{ad}\right)=h_{react}\left(T_{react}\right). \tag{6.13}$$

The above expression can be re-written in extensive properties as

$$H_{prod}\left(T_{ad}\right)=H_{react}\left(T_{react}\right). \tag{6.14}$$

Taking a stoichiometric complete ideal combustion by way example (Eq. (6.1)) and applying the definition of the absolute enthalpy (Eq. (6.8)), the adiabatic temperature can be roughly evaluated by equating

$$H_{react}=1h_{C_xH_y}+\varepsilon h_{air}, \tag{6.15}$$

where

h_i - absolte molar specific enthalpy of species i, $\left[{}^{J}\!/\!{}_{kmol}\right]$

with

$$\begin{aligned}H_{prod}=&\,n_{CO_2}\left[h^0_{f,CO_2}\left(T_{ref}\right)+c_{p,CO_2}\left(T_{ad}-T_{ref}\right)\right]+n_{H_2O}\left[h^0_{f,H_2O}\left(T_{ref}\right)+c_{p,H_2O}\left(T_{ad}-T_{ref}\right)\right]\\&n_{N_2}\left[0+c_{p,N_2}\left(T_{ad}-T_{ref}\right)\right]+n_{Ar}\left[0+c_{p,Ar}\left(T_{ad}-T_{ref}\right)\right],\end{aligned} \tag{6.16}$$

where

$c_{p,i}$ - molar cpecific heat at constant pressure of species i, $\left[{}^{J}\!/\!{}_{kmol\text{ - }K}\right]$.

The unknown stoichiometric coefficients n_{CO_2}, n_{H_2O}, n_{N_2}, n_{Ar} can be found from the conservation of atoms:

$$\begin{aligned}C:&\quad n_{CO_2}=x+\varepsilon X_{O_2}\\H:&\quad n_{H_2O}=\frac{y}{2}\\N:&\quad n_{N_2}=\varepsilon X_{N_2}\\Ar:&\quad n_{Ar}=\varepsilon X_{Ar}.\end{aligned} \tag{6.17}$$

Values of the absolute enthalpies of the reactants in expression (6.15) can be evaluated from thermodynamic databases, [5, 6] or [8], for the right temperatures: $T_{C_xH_y}$ and T_{air} in our case.

The careful reader may notice that such a temperature estimation may only be approximate as, effectively, the absolute enthalpies of the products are being evaluated with constant specific heats. To be more precise, the expression $\int_{T_{ref}}^{T_{ad}} c_{p,i}dT$ is substituted with $c_{p,i}\int_{T_{ref}}^{T_{ad}} dT$ for each species i, which results in $c_{p,i}\left(T_{ad}-T_{ref}\right)$. To make an estimate fair, one may guess the adiabatic flame temperature and evaluate each $c_{p,i}$ at $0.5\left(T_{ref}+T_{ad}\right)$ using thermodynamics.

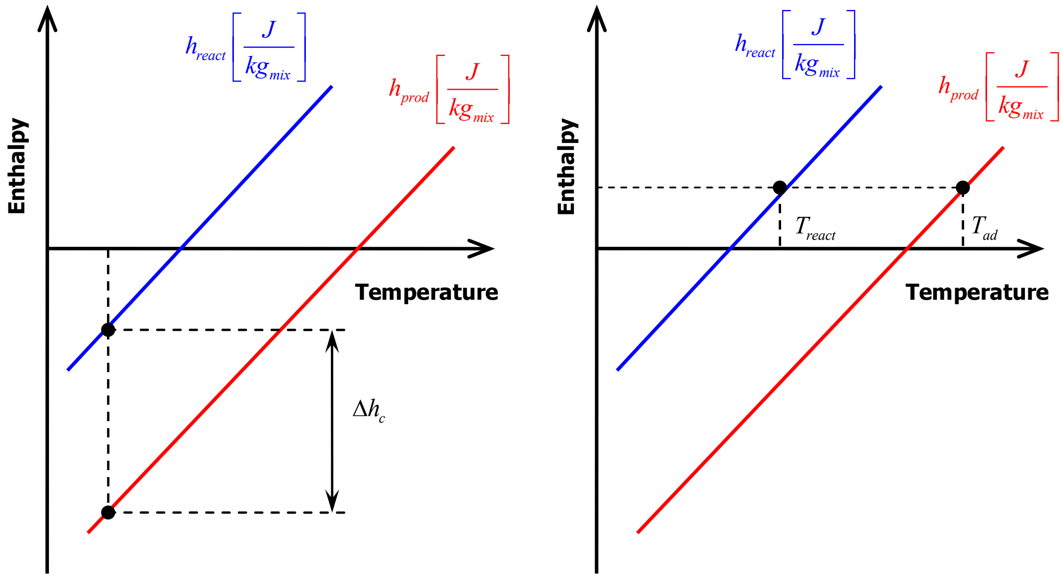


Figure 6-5 Definition of enthalpy of combustion (left) and adiabatic flame temperature (right)

Full Combustion Equilibrium

If combustion reactions are given the time and environment to complete adiabatically at constant pressure, the temperature and composition of the products will develop to a final condition. These will be the constant-pressure adiabatic flame temperature and the full equilibrium composition.

To accurately quantify the equilibrium state, we shall abandon the assumption of ideal combustion. In gas turbine combustors, the products are not a simple mixture of CO_2 and H_2O with nitrogen and argon from the air. Species dissociate and react with each other.

Besides, combustion is necessarily staged at stoichiometric conditions. Overall, the fuel-to-air mixtures are quite lean for gas turbine engines (*Table 6-1*). However, the initial mixing of the fuel and air in a combustion chamber can well result in a rich mixture.

The composition of post-combustion products will strongly differ for different mixture strengths. CO_2, H_2O and nitrogen will always be present as the so-called major products of combustion. Argon can be related to the same group as an inert species coming from the air. O_2 will join the group of major combustion products at lean conditions; while CO and H_2 would become major combustion products at rich conditions. In addition to that, species like O, H, OH and others will emerge as minor products.

Figure 6-6 illustrates the mole fractions of major (in %) and minor (in ppm, $X_i\left[ppm\right]=10^6 X_i\left[-\right]$) products of equilibrium combustion of $C_{12}H_{23}$ with air over a range of equivalence ratios (conditions from lean to rich).

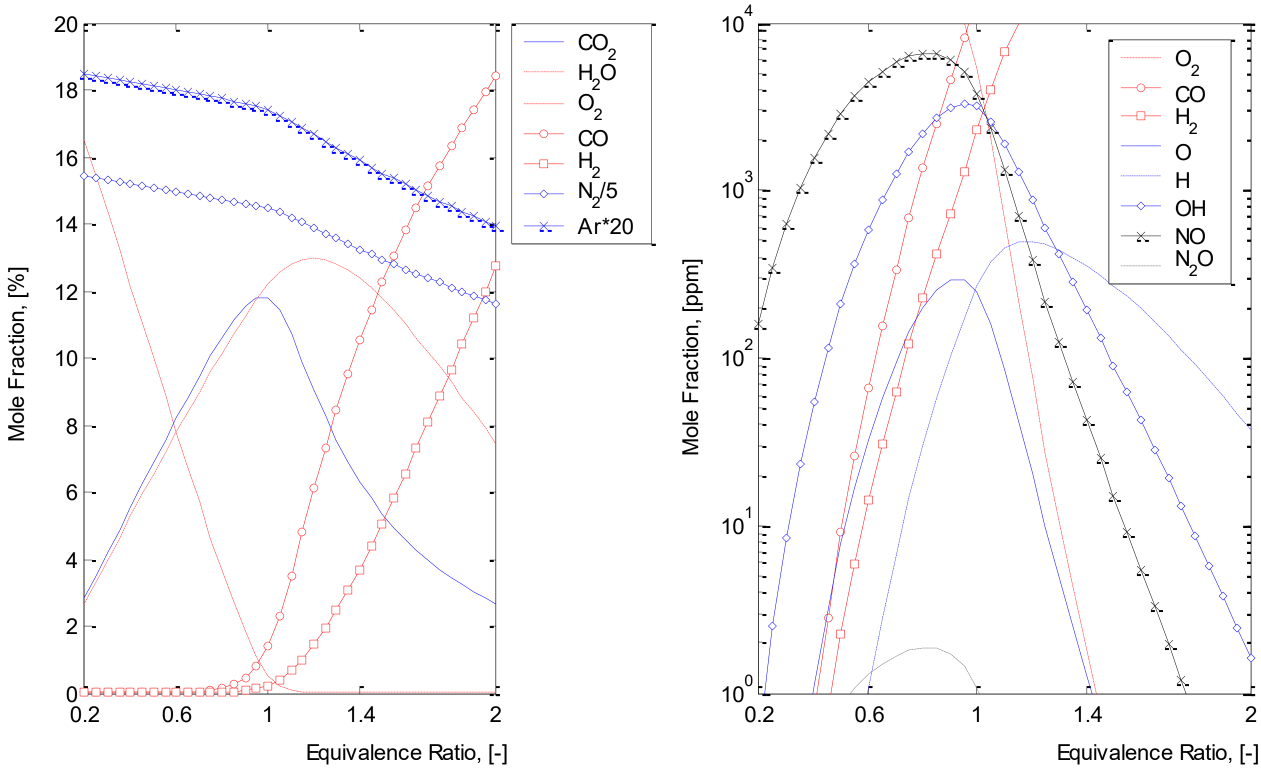


Figure 6-6 Products of equilibrium $C_{12}H_{23}$ in air at p = 28 [atm], T_{air} = 700 [K], T_{fuel} = 298.15 [K]

The combustion reaction equation for cases illustrated in Fig. 6.6 would then evolve from Eq. (6.1) to

$$\frac{\dot{m}_f}{M_{C_xH_y}} C_xH_y + \frac{\dot{m}_{air}}{M_{air}}\left(X_{O_2}O_2 + X_{N_2}N_2 + X_{CO_2}CO_2 + X_{Ar}Ar\right)$$
$$\rightarrow \frac{\dot{m}_{prod}}{M_{prod}}\left(\begin{array}{l}\underbrace{X_{CO_2}CO_2 + X_{H_2O}H_2O + X_{N_2}N_2 + X_{Ar}Ar}_{\text{ideal combustion}} \\ \underbrace{+X_{CO}CO + X_{H_2}H_2 + X_{O_2}O_2 + X_O O + X_H H + X_{OH}OH + X_{NO}NO + X_{N_2O}N_2O}_{\text{products of dissociation and other species}}\end{array}\right). \quad (6.18)$$

Equation (6.18) can be re-written in the form of one mole of fuel. It would then reflect the mixture strength and be independent of mass flows. To that end, both sides of the equation should be divided by the molar flux of fuel, $\frac{\dot{m}_f}{M_{C_xH_y}}$. Remembering that (from Eq. (6.4)) $\varphi = \varepsilon \frac{\dot{m}_f}{\dot{m}_{air}} \frac{M_{air}}{M_{C_xH_y}}$,

we obtain

$$\begin{aligned}&C_xH_y+\frac{\varepsilon}{\varphi}\left(X_{O_2}O_2+X_{N_2}N_2+X_{CO_2}CO_2+X_{Ar}Ar\right)\\&\rightarrow\underbrace{n_{CO_2}CO_2+n_{H_2O}H_2O+n_{N_2}N_2+n_{Ar}Ar}_{\text{ideal combustion}}\\&+\underbrace{n_{CO}CO+n_{H_2}H_2+n_{O_2}O_2+n_OO+n_HH+n_{OH}OH+n_{NO}NO+n_{N_2O}N_2O}_{\text{products of dissociation and other species}},\end{aligned}\tag{6.19}$$

where

n_i - number of moles of species i per mole of fuel.

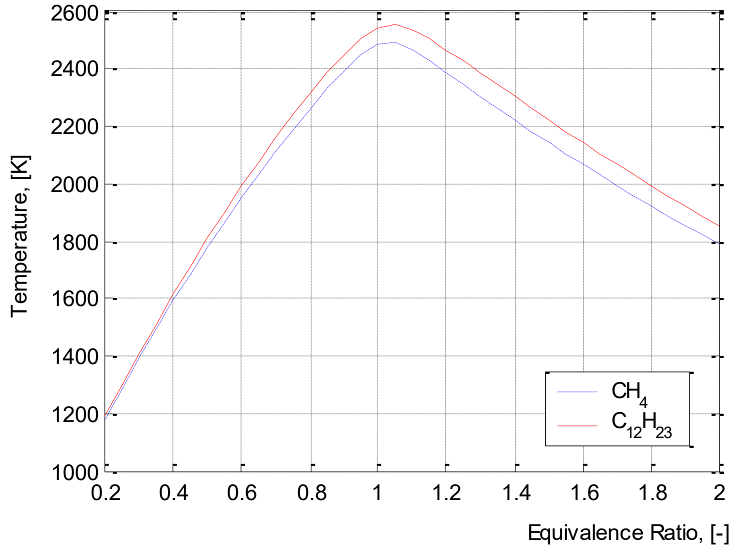


Figure 6-7 Adiabatic temperatures for $C_{12}H_{23}$ and CH_4 combustion in air at p = 28 [atm], T_{air} = 700 [K], T_f = 298.15 [K]

The adiabatic flame temperature for each combustion case described by Eq. (6.19) can be found again by equating the absolute enthalpies of the reactants and the products, Eq. (6.13, 6.14). However, the composition of combustion products can no longer be solely determined from the conservation of atoms: the number of species (12) exceeds the number of elements (5).

There are several ways to approach the calculation of equilibrium composition, which are described in texts on thermo chemistry. This chapter will be limited to illustrating temperatures of $C_{12}H_{23}$ and methane combustion, Figure 6-7. For details, the interested reader is advised to address references [5 - 7, 11 - 14].

6.3 Simplified Combustor Heat Balance

The insight into combustion thermo chemistry given above should help the reader to better understand the simplified combustion heat balance equation provided in Chapter 3 for gas turbine cycle calculations. We shall now explore its assumptions and limitations.

An exercise in thermodynamics may show that the enthalpy balance equation (Eq. 6.14) may look as follows when applied to a combustion chamber:

$$\left[\dot{m}h_f^0\right]_{air}+\left[\dot{m}h_f^0\right]_f-\left[\dot{m}h_f^0\right]_{prod}=\left[\dot{m}\int_{T_{ref}}^{T_{ad}}c_p dT\right]_{prod}-\left[\dot{m}\int_{T_{ref}}^{T_a}c_p dT\right]_{air}-\left[\dot{m}\int_{T_{ref}}^{T_f}c_p dT\right]_f, \tag{6.20}$$

where

$c_{p,i}$ - mass specific heat at constant pressure of species i.

The left-hand side of the equation can be multiplied and divided by $\dot{m}_f$ to obtain:

$$\dot{m}_f\underbrace{\frac{\left[\dot{m}h_f^0\right]_{air}+\left[\dot{m}h_f^0\right]_f-\left[\dot{m}h_f^0\right]_{prod}}{\dot{m}_f}}_{I}=\left[\dot{m}\int_{T_{ref}}^{T_{ad}}c_p dT\right]_{prod}-\left[\dot{m}\int_{T_{ref}}^{T_a}c_p dT\right]_{air}-\left[\dot{m}\int_{T_{ref}}^{T_f}c_p dT\right]_f, \tag{6.21}$$

where term *I* could have been the heat of combustion at reference temperature as defined above, were the combustion ideal. As it is unlikely for the water vapour to condense at the exit of a gas turbine combustor, we can substitute term *I* with the fuel lower calorific value (LCV) corrected by a certain parameter η_{comb}. Then

$$\dot{m}_f\eta_{comb}LCV=\left[\dot{m}\int_{T_{ref}}^{T_{ad}}c_p dT\right]_{prod}-\left[\dot{m}\int_{T_{ref}}^{T_a}c_p dT\right]_{air}-\underbrace{\left[\dot{m}\int_{T_{ref}}^{T_f}c_p dT\right]_f}_{II}. \tag{6.22}$$

As can be inferred from Table 6-1 the fuel flow does not exceed few percentage points of the airflow in gas turbine combustion. This justifies the simplification to omit the absolute enthalpy contribution due to fuel in Eq. (6.22), term *II*.

An important aspect that should be taken into account in cycle calculations is the appreciable difference between total and static properties. To conserve the total enthalpy of the system, we re-write Eq. (6.22) in total properties (with the fuel enthalpy being eliminated):

$$\dot{m}_f\eta_{comb}LCV=\left(\dot{m}c_pT\Big|_{T_{ad}}-\underbrace{\dot{m}c_pT\Big|_{T_{ref}}}_{III}+\dot{m}\frac{V^2}{2}\right)_{prod}-\left(\dot{m}c_pT\Big|_{T_{air}}-\underbrace{\dot{m}c_pT\Big|_{T_{ref}}}_{IV}+\dot{m}\frac{V^2}{2}\right)_{air}. \tag{6.23}$$

Another assumption we are going to make is to neglect the difference between the enthalpy of the products and the air at their reference states. In the other words, terms *III* and *IV* are allowed to cancel each other in Eq. (6.23). Recalling that $T_0=T+\frac{v^2}{2c_p}$, we may apply Eq. (6.23) to find the total temperature rise in the gas turbine combustion chamber. Therefore, only the air mass flow and specific heat of the products are accounted for in simplified calculations:

$$\dot{m}_f\eta_{comb}LCV=\dot{m}_{air}c_{p,g}\left(T_{0,4}-T_{0,3}\right). \tag{6.24}$$

The products of combustion are commonly referred to as combustion gases in gas turbine literature and designated with a subscript $_g$. It is also common to set the value for the combustion gases specific heat to $1150\left[\frac{J}{kg\,K}\right]$. Parameter η_{comb}, loosely speaking, accounts for the heat consumed during dissociation. In the other words, it's a ratio between the theoretical heat release in ideal combustion and the actual heat release. η_{comb} can be therefore called combustion efficiency.

A quick analysis of Eq. (6.24) would suggest that we can introduce the fuel-to-air equivalence ratio into it, making the equation independent of mass flows, namely

$$\frac{\dot{m}_f}{\dot{m}_{air}}\frac{1}{FAR_{stoich}}FAR_{stoich}\eta_{comb}LCV = c_{p,g}\left(T_{0,4}-T_{0,3}\right) \quad \Rightarrow \quad T_{0,4} = T_{0,3} + \varphi\frac{1}{c_{p,g}}FAR_{stoich}\eta_{comb}LCV. \tag{6.25}$$

This re-arrangement makes us see that the combustor outlet temperature would be continuously increasing with increasing equivalence ratio, provided we do not re-evaluate the combustion gases specific heat value. We have learnt from full equilibrium solutions (Fig. 6.8) that this is not the case.

Numerical experiments may demonstrate however that Eq. (24-25) gives reasonable estimates of combustor outlet temperatures for the range of overall equivalence ratios commonly encountered in gas turbine engines. However, these equations may not be applied to evaluating temperature profiles across the combustor length, unless the values of η_{comb} and $c_{p,g}$ are modified. This is because local mixture strengths can vary between rich and lean across the space of the gas turbine combustion chamber. The application of Eq. (6.24, 6.25) is benchmarked against the full equilibrium solution in Figure 6-8.

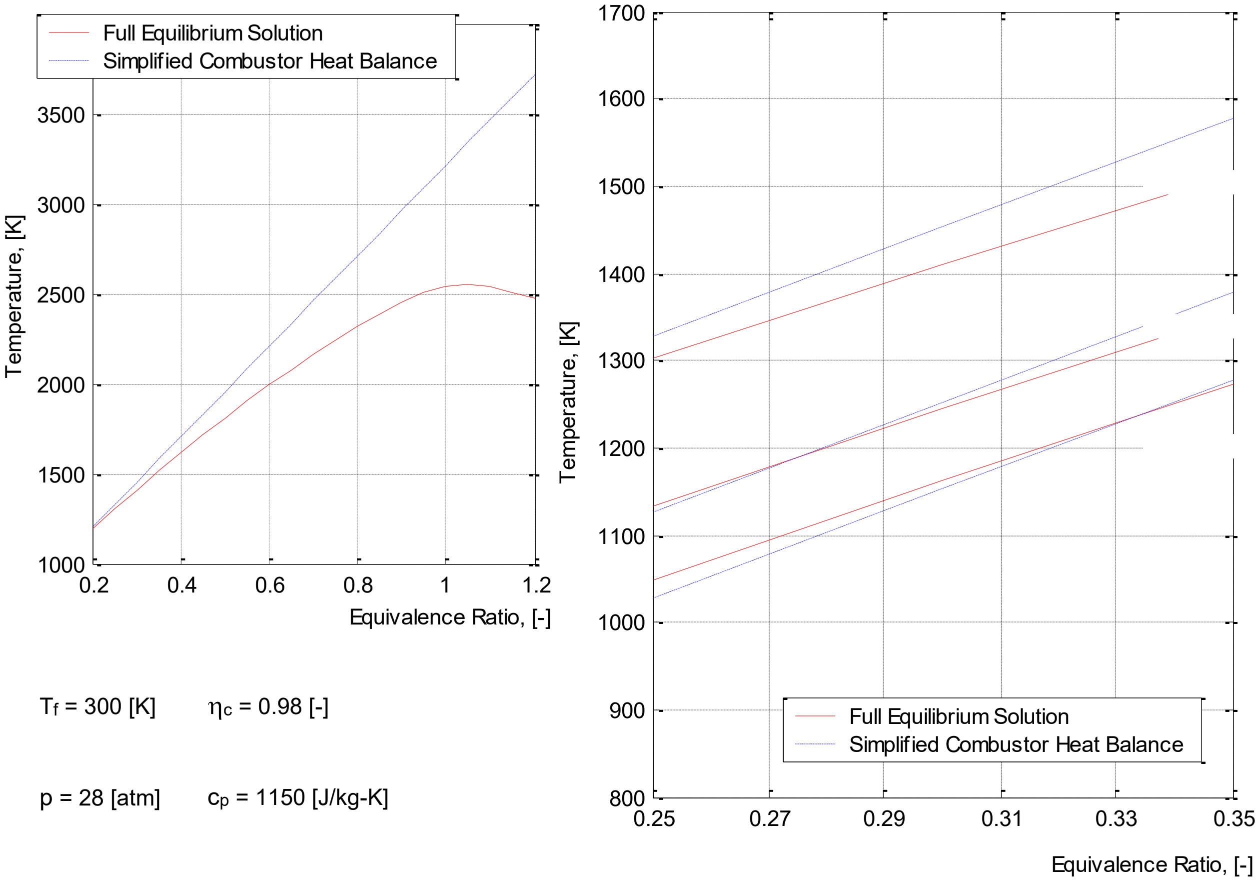


Figure 6-8 Applicability of simplified combustor heat balance equation (Equ. 6.24 - 6.25) to $C_{12}H_{23}$ combustion in air; right: Discrepancy between Equ. (6.24 - 6.25) and full equilibrium solutin; left: Applicability range for Equ. (6.24 - 6.25)

Equilibrium or Not Equilibrium

Both Eq. (6.24, 6.25) and the full equilibrium approach exemplified in Figure 6-6 imply that the combustion system is at its final state, which is not subjected to any change. It can be said either that the system has reached equilibrium instantaneously or that it has been given an indefinite time to do so. However, combustion reactions progress in a discrete-time span. The end reaction temperature and composition evolve in time. Reactions generally start with the thermal degradation of the fuel: intermediate lower hydrocarbons are formed from the parent fuel. Then, those intermediate species are oxidized to CO and H_2. The carbon monoxide and molecular hydrogen, in their turn, oxidize to CO_2 and H_2O in elementary reactions. As the reaction temperature builds up, the species dissociate and react with each other.

If a chemical reaction is not restricted in time, the temperature and product concentrations converge to certain steady values. These values are called equilibrium values. In the other words, the system approaches equilibrium. How fast this state would be achieved depends on the system parameters and the speed or rate of the chemical reactions involved. The equilibrium condition is however unique for a system and does not depend on the path taken to achieve it.

The process of heat release associated with hydrocarbon fuel combustion is usually very rapid. It is rapid enough to say that the equilibrium temperature will be achieved before the fuel-and-air mixture has escaped a gas turbine combustion chamber. This justifies the use of a simplified combustor heat balance (Eq. 6.24) determining combustor exit temperature.

The full equilibrium approach would give a better temperature estimate and is well applicable to determine the composition of major combustor products. However, the minor products of combustion would be better estimated by studying the combustion mechanism itself. The areas where equilibrium calculations are not applicable at all are for example ignition, flame extinction, pollutant studies.

The analysis of chemical reactions and their rates is dealt with in a specialized field of physical chemistry called chemical kinetics. The most widely studied combustion reaction mechanism is probably methane combustion. Kaufman indicated in his review [15] that the methane combustion mechanism evolved from the period from 1970-1982 from less than 15 elementary steps with 12 species to 75 elementary steps, plus 75 reverse reactions, with 25 species. The mechanism has further evolved through the 90s to 158 reversible reactions with 43 species [16]. The number of species and reactions describing combustion kinetics increase drastically with an increase in the hydrocarbon molecular weight. The Computing Centre of the Vrije Universiteit Brussel advertises the Konnov's combustion mechanism for hydrocarbons including methanol, acetaldehyde, ethanol, and ethylene oxide, which consist of 1200 reactions among 127 species (http://www.vub.ac.be/BFUCC/). Combustion kinetics of high hydrocarbons simply remains poorly understood.

In developing and improving gas turbine combustion systems, heat release and composition calculations should also take into account the turbulent nature of combustor flows. Flow Reynolds numbers can be on the order of 10^5 to 10^6 in the flame zone. The turbulent fluctuations in the reactant and product fluids strongly influence mean chemical rates, particularly when reaction time scales are on the order of the turbulent mixing time scale or less. The turbulence and combustion interact in the phenomenon called turbulent combustion. This interaction is bi-directional. On one hand, turbulence is modified by combustion because of the strong acceleration and changes in the flow properties due to heat release. On the other hand, turbulence alters the flame structure: chemical reactions may enhance or inhibit leading, in extreme cases, to extinction.

Turbulent combustion studies with detailed combustion kinetics require tremendous amounts of computing power. Such studies are hardly possible to handle analytically for gas turbine combustors. This is why numerical modelling of combustion for turbulent flames is a fast-growing engineering discipline.

6.4 Combustor Components

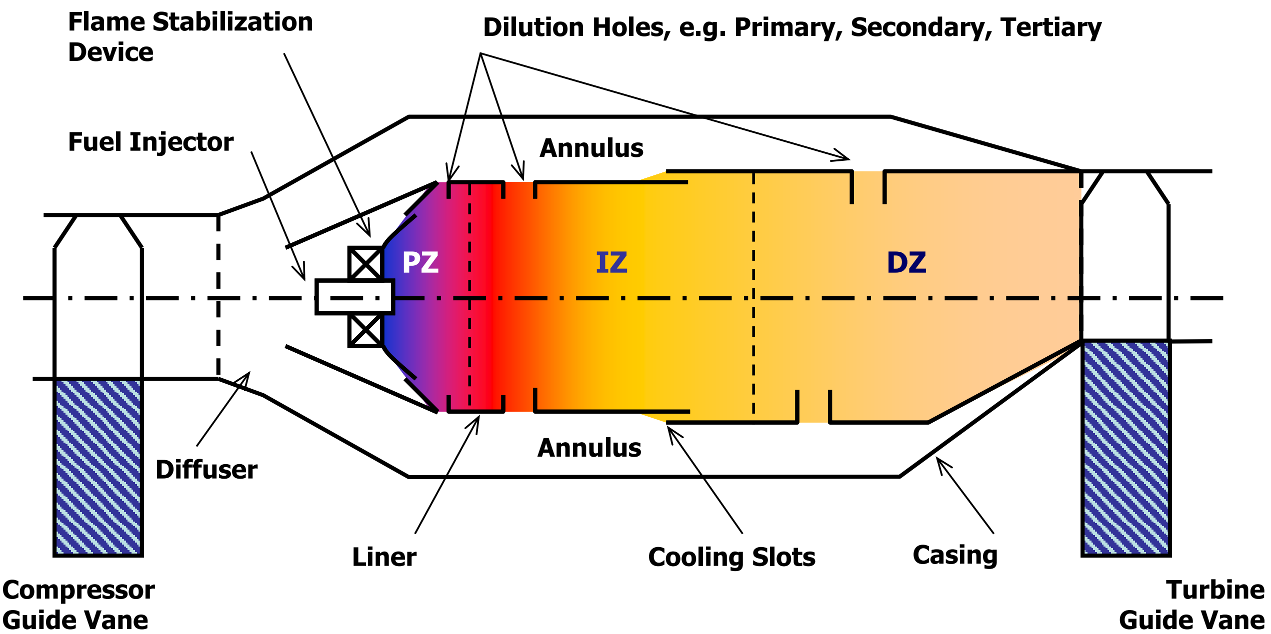


Figure 6-9 Generic gas turbine combustor components [17]

The type and layout of a gas turbine combustor depend on many factors and are greatly affected by the gas turbine application and specifications. However, all combustors incorporate a list of the main or, so-called, generic components which are always responsible for the same functions despite the diversity in their design. These components are a diffuser, casing, liner, fuel injector, some type of flame stabilization device and a cooling arrangement. Schematically, they are assembled as shown in Figure 6-9.

Diffuser

As it has been emphasized above "no process can be staged in an engineering system without a loss in pressure". This loss is, partially, incurred in a combustion chamber by simply pushing the air through it. Quantitatively, the drop in total pressure associated with this process – cold loss, in the other words – is proportional to the dynamic head of the flow. Therefore, the compressor discharge velocity should be reduced to minimize the cold loss. It is customary to do so by incorporating a diffusing channel – simply, diffuser - at the combustor inlet. Additional functions of the diffuser are to recover the flow dynamic pressure by raising its static pressure and smoothen flow instabilities.

Until recent days, there were two different philosophies in regard to diffuser design; both are illustrated in Figure 6-10. One is to employ a relatively long aerodynamic duct to achieve a gradual flow deceleration without stall. The other main diffuser type is the so-called "dump" diffuser. It consists of a short aerodynamically smooth pre-diffuser where the air velocity is reduced to about half its inlet value. At the exit, the air is literally dumped into the combustor casing.

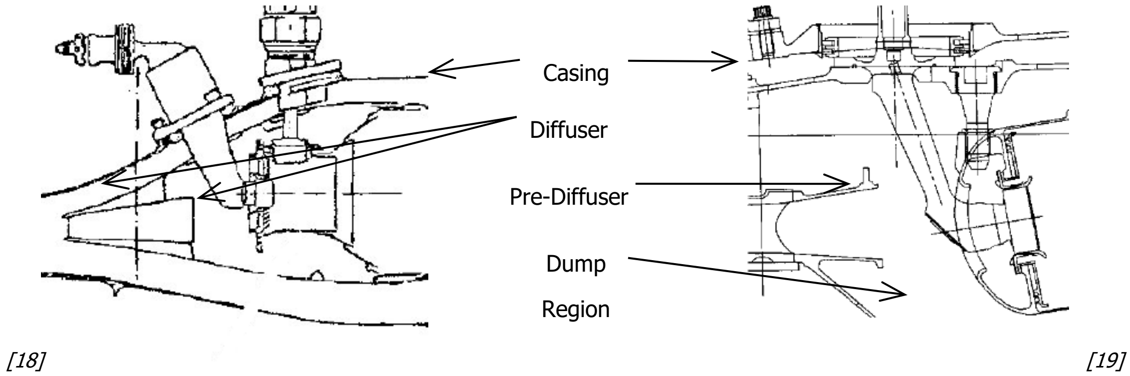


Figure 6-10 Two basic types of diffusers: Aerodynamic (left) and dump (right)

The aerodynamic diffuser traditionally has smaller pressure losses but results in a considerable length. The dump diffuser is exactly the opposite. Its shorter length made this type the automatic design choice for modern aircraft engines.

Other types of diffuser design traditionally aim at achieving a greater reduction in air velocity at a smaller pressure loss.

Casing and Liner

The simplest possible form of combustor would be a straight-walled duct connecting the compressor to the turbine. Unfortunately, this arrangement is impractical – at least, due to the fact, the pressure loss incurred would be excessive. Simply fitting a diffuser, however, is not sufficient to obtain a viable combustion system. First of all, the air velocity remains too high for a flame to sit. Secondly, poring fuel directly into the available airflow would result in no flame, as the air-to-fuel ratios are too high for the mixture to ignite. The airflow should therefore be partitioned and the flame stabilized. Arrangements for that are provided inside the so-called liner located inside the casing and downstream the diffuser (Figure 6-11). The space between the liner and casing is called the annulus (Figure 6-9).

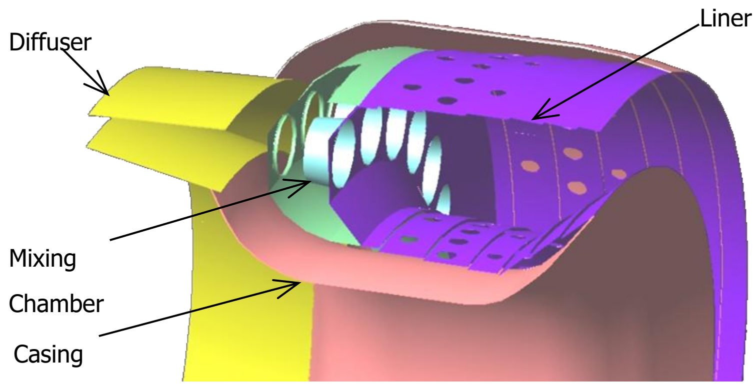


Figure 6-11 Combustor liner inside casing

The liner is virtually divided into primary (PZ), secondary (SZ) and tertiary or dilution (DZ) zones. Each zone is commonly provided with a band of air admission holes, which bear the name of the zone: primary holes (PH), secondary (SH), dilution (DH) holes (Figure 6-12).

A fuel injector and a flame stabilization device mark the front boundary of the primary zone. In some cases, the PZ may also be preceded by an additional mixture preparation device: a mixing chamber, for example (Figure 6-9). The main function of the PZ is to anchor the flame and provide sufficient time, temperature and mixing to achieve essentially complete combustion of the incoming fuel-air mixture. If the PZ temperature is around or above 2000 [K], dissociation reactions will result in significant amounts of CO in the out-coming gases. If the residence time of the fuel-air mixture in the PZ is too short, appreciable amounts of UHC and CO will be present due to incomplete combustion. CO and UHC are first of all polluting species. Secondly, their presence in post-combustion gases equals wasting the heat, which could have been otherwise released during oxidation.

To provide additional time and space for the burnout of CO and UHC, the secondary zone succeeds the PZ. The right conditions are created in the SZ by adding extra air to, on one hand, reduce the temperature and, on the other hand, not to quench combustion reactions.

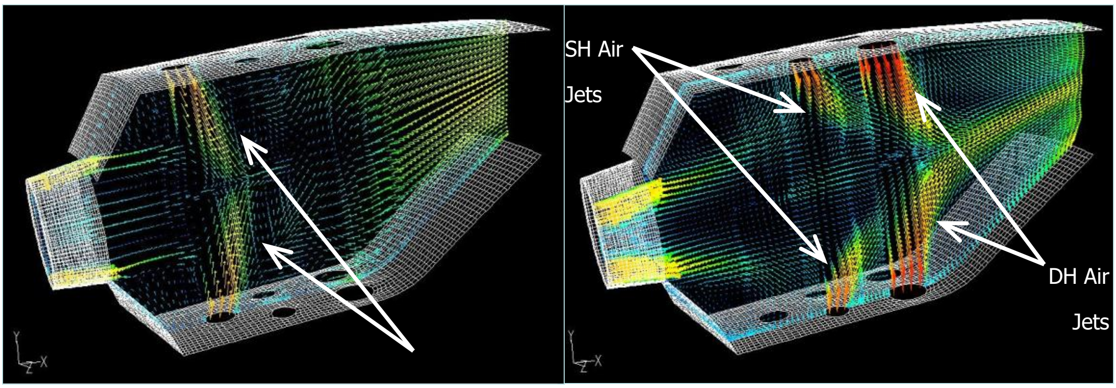


Figure 6-12 Air-admission through liner holes [11]

After the combustion and other flow requirements (cooling, as will be described below) have been met, the remaining air is admitted into the dilution zone. This is done to reduce the temperature of the outlet stream and make its temperature pattern acceptable to the turbine.

In early combustor designs, all three zones were distinctly present. The zonal air distribution was often quoted in the literature as about 28 % being admitted into the PZ and the remaining air to the SZ and DZ [18, 20]. In today's combustors, the distinction between the zones is rather vague. The choice of air distribution is made based on desired combustion performance. This choice greatly affects the temperature profile of a gas turbine combustion chamber with implications on combustion stability, exhaust emissions and other important parameters. By way of example, Figure 6-13 illustrates the temperature profile of the combustor liner for a large turbofan. Its airflow partition is as such that about 24 % is admitted into the liner through the mixing chamber;

about 11, 29 and 13 % are done through the primary, secondary and dilution holes, respectively. The rest of the air is spent on cooling purposes.

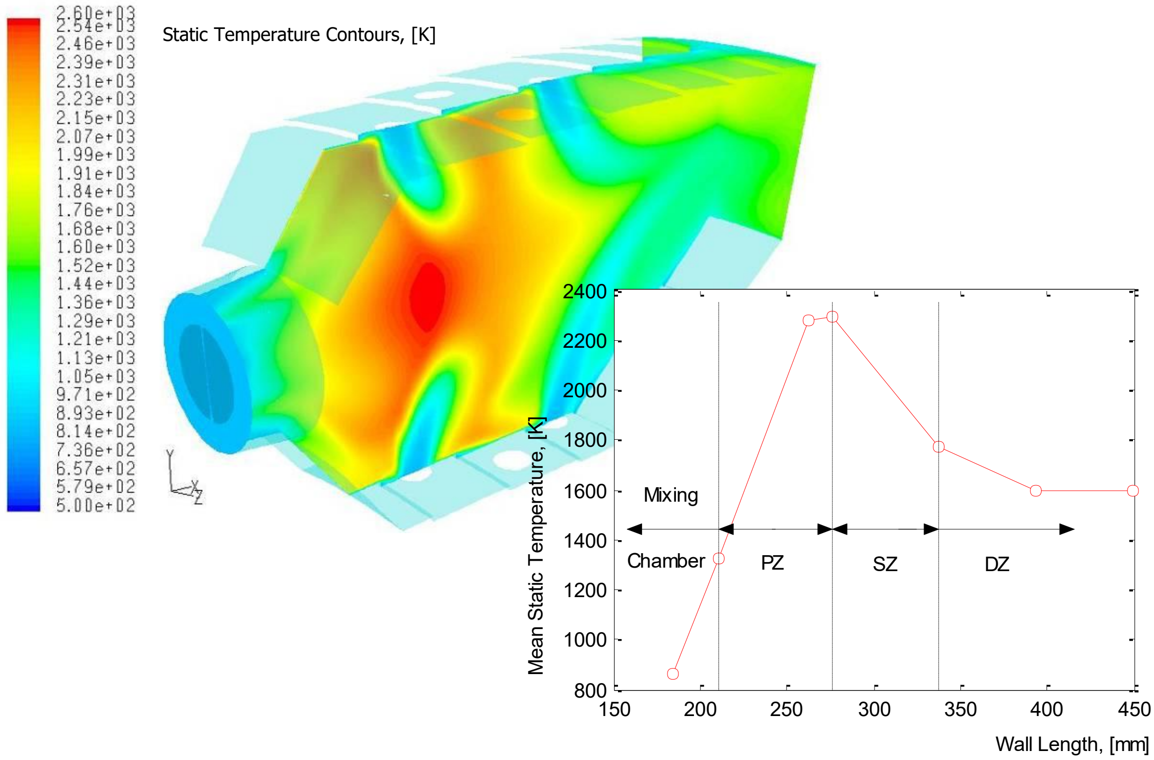


Figure 6-13 Liner temperature profile [21]

Fuel Injector

The flames encountered in gas turbine combustors can be classified into two types – diffusion flames and premixed flames – depending on whether the fuel and air are mixed by diffusion in the flame zone or premixed before combustion. This classification can be best applied to gaseous fuels. A common method of supplying gaseous fuel into a combustion chamber is forcing it through a specially designed orifice.

Liquid fuels are usually not sufficiently volatile to produce vapour in the amounts sufficient for combustion. This significantly complicates the combustion mechanism. If the fuel is not completely vaporized before entering the flame zone, heterogeneous spray combustion occurs. A diffusion flame burning individual evaporating droplets then superimposes on a premixed turbulent flame zone.

To promote the vaporization of liquid fuel, it should be atomized, i.e. the bulk liquid should be converted into small drops to increase the specific surface area of the fuel. This process gives name to the class of widespread fuel injectors called atomizers.

Essentially, good atomization requires a high relative velocity between the fuel and the surrounding air. Some atomizers accomplish this by discharging the liquid at high velocity into a slower-moving air supplied into the liner. A notable example is the pressure-assist atomizer, which converts the pressure in the fuel manifold into kinetic energy. An alternative approach is to expose the relatively slow-moving fuel to a high-velocity air stream. Herein, a typical example is the air blast atomizer.

A practical design solution of the pressure-assist atomizer is a dual-orifice atomizer (Figure 6-14). It allows satisfactory atomization to be achieved over a wide range of fuel flows encountered in gas turbines, especially in aircraft engines.

Figure 6-14 Dual-orifice atomizer [22]

A dual-orifice atomizer incorporates two concentrically located discharge nozzles. The outer nozzle is much larger than the inner one. At low fuel flows and fuel-system pressures, all the fuel is supplied through the inner nozzle. The atomization quality is good because the delivery pressure, although not high, is adequate for a small orifice. As increasing the system pressure increases fuel flow, fuel is also passed to the outer nozzle. The larger orifice diameter can satisfy high fuel flow demands without excessive fuel pressures.

A practical design concept for the air blast atomizer is to supply fuel at low pressure through a lip located in a high-velocity airstream (Figure 6-15). As the fuel flows over the lip, it is atomized by the air, which then enters the combustion zone carrying fuel droplets along with it.

A concept of liquid fuel injection different to atomization is vaporization. Historically, vaporizing systems were developed before atomizers. The fuel may be heated up in tubes located in the flame zone and released from the injection system in the form of vapour. Besides, air can be allowed into the vaporization system to mix with the fuel vapour before combustion (Figure 6-15). Such a system is commonly referred to as the premix-prevaporize.

FRONT VIEW **REAR VIEW**

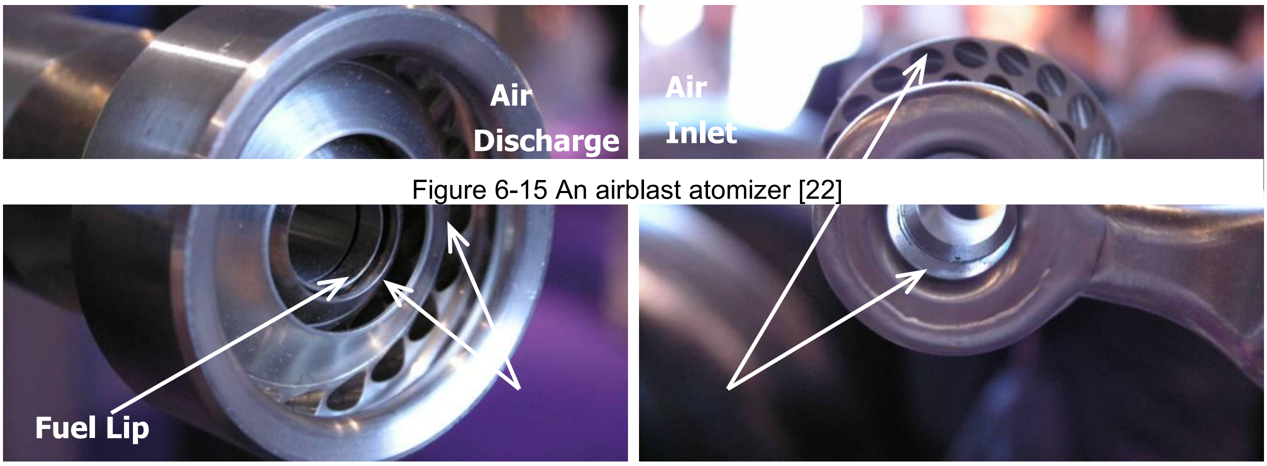


Figure 6-15 An airblast atomizer [22]

6.5 Flame Stabilization

Air leaves the compressor at a speed on the order of 150 to 200 [m/s] for the case of an industrial gas turbine or a large aircraft engine. The velocity is reduced about twice in the combustor diffuser. It is further decelerated down to 25 to 35 [m/s] in the primary zone of the liner. However, the flow velocity remains far greater than the speed of the flame. To prevent the flame from being blown away, it is, therefore, necessary to set up local regions with much smaller velocities. The most common solution is to generate a flow reversal in the liner PZ.

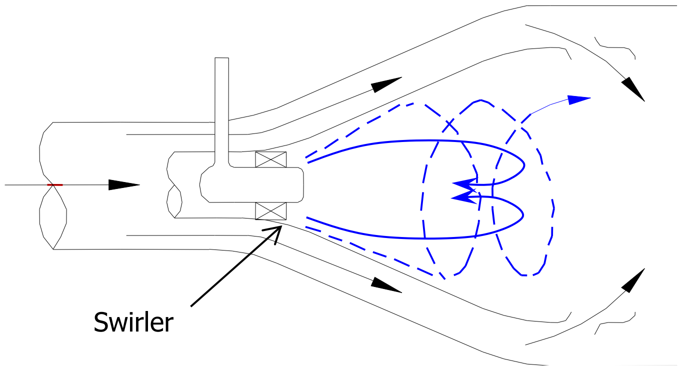


Figure 6-16 Swirl-induced flow recirculation

An efficient way to reverse the flow is to impart a swirl to it. Recirculation is created in the core region of the swirl where the amount of rotation is high (Figure 6-16). The flow velocity changes its sign on the boundary of the recirculation zone. Somewhere near that boundary, the velocity assumes the value that matches the value of the flame speed. This is the region, where the flame "anchors".

Apart from that, the flow reversal entrains and recirculates a portion of the hot combustion gases to provide continuous ignition to the incoming fuel-and-air mixture.

A common arrangement for imparting a swirl to the combustor flow is fitting a swirler at the liner front around the fuel injector (Figure 6-16). Swirlers also contribute to fuel-air mixing and provide effective ways to control the stability and intensity of combustion and the size and shape of the flame region.

The two main types of swirler are axial and radial, as shown in Figure 6-18. They are often fitted as single swirler, but sometimes are double swirler mounted concentrically to supply either co-rotating or counter-rotating flows. In some cases, swirlers are designed as part of an integral mixture preparation unit (Figure 6-17).

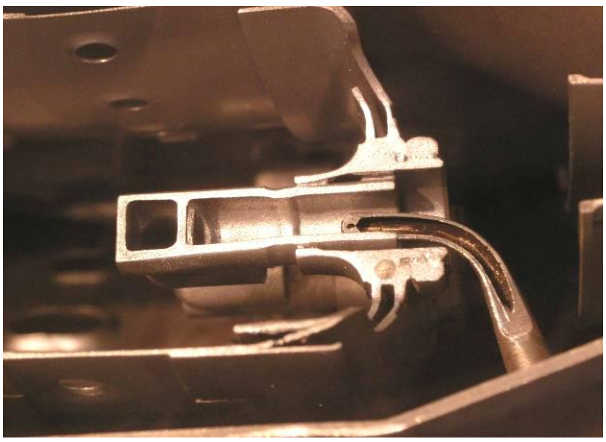


Figure 6-17 A premix-prevaporizer [22]

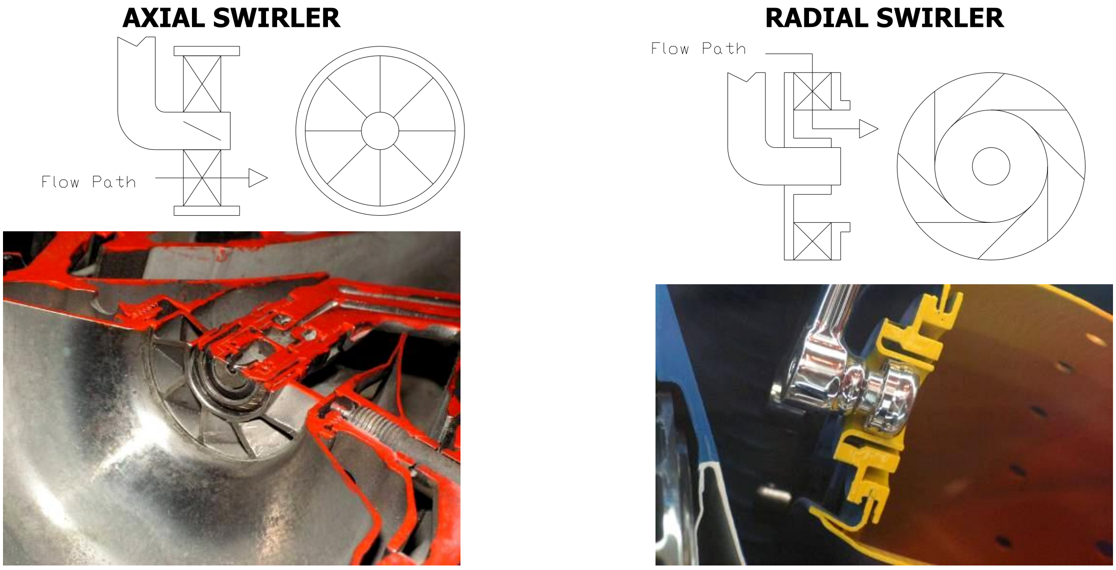


Figure 6-18 Two main swirler types [22]

6.6 Cooling

The combustor liner is exposed to very high temperatures during gas turbine operation. In the case of large industrial gas turbines and aircraft engines, the flame temperature may simply rise above the melting point of the liner material. To ensure the life of the liner, it is necessary to remove the heat transferred to its walls and prevents contact with the hot combustion gases.

An efficient way to cool and protect the liner is film cooling. This technique employs stacks of holes or annular slots through which air is injected axially along the inner surface of the liner wall to provide a protective cooling film (Figure 6-19). This film is being gradually destroyed downstream of its injection region by mixing with the hot combustion gases. Therefore, normal practice is to provide a succession of cooling-hole bands or slots along the length of the liner. There are many variations in design solutions for the film cooling arrangement. However, a common limitation of the method is that it does not allow a uniform wall temperature. The wall is inevitably cooler near the injection region and hotter further downstream.

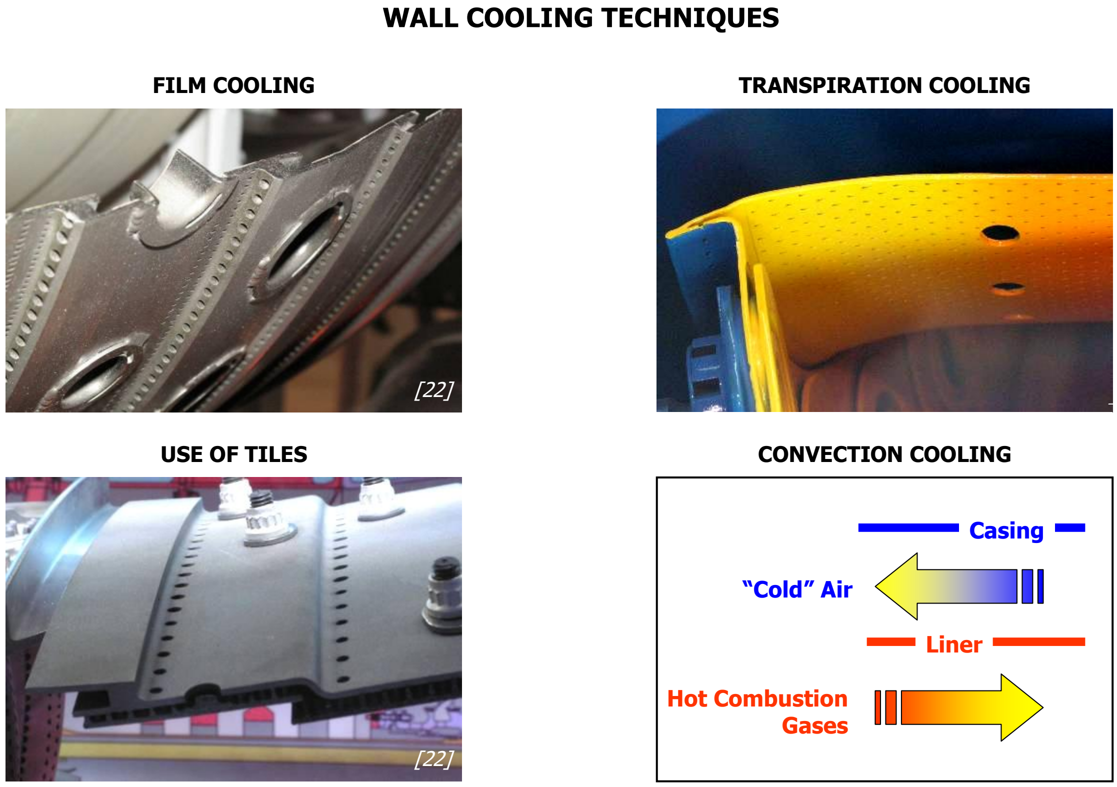


Figure 6-19 Liner wall cooling techniques

An ideal wall-cooling system would be one in which the entire liner was maintained at the permitted temperature. The technique that comes closest to this ideal is known as transpiration cooling. Herein, the liner wall is constructed from a porous material allowing the air to pass through it. Although this is potentially the most efficient method of liner cooling, its practical implementation has always been hampered by the limitations of available porous materials. A practical form of transpiration cooling is a wall perforated by a large number of small holes, as shown in Figure 6-19. The holes are being drilled at a shallow angle to the surface, increasing the contact area for heat removal. Besides, the emerging jets impinge and once again form a film to protect the liner. To reduce the heat flux to the wall, one can imagine simply lining it with refractory bricks. This practice is indeed well established for large industrial gas turbines. Since recent days, this method has been also favoured in aircraft applications. The heavy bricks though are replaced with metallic tiles, as illustrated in Figure 6-19.

Heat removal from the liner can be also provided by the convective effects of the air flowing on its back cold side (Figure 6-19). The area of convective heat transfer can be increased by the use of fins, ribs or any other form of secondary surface that increases the effective area for heat exchange. This technique is simple in design. However, it is potentially less effective compare to those described above. Besides, it does not protect the inner surface of the liner against contact with post-combustion gases. The application of pure backside convective cooling is limited to combustion systems with lower flame temperatures.

6.7 Combustor Types

The generic components discussed above can be assembled into two fundamental types of combustors: can/tubular and annular.

Tubular Type

Can-type or tubular type combustors are composed of cylindrical liners concentrically mounted in cylindrical casings (Figure 6-20). A gas turbine can feature between 1 and 16 of such tubular combustors (cans). The compressor-delivered air then has to be distributed between these chambers. The post-combustion gases have to be collected into a single flow again prior to entering the turbine.

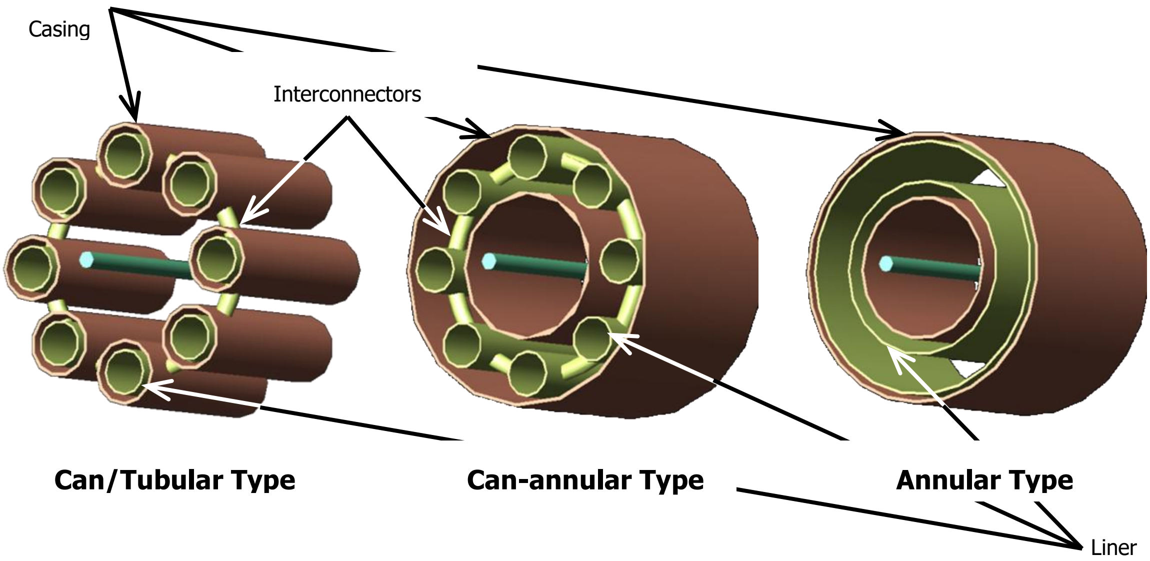


Figure 6-20 Combustor types

The separate combustor cans are however interconnected. This allows each can to operate at the same pressure and also allows combustion to propagate around during engine start-up.

The main advantage of tubular systems is the relative ease of development and testing due to the possibility of splitting the flow. However, their excessive dimensions and therefore weight prohibit their use in modern aircraft engines. The main application of can-type combustors is limited to industrial power plants.

Annular Type

Combustors of this type have a single annular liner mounted inside a single annular casing (Figure 6-20). This arrangement results in a compact unit of lower length, frontal area and therefore weight compare to tubular combustors. Its other advantages are low-pressure loss and reduced liner-wall area, which minimizes the amount of cooling air required. The annular design does not require interconnectors and simplifies flame propagation in the circumferential direction.

Qualities like that – and specifically the low weight and compact dimensions - made the annular type the choice for modern aircraft engines. The annular liner is however subject to heavy buckling loads. This is the drawback that confined the application of annular combustors in early aero-engines to low-pressure ratio designs.

Another apparent disadvantage of the annular design is the necessity of supplying the full engine mass flow at rig testing. Achieving a uniform distribution of fuel around the annular space using a fixed number of fuel injectors is also difficult.

Can-Annular Type

Can-annular combustors are hybrids of the previous two types. They are specified by a group of tubular liners arranged inside a single annular casing (Figure 6-20). The can-annular type shares the advantages and disadvantages of can and annular combustors.

A short summary of combustor types is provided in Table 6-3.

Table 6-3 - Relative advantages and disadvantages of combustor types (based on Lefebvre, [23])

Combustor Type	Advantages	Disadvantages
Can (Tubular)	Mechanically robust; Fuel and air flows are easily matched; Rig testing necessitates only a small fraction of total engine air mass flow.	Bulky and heavy; Large wall and frontal area; High pressure loss; Requires interconnectors; Incurs problem of light-around[3)].
Annular	Minimum length and weight; Minimum wall and frontal area; Minimum pressure loss; Easy light-around.	Serious buckling problems; Rig testing necessitates full engine air mass flow; Difficult to match fuel and air flows.
Can-annular (Tubo-annular)	Mechanically robust; Fuel and air flows are easily matched; Rig testing necessitates only a small fraction of total engine air mass flow; Lower pressure loss than in can type; Shorter and lighter than can type.	Less compact than annular type; Larger wall and frontal area than in annular type; Requires connectors; Incurs problem of light-around.

3) Circumferential flame propagation.

6.8 Flow Direction

All combustion chambers discussed so far can be classified as "straight-through", i.e. the flow system has no bends or passages that reverse the direction of the flow. However, either can-type, annular or can-annular combustors can also be of the so-called "reverse-flow design".

Reverse-flow combustors are commonly chosen for small gas turbine engines. As can be inferred from Figure 6-21, they provide a more compact unit and a closer coupling of the compressor and turbine. Apart from space savings, this layout allows for a shorter engine shaft. As small gas turbines are known for their high rotational speeds, this is an important advantage, which eases shaft whirling problems.

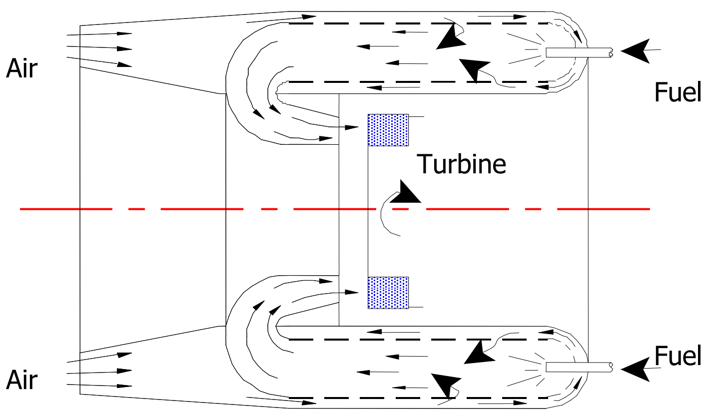


Figure 6-21 A reverse-flow combustor for small gas turbine engines

It is apparent that the combustor shown in Figure 6-21 has a very complex flow system. It suffers from high-pressure losses and "awkward" air admission into the liner. Besides, the surface-to-volume ratio is inherently high for reverse flow combustors, which adds to the problem of wall cooling. Therefore, this design is avoided where the engine specifications permit so.

The other extreme (concerning small engines) of the application of reverse-flow combustors concerns large industrial gas turbines. Low weight and compact layout are not so important in their case. On the contrary, industrial gas turbine combustors tend to be large in size. A resulted advantage is, apparently, longer times available for fuel burnout. This is often becoming critical when gas turbines have to operate on poor quality heavy fuels under strict environmental regulations. Large combustors also have lower pressure losses as flow velocities become slower. They also offer such advantages as ease of accessibility and maintenance.

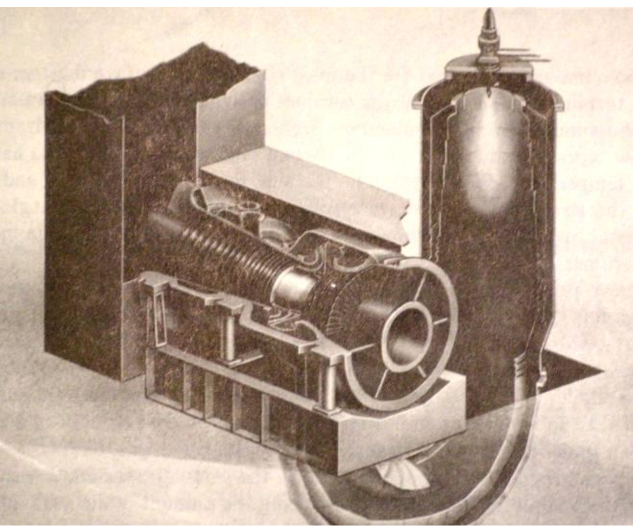


Figure 6-22 Silo-type combustor [18]

A common design solution for reverse flow industrial combustion chambers is a single-can design, as illustrated in Figure 6-22. This arrangement is commonly referred to as a silo-type combustor.

6.9 Combustion Performance

6.9.1 Ignition

Combustion in gas turbines is a continuous process. At the start-up, while the engine is being cranked up to its self-sustaining speed, a light-up is though required. A common and efficient way to ignite a combustible mixture is an electric spark produced by an igniter plug. As concisely stated by

Lefebvre [24], the spark "must supply to the combustible mixture sufficient energy to create a volume of hot gas that just satisfies the necessary and sufficient condition for [flame] propagation – namely that the rate of heat generation just exceeds the rate of heat loss".

The process of ignition and flame development is illustrated in Figure 6-23 in a series of snapshots from the numerical study by Pascaud [25].

In a fully operative combustion chamber, no flame extinction should occur under a wide range of operating conditions. In an adverse climatic environment or, for an aircraft gas turbine, on takeoff from a wet runway where there is a risk of excessive water or ice ingestion, the ignition system must however be capable of continuous operation. This is to ensure immediate relighting in the event of flame extinction. Besides, aircraft combustors are subject to the requirement of rapid relighting after a flameout in flight.

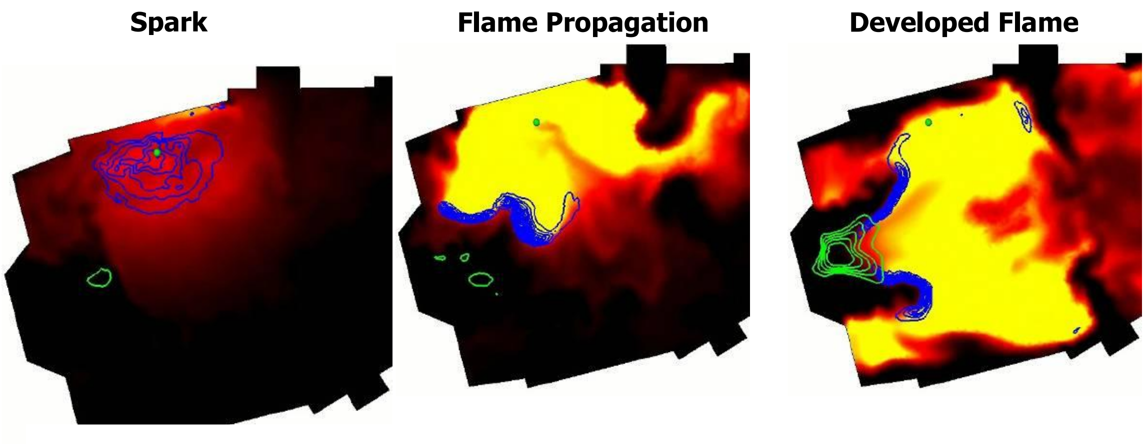


Figure 6-23 Ignition sequence in one sector of an annnular aeronautical combustor [25]

6.9.2 Combustion Stability

As mentioned before, gas turbine combustion is envisaged to be a self-sustaining continuous process under broad operating conditions. In the other words, the flame should remain stable at varying mixture strengths, inlet temperatures and pressures, turbulence levels, flow speeds and so on.

Combustion stability is often described by a range of fuel-to-air ratios that circumscribe the combustor stability loop. Its main features are qualitatively demonstrated in Figure 6-24. The region of stable combustion should be seen as being bounded by two limiting converging lines.

Falling below the lower line at a given mass flow would result in the “weak extinction” of flame. Climbing above the upper line would cause the “rich extinction”.

Apparently, the flow velocity increases with an increase in the mass flow rate for a given combustor. Should the velocity of the mixture flow become too high, the flame front will move downstream and eventually “blow out”. This explains the converging nature of the “rich” and “lean” borderlines in Figure 6-24. Combustion is unattainable beyond their convergence points at any fuel-to-air ratio. It should be however emphasized that combustion chambers may still operate stably with the air stream flowing at a speed many times greater than the normal burning velocity of the fuel employed. The flame is then anchored behind a stabilization device as mentioned in Section 6.4. In general, combustor designs, which are capable of maintaining flames at high flow speeds, are characterized by high blowout velocities. This quality is commonly referred to as “good stability performance”.

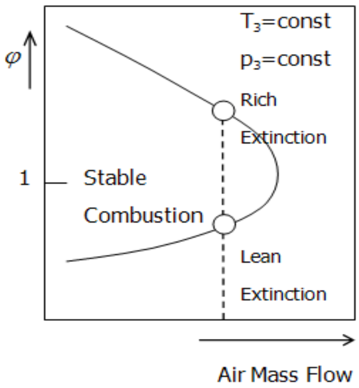


Figure 6-24 Typical combustor

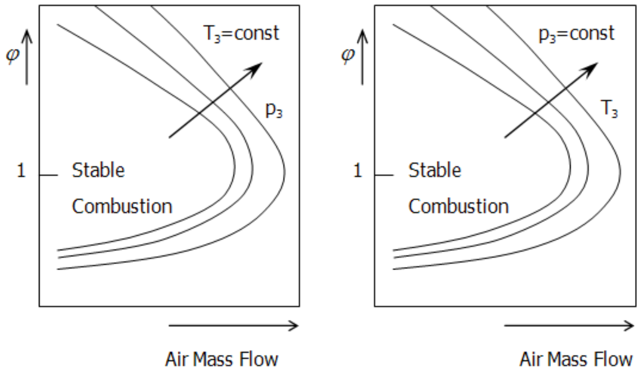


Figure 6-25 Influence of inlet pressure and temperature on stability loops

Loops similar to the one shown in Figure 6-24 are being obtained in the development testing of a new combustion chamber. Carrying out sufficient extinction tests at different levels of inlet pressure and temperature allows obtaining a number of stability loops as shown in Figure 6-25. Such performance characteristics are particularly important for aero-engine combustors – they help define the range of flight conditions over which stable combustion is possible.

On the contrary, should the flow velocity drop below the flame speed, the flame will propagate upstream. It may then stabilize inside the mixture preparation unit and, in extreme cases, burn through it. This phenomenon is commonly called a flashback. It is an inherent feature of premixed combustion systems. Figure 6-26 shows snapshots by Légier [26] where the flame travels from the combustion zone into the premixing section.

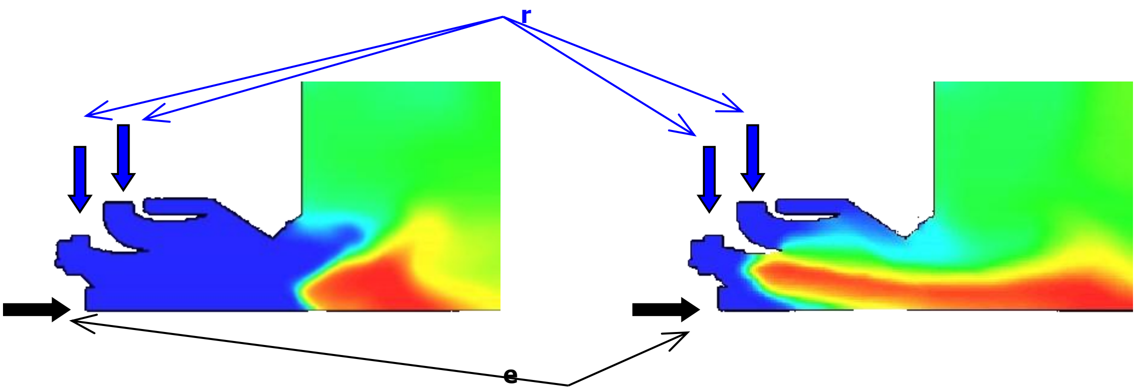


Figure 6-26 Flashback in a premixed combustion system [26]

6.9.3 Heat Losses and Incomplete Combustion

The process of combustion converts the latent chemical energy of the fuel into heat. In practice, the heat released from the ideal combustion of fuel – described by its heating value – cannot be fully utilized due to unavoidable losses.

Above, we have defined the combustion efficiency η_{cc} (Eq. (6.24) to account for heat losses due to dissociation. Additionally, small amounts of heat are being spent on evaporating the fuel, in the case of liquid fuel. Besides, the heat is being spent on heating the combustor itself. This loss has a significant effect on combustor outlet temperature during engine start-up – before the temperatures of combustor structures have come to equilibrium with the flow temperatures.

Thus, the inner wall of the liner is being heated by radiation and convection from the hot gases inside. The received heat is then conducted to the outer wall. From the outside, the liner is cooled by convection to the annulus air and radiation to the casing. This basic heat transfer process is sketched in Figure 6-27. Please note that the use of cooling will only complicate this process, but not add or eliminate any heat transfer mechanisms.

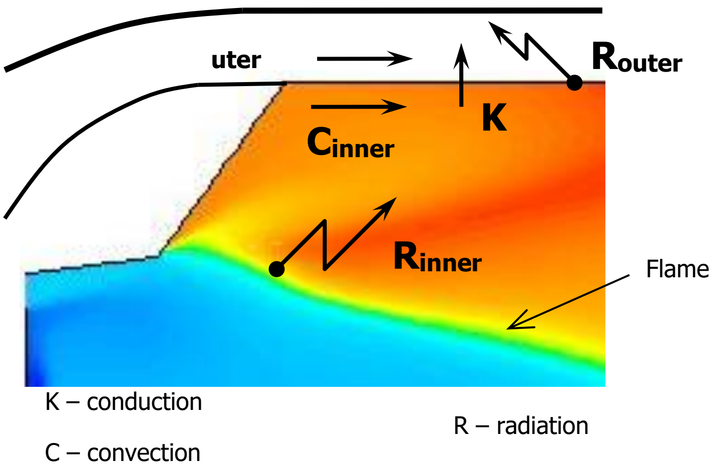


Figure 6-27 Basic heat transfer process

The heat transferred by convection mainly affects the flow regions adjacent to the wall. Radiation, however – and particularly the inner radiation – can cause a significant heat flux out of the combustion zone. This would have the most pronounced effect on the primary zone temperature. As part of the lost heat is spent on heating the annulus air, it is being recuperated downstream the combustion chamber as the flow gets admitted into the liner.

Another cause for a disagreement between theoretical and actual heat release may lie in incomplete combustion. Its occurrence can be best explained by looking at the time required to burn the fuel in a combustion chamber. For the case of a gaseous fuel, this time is the sum of the times needed to:

- Mix the fuel and air to produce a combustible mixture (which would fall within the burning region shown in Figure 6-24, Figure 6-25 under given conditions); and
- Complete combustion reactions.

For the case of liquid fuel, this total time should also account for the period required to evaporate the fuel.

The mixture residence time may fall short of the time required for complete combustion. In a case like that, the amount of fuel that has been given the chance to fully release its chemical energy will be different from the amount of fuel supplied by the injectors. The ratio between the two can be defined as a factor of complete combustion, say $\eta_{complete}$. In other words,

$$\eta_{complete} = \frac{\dot{m}_f^{evaporated}}{\dot{m}_f^{injected}} \frac{\dot{m}_f^{mixed}}{\dot{m}_f^{evaporated}} \frac{\dot{m}_f^{reacted}}{\dot{m}_f^{mixed}} = \frac{\dot{m}_f^{reacted}}{\dot{m}_f^{injected}}. \quad (6.26)$$

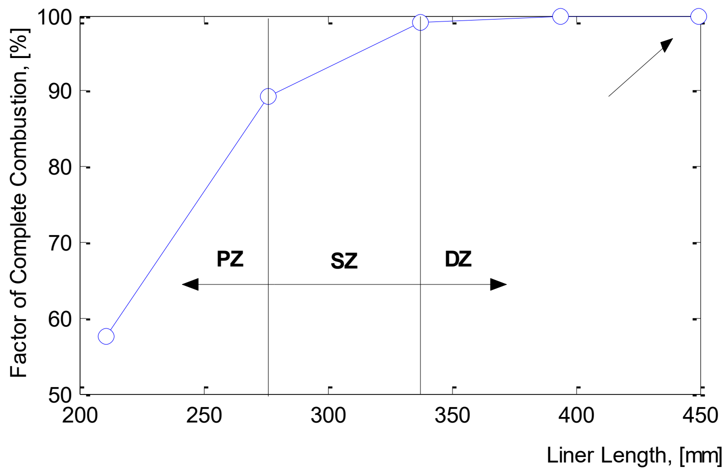


Figure 6-28 Factor of complete combustion in a generic combustor at full power [11]

In modern gas turbines, combustion is almost 100 % complete at full power settings or thrust ratings. That means almost 100 % of the fuel has completely reacted by the time the mixture is leaving the combustor. However, this level is not being attained instantaneously, but rather develops through the length of a liner. By way of example, Figure 6-28 illustrates how the factor of complete combustion reaches the value of around 90 % in the primary zone, climbs above 99 % in the secondary zone and attains roughly 100 % at combustor exit.

Even though combustion is usually almost complete at combustor exit at full-power operation, that level drops at lower power settings. Figure 6-29 shows the factors of complete combustion evaluated by approximate methods [11] for the aircraft engines from Table 6-1 at different thrust ratings.

Eventually, incomplete combustion can be treated as a loss. Together with other losses - such as dissociation, heat losses on fuel evaporation and to combustor walls - incomplete combustion introduces a difference between the theoretically expected and actual heat release. We can therefore revisit the concept of combustion efficiency, η_{cc}, employed in Eq. (6.24, 6.25) and define it as a function of losses:

$$\eta_{cc} = \frac{\text{Actual Heat Release}}{\text{Theoretical Heat Releasse}} = f\begin{pmatrix}\text{dissociation losses,} \\ \text{heat transfer,} \\ \text{incomplete combustion}\end{pmatrix},\ \eta_{cc}<1. \qquad (6.27)$$

Pressure Losses

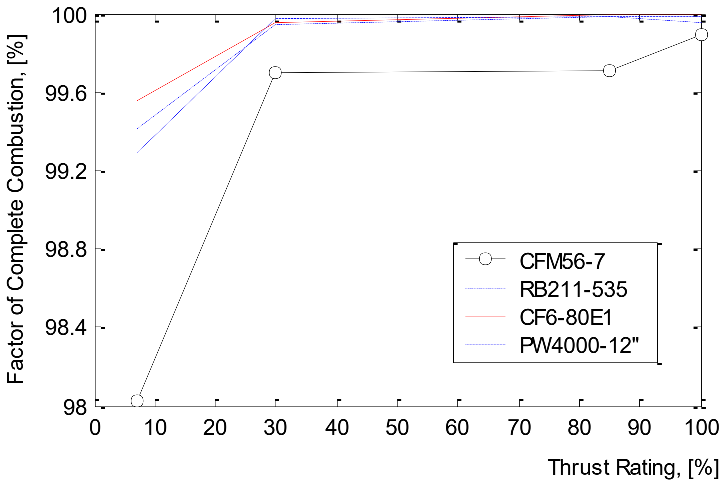


Figure 6-29 Combustion completeness in commercial turbofans at different thrust ratings

Fundamentally, the loss in total pressure sustained in a combustion chamber can be split into two components named cold and hot pressure losses:

The cold pressure loss is incurred by skin friction and large-scale turbulence taking place in a combustor. The turbulence of this kind is created, first of all, in the diffuser dump region (Figure 6-10) and by the flame stabilization device, e.g. a swirler (Figure 6-18). In addition, there is turbulence induced by the air admission jets into the liner (Figure 6-12). Therefore, the cold pressure loss can be apportioned between the diffuser, flame stabilization device and liner:

$$\Delta p_{0,3-0,4} = \left(\Delta p_{0,3-0,4}\right)_{cold} + \left(\Delta p_{0,3-0,4}\right)_{hot}. \qquad (6.28)$$

$$\left(\Delta p_{0,3-0,4}\right)_{cold} = \left(\Delta p_0\right)_d + \left(\Delta p_0\right)_{sw} + \left(\Delta p_0\right)_l. \qquad (6.29)$$

This loss tends to be 3 to 5 per cent of the compressor-delivered pressure in modern gas turbine engines. Up to 30 % of it can be sustained in the diffuser, and the remaining part is shared between the swirler and liner. The fluid is also experiencing a slight reduction in total pressure, as it flows down the annulus and gets admitted into the liner.

The hot loss (Eq. 6.28) arises due to a change in the momentum of the flow as it passes through the combustion zone: an increase in temperature implies a decrease in density followed by an increase in velocity and, consequently, in momentum. This loss is commonly referred to as fundamental.

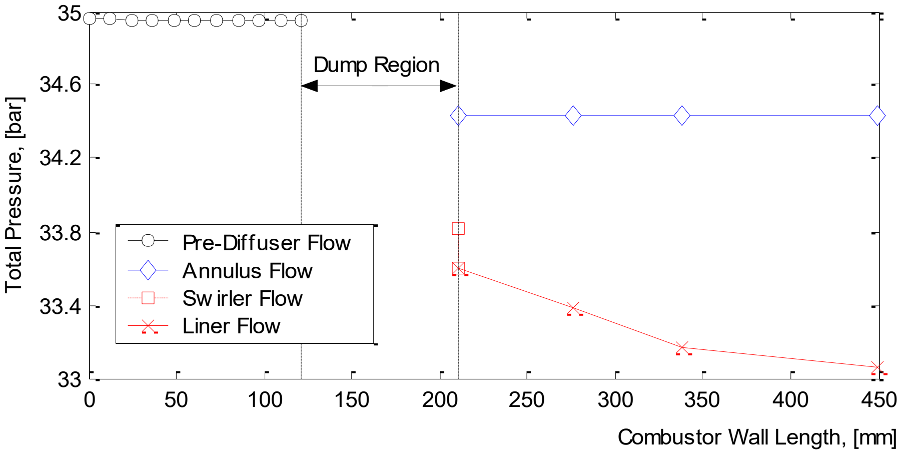


Figure 6-30 Longitudinal total pressure distribution in a generic combustor [11]

When the flow velocity is low and the fluid can be treated as incompressible, $\left(\Delta p_{0,3-0,4}\right)_{hot}$ can be estimated from the conservation of momentum as

$$\frac{\left(\Delta p_{0,3-0,4}\right)_{hot}}{\rho_3 V/2} = \left(\frac{T_4}{T_3} - 1\right), \tag{6.30}$$

where

$$V = \frac{\dot{m}_3}{\rho_3 A_{\text{casing}}}$$

The derivation of Eq. (6.30) is exemplified in Cohen [28] for a constant cross-section abstract duct. In our case, the duct is substituted with a combustor casing. Although the assumptions of incompressible flow and constant area are crude, Eq. (6.30) can produce a fairly accurate first-order estimate of the fundamental pressure loss.

6.10 Pollutant Emission

Hydrocarbon-fueled gas turbines emit exhaust products, which are characteristic to fossil fuel combustion. The exhaust species can be loosely classified as products of complete combustion, products of dissociation and pollutants or undesirable chemicals.

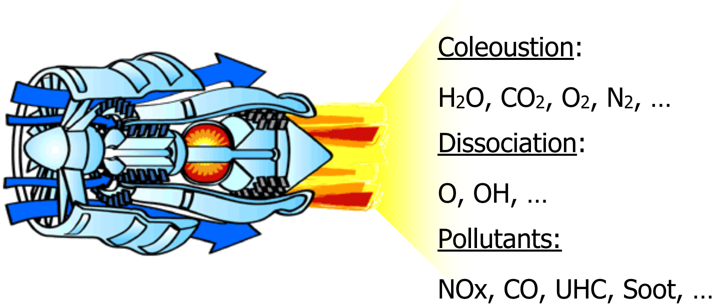


Figure 6-31 Gas turbine exhaust

Considering the overall fuel-to-air ratios (Table 6-1) on which gas turbines commonly operate, the most abundant exhaust products are oxygen and nitrogen coming from the air. As combustion is essentially complete within the standard operational envelope of a gas turbine, carbon dioxide and water vapour are also present in abundance in post-combustion gases. These four (H_2O, CO_2, O_2, N_2) so-called products of complete combustion usually make up to 99 % of a gas turbine exhaust as illustrated in Figure 6-31. The remainder of the exhaust is shared between dissociation products and a group of chemicals known as pollutants (Figure 6-31) or, simply, emissions. The latter can be divided into gaseous pollutants and smoke.

Gaseous pollutants include a range of nitrogen oxides (mainly NO and some NO_2), jointly designated NOx, carbon monoxide and a variety of unburned hydrocarbons (UHC). Smoke is a particulate pollutant composed of soot particles, which are made up of about 96 % of carbon. Smoke is often referred to as soot.

For the sake of analysis, gaseous pollutants can be expressed in mole (volume) fractions, mass fractions or the so-called emission indexes. The emission index (EI) of a gaseous pollutant *i* is defined as:

$$EI_i = \frac{\text{mass of } i \text{ produced in } g}{\text{mass of fuell used in } kg}. \tag{6.31}$$

A common measure for smoke or soot content is the smoke number [28]. Loosely speaking, it characterizes the transparency of the exhaust plume. High smoke numbers are pertinent to black opaque exhausts. Modern gas turbines have almost transparent plumes characterized by low smoke numbers (Figure 6-32).

In the early days of gas turbine engines, combustion engineers were making attempts to correlate pollutant production with thrust or power settings. Indeed, looking at Figure 6-32, we may notice that pollutant concentrations change far greater with changes in power settings compare to the products of complete combustion. A better insight into variation trends of pollutant emission from the generic aero-engine is shown in Figure 6-33. Both mathematical modelling [11] and measurement results [29] agree that NOx and soot production generally diminishes with a

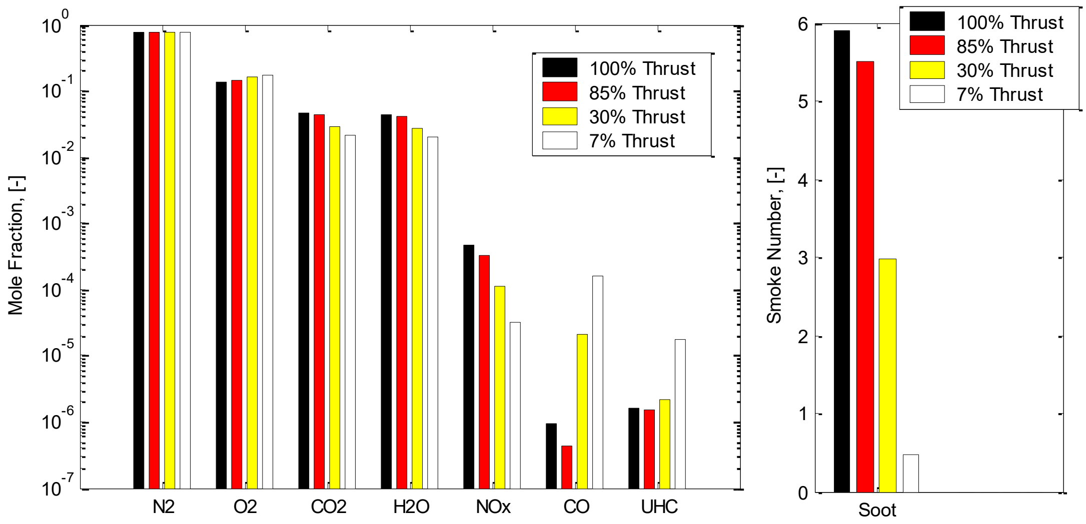


Figure 6-32 Exhaust composition from a generic aero-gasturbine [11]

decrease in engine thrust. On the contrary, CO and UHC emissions rise. Even though the trends in Figure 6-33 have been obtained for a specific engine, they hold qualitatively true for most gas turbines being in industrial and aircraft operation today.

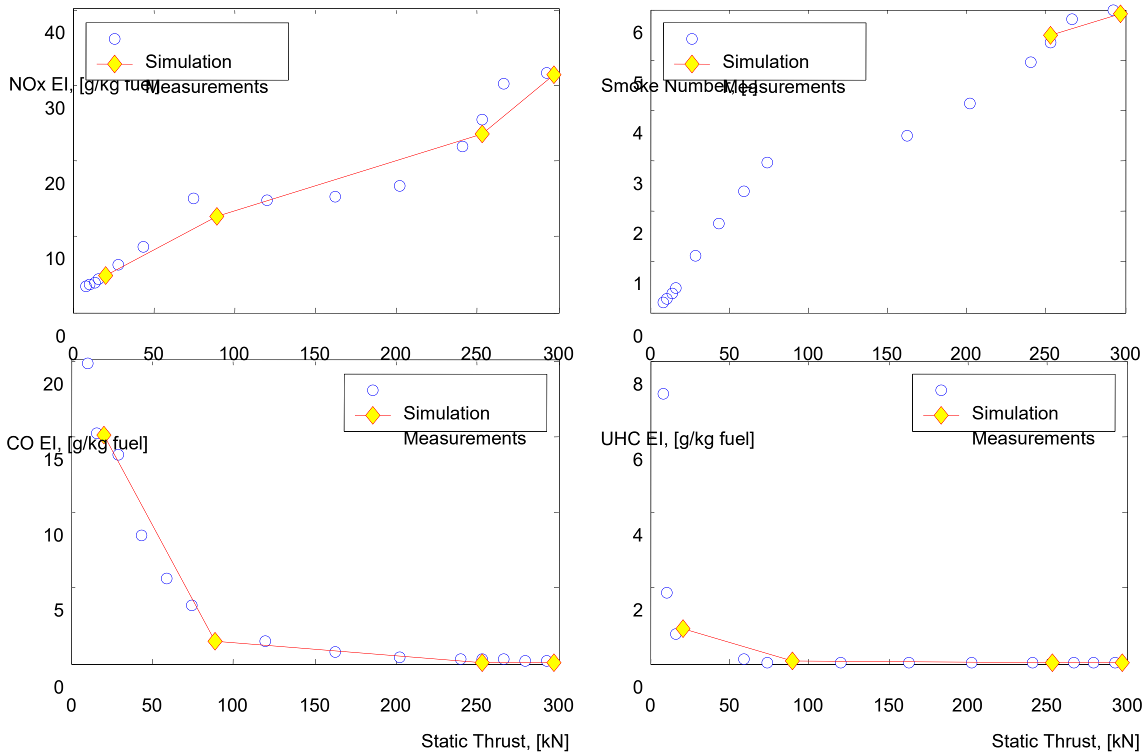


Figure 6-33 Pollutant emission from a generic aero-engine

However, the trends shown in Figure 6-33 may drastically change, should non-trivial measures be applied to the engine combustor design. This is because the rates of pollutant formation depend on the internal conditions in the combustion chamber. Slight changes in combustor operational variables are capable of causing large changes in pollutant concentrations.

A good overview of pollutant formation versus combustion characteristics can be obtained by plotting emission concentrations versus fuel-to-air equivalence ratio, as shown in Figure 6-34. The graph illustrates the conflicting behaviour of emission production mechanisms in response to changes in φ and temperature. The exact concentration values and the extreme locations shown in Figure 6-34 strongly depend on available residence time, temperature, pressure and other system parameters. These dependencies are briefly discussed below for the four polluting species.

Oxides of Nitrogen

In gas turbine combustors, NOx is produced by four different mechanisms or pathways:

- Thermal NOx;
- Prompt NOx;

- NOx due to Nitrous Oxide (N_2O); and
- Fuel NOx.

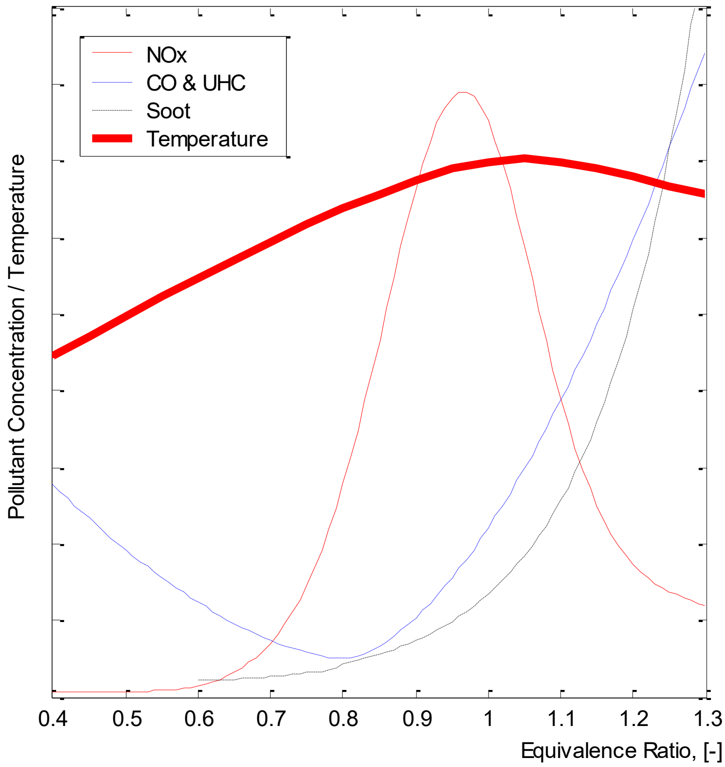


Figure 6-34 Pollutant production

Thermal NOx is responsible for most of the nitrogen oxides emissions from conventional large gas turbines operating on high-calorific value fuels. This is because the mechanism is endothermic and proceeds at a significant rate only at temperatures above around 1800 [K] – 1850 [K]. As soon as flame temperatures climb towards and above 2200 [K], thermal NOx production almost doubles for every temperature increase of few dozen K.

Thermal NOx is produced by the "slow" oxidation of atmospheric nitrogen in high-temperature post-combustion gases. This pathway is also called the Zeldovich mechanism, as its reaction chain has been first postulated by the Soviet scientist Yakov Zeldovich in 1946.

Fundamentally, thermal NOx formation is largely controlled by temperature. However, it is found to peak on the fuel-lean side of stoichiometry, while the combustion temperature is higher on the slightly rich (Figure 6-34). This is a consequence of the competition between fuel and nitrogen for available oxygen.

In gas turbine combustors, where the residence times are measured in milliseconds, NOx does not attain its equilibrium values (Figure 6-6). Nitrogen oxide emissions, therefore, increase, if the combustor design implies a longer residence time.

Prompt NOx. Under certain conditions, NOx is found very early in the flame region in conflict with the idea of a "slow" nitrogen oxidation pathway. The GE researcher Fenimore argued in the 1970s that reactions other than the Zeldovich mechanism were playing role in the flame. It was later proved by Fenimore and others that some NO was being indeed formed in the flame region. This "fast" formation mechanism was called prompt NOx or Fenimore NOx.

The literature information about the prompt pathway is often controversial. Generally, studies on Fenimore NOx for gas turbine combustors are less accurate than thermal NOx studies. According to Warnatz [29], prompt NOx can be a significant contributor to the NOx emissions produced in rich combustion. Prompt NOx is not found in non-hydrocarbon combustion.

NOx due to N_2O formation is analogous to the thermal mechanism as oxygen attacks nitrogen molecules to form N_2O. The formed nitrous oxide though rapidly oxidizes to NO. Therefore, N_2O is usually an intermediate species. Its contribution to the total nitrogen oxide emissions is often overlooked and considered insignificant. However, the N_2O route can become the major source in high-pressure lean premixed combustion, when prompt NOx formation is low [29] and thermal NOx is suppressed by low temperatures.

Fuel NOx is usually the major contributor to NOx emissions in the low-temperature combustion of some low calorific value gasification products. Such fuels may contain up to 40 percent of nitrogen and nitrogen compounds, for example, amines (NHx). During combustion, some of this nitrogen and its compounds oxidize to form the so-called fuel NOx.

Fuel NOx formation appears to be only slightly dependent on temperature with high concentrations obtained at lean and stoichiometric conditions and relatively lower yields found in fuel-rich combustion.

Fuel NOx can be still significant in the combustion of high-distillate fuels containing about 1.8 [24] percent of organically bound nitrogen. It becomes fairly insignificant when burning light distillates with less than 0.06 percent of nitrogen [24] and, especially, aviation kerosene with an N_2 content of around 0.01 percent [9].

Carbon Monoxide and Unburned Hydrocarbons

When the combustion zone is operating fuel-rich, large amounts of carbon monoxide are formed owing to the lack of enough oxygen to complete the oxidation reactions to CO_2 (Figure 6-34). If, however, the mixture strength is stoichiometric or moderately lean, significant amounts of CO will also be present owing to the dissociation of CO_2 (Figure 6-34).

In practice, CO emissions found in gas turbine exhaust conflict with the predictions of equilibrium theory (Figure 6-6). It is commonly suggested that much of the CO arises from incomplete combustion of the fuel: the parent fuel pyrolyzes to carbon monoxide and then lacks the time and conditions to oxidize to CO_2. The fuel that happens to escape the combustor in the form of drops or, more commonly, vapour, as well as species of a lower molecular weight is classified as unburned hydrocarbons.

Therefore, both CO and UHC are referred to as products of incomplete combustion. The factors that are commonly responsible for their production in a gas turbine combustor are one or more of the following:

- Inadequate burning rates due to a fuel-to-air ratio that is too high or excessively low.
- Inadequate mixing of fuel and air, which produces local fuel-rich pockets that give rise to high local CO concentrations.
- Insufficient residence times in the combustion zone.
- Quenching of the combustion reactions by the liner wall-cooling air and dilution jets.

Soot / Smoke

Soot particles are produced in excessive quantities in fuel-rich combustion (Figure 6-34) of carbon-containing fuels. In conventional gas turbine combustors, this usually takes place close to the fuel spray. Most of the produced soot then burns out in the diluted high-temperature combustion gases downstream. Improved fuel-air mixing that minimizes the occurrence of local over-rich pockets drastically reduces the sooting tendency of combustion chambers.

Soot production is impossible to analyze by equilibrium methods. Even extended kinetic schemes often fail to deliver plausible results. In practice, the rate of soot formation is rather governed by combustion physics and component performance. The following factors are of primary importance as summarized by Lefebvre [23, 24]: pressure, inlet/outlet temperature, quality of fuel injection, mode of fuel injection, fuel-to-air ratio, fuel type.

Emissions and the Environment

Pollutant emissions cause a detrimental impact on the environment. It can be divided into global and local effects. Also, health impacts have to be considered.

Thus, oxides of nitrogen are very influential in atmospheric chemistry, and they are remarkable in both ozone production and destruction processes. In the lower atmosphere, NOx emissions cause increased ozone amounts and contribute to global warming. On the contrary, NOx emissions at 18 [km] or above tend to deplete ozone, jeopardizing the ozone layer that protects the Earth from ultraviolet fluxes from outer space. This should be considered when operating high-altitude military aircraft and possibly civil supersonic aircraft in the long-term future. Besides, nitrogen oxides cause acidification that may result in acid rains. In the presence of sunlight, NOx can react to produce smog, which would be seen as a brownish cloud above power plants, airports and the local communities. From the viewpoint of direct health effects, oxides of nitrogen are toxic - an excessive exposure can particularly cause damage to the lung tissue.

Carbon monoxide is extremely toxic. Chronic exposure to CO can affect the brain structure and complex task performance. It binds haemoglobin, thereby reducing the oxygen-carrying capacity of the blood. As for the global environment, CO has a minor contribution to ozone production.

Unburned hydrocarbons are known for a range of harmful effects on humans and the local environment. They are toxic and can be the main source of the odours prevalent around power plants and airports. UHC also combine with NOx to form photochemical smog. Contemporary studies show that UHC may contain carcinogenic species. Some hydrocarbons strongly affect the Earth-atmosphere energy balance and largely contribute to global warming.

Eventually, smoke emissions are undesirable because they soil the atmosphere and reduce its transparency. This has a "coating" effect on the Earth and causes warming. Recent studies indicate a strong association between soot particles and respiratory diseases and cancer.

The products of complete combustion, carbon dioxide and water vapour, also deserve to be mentioned in the environmental context. They are not referred to as pollutants, as they hardly have any direct effect on the local communities living close to a facility that operates gas turbines. However, CO_2 and H_2O participate in atmospheric chemistry and impact the climate. Particularly, CO_2 molecules absorb outgoing infrared radiation emitted by the Earth's surface and cause warming of the atmosphere. H_2O emissions by gas turbines are, are one hand, less than fluxes within the natural hydrological cycle. However, they also modify the energy balance between the Earth and the atmosphere and contribute to global warming. Besides, water vapour emitted by aero-engines in-flight resides in the atmosphere in the form of concentration trails and enhances cirrus formation. This has a warming effect on the climate.

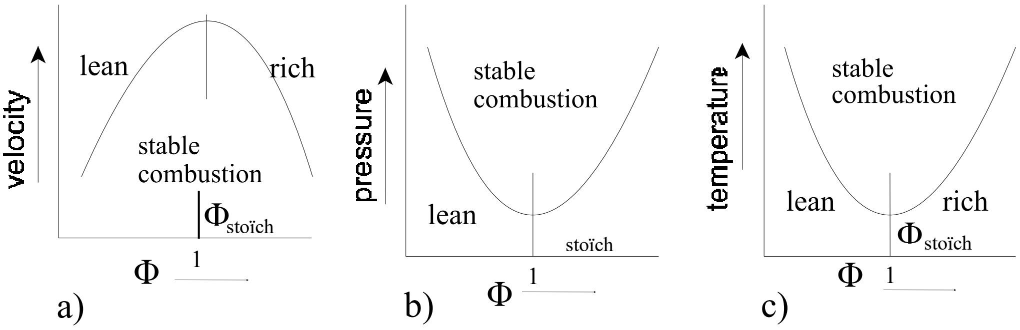


Figure 6-35 Effect of flow velocity, pressure and temperature on combustion stability

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7. Turbomachinery

(Prof. Ir. Jos P. van Buijtenen, Francesco Montella)

Please note that the notation in this chapter is different from the rest of the reader for historical reasons and to achieve analogy to other literature on turbomachinery. The used symbols are summarized below for convenience.

7-1 - Nomenclature

Symbol	**Explanation**	**Unit**
c	Absolute velocity	[m/s]
M	Torque	[Nm]
P	Power	[W]
R	Radius	[m]
u	Peripheral velocity	[m/s]
w	Relative velocity	[m/s]
ω	Rotational velocity	[rad/s]

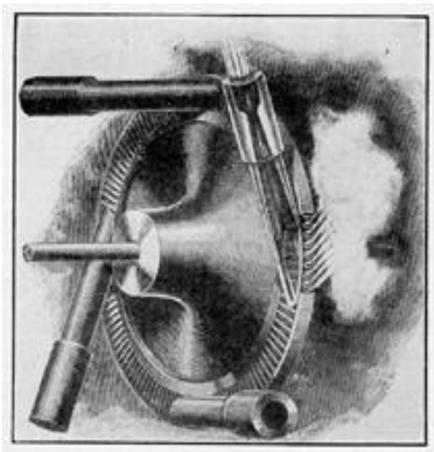
7.1 History

Historically there have been a lot of attempts to convert the energy of a hot flow in mechanical energy as can be seen in the first figure on the right.

Fig. 2. Hero's Reaction Steam Wheel.

In the next figure, the heat of the flame warms up the water that becomes steam. This steam gives rotational speed and torque to the device

The first industrial application is shown here: The Laval's turbine of 1884. High-velocity steam hits the blades, producing work.



7.2 Change of Velocities in a Turbo-Machine

There is a shaft rotating at ω with a certain torque M. There is a rotating body (in this case a compressor) through which the fluid has its motion. The particles go from R1 to R2. Looking at the rotational speed vectors of the in-flow point and out-flow point, it is possible to see that the flow path is a spiral. Actually, there are also axial and radial components of the flow velocity, but we are interested in the tangential component to obtain the torque.

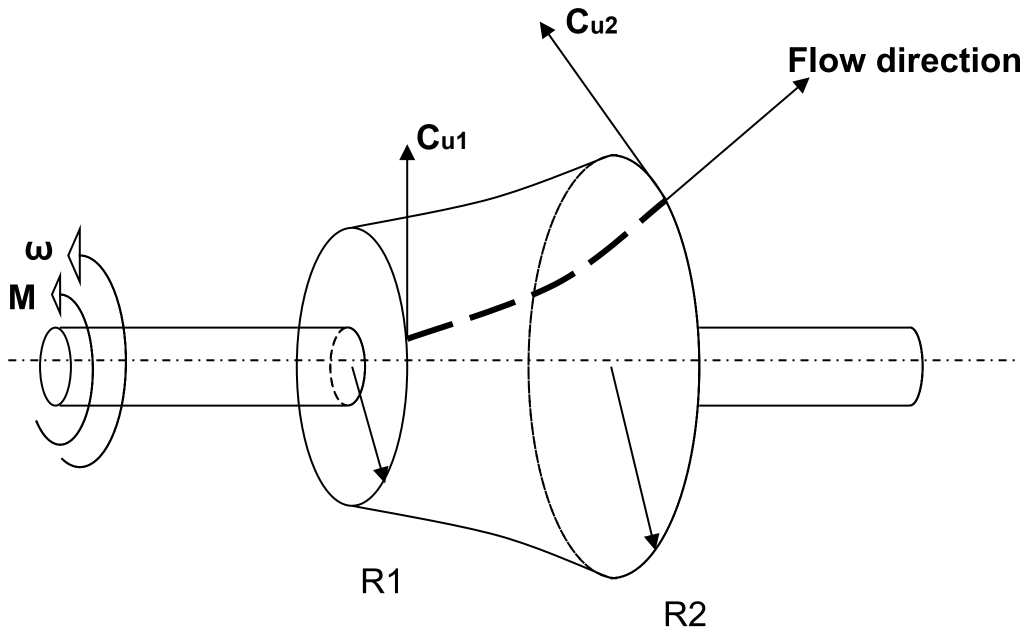


Figure 7-1 - Control volume for a generalized turbomachine

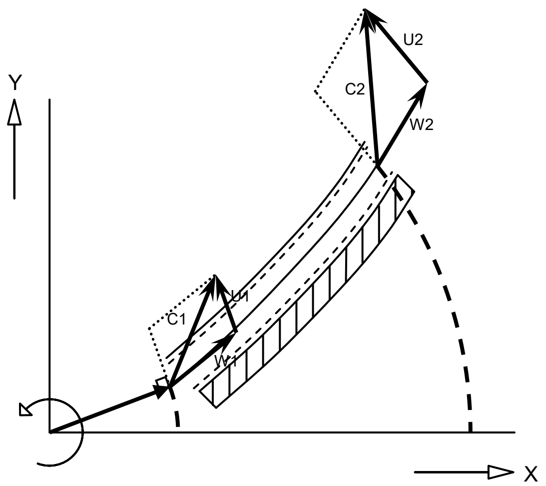


Figure 7-2 - Flow in a compressor

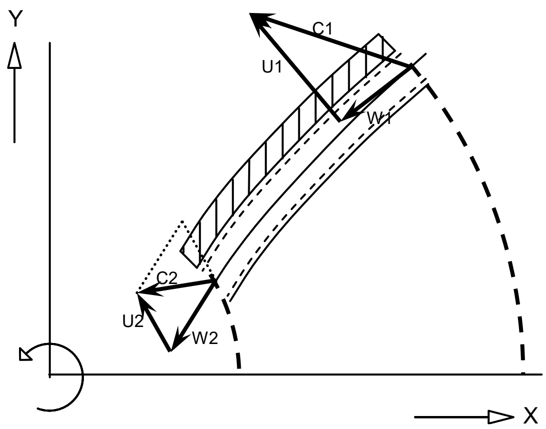


Figure 7-3 - Flow in a turbine

7.3 Euler's Equation for Turbomachinery

Because of the blading arrangements, the flow inside a turbo-machine rotor is both unsteady and asymmetric relative to a control surface fixed in space. However, the unsteadiness is periodic (and of high frequency) so that, on average, we may omit the time contribution of the velocity vector to the torque equation.

We consider now a compressor and a fluid element along its path as shown in Figure 7-4.

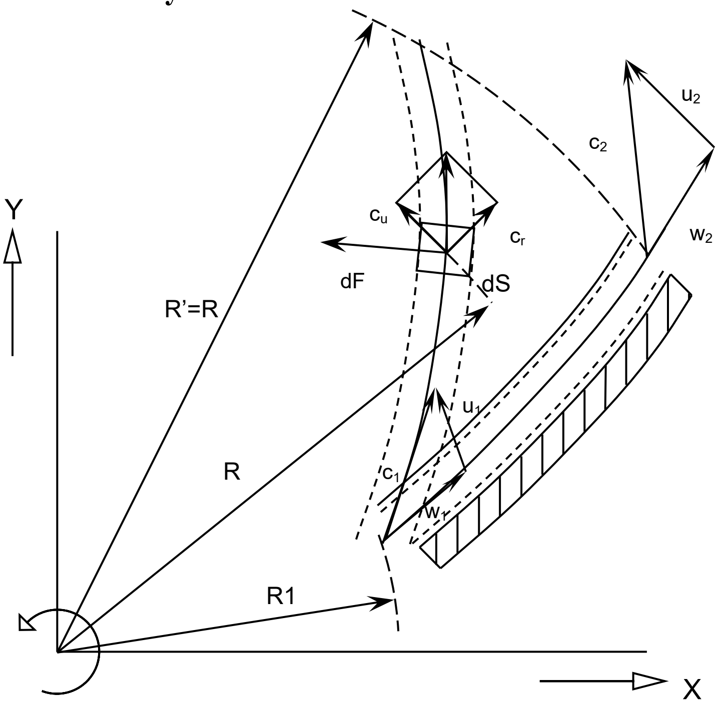


Figure 7-4 - Fluid element along a compressor

For the whole compressor holds:

$$\text{Power} = P = M \cdot \omega \tag{7.1}$$

$$F\ dt = d\left(m\ c\right) \tag{7.2}$$

Where c stands for the absolute velocity of a fluid particle. From this follows

$$F = \frac{d}{dt}\left(m\ c\right) = \dot{m}\ c + m\ \dot{c} \tag{7.3}$$

Assuming $\dot{c} = 0$ it follows

$$F = \dot{m}\,c \tag{7.4}$$

For the fluid element holds:

$$dF = \dot{m}\,dc \tag{7.5}$$

$$dF_r = \dot{m}\,dc_r \tag{7.6}$$

$$dF_u = \dot{m}\,dc_u \tag{7.7}$$

As the radial component gives no contribution to the torque *M*, we obtain:

$$dM = r\,dF_u \tag{7.8}$$

$$dM = r\,\dot{m}\,dc_u \tag{7.9}$$

The torque required for the compression is:

$$M = \dot{m}\int_1^2 r\,dc_u \tag{7.10}$$

From which follows that

$$M = \dot{m}\left(c_{u2}r_2 - c_{u1}r_1\right) \tag{7.11}$$

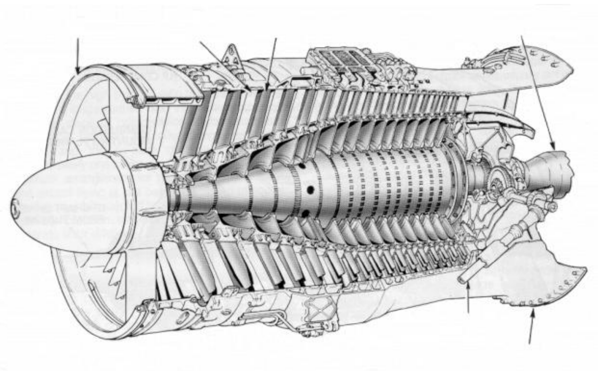
$$P = M\omega = \dot{m}\left(c_{u2}r_2 - c_{u1}r_1\right)\omega \tag{7.12}$$

Using $\omega = u_1/r_1 = u_2/r_2$, we obtain the **Euler's Equation for Turbomachinery:**

$$\boxed{P = \dot{m}\left(c_{u2}u_2 - c_{u1}u_1\right)} \tag{7.13}$$

7.4 The Axial Compressor

In an axial compressor, there are a lot of stages to achieve a high-pressure ratio gradually.



The mechanical power of the rotor is used to increase the kinetic energy of the fluid. Passing through the stator, this kinetic energy is converted into pressure by a diffusion process. The stator (or diffuser) is also used to modify the direction of the fluid and make it as it was before entering the next rotor as can be seen in Figure 7-5.

From the scheme in Figure 7-6 it is possible to see how the distance perpendicular to the flow path between the stator blades increases, allowing in this way the diffusion process.

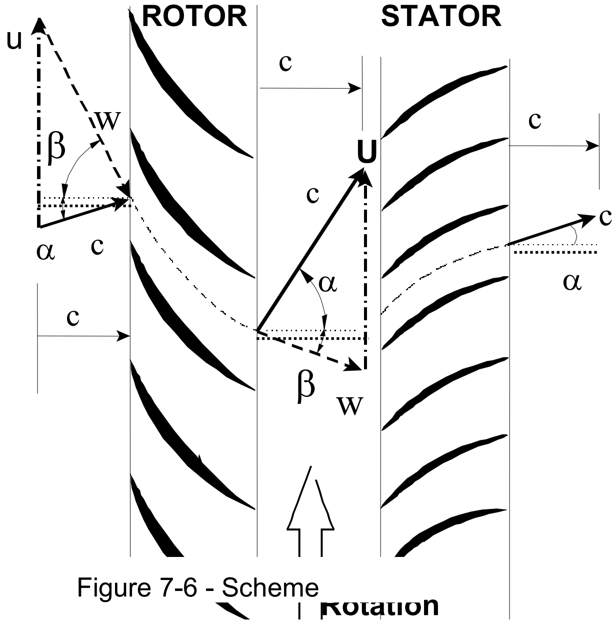
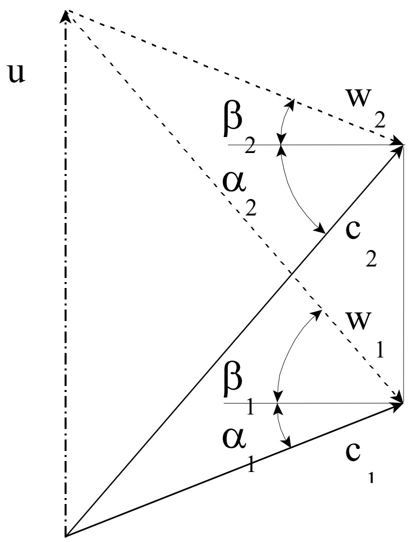


Figure 7-6 - Scheme

Figure 7-5 - Axial compressor



In particular, looking at velocity triangles of the rotor and assuming that the axial velocity has the same value, we obtain:

$$\begin{aligned} &u = c_{u1} + c_{w1} = c_a\left(\tan\alpha_1 + \tan\beta_1\right) = c_{u2} + c_{w2} = c_a\left(\tan\alpha_2 + \tan\beta_2\right) \Rightarrow \\ &\Rightarrow \left(\tan\alpha_1 + \tan\beta_1\right) = \left(\tan\alpha_2 + \tan\beta_2\right) \end{aligned} \tag{7.14}$$

Using the Euler's formula divided by the mass flow and with $u_1 = u_2 = u$ we obtain the specific power:

$$P_s = \dot{W}_S = u\left(c_{u2} - c_{u1}\right) \tag{7.15}$$

Because of

$$c_{u1,2} = c_a \tan\alpha_{1,2} \Rightarrow \dot{W}_s = u \cdot c_a\left(\tan\alpha_2 - \tan\alpha_1\right) \tag{7.16}$$

and

$$\left(\tan\alpha_1 + \tan\beta_1\right) = \left(\tan\alpha_2 + \tan\beta_2\right) \tag{7.17}$$

It follows that (7.18)

$$P_s = \dot{W}_s = u \cdot c_a\left(\tan\beta_1 - \tan\beta_2\right)$$

All the considerations that have been made referring to an imaginary plane of the blade. The velocity triangles have been drawn in a tangential plane at the mean radius. This simplified analysis is reasonable for the later stages where the blade speeds at root and tip are similar. At the front end of the compressor, however, the blades are much longer, there are marked variations in blade speed from root to tip, and it becomes essential to consider three-dimensional effects in analyzing the flow. Moreover, in the real condition, the axial velocity changes between the root and tip of the blades because of the presence of the boundary layer along the annulus walls. This effect becomes more evident in the later stages because the boundary layer thickens as the flow progresses and the area available for the flow is reduced below the geometric area, as it is shown in Figure 7-7:

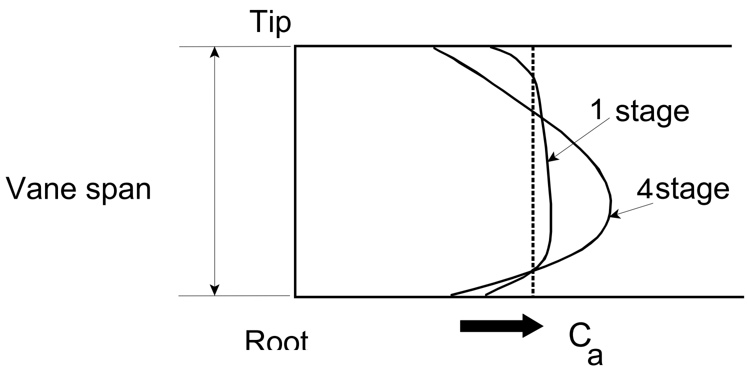


Figure 7-7 - Boundary layers on different blades

A so-called "Work-Done" factor is introduced to account for the reduction in work capacity caused by the change in axial velocity. Integrating the specific power along with the blade height and using the "Work-Done" factor λ, we obtain:

$$P = \dot{W} = \lambda\, \dot{m}\, W_s \tag{7.19}$$

This input energy will be absorbed usefully in raising the pressure of the air and wastefully in overcoming various frictional losses. But regardless of the losses, or in other words of the efficiency of compression, the whole of this input will reveal itself as a rise in stagnation temperature of the air.

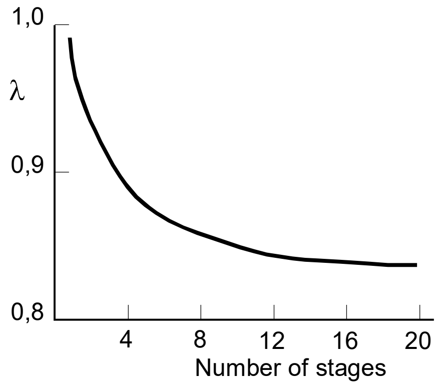


Figure 7-8 - The "work-done" factor

The stagnation temperature rise in one stage is given by

$$\Delta T_{0,s} = T_{0,3} - T_{0,1} = T_{0,2} - T_{0,1} = \frac{W_s}{c_p} = \frac{uc_a}{c_p}\left(\tan\alpha_2 - \tan\alpha_1\right) \tag{7.20}$$

The pressure rise obtained will be strongly dependent on the efficiency of the compression process. Denoting the isentropic efficiency of the stage by $\eta_{is,stage}$, the stage pressure ratio is then given by

$$\frac{p_{0,3}}{p_{0,1}} = \left(1 + \eta_{is,stage}\frac{\Delta T_s}{T_{0,1}}\right)^{\frac{k}{k-1}} = \left(1 + \frac{\eta_{is,stage}\lambda uc_a\left(\tan\alpha_2 - \tan\alpha_1\right)}{c_p T_{0,1}}\right)^{\frac{k}{k-1}} \tag{7.21}$$

We obtain a relation between pressure ratio and mechanical parameters: We would like to have high u to achieve high pressure ratios, but the centrifugal stresses do not allow too many high rotational velocities; we would like to have high c_a, but flow separation and losses problems limit the axial velocity; we would like to have high $\left(\tan\alpha_2 - \tan\alpha_1\right)$, but there is a limit to the blade curvature caused by flow separation problems.

In Figure 7-9 a sketch of the typical stage is shown together with the h-s diagram, in which it is possible to see total and static conditions. All the power is absorbed in the rotor, and the stator merely transforms kinetic energy to an increase in static pressure with the stagnation temperature remaining constant. The increase in stagnation pressure is accomplished wholly within the rotor and, in practice, there will be some decrease in stagnation pressure in the stator due to fluid friction. Losses will also occur in the rotor and the stagnation pressure rise will be less than would be obtained with an isentropic compression and the same power input.

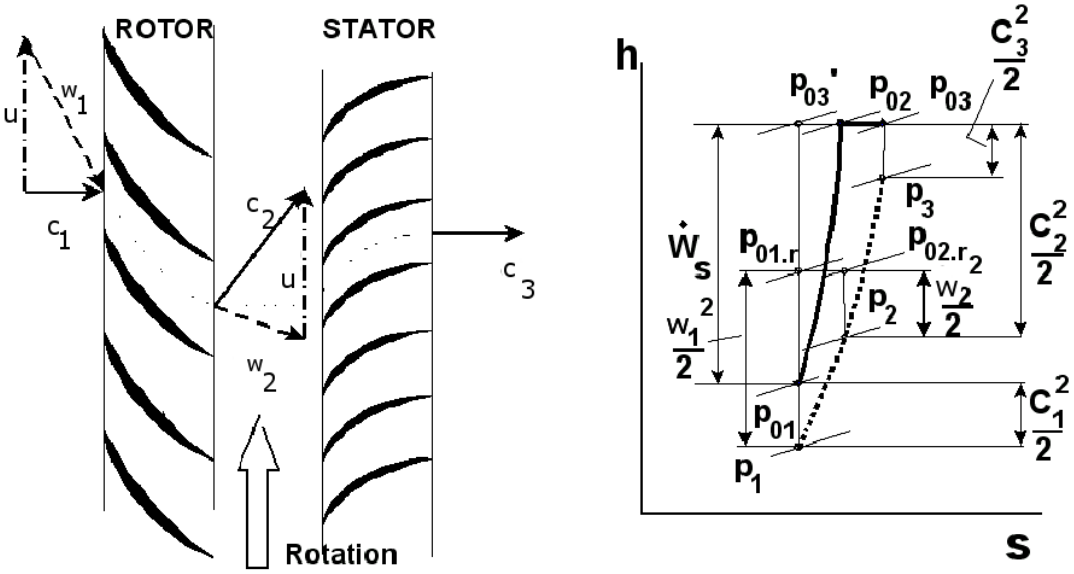


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Figure 7-9 – A typical compressor stage and the accompanying h-s diagram

As diffusion takes place in both rotor and stator and there will be an increase in static pressure, the *degree of reaction* Λ provides a measure of the extent to which the rotor contributes to the overall static pressure rise in the stage. It is normally defined as

$$\Lambda = \frac{\Delta T_{rotor}}{\Delta T_{rotor} + \Delta T_{stator}} \tag{7.22}$$

and for an axial compressor

$$\Lambda = \frac{C_a}{2U}\left(\tan\beta_2 + \tan\beta_1\right) = \frac{C_a}{2U}\left(\tan\alpha_1 + \tan\alpha_2\right) \tag{7.23}$$

The higher is the degree of reaction, the more is the amount of energy increased in the rotor and so the higher is the temperature rise in that part of the compressor.

Different values of the degree of reaction are responsible for a different shape of the vanes, due to the velocity triangles that are different, as it is shown in Figure 7-10.

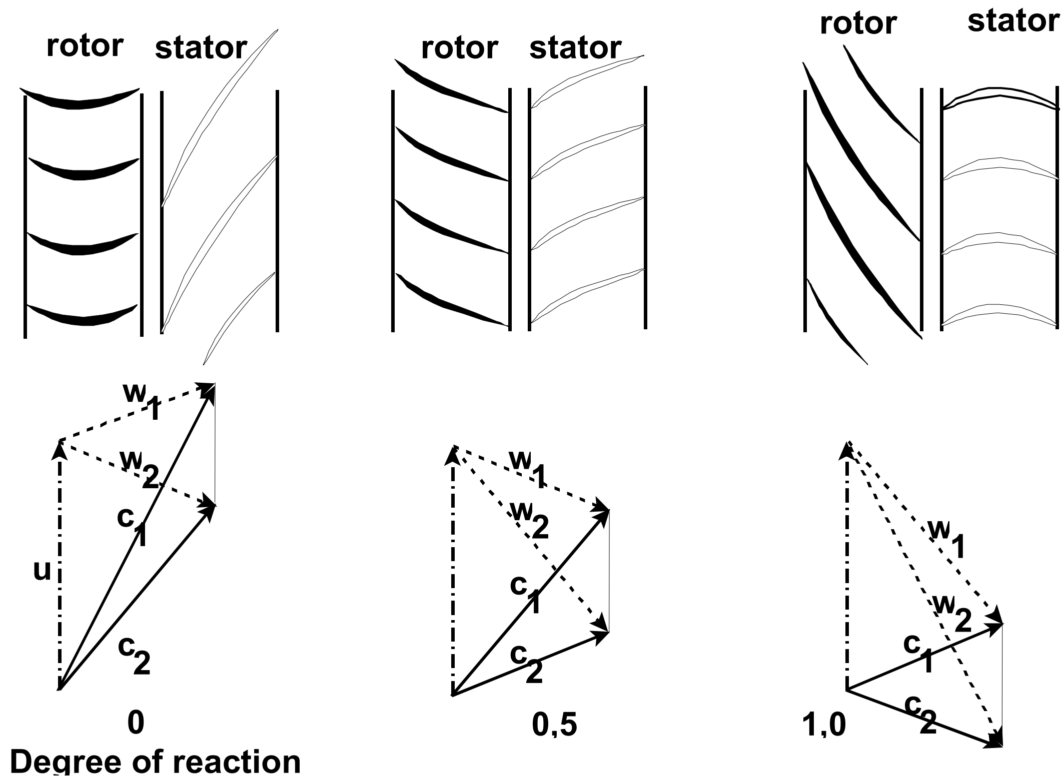


Figure 7-10 - Degree of reaction

The elementary theory that has been presented in this chapter is two-dimensional, meaning that any effect due to the radial movement of the fluid is ignored. This assumption is quite reasonable for stages in which the value of the hub-tip ratio is greater than about 0.8, which would be typical of the later stages of a compressor. The front stages of aero-engines, however, have values lower than 0.4 to cope with the high mass flow. In this case, the annulus will have a substantial trapper, and this will give a radial component of velocity. Moreover, because the flow has a whirl component, the pressure must increase with radius to provide forces associated with the centripetal acceleration of the fluid.

To take into account these effects, a radial equilibrium of the fluid element can be written

$$\frac{1}{\rho}\frac{dp}{dr} = \frac{c_w^2}{r} \tag{7.24}$$

Using the thermodynamic relation $Tds = dh - dp/\rho$ and the stagnation enthalpy definition

$$h_0 = h + \frac{c^2}{2} = h + \frac{1}{2}\left(c_a^2 + c_u^2\right), \tag{7.25}$$

the final form of the equation gives the **Vortex Energy Equation** (if the entropy gradient term is ignored):

$$\boxed{\frac{dh_0}{dr} = c_a\frac{dc_a}{dr} + c_u\frac{dc_u}{dr} + \frac{c_u^2}{r}} \tag{7.26}$$

Assuming that the enthalpy doesn't change along the blade and the axial velocity is constant, we obtain the **Free Vortex Condition**:

$$C_u\, r = const \tag{7.27}$$

This condition has been used to design compressors, because it gives information about the blade parameters along the radius that are useful for the blade twisting, like attack angles and velocity triangles, as is shown in Figure 7-11.

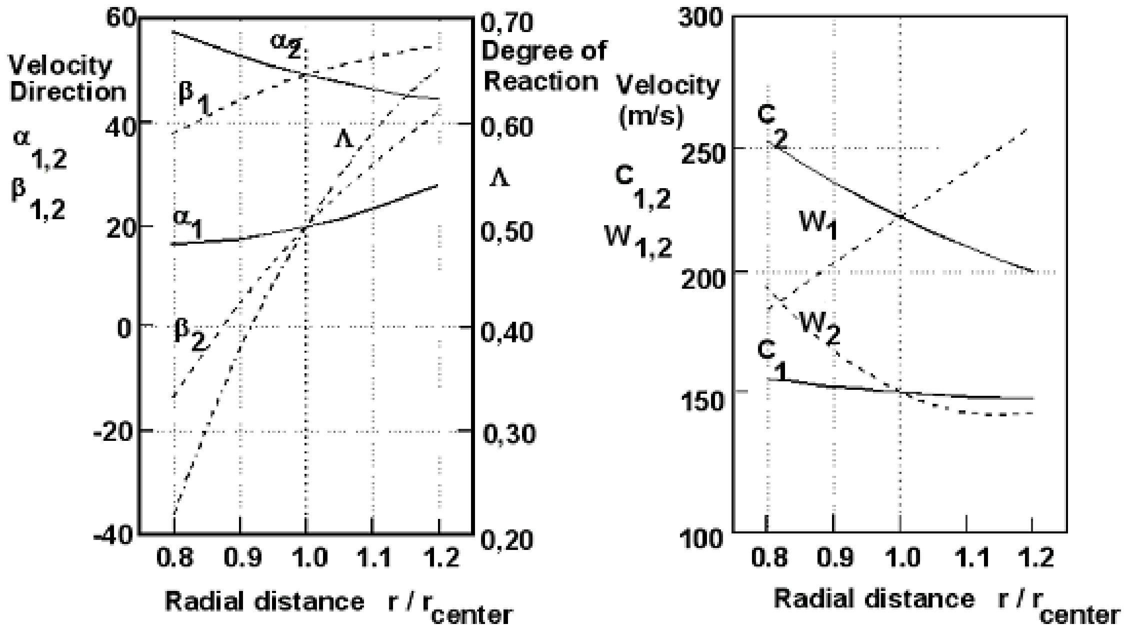


Figure 7-11 – Blade design parameters

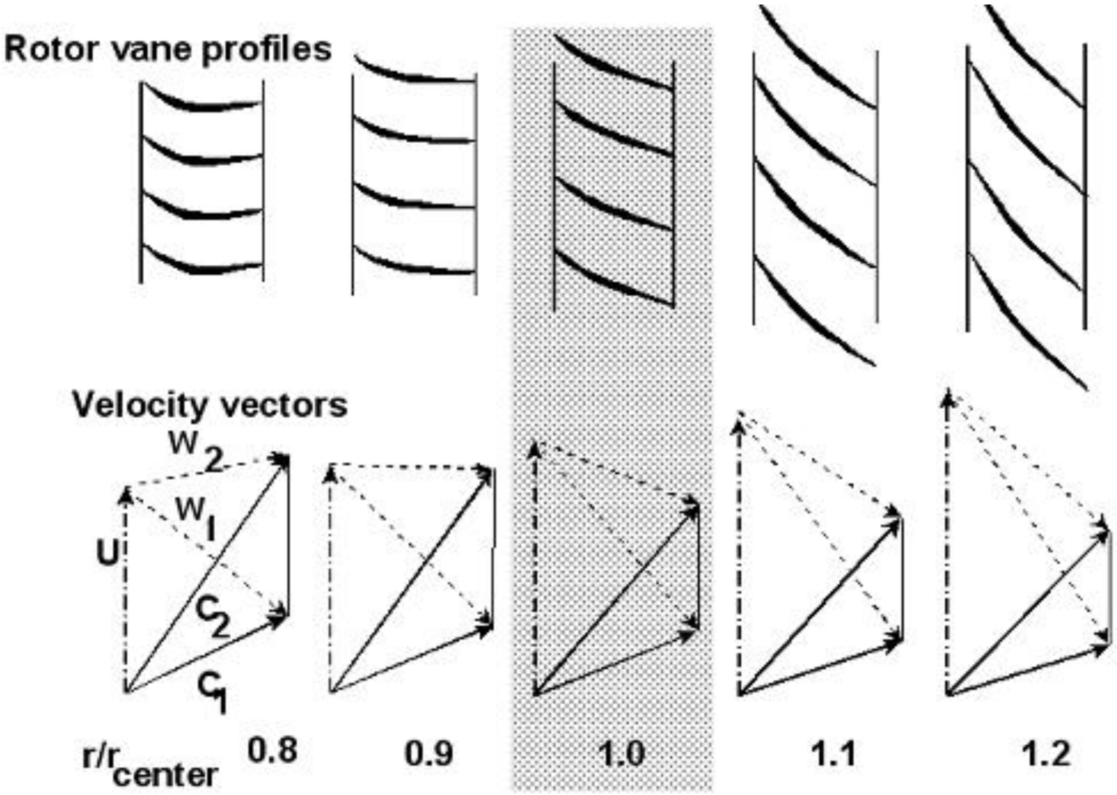


Figure 7-12 - Rotor vane profiles

7.5 The Radial Compressor

For large turbojet or turbofan engines, the axial compressor has the great advantage of a smaller cross-sectional area per unit of airflow rate. The axial compressor may be said amenable for multistage and thus to the large overall pressure ratios typically needed for large engines. For "small" gas turbines used to drive propellers or helicopter rotors, for example, the overall diameter may be no such an important consideration as for large engines. The mechanical power of the rotor is used to increase the kinetic energy of the fluid. The velocity of the air leaving the rotor or impeller has no axial component and has different attack angles to the tip and the root of the impeller, due to the different rotational velocities.

A "Slip Factor" can be introduced to take into account the real behaviour of the fluid at the outlet of the rotor.

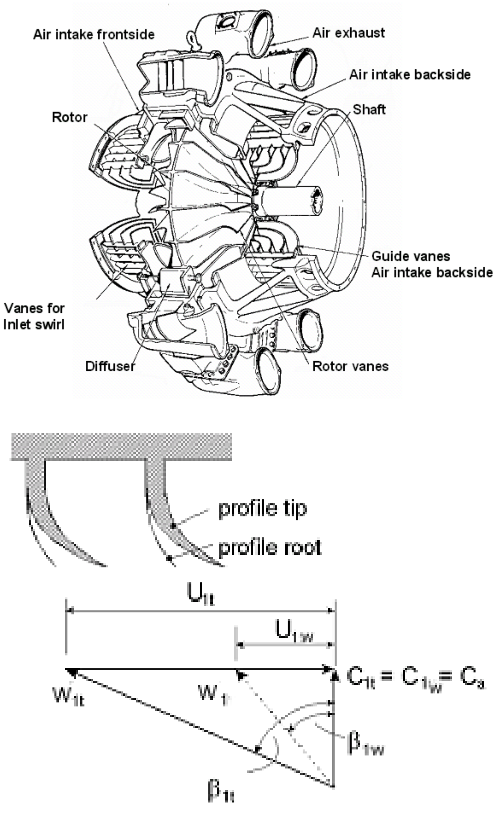


Figure 7-13 - Caption

After leaving the impeller, the air passes through a radial diffuser, consisting of vaned flow passages in which momentum is exchanged for pressure.

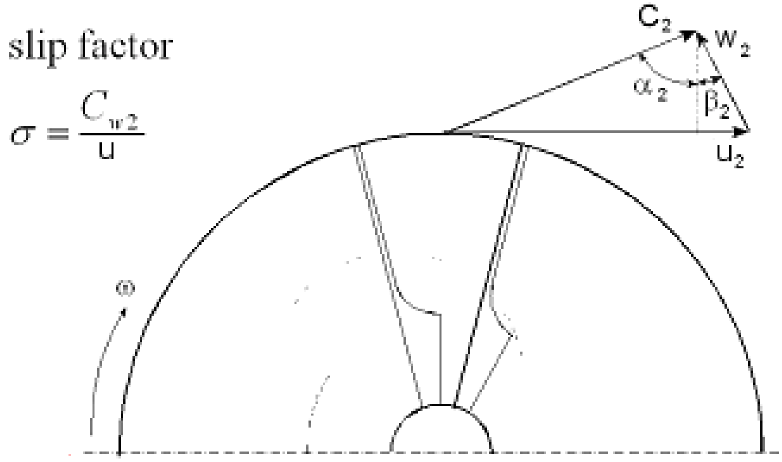


Figure 7-14 - Slip factor

7.6 The Axial Turbine

An axial turbine stage consists of a row of stationary blades, called nozzles or stators, followed by the rotor, as is shown in Figure 7-15.

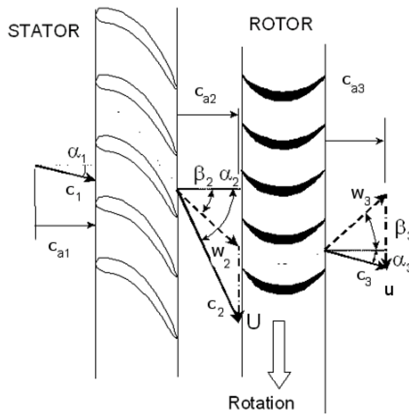


Figure 7-15 - Axial turbine stage

The nozzles accelerate the flow, imparting an increased tangential velocity component. The velocity diagram of the turbine differs from that of the compressor in that the change in tangential velocity in the rotor is in the direction opposite to the blade speed U. The reaction to this change in the tangential momentum of the fluid is a torque on the rotor in the direction of the motion. Hence the fluid does work on the rotor.

The "Degree of Reaction" may be defined for a turbine as the fraction of the overall enthalpy drop occurring in the rotor. Turbine stages in which the entire pressure drop occurs in the nozzle are called "impulse stages". Stages in which a portion of the pressure drop occurs in the nozzle and the rest in the rotor are called "reaction stages". An impulse turbine would therefore be a zero reaction machine because the energy comes only from the changing of the velocity direction in the rotor. In a 50% reaction machine, the enthalpy drop in the rotor would be half of the total for the stage and the energy comes from the changing of both the velocity module and direction.

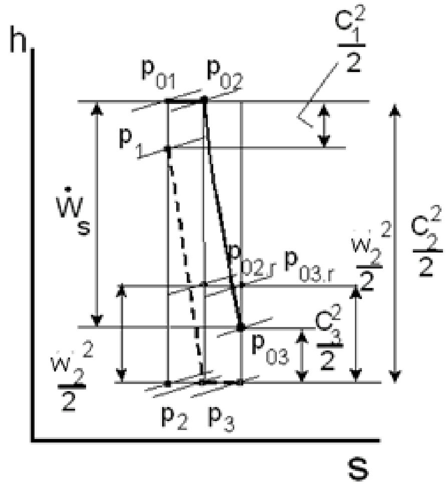
A low degree of reaction is also applied to obtain a low temperature of the gases on the rotor blades because a great part of the expansion process takes place in the nozzle vanes.

7.7 Characteristic Performance of a Compressor

For any gas compressor we could express the dependence of the stagnation pressure at the compressor outlet total pressure p_{02} and adiabatic efficiency η_{comb} on the other important physical variables, in the following form:

$$\Delta h_{is}, P = f\left(\dot{m}, \Omega, k, R, \nu, \rho, design, D, a\right) \tag{7.28}$$

The mass flow rate is denoted by $\dot{m}$. The symbol Ω that denotes the rotational speed of the shaft. The gas properties significant to the compression process are specified by stating the kinematic viscosity ν, the density ρ, the specific heat ratio *k*, and the gas constant R that appears in the perfect gas law. *Design* is the complete specification of the geometric shape of the machine and D is the characteristic size. *A* is the speed of sound. Δh_{is} is the variation of enthalpy and *P* is the output of the compressor.



Any attempt to allow for full variations of all these quantities over the working range would involve an excessive number of experiments and make a concise presentation of the results impossible. Much of this complication may be eliminated by using the technique of dimensional analysis, by which the variables involved may be combined to form a smaller and more manageable number of dimensionless groups.

At the first step, we check that all the variables on the right-hand side of the relation are independent, which means that each variable in the set can be varied without necessarily altering the value of any other member of the set.

At the second step, we have to reduce the statement of the physical dependence to one in which each (dimensionless) dependent variable is seen to be a function of a low number of independent dimensionless variables ("low" compared with the original number of independent dimensional variables).

$p_{0,1}, T_{0,1}, \rho_{0,1}$ denote the inlet stagnation pressure, temperature and density. $p_{0,2}, T_{0,2}, \rho_{0,2}$ are the outlet values.

$$\Delta h_{is} = c_p\left(T_2 - T_1\right) = c_p T_1\left[\left(\frac{p_2}{p_1}\right)^{\frac{k-1}{k}} - 1\right] \tag{7.29}$$

$$a_{0,1} = \sqrt{kRT_{0,1}} \tag{7.30}$$

As the variables k and *design* are already dimensionless, we look for other dimensionless variables:

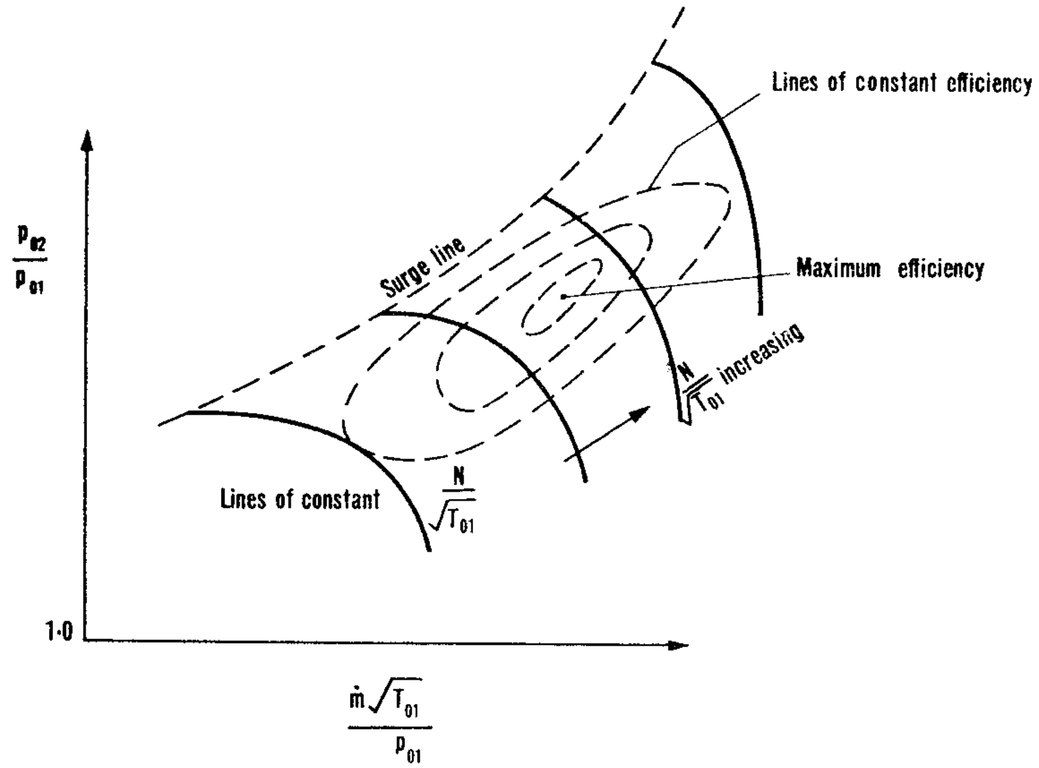
$$\frac{\Delta h_{is}}{a_{0,1}^2}, \frac{P}{\rho_{0,1}\Omega^3 D^5} = f\left(\frac{\dot{m}}{\rho_{0,1}a_{0,1}D^2}, \frac{\rho_{0,1}\Omega D^2}{\nu}, \frac{\Omega D}{a_{0,1}}, k\right)$$

$$\frac{p_2}{p_1}, \frac{\Delta T}{T} = f\left(\frac{\dot{m}\sqrt{RT_{0,1}}}{D^2 p_{0,1}}, \frac{\Omega D}{\sqrt{RT_{0,1}}}, \mathrm{Re}, k\right)$$

Considering that D = const, k = const, R = const and $Re \sim$ const, we find

$$\frac{p_2}{p_1}, \frac{\Delta T}{T} = f\left(\frac{\dot{m}\sqrt{RT_{0,1}}}{D^2 p_{0,1}}, \frac{\Omega D}{\sqrt{RT_{0,1}}}\right)$$

These relations are shown in the following diagram, remembering that the isentropic efficiency is related to the real difference of temperature in the process, ΔT



8. **Performance Characteristics**

(Ir. Wilfried P.J. Visser)

8.1 Component characteristics

8.1.1 Dimensionless parameter groups

The relation between pressure ratio, mass flow, shaft speed and efficiency of turbomachinery components (compressor and turbine) can be captured in a characteristic. Figure 8.1 shows an example characteristic of an axial compressor.

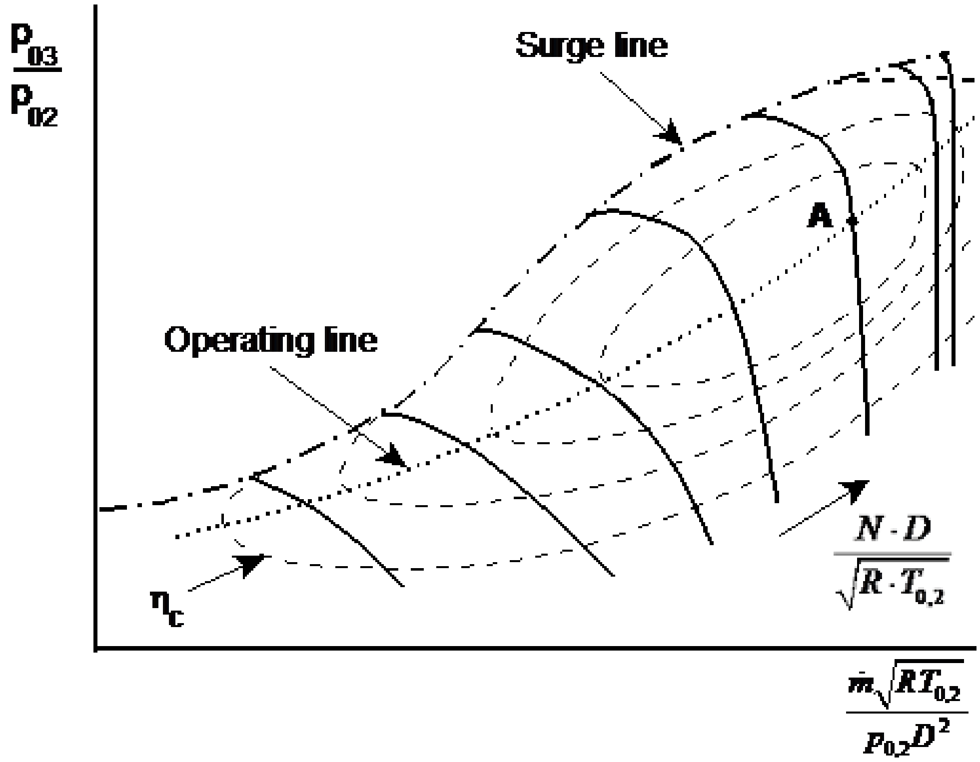


Figure 8-1 - Axial compressor characteristic

The compressor characteristic shows the most important quantities that influence the performance of the compressor. These quantities are grouped in dimensionless parameter groups:

$$
\begin{aligned}
a)&\quad \frac{\dot{m}\sqrt{RT_{0,2}}}{p_{0,2}D^2} &&\text{; mass flow parameter group}\\
b)&\quad \frac{p_{0,3}}{p_{0,2}} &&\text{; the pressure ratio}\\
c)&\quad \frac{N\,D}{\sqrt{RT_{0,2}}} &&\text{; shaft speed parameter group}\\
d)&\quad \eta &&\text{; efficiency (isentropic or polytropic)}
\end{aligned}
\tag{8.1}
$$

These parameter groups are the result of a dimension analysis on the parameters $T_{0,2}$, $T_{0,3}$, $p_{0,2}$, $p_{0,3}$, R, N and D. These eight parameters consist of four basic units: mass, length, time and temperature. The dimension analysis leads to four parameter groups, a till c and the temperature ratio, $T_{0,3}/T_{0,2}$. The temperature ratio, $T_{0,3}/T_{0,2}$, and the pressure ratio, $p_{0,3}/p_{0,2}$ (which is parameter group c) determine the efficiency η, meaning that parameter group d ($T_{0,3}/T_{0,2}$) is an alternative for the efficiency. The advantage of the dimensional analysis is that the number of parameters that describe the characteristics can now be reduced to four parameters. Presenting characteristics in this way now enables displaying geometrically identical compressors in one single characteristics map. The characteristic can be used for any balanced unity system, provided that it is used consequently.

The characteristic can now be used to solve the following questions:

- Suppose the compressor has an operating point in point A of Figure 8-1. Now consider a similar compressor having the same efficiency value and pressure ratio, but requires compressing twice as much air. How much bigger does the engine will have to be to fulfil the larger airflow? The answer can directly be read from the compressor characteristic. Since the pressure ratio and efficiency remain the same, the position of the operating point in the characteristic will not change. The values for the mass flow parameter group and the shaft speed parameter group remain the same. A mass flow twice as many as the original compressor requires the diameter to increase with $\sqrt{2}$ (see equation 8.1 a)) and the shaft speed to be multiplied by a factor $1/\sqrt{2}$ (see equation 8.1 c)).
- What will be the effect on mass flow and shaft speed if a rise in temperature of $T_{0,2}$ is experienced for an operating point remaining its current position? From the characteristic follows that the mass flow reduces with $\sqrt{T_{0,2}}$ (see equation 8.1 a)) and the shaft speed increases with $\sqrt{T_{0,2}}$ (see equation 8.1 c)).

The shape of the speed- and efficiency lines in the characteristic is dependent on the compressor type (axial, radial, high/low pressure ratio per stage and the number of stages).
The surge- and operating line in Figure 8-1 will be explained further in the next sections.

Derivatives from the dimensionless parameter group are commonly used. Those derivatives frequently exclude the universal gas constant, because the fluid entering the gas turbine (usually air) is fixed. The characteristic diameter, D, sometimes is excluded as well, which can be justified if the same compressor types are being compared. This results in quasidimensionless parameter groups as e.g.:

$$a) \quad \frac{\dot{m}\sqrt{T_{0,2}}}{p_{0,2}} \quad \text{; mass flow parameter group}$$
$$b) \quad \frac{N}{\sqrt{T_{0,2}}} \quad \text{; shaft speed parameter group} \qquad (8.2)$$

Note that these parameter groups are **not** dimensionless!
The component entry pressure and temperature can be normalised using reference values for pressure and temperature from the ISA standard sea-level static conditions.

$$\delta = \frac{p_{inlet}}{101.325\,kPa} \quad \text{; delta}$$
$$\theta = \frac{T_{inlet}}{288.15K} \quad \text{; theta} \qquad (8.3)$$

Filling in the delta and theta in the dimensionless parameter groups referred or corrected parameter groups are obtained that are proportional to the quasidimensionless parameter groups:

$$a) \quad \frac{\dot{m}\sqrt{\theta}}{\delta} \quad \text{; referred mass flow parameter group}$$
$$b) \quad \frac{N}{\sqrt{\theta}} \quad \text{; referred shaft speed parameter group} \qquad (8.4)$$

An advantage of these referred parameter groups is that the dimensions of the groups are respectively [kg/s] and [rev/min.]
Another way of correcting the parameter groups is to normalise them to the design point. An advantage of this representation is that the shape of different compressor characteristics can be compared to display differences.

A similar analysis for turbines can be considered. The following dimensionless parameters are obtained:

$$a) \quad \frac{\dot{m}\sqrt{RT_{0,4}}}{p_{0,4}D^2} \quad \text{; mass flow parameter group}$$
$$b) \quad \frac{p_{0,4}}{p_{0,5}} \quad \text{; the pressure ratio}$$
$$c) \quad \frac{N\,D}{\sqrt{RT_{0,4}}} \quad \text{; shaft speed parameter group} \qquad (8.5)$$
$$d) \quad \eta \quad \text{; efficiency (isentropic or polytropic)}$$

Figure 8-2 shows an example characteristic of a turbine.

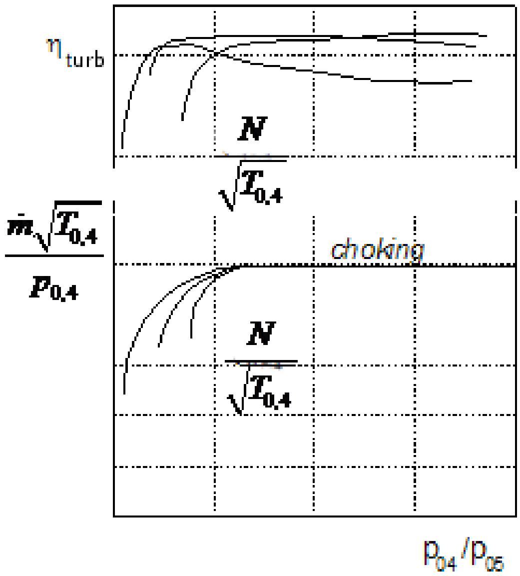


Figure 8-2 - Axial turbine characteristic

8.1.2 Operational Limits

8.1.2.1 Stall

Airflow separation occurs when the angle of incidence (*i,* see Figure 8-3) of the inflow in blades becomes too large for the airflow to follow the blade profile. The magnitude of change in angle of incidence is highly dependent on the direction the incidence is changing. Typical values are about 5° to the positive direction and about 15° to the negative direction (see Figure 8-3 for the positive and negative definition). The rather low value for the positive direction is caused by the ease of separation of the flow at the convex side of the blade in combination with the lower pressure. Separation of flow at the convex part of the blade is referred to as "**stall**".

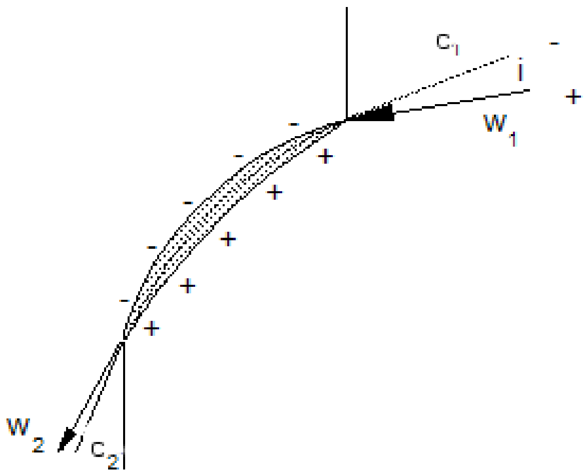


Figure 8-3 - Blade profile topology

When a disturbance in the flow or geometry is observed a breakdown of flow in e.g. channel B (see Figure 8-4) results in deflection of the inflow incidence at channels A and C. Channel A receives the fluid at a reduced angle of incidence (positive direction), while channel C receives the fluid at an enlarged angle of incidence (negative direction). The reduced angle of inflow at channel A will result in a breakdown of flow in that channel. The flow to channel B will now be more favourable due to the deflection around the blockage in channel A. It appears that the blockage ("stall") has moved opposite to the direction of rotation. This phenomenon is called "rotating stall".

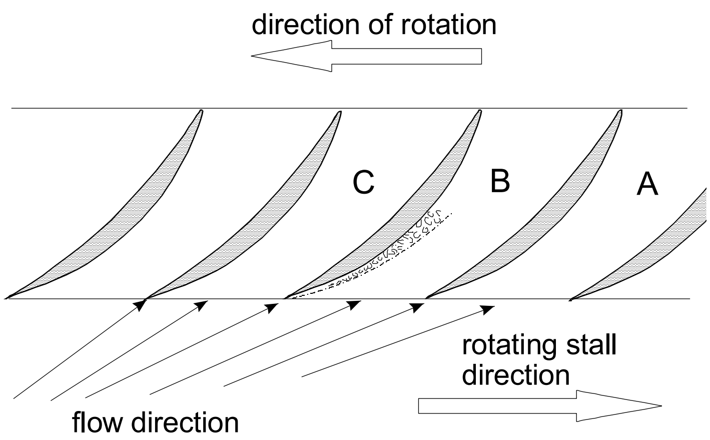


Figure 8-4 - Rotating stall

"Rotating stall" appearing at the blade tips (since the speeds are at maximum), often is referred to as "part span stall". In case the flow worsens the stall areas will expand to grow to the length of the blade, which is called "full span stall" (see Figure 8-5).

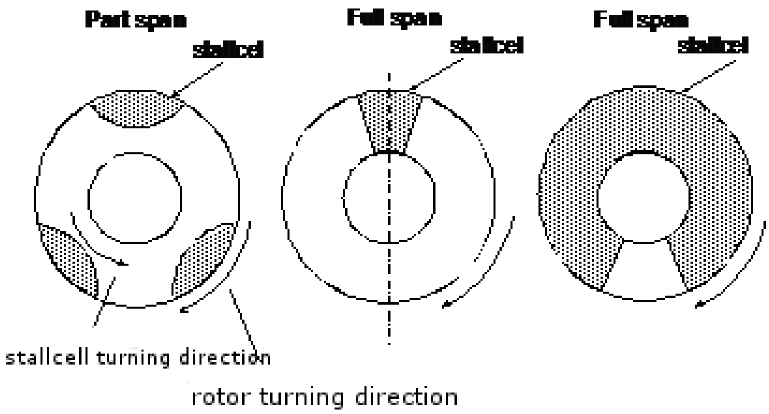


Figure 8-5 - Rotating stall cells

To get an impression in the operating conditions at which “stall” is likely to occur the following cases are discussed:

Case I

Assume a gas turbine running at a certain fixed point **A** in the upper part of the compressor characteristic of Figure 8-6. If the operating point changes along the constant speed line to point **B**, e.g. as a result of acceleration, the pressure ratio increases and the mass flow decreases, resulting in a decrease of axial velocity c_a. A fair assumption is that the angle of the air leaving the stator vane remains constant. This assumption is valid because the stator channels guide the airflow entering with different angles of incidence to an outflow angle (nearly) equal to the exit angle of the stator vane. This causes a change of the velocity triangles as can be seen in Figure 8-7(a). In Figure 8-7(a), c, denotes the absolute velocity of the entering flow, c_a, the axial velocity, u, the rotational speed and w the relative velocity. Figure 8-7(a) clearly shows that the angle of incidence of the relative velocity with a shift of operating point **A** to point **B** increases, thus increasing the possibility of stalling. The stalling effect will be first noticeable at the last stages of the compressor because the change in pressure has more effect on the axial velocity in the last stages than on the first stages.

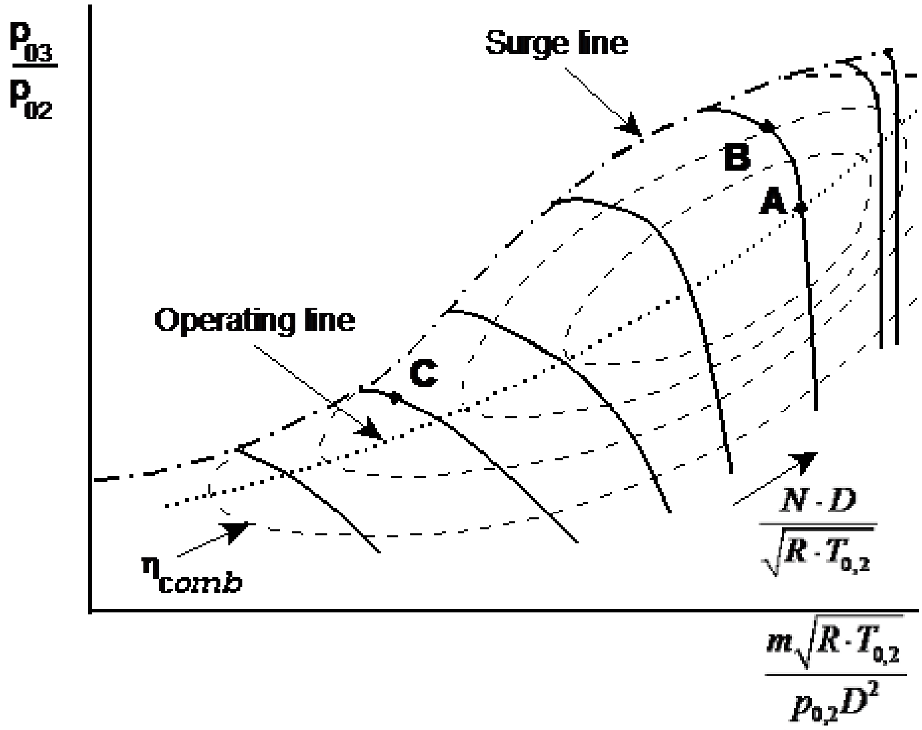


Figure 8-6 - Axial compressor characteristic operating point shift

Case II

Assume that the gas turbine is running at a certain fixed point **C** in the lower part of the compressor characteristic of Figure 8-6, which is a typical operating point during start-up. This point is characterised by a lower mass flow and a lower spool speed and will be compared to operating point **A**. If the figure contained numerical values for the parameter groups, the figure would show that the mass flow decreased much more rapidly than the spool speed decreased. The velocity triangles will undergo a similar change as described in "Case I", except for the last compressor stages. Due to the low pressure (and therefore low density), the axial velocity remains the same or could even increase! The increase in axial velocity causes the angle of inflow incidence to shift in the negative direction (see Figure 8-7(b)).

It seems that the possibility of stalling the compressor increases the further the operating points shift to the left in the characteristic. For high spool speeds, the stall can be expected to occur in the last stages, while for lower spool speeds the stall will occur in the first stages. The phenomenon "stall" is important for the understanding of an associated effect called "surge".

8.1.2.2 Surge

To explain "surge", assume a compressor running at a fixed spool speed having an adjustable valve at the exit channel. By adjusting the position of the valve, the mass flow through and the pressure ratio over the compressor can be controlled. Plotting the pressure over the compressor as a function of the mass flow through the compressor by slowly adjusting (opening) the valve positions a curve similar to the "curve A" in Figure 8-8 will be obtained. The characteristic of the valve itself can be plotted in this figure as well. Note that for three valve positions the

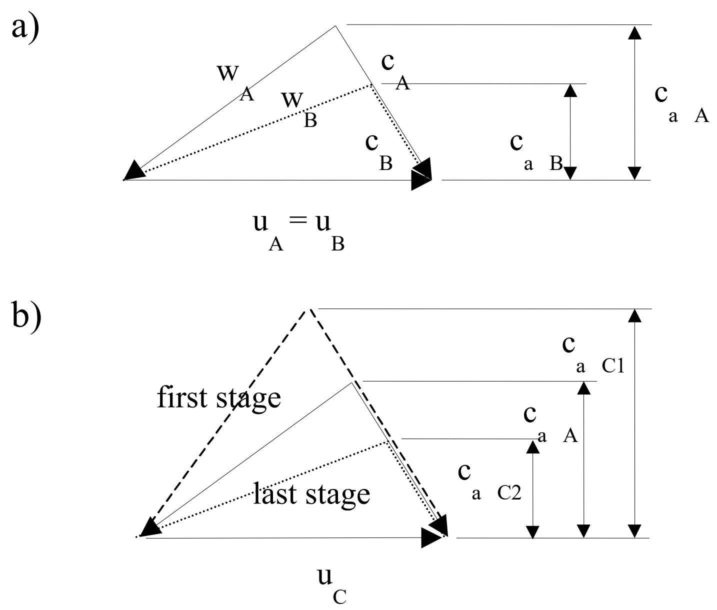


Figure 8-7 - Effect of operating point shift for velocity profiles
a) shift A => B
b) shift A => C

characteristics are drawn in the figure (vp_1, vp_2 and vp_3,). For a closed valve, there will be no mass flow through the compressor, but still, it would have a pressure ratio due to the pumping effect of the rotor of the trapped air in the compressor (note that the stator does not contribute to this pressure ratio since the mass flow through the compressor is zero). When the valve is opened slowly, the pressure ratio will increase at first since the stator vanes contribute to the pressure ratio and inflow to the blades will be better (more efficient since the inflow angles tend to shift from the positive region to zero).

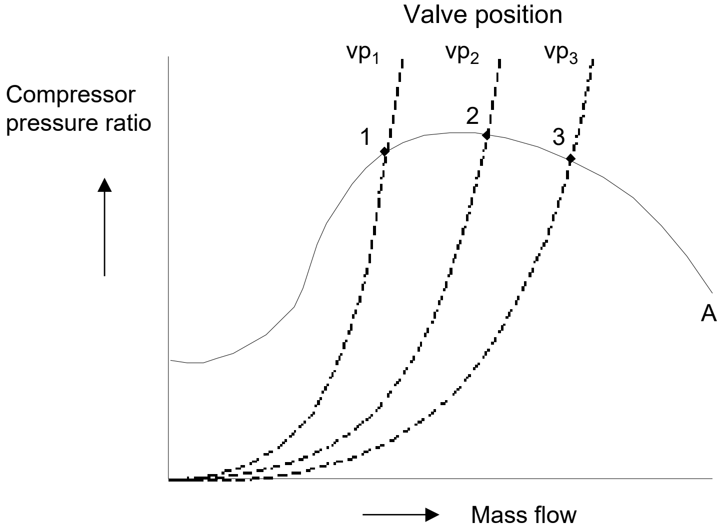


Figure 8-8 - Fixed spool speed and valve controlled compressor characteristic

From a certain valve position, the pressure ratio will decrease because the efficiency decreases as a result of an increasing (more negative) inflow angle of incidence. The points in the figure denote the equilibrium points for valve positions vp_1 through vp_3. The curve enclosed by points 2 and 3 is considered to represent the stable operation of the compressor in contrast to the "unstable" curve up to point 1. The curve before point 1 is unstable because a disturbance causing the mass flow to decrease results in a pressure drop. If the downstream air pressure does not fall quickly enough, the air will tend to flow in the direction of the resulting pressure gradient (to the lower pressure part of the compressor), causing reverse flow in the compressor. If this occurs, the pressure will drop suddenly. In the meantime, the pressure downstream of the compressor also drops, facilitating the compressor to pick up again to repeat the whole cycle. The whole cycle of events taking place at high frequency is called "surge".

The mechanism of the surge phenomenon is very difficult to describe precisely. The difference with the stall phenomenon is that the stall is a local disturbance of airflow with local flow reversal, while the surge is characterised by a flow reversal of the entire compressor. During the surge, the complete flow system participates in the vibration, in which the volume (like the mass in a spring-mass system) and the duct lengths (the springs) are parts of this system. The relation between stall and surge is such that stall can excite the compressor to reach the surge state.

Compressor characteristics as depicted in Figure 8-1 and Figure 8-6 are limited on the left side by the surgeline. By definition, the surge line does not run through the local maximum of the speed lines, as could be expected from the previous example of a compressor with a valve in the exit, but is located left of the speed line maximums. The position of the surge line is highly dependent on the highly dynamic character of the surge phenomenon and is not only compressor dependent but also dependents on the whole engine system.

The surge in compressors has to be avoided because surge causes periodically oscillating forces of high magnitude, which can set individual blades into vibration. The vibration of the blades will ultimately result in a fracture (fatigue). Since the flow is reverted continuously during the surge, compressor air of the end of the compressor gets to the front (which is then compressed again) causing the compressor to warm up. The warming up also contributes to the fracture in the form of thermal fatigue. There are several possible solutions to prevent surge:

1. Blow-off bleed air;
   Typical bleed ports are located halfway through the compressor or between the low-pressure and high-pressure compressor. Blow-off reduces the flow resistance downstream and increases the airflow in front of the blow-off. During the start-up of the engine, blow-off of a considerable portion of the airflow is commonly used to prevent the compressor to exceed the stall line in the characteristic.
2. Variable stator vanes (VSV's);
   The front stages are fitted with variable stator vanes to provide optimal angles of airflow incidence for low spool speeds.
3. Splitting up compressor;
   Each compressor will be given a separate spool speed (thus an individual tangential velocity u) to guarantee an axial velocity high enough for a correct angle of inflow incidence.

In contrast to compressors, turbines have no stall or surge phenomena caused by the pressure gradient in the turbine. The turbine characteristic therefore shows that the turbine efficiency is constant for a large portion of the mass flow (see Figure 8-9).

8.1.2.3 Choking

The turbine characteristic shows that the mass flow parameter group is constant from a certain pressure ratio. It seems that increasing the pressure ratio further from a certain point does not increase mass flow. It appears that somewhere in the turbine the speed of sound is reached during expansion. Usually, the speed of sound is reached at the exit of a stator passage. Note that choking can take place in the compressor as well, which can be seen in the compressor characteristic as the vertical (straight) parts of the higher speed lines (located at the top/right of the compressor characteristic of Figure 8-1), implying that the pressure ratio loses the ability to control the mass flow, parameter group.

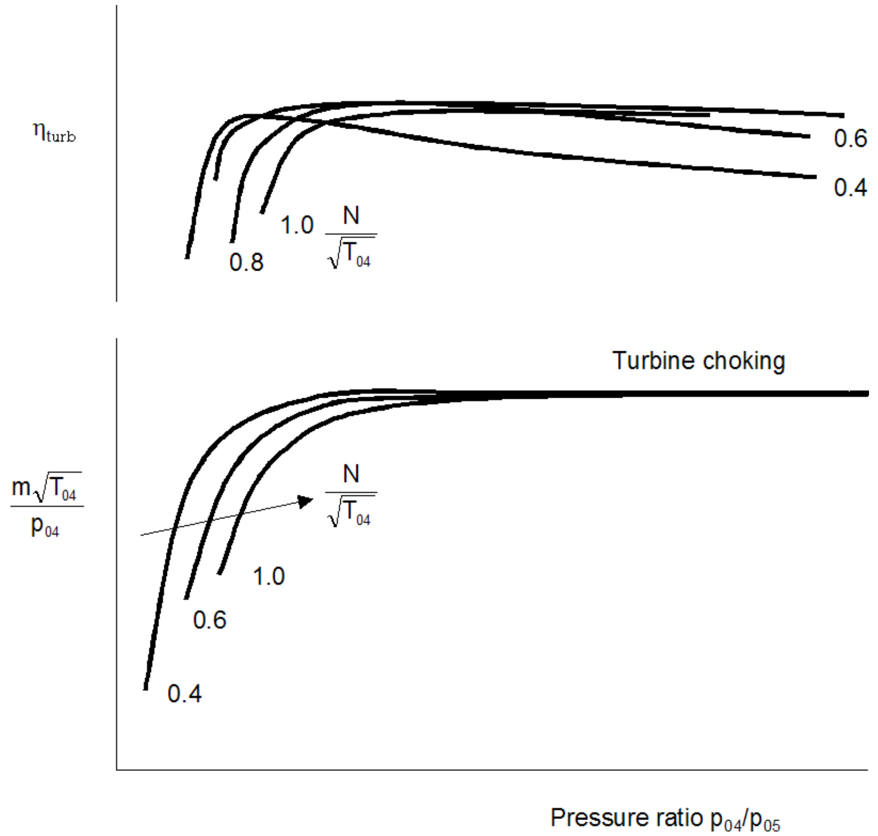


Figure 8-9 - Turbine characteristics

8.2 Gas Turbine System Characteristics

8.2.1 Gas Generator Characteristics

Assume we construct a gas turbine of the following components each having its characteristic: inlet, compressor, combustor and turbine (for the sake of simplicity the characteristics of the inlet and the combustor are reduced to a single efficiency and a single efficiency for constant pressure loss respectively). This assembly resembles a typical arrangement of a gas generator. An interesting question for this assembly would be if the components match up to form the actual gas generator, in other words: which points of the individual characteristics match up to form the operating (equilibrium) running line of the gas generator. Find for all points in all characteristics a matching value of $T_{0,4}/T_{0,2}$. The following equilibrium equations have to be satisfied:

1. Rotational speed balance;
   Compressor speed equals turbine speed (assuming that no gearbox is between these components, otherwise a gearbox ratio must be applied).
2. Mass flow balance;
   The mass flow through the turbine (including cooling air) equals the sum of mass flow

through the compressor, the fuel flow and the extraction of bleed flow. For some applications, the fuel flows and/or bleed flow may be disregarded if mentioned.

3. Power balance;
   The power the turbine delivers equals the sum of power required by the compressor and the mechanical power loss.

The actual matching procedure will not be discussed in this syllabus. The matching procedure is an iterative process for which often computer programs are used. The matching analysis results in gas generator characteristics as depicted in Figure 8-10. The basic outline shows the compressor characteristic in which lines of constant $T_{0,4}/T_{0,2}$ are drawn to represent the connection to the turbine.

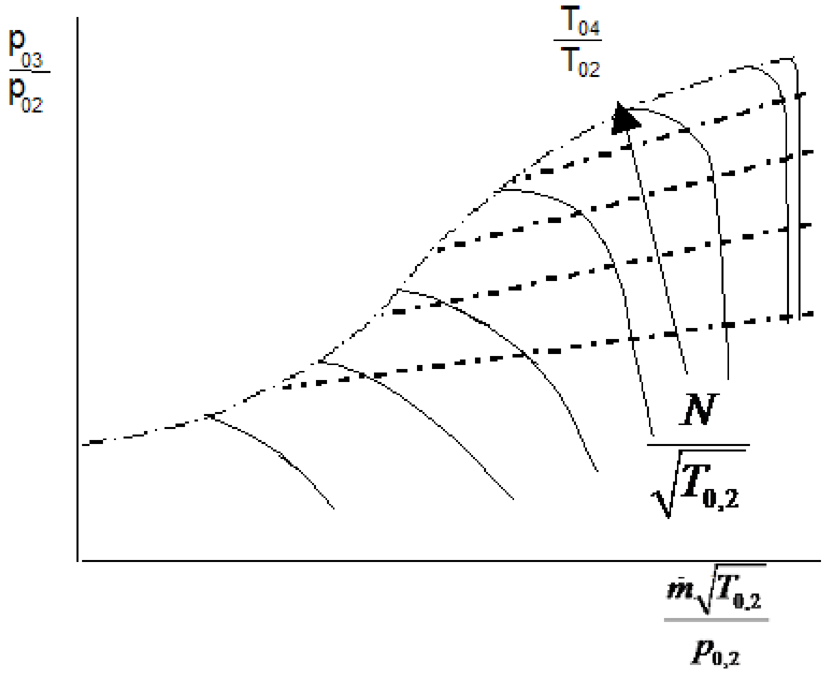


Figure 8-10 - Gas generator characteristic

8.2.2 System Characteristics of Different Applications

The gas generator characteristic of Figure 8-10 shows a collection of various possible operating points. The collection drastically decreases when the gas generator is completed to a gas turbine application by adding a (jet) nozzle or a power turbine. The addition of the last component implies that another equilibrium equation has to be taken into account:

4. Mass flow balance;
   The mass flow through the gas generator equals the mass flow through the jet nozzle or power turbine.

The value of $T_{0,4}/T_{0,2}$ that complies with all four equilibrium equations can be found using e.g. software tools as described in the previous section. The result of such an analysis is shown in Figure 8-11.
The characteristic of a turbojet engine and a turboshaft engine with a free power turbine seems to have the same shape. For the gas generator, the jet nozzle and the power turbine are treated as certain flow resistance. The addition of the fourth equilibrium equation reduces the collection of the gas generator operating points to one single line; the (equilibrium) operating line.

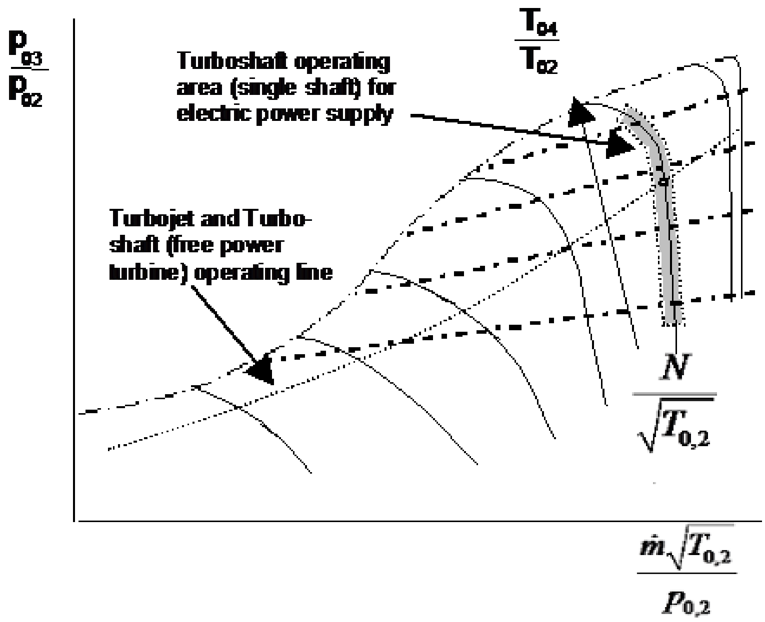


Figure 8-11 - Gas turbine operating lines

When the matching of the components is successful, i.e. operating points of gas generator compressor, -turbine and power turbine or jet nozzle coincide, the power characteristic of the complete system can be determined. Figure 8-12 shows an example of such a power characteristic. The figure shows the shaft power and specific fuel consumption as a function of power turbine and gas generator turbine. Figure 8-13 shows the course of the maximum torque course (i.e. the torque that corresponds to $N_{HP}/N_{HP\ des} = 1$ from Figure 8-12). Figure 8-13shows also torque curves for a diesel reciprocating engine and a single shaft turboshaft engine for comparison reasons. It appears that the torque curve for the turboshaft engine with multiple spools (shafts) is favourable with respect to the diesel engine and single spool turboshaft engine. A well-appriciated characteristic for engines that drive e.g. pumps, vehicles, fixed marine propellors, etc is the increase in torque for decreasing spool speed.

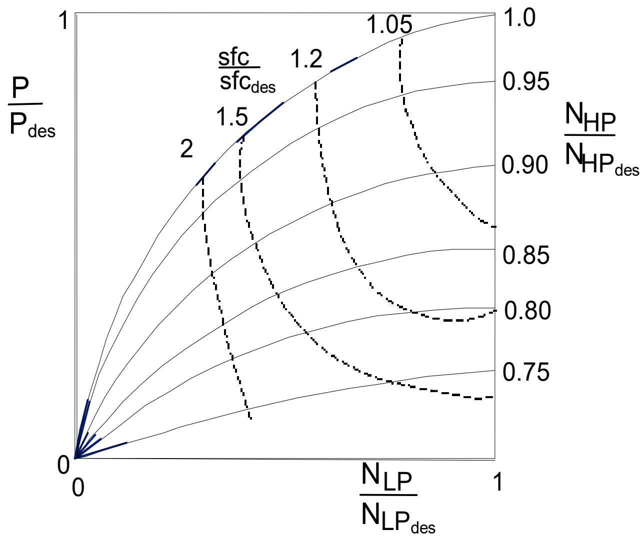


Figure 8-12 - Power characteristic of a turboshaft engine with free power turbine (2 spool engine)

The reason for the unfavourable torque curve of the single spool turboshaft is the direct relation between the spool speed at which the power is extracted and the pressure ratio of the process.

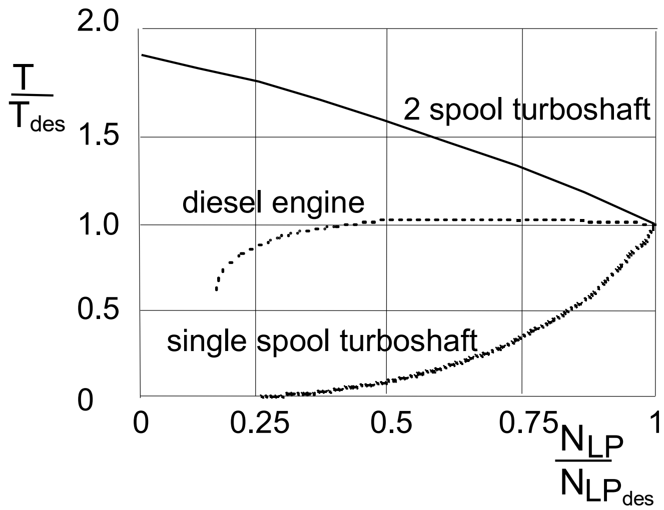


Figure 8-13 - Torque characteristics of turboshaft engines (single spool and free power turbine: two-spool engine) and reciprocating diesel engine

Due to the unfavourable torque curve, single spool turboshaft engines are not suitable to deliver power for different spool speeds (e.g. pumps, vehicles, fixed marine propellors, etc). Applications in which single spool turboshaft engines are used are e.g. the electricity power supply (steady grid frequency) or in combination with continuous variable transmissions.

Appendix A Station numbering

This section discusses the international standard for engine station numbering. This topic seems unnecessary and of secondary importance, but in practice reduces the misinterpretations and increases cost savings due to efficiency gains. Since the development of gas turbine engines is done by alliances of gas turbine companies, it is essential to unambigiously define the stations to improve the transfer of performance data, or performance software. ARP 755A [4] (ARP stands for Aerospace Recommended Practice) is the internationally recognised standard for gas turbine engine station numbering and nomenclature.

The station numbers are appended to symbols, such as temperature or pressure, identifying that gas condition to a certain position in the engine. The first sub-section discusses the fundamantal station numbering. More detailed station numbering information for specific gas turbine applications can be found in subsequent sub-sections.

A.1 Fundamental station numbers

The fundamental station numbers for the core stream of the gas turbine are listed below.

amb	Ambient conditions
0	Ram conditions in free streem
1	Engine intake front flange
2	First compressor/fan front face
3	Last compressor exit face
4	Combustor exit plane
5	Las turbine exit face
6	Front face of mixer, afterburner, etc.
7	Propelling nozzle inlet
8	Propelling nozzle throat
9	Propelling nozzle or exhaust diffuser exit plane

A.2 Intermediate station numbering

Any station between the fundamental stations is numbered using a second digit number suffixed to the upstream fundamental station number. This is not generally regulated by the ARP 755A standard, hence many companies will have their own practices. In case more then ten intermediate station numbers are required, a third digit will be suffixed to the prior two digits.

Since many gas turbines more or less have an overall fixed configuration, commonly used station indices will be discussed in the following sub-sections.

[4] SAE (1974) *Gas Turbine Performance Station Idetification and Nomenclature*, Aerospace Recommended Practice, ARP 755A, Society of Autimotive Engineers, Warrendale, Pennsylvania

A.3 Turbojets

Station numbers most commonly used for two spool turbojets are listed below. Note that additional station numbers would be created to deal with the mixing of cooling air flow back into the main stream.

24	First compressor exit
26	Second compressor front face
31	Compressor exit diffuser exit/combustor inlet
405	First turbine nozzle guide vane throat
41	Stator exit trailing edge
44	First turbine exit
45	Second turbine nozzle guide vane leading edge

A.4 Turbofans

The fundamental station numbers are prefixed with a 1 to identify the bypass stream. The core station numbering will be defined as in sub-sections A.1 – A.3. Turbojets having separate jets for cold and hot flow, common bypass duct station numbers are listed below:

12	Fan tip front face, if conditions are different from the fan root front (station 2)
13	Fan exit
17	Cold propelling nozzle inlet
18	Cold propelling nozzle throat

If the bypass flow is mixed to the hot core flow or afterburner flow, typical station numbering would be as listed below.

16	Cold mixer inlet
6	Hot mixer inlet
65	Mixer exit/afterburner inlet
7	Afterburner exit/propelling nozzle inlet

In case of tubofans having three spools, common stations for the second compressor entry are 24, and 26 for the third compressor entry.

A.5 Shaft power gas turbines

Simple shaft power gas turbines have station numbering as described in sub-sections A.1 – A.3. Normally, stations 6, 7 and 8 are redundant since there will only be an exhaust diffuser between station 5 and 9. For an industrial gas turbine, the entry at the intake flange would get station number 1, and the exhaust flange exit would get number 9. The numbers 0 and 10 are reserved for the plant intake flange and the plant exhaust flange respectively.

For intercooled and/or recuperated shaft power cycles, the typically employed station numbering is listed below.

21	First compressor exit face
23	Intercooler inlet face
25	Intercooler exit face
26	Second compressor inlet face
307	Recuperator air side inlet face
308	Recuperator air side exit face
31	Combustor inlet
6	Recuperator gas side inlet
601	Recuperator gas side exit

A.6 Spool rotational speeds, inertias, etc.

The inlet station number of the first component on a certain spool is used as suffixes for quantities related to that spool.

Appendix B Glossary

Afterburner: a device common in military engines where fuel is burned downstream of the turbine and upstream of the final propelling nozzle. Also known as reheat or an augmenter.

Aspect ratio: the ratio of span to chord.

Ambient: The condition of the atmosphere existing around the engine, such as pressure or temperature.

Blades: The compressor and the turbine are composed of many rows of small airfoil shaped blades. Some rows are connected to the inner shaft and rotate at high speed, while other rows remain stationary. The rows that spin are called rotors (Buckets) and the fixed rows are called stators (Nozzle guide vanes)

Bypass engines: an engine in which some of the air (the bypass stream) passes around the core of the engine. The bypass stream is compressed by the fan and then accelerated in the bypass stream nozzle. They are also called turbofan engines

Bypass ratio: the ratio of mass flow rate in the bypass stream to the mass flow rate through the core of the engine.

Chord: the length of a turbomachine blade in the direction of flow.

Combustor: also known as Combustion chamber, A chamber in which the fuel is combined with high-pressure air and subsequently the fuel-air mixture is combusted to provide a stream of hot gas that releases its energy to the turbine and nozzle sections of the engine. There are mainly three types of combustion chambers: can type chamber, Can-annular chamber, an annular chamber.

Compressor: the part of the engine, which compresses and consequently increases the pressure of the incoming air before it enters the combustor. There are mainly two types of compressors - centrifugal flow compressor and axial flow compressor.

Core: the compressor, combustion chamber and turbine at the centre of the engine. The core turbine drives only the core compressor. A given core can be put to many different applications, with only minor modifications. The core is sometimes called as the Gas generator.

Drag: The force that resists the motion of the aircraft through the air. In other words, the force in the opposite direction of the motion (travel)

Engine cycle: inlet, compression, combustion, and expansion of air with the result of work (thrust) being created.

Fan: the compressor operating on the bypass stream; normally the pressure ratio of the fan is small, not more than about 1.8 for a modern high bypass civil engine (in a single stage without inlet guide vanes) and not more than about 4.5 in a military engine in two or three stages

Gas Generator: Refer to "Core" above

Gross thrust: The thrust created by the exhaust stream without allowing for the drag created by the engine inlet flow; for a stationary engine the gross thrust is equal to the net thrust.

HP: the **high-pressure** compressor or turbine is part of the engine core. They are mounted on either end of the HP shaft. In a two-spool engine, they form the core spool.

Impeller: The impeller is a part of the compressor. It is designed to impart motion to the airflow within the compressor.

Incidence: sometimes called the angle of attack, is the angle at which the inlet of compressor or turbine blade is inclined to the inlet flow direction.

Inertia: The opposition of a body to have its state of rest or motion changed.

IP: the **intermediate pressure** compressor or turbine, mounted in the IP shaft. There is only an IP shaft in a three-shaft engine.

Jet pipe: the duct or pipe downstream if the LP turbine and upstream of the final propelling nozzle.

Lift: the force created by the wing (airfoil) perpendicular to the direction of flow.

LP: the **low-pressure** compressor and turbine are mounted on either end of the LP shaft. Combined they form an LP spool.

Mach number: representation of the speed of an aeroplane (It is indicated by the number of times faster than the speed of sound)

Mixer: The nozzle may be preceded by a mixer, which combines the high-temperature air coming from the engine core with the lower temperature air that was bypassed in the fan. This result in a quieter engine than if the mixer was not present.

Nacelle: the surfaces enclosing the engine, including the intake and the nozzle.

Net thrust: the thrust created by the engine available to propel the aircraft after allowing for the drag created by the inlet flow to the engine (Net thrust = Gross thrust minus the ram drag)

NGV: the nozzle guide vane, another name of the stator row in a turbine

Nozzle: a contracting duct used to accelerate the hot gas stream to produce a jet. In some cases for high-performance military engines, a convergent-divergent (CD) nozzle may be used.

Payload: the part of the aircraft weight, which is capable of earning revenue to the operator (can be freight/passengers)

Pylon: the strut that connects the engine to the wing

Ram drag: The amount of pressure buildup above ambient pressure at the engine's compressor inlet, due to the forward motion of the engine through the air - air's initial momentum.

Ram ratio: The ratio of ram pressure to ambient pressure.

Ram recovery: The ability of an engine's air inlet duct to take advantage of ram pressure.

Regime: An aircraft speed category i.e. subsonic, supersonic, hypersonic.

Steady-state operation: A condition where no appreciable fluctuation, intentional or unintentional, is occurring to any of the engine's variables, such as rpm, temperature, or pressure. Sometimes called stable operation

Stationary mode: A time where engine parameters do not change (for example cruising flight)

SFC: specific fuel consumption (actually the thrust SFC) equal to the mass flow rate of fuel divided by the net thrust. The unit should be in the form (Kg/s) / KN, but are often given as lb/h/lb or kg/h/kg.
Specific thrust: the net thrust per unit mass flow rate through the engine, unit /s
Spool: used to refer to the compressor and turbine mounted in a single shaft, so a two-spool engine is synonymous with a two-shaft.
Stagnation: Stagnation temperature is the temperature that fluid would have if brought to rest adiabatically. The stagnation is the pressure if the fluid would have if brought isentropically to rest. Stagnation quantities depend on the frame of reference.
Static: static temperature and pressure are the actual temp and pressure of the fluid in contrast to the stagnation quantities
Thrust reverser - A device used to partially reverse the flow of the engine's nozzle discharge gases and thus create a thrust force in the opposite direction.
Transient mode - Conditions that may occur briefly while accelerating or decelerating, or while passing through a specific range of engine operation. A time of rapid change.
Turbines: a component that extracts work from a flow. It consists of rotating and stationary blades. The rotating blades are called rotor blades and the stationary ones are called stator blades or nozzle guide vanes
Turbofan: a jet engine with a bypass stream
Turbojet: a jet engine with no bypass stream – these were the earliest types of jet engines and are still used for very high-speed propulsion

Appendix C Suggested Readings

Chapter 1- Introduction
The Theory and Design of Gas Turbine and Jet Engines**,** Vincent E.T.
Gas Turbine Theory**,** Cohen, H and Rogers, GFC and Saravanamuttoo
An Introduction to Aerospace Propulsion**,** D. Archer & M. Saarlas

Chapter 2 - Ideal Cycles
Gas Turbine Theory**,** Cohen, H and Rogers, GFC and Saravanamuttoo
Engineering Fundamentals of the Internal Combustion Engine*,* Pulkrabek, William W.

Chapter 3 - Real Cycles
Mechanics and Thermodynamics of Propulsion, P. Hill and C.P. Patterson
Gas Turbine Theory**,** Cohen, H and Rogers, GFC and Saravanamuttoo
Aero Thermodynamics of Gas Turbine and Rocket Propulsion, Oates, G. C.
The Internal Combustion Engine in Theory and Practice*,* ***Volume 1: Thermodynamics, Fluid Flow, Performance,*** Taylor, C. F.

Chapter 4 - Shaft power Gas Turbines
Gas Turbine Theory, Cohen, H and Rogers, GFC and Saravanamuttoo

Chapter 5 - Aircraft Gas Turbines
Elements of Gas Turbine Propulsion, Mattingly, J. D.
Aircraft Engines and Gas Turbines*,* Kerrebrock, J. L.
Aircraft Propulsion, PJ McMohan
Aircraft and Missile Propulsion**,** Zucrow, M.J.
Power plant for Aircraft**,** Liston Joseph
Gas Turbine Theory**,** Cohen, H and Rogers, GFC and Saravanamuttoo

Chapter 6 - Combustors:
An Introduction to Combustion, McGraw Hill, Inc, Turns, S.R., 1996
Design of Modern Gas Turbine Combustors, Academic Press, Mellor, A.M., ed, 1990
Combustion, Academic Press, Glassman, I., 1996
Principles of Combustion, Kuo, K.K. 1986, John Wiley & Sons, Inc.
Gas Turbine Combustion, Taylor & Francis, Lefebvre, A.H. 1999
Combustion, Springer. Warnatz, J. Maas, U. Dibble, R.W., 2001

Chapter 7 - Turbo machinery

The Theory and Design of Gas Turbine and Jet Engines, Vincent E.T.
Compressor Aerodynamics, Cumpsty, N.A.
Axial Flow Compressors, **Horlock,** J.H.
Axial Flow Turbines, Horlock, J.H.
***Centrifugal and Axial Flow Pumps; Theory, Design and Application**,* Stepanoff, A.J.
Gas Turbine Theory**,** Cohen, H and Rogers, GFC and Saravanamuttoo

Chapter 8 - Performance characteristics
The Theory and Design of Gas Turbine and Jet Engines, Vincent E.T.
Jet Engines, Fundamentals of Theory, Design and Operation: Klaus Hunecke
Gas Turbine Theory**,** Cohen, H and Rogers, GFC and Saravanamuttoo
Aircraft Propulsion**,** PJ McMohan
Elements of Gas Turbine Propulsion, Mattingly, J. D.

Chapter 9 - Loads and Materials
Gas Turbine Theory**,** Cohen, H and Rogers, GFC and Saravanamuttoo
Fracture Mechanics**,** M. Janssen, J. Zuidema, R.J.H. Wanhill
Heat Transfer in Gas Turbines, Suden, B and Faghri, M.
***The Internal Combustion Engine in Theory and Practice**,* Volume 2, Combustion, Fuels, Materials, Design. Taylor, C. F.
Metaalkunde (deel 1), Den Ouden, G and Korevaar, BM